

Dr Dinu Dragan



- SIIT -

INTERAKCIJA ČOVEK RAČUNAR

KO IZVODI NASTAVU



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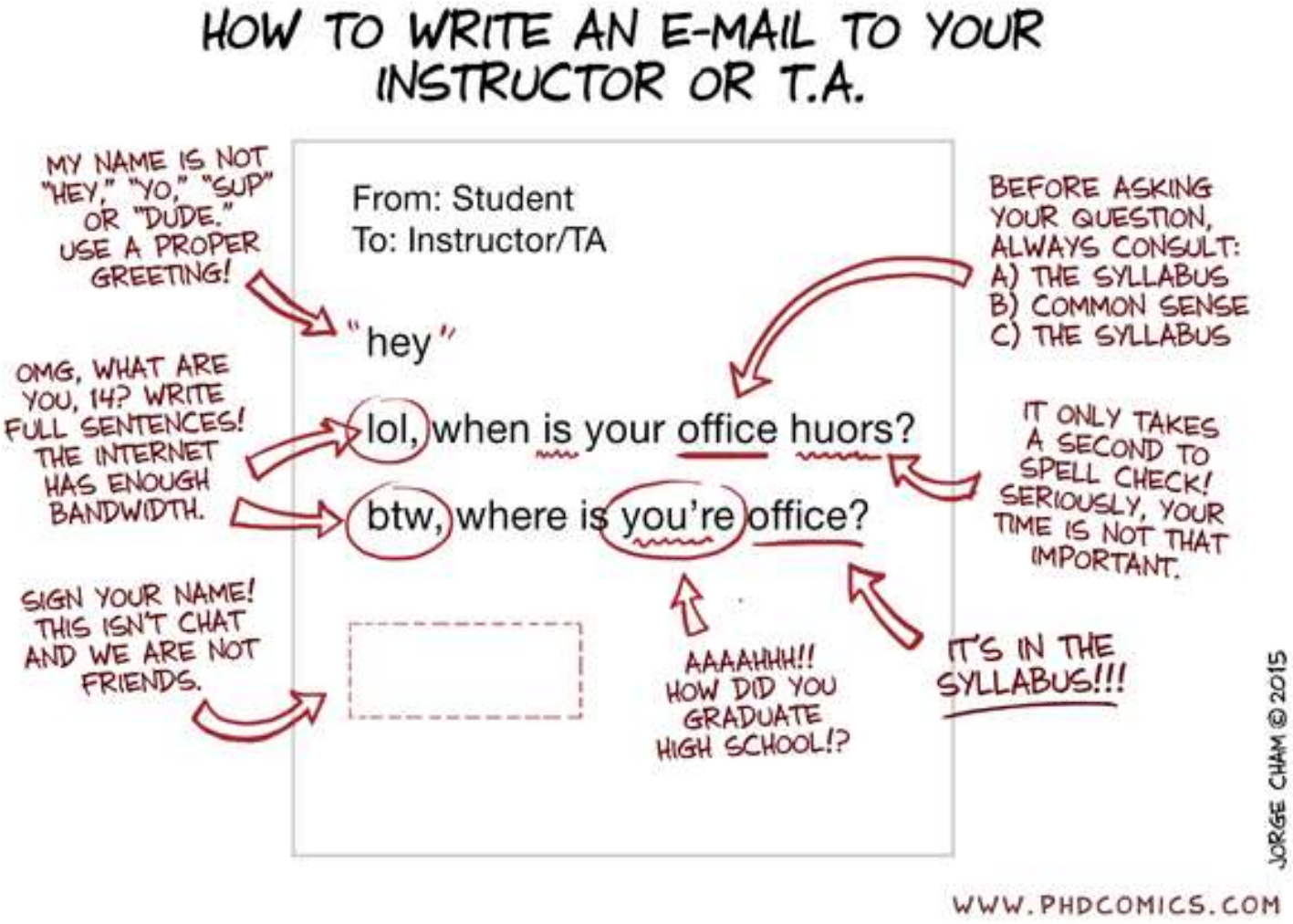
Canvas (http://acs.uns.ac.rs/)

Ako postoje pitanja, iskoristiti za to:

- Slobodno vreme na predavanjima i vežbama
- Individualne konsultacije (dogovor putem mejla)
- E-mejl

Asistent nije dibager !

O PISANJU E-MEJLOVA



O NASTAVNIKU - *Dr Dinu Dragan*



- 1998. - 2003. – Elektrotehnika, smer Računarstvo i upravljanje sistemima, usmerenje Računarska nauka i informatika, FTN, Novi Sad
- 2003. – Diplomski rad, “Primena OMR algoritama u DMS sistemima,” FTN
- 2008. – Magistarska teza, “Enkapsulacija JPEG2000 kompresione tehnike u DICOM standard,” FTN
- 2013. – Doktorska disertacija, “Metrika prihvatljivosti kompresione tehnike mirne slike u implementaciji PACS sistema,” FTN
- Od februara 2014. godine docent na Departmanu za računarstvo i automatiku Fakulteta tehničkih nauka
- Na predmetu Interakcija čovek računar angažovan od njegovog uvođenja 2003./2004. šk. godine
- **Važnije: praktični projekti sa velikim osvrtom na HCI**

NAČIN POLAGANJA



1. Predispitne obaveze **70 poena** (**75** za izuzetan trud)
   - 2 zadatka (WPF i C#) : **50 poena**
     - **Mini zadatak – 15 poena**
     - **Završni projekat – 35 poena**
   - Modelovanje interakcije – **10 poena**
   - Evaluacija rešenja – **10 poena**
   - aktivnost na predavanjima – **5 poena**

2. Usmeni ispit (možda i pismeni), ukupno **30 poena**

- Za potpis potrebno minimum 30 **poena** sa vežbi

Broj bodova	Ocena
51 – 60	Šest
61 – 70	Sedam
71 – 80	Osam
81 – 90	Devet
91 – 100	Deset

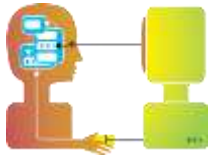
LITERATURA



1. SKRIPTA + BELEŠKE SA PREDAVANJA
2. A. Dix et al., HUMAN-COMPUTER INTERACTION. Harlow, England: Prentice Hall, 2004, ISBN-10: 0130461091
3. W.O. Galitz, THE ESSENTIAL GUIDE TO USER INTERFACE DESIGN: AN INTRODUCTION TO GUI DESIGN PRINCIPLES AND TECHNIQUES, 3rd Edition. Indianapolis, USA, Wiley Publishing, Inc., 2007, ISBN: 978-0-470-05342-3
4. D.L. Stone, USER INTERFACE DESIGN AND EVALUATION, San Francisco, USA, Morgan Kaufmann, 2005, 0-12-088436-4
5. I.S. MacKenzie, HUMAN-COMPUTER INTERACTION: AN EMPIRICAL RESEARCH PERSPECTIVE, Waltham, USA, Morgan Kaufmann, 2013, ISBN: 978-0-12-405865-1
6. D. Norman, THE DESIGN OF EVERYDAY THINGS, New York, USA, Basic Books, 2013, ISBN 978-0-465-05065-9
7. B. Shneiderman, C. Plaisant, DIZAJNIRANJE KORISNIČKOG INTERFEJSA, Beograd, CET, 2005, ISBN: 86-7991-282-4

UVOD

HCI KAO DISCIPLINA ...



- SIGCHCI (Special Interest Group on Computer-Human Interaction):

  Disciplina koja se bavi projektovanjem, evaluacijom i implementacijom interaktivnih računarskih sistema namenjenih čovekovoj upotrebi i fenomenima koji ih okružuju.

- Za cilj ima da poboljša interakciju između korisnika i računara tako što će računare učiniti prikladnijim i lakšim za korišćenje (približiti računar svakom korisniku)
  - procena potreba korisnika i zadataka koje interfejs treba da izvrši (loš interfejs, sa neodgovarajućom funkcionalnošću, frustrira korisnika)
  - obezbeđivanje pouzdanosti (akcije funkcionišu na predviđeni način)
  - element interfejsa mora da asocira korisnika na povezanu akciju
  - kontekst upotrebe korisničkog interfejsa

... HCI KAO DISCIPLINA ...

- Izrazita multidisciplinarnost



... HCI KAO DISCIPLINA ...

- HCI ne proučava samo interakciju čovek računar
  - Human–Machine Interaction (HMI)
  - Man-Machine Interaction (MMI)
  - Computer-Human Interaction (CHI).





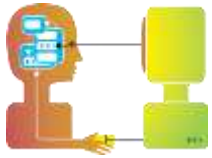








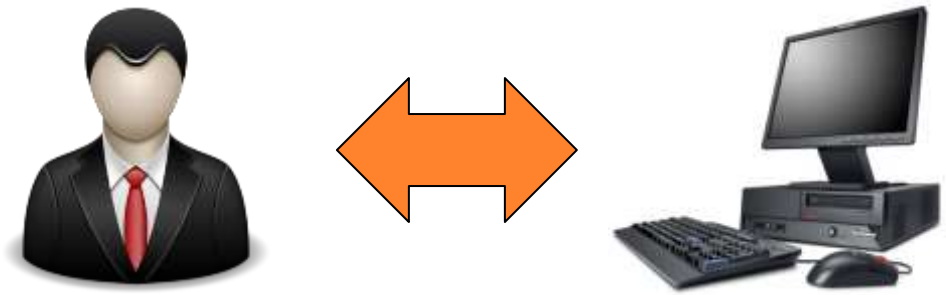
... HCI KAO DISCIPLINA



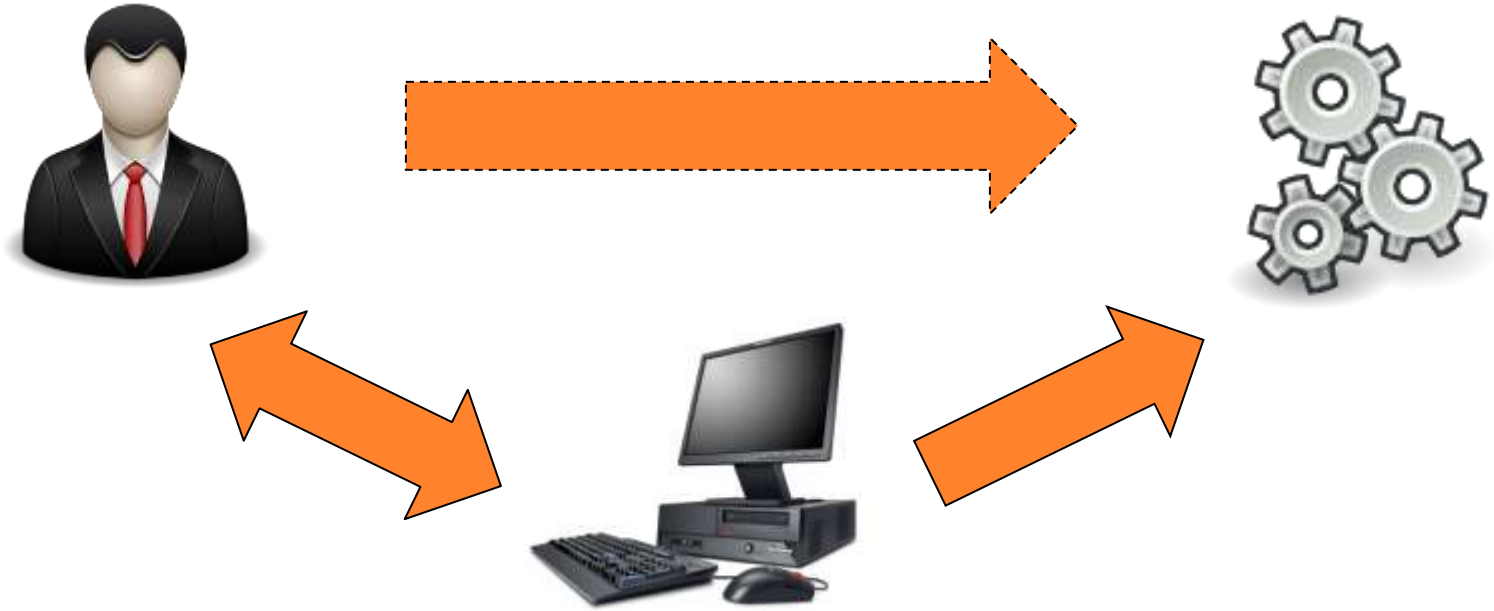
- Oblast proučavanja HCI-a obuhvata i sledeće:
  - karakteristike zajedničkog izvršavanja zadataka korisnika i računara
  - strukturu komunikacije između korisnika i računara
  - čovekovu mogućnost da koristi računar (uključujući učenje upotrebe interfejsa)
  - razvoj algoritama i programiranje samih interfejsa
  - proces specifikacije, projektovanja i implementacije interfejsa
- HCI nije samo korisnički interfejs
- Korisnički interfejs se ne odnosi samo na softver
- Korisnici rešavaju svoje zadatke nezavisno od sistema (softvera/hardvera)

UČESNICI INTERAKCIJE

- HCI proučava interakciju između čoveka i računarskog sistema (računar/softver)
- Dve strane interakcije:
  - ČOVEK
  - RAČUNAR



- Cilj je da čovek pomoću računara obavi željeni zadatak/radnju



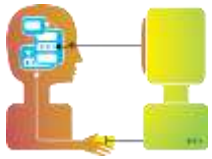
UČESNICI INTERAKCIJE – ČOVEK ...





- Ljudi imaju ograničene sposobnosti da procesiraju informacije (uvek voditi računa o tome tokom projektovanja interakcije)
- Informacije se prihvataju i na njih se reaguje preko različitih ulaznih i izlaznih kanala
  - Vizualni kanal (vizualni efekti i percepcija, boja, osvetljenost, veličina, dubina, pozicioniranje)
  - Audio kanal (za skretanje pažnje, ali i uputstva)
  - Haptički kanal (virtualna realnost, ekrani osetljivi na dodir)
  - Pokret (virtuelna realnost, mobilni telefon?)
- Informacija se smešta u ljudsku memoriju
  - Senzorska memorija (inicijalna informacija)
  - Kratkotrajna (radna) memorija (ATM primer – kartica)
  - Dugoročna memorija (zapamćena akcija, ATM primer – tastatura)

… UČESNICI INTERAKCIJE – ČOVEK …



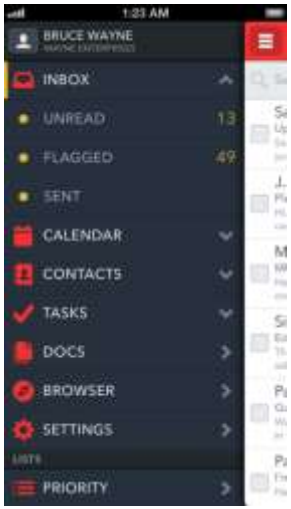
- Prepoznavati ili pamtiti

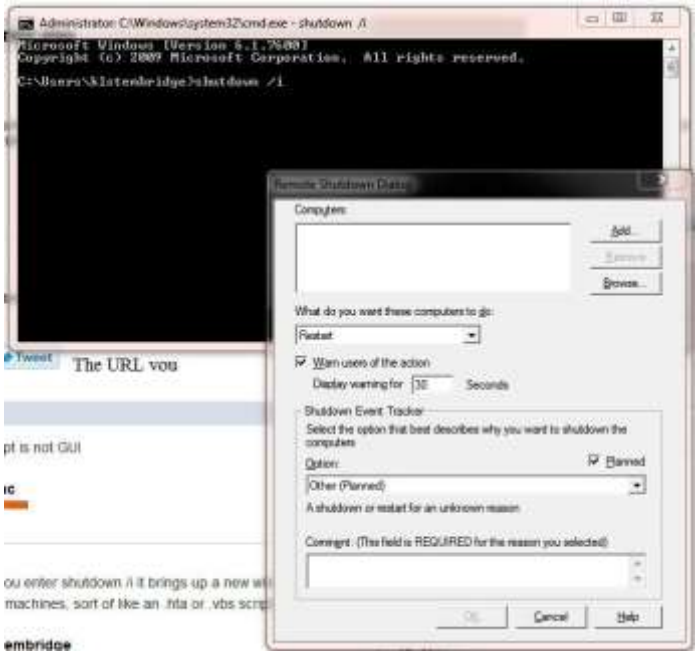
  PREPOZNAVANJE

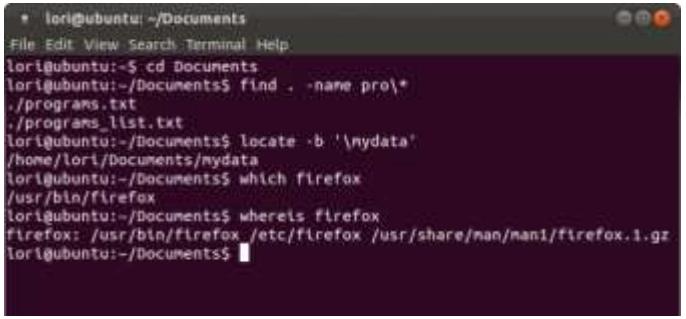
  – Podsetiti se kako se nešto radi na osnovu vizuelnih sugestija (pomoć)
  – „Knowledge in the world"

  PAMĆENJE

  – Pamtiti kako se nešto radi bez dodatnog podsećanja (bez pomoći)
  – "Knowledge in the head„

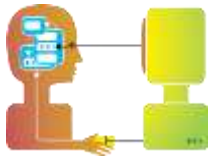






- Prepoznavanje je lakše

... UČESNICI INTERAKCIJE – ČOVEK



- Informacije se obrađuje i primenjuje
  - Rezonovanje (upotreba znanja kako bi se došlo do zaključaka i stvorila nova saznanja – ako se nešto radilo na jedan način, a ovo novo izgleda slično, da li se radi na isti način?)
  - Rešavanje problema (na osnovu znanja rešavaju se problemi, nepoznati zadaci)
  - Usvajanje veština (rešavanjem domenskih problema, stiče se veština, čime rešavanje novih problema iz istog domena postaje jednostavnije)
  - Greške (zato što je nešto novo, ali i zato što je nešto pogrešno naučeno – mentalni modeli)
- Emocije utiču na ljudske sposobnosti
- Ljudi dele slične sposobnosti i karakteristike, ali svaka osoba je jedinstvena i različita od ostalih, što se nikako ne sme ignorisati





UČESNICI INTERAKCIJE – RAČUNAR ...

- Računari su kompleksni sastavljeni iz više delova, pri čemu svaki utiče na rad sistema (**ubiquitous**)
  - Kombinacija hardvera i softvera



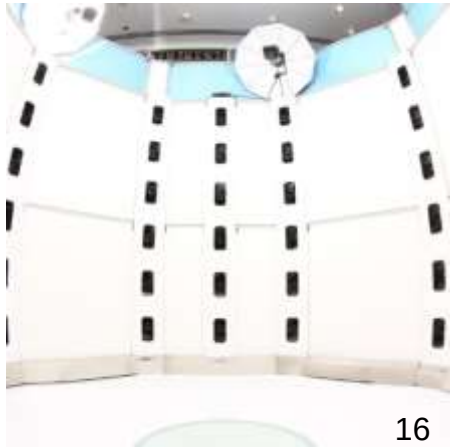


- Ulazni uređaji za interaktivnu upotrebu, za unos teksta, crtanje, i selekciju elemenata na ekranu
  - Unos teksta – tradicionalne tastature, ekrani osetljivi na dodir, govor, rukopis, skeniranje
  - Pointing – tradicionalno miš, ali i stilus, touchpad, trackball, i ostalo
  - 3D interakcioni uređaji









... UČESNICI INTERAKCIJE – RAČUNAR ...

- Izlazni uređaji za interaktivnu upotrebu
  - Monitori različitih dimenzija i rezolucija
  - Štampači
  - Zvučne kartice
  - Projektori
  - različiti mehanički uređaji (servo motori, hidraulika)
  - sl.

















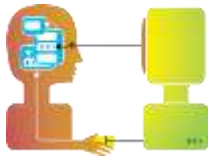






- Specijalna VR okruženja i uređaaji

... UČESNICI INTERAKCIJE – RAČUNAR



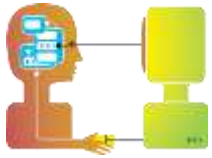


- Memorija (njen kapacitet i brzina – ograničenja)
- Procesna moć
  - Efekti prebrzog ili presporog rada (mit o mašini neograničene brzine)
  - Računarska mreža i njen uticaj na rad čitavog sistema
- Sukob između mogućnost računara i očekivanja korisnika
  - Procesiranje
  - Čitanje iz memorije
  - Render grafike
  - Prenos preko rač. Mreže



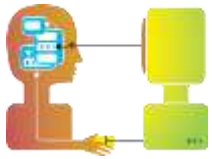
- Subjektivni osećaj šta je brzo i koliko su korisnici spremeni da čekaju
  - Primer preuzimanja slika sa 66 kamera

KORISNIČKI INTERFEJS ...



- Interakcija između korisnika i računara odvija se putem korisničkog interfejsa
- Korisnički interfejs je podržan hardverom i softverom
  - Ulazni deo interfejsa koji omogućava korisniku da unosi podatke, definiše poslove, želje, ...
  - Izlazni deo interfejsa preko kojeg računar informiše korisnika o rezultatima, potrebnim podacima, željama, ...
- Omogućava korisniku da obavi svoj zadatak (korisnik je preko njega u interakciji sa svojim zadatkom)
- Iako je korisnički interfejs deo računarskog sistem, **za korisnika je on često sam sistem**
- **Korisnikov doživljaj rač. sistema je često ograničen na korisnički interfejs i baziran isključivo na njemu**

... KORISNIČKI INTERFEJS ...



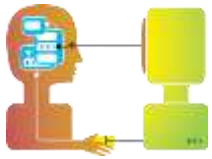
- Dobar korisnički interfejs
  - podrazumeva miks dobro dizajniranog ulaza i izlaza
  - uključuje adekvatne tehnike interakcije
  - efikasno ispunjava korisnikove potrebe, kapacitete i limite na
  - je neprimetan
  - se bolje prodaje
  - omogućava korisniku da se orijentiše na informacije i zadatak, a ne na mehanizme koji se koriste za prezentaciju informacij i realizaciju zadatka





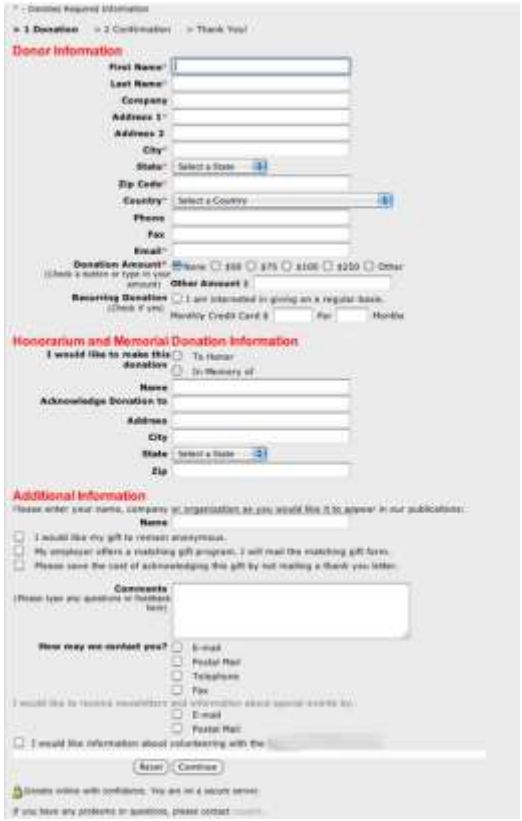


... KORISNIČKI INTERFEJS…



- Lako je naći primere lošeg interfejsa, daleko teže naći primer dobrog
- Razlozi za loš interfejs:
  1. Interfejs je nevažan
  2. Nedostatak zdravog razuma
  3. Nedostatak vremena
  4. Ne postoji jasna definicija (pravilo) šta je to dobar interfejs

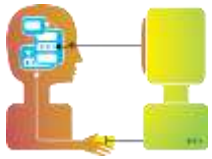
  U većini slučajeva 3. i 4.

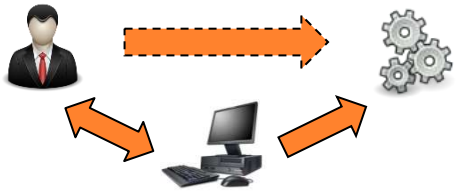




- Dobar, loš, očajan korisnički interfejs u odnosu na njegovu upotrebljivost (usability)

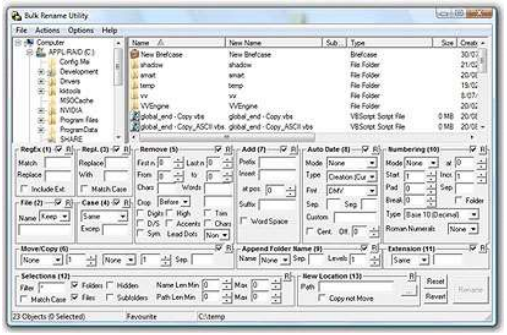
... KORISNIČKI INTERFEJS





- Zašto je teško doći do dobrog usability-a?
  - Setiti se svih aktera
  - Korisnici (različiti, nepredvidljivi, ...)
  - Hardver/softver (ograničen, ne podržava željene interakcione tehnike, ...)
  - Programeri (obično nisu dobri u dizajniranju kor. interfejsa)
    - ne razmišljaju kao normalni ljudi
    - znaju kako softver radi, tehnologija na prvom mestu
    - vole rad sa softverom/hardverom više od drugih ljudi





  - Dobar dizajn nije dovoljan za dobar kor. interfejs (dizajneri često "samo" umetnici, ne poznaju HCI, ...)
  - Ponos programera/dizajnera
  - Kor. interfejs za korisnike, a ne za programere



LOŠ KORISNIČKI INTERFEJS ...

- Korisnikova frustracija i nezadovoljstvo
  - Rage against the Machine – nezadovoljstvo tehnologijom na poslu raste
  - Izlivi verbalnog nasilja nad računarom
  - Izlivi fizičkog nasilja nad računarom
  - *The cost to business of this increase in stress levels of employees is not only based on sick days or under-performance, but also the working time lost through waiting for IT problems to be solved. Confederation of British Industry (CBI) statistics currently evaluate this at a staggering £25,000 ($40,000) per person in lost business each year (based on one hour a day being spent sorting out IT problems*
  - "Računari ne znaju šta korisnici žele"





... LOŠ KORISNIČKI INTERFEJS ...

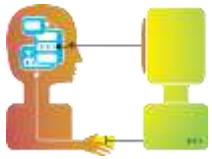
- Gubitak produktivnosti, efikasnosti i para

  Primer: *UK Passport Agency: The passport delays of Summer 1999*

  – Sistem nije bio dovoljno dobro testiran, nije bio u skladu sa zadacima i očekivanjima, korisnici nisu bili dovoljno obučeni, pojavio se neočekivano veliki broj zahteva za pasošima
  – Procedura je produžena sa očekivane 2 na 7 – 10 nedelja
  – Čitav fiasko je koštao $20M:
    - $9M na prekovremeni rad
    - $242 000 na to što su ljudi propustili godišnji odmor
    - $24 000 na kišobrane
  – Zbog svega je skočila cena kišobrana
  – Developeri su platili $3,7M, ostalo poreski obveznici



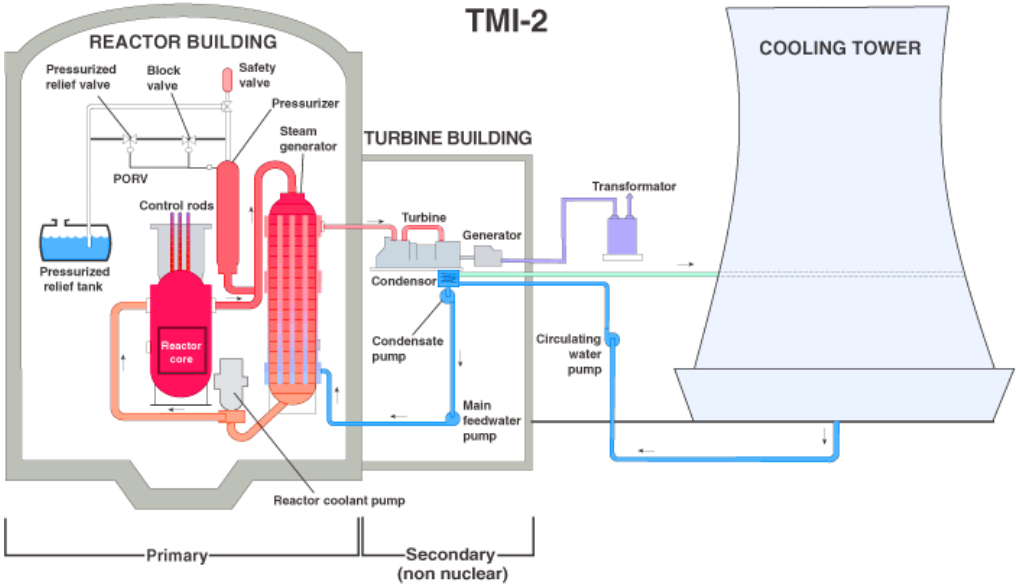
... LOŠ KORISNIČKI INTERFEJS ...



- Uticaj na bezbednost, kako samih korisnika, tako i drugih ljudi u okolini
  - Veliki broj primera u kojima je loš interfejse doveo do gubitka života i ugrožavanja životne sredine

Primer: *The Three Mile Island Nuclear Power Plant Disaster*

- Došlo do topljenja reaktora kombinacijom ljudske greške, ali i propusta u interfejsu
- Niko nije bio povređen
- Pumpe za snabdevanje vodom za hlađenje su prestale da rade
- Sistem signalizirao da je poklopac zatvoren, a u stvari nije bio

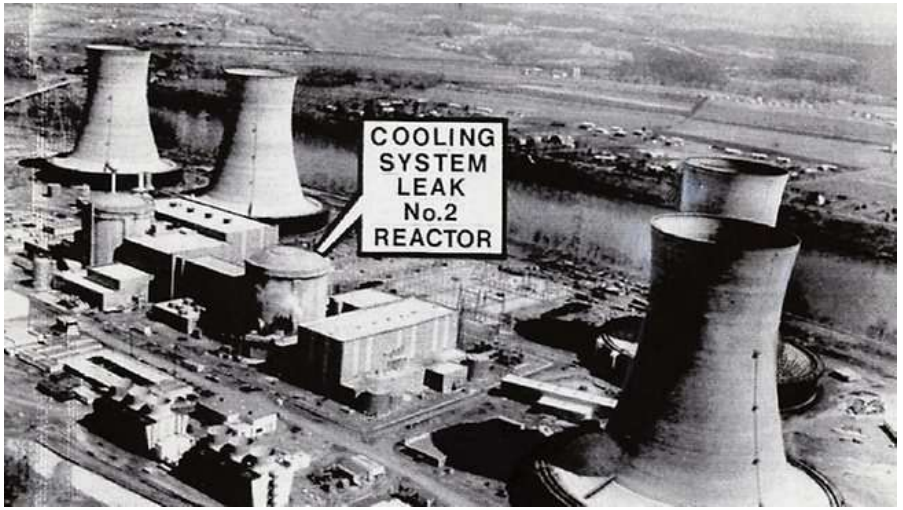


- Zbog toga je voda iscurela, reaktor se pregrejao i došlo je do topljenja

... LOŠ KORISNIČKI INTERFEJS ...

Primer: *The Three Mile Island Nuclear Power Plant Disaster (nastavak)*

- Uočeni propusti u korisničkom interfejsu:
- *A light indicated that a valve had been closed when in fact it had not.*
- *The light indicator was obscured by a caution tag attached to another valve controller.*
- *The control room alarm system provided audible and visual indication for more than 1500 alarm conditions. The layout and grouping of controls on the control panel had not been well thought out and so enhanced, rather than minimized, operator error*
- *A single "acknowledge" button silenced all the alarms at the same time, but it was not used because the operators knew they would lose information if they silenced some of the alarms. There was simply no way for the operators to cancel the less important signals ...*

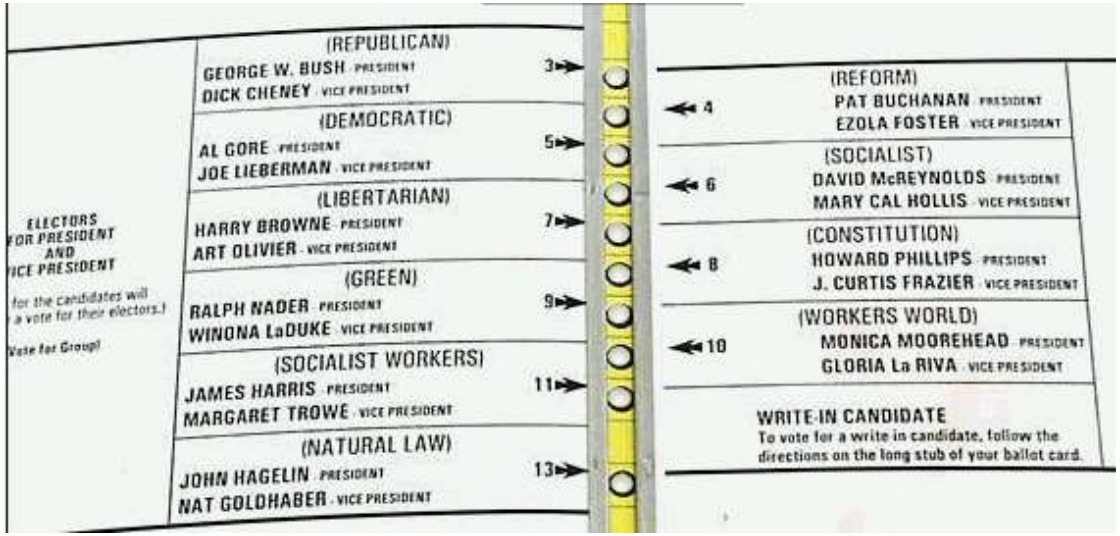


... LOŠ KORISNIČKI INTERFEJS ...

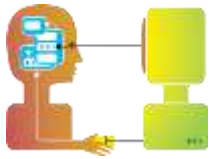
Primer: *Izbori u Americi 2000. godine*

- Kontraverze zbog lošeg interfejsa u kabini za glasanje
- Iako su želeli da zaokruže (označe) drugog po redu kandidata (demokrate), drugo polje nije odgovaralo drugo po redu kandidatu, već prvom kandidatu na sledećoj stranici





... LOŠ KORISNIČKI INTERFEJS

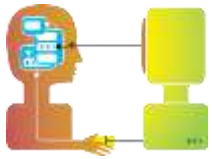


Primer: *National Cancer Institute, Panama City*

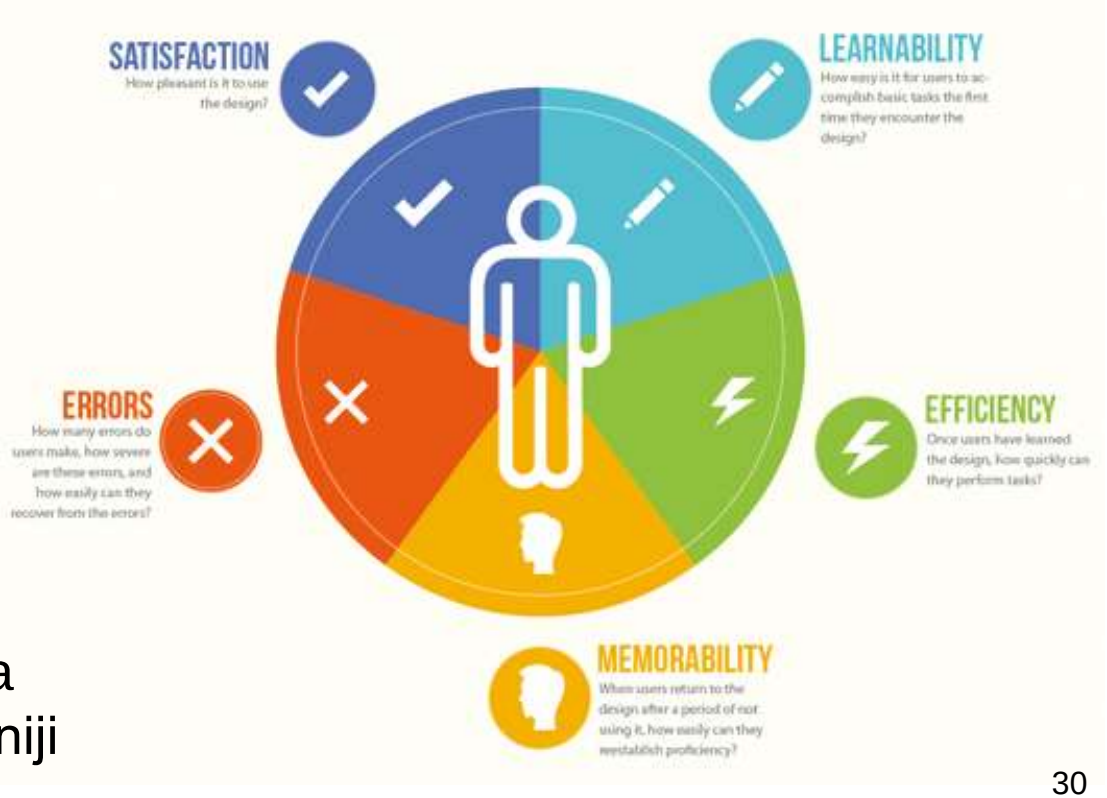
- Softver za planiranje terapije zračenjem Multidata Systems Intern. Corp. doveo do toga da su pacijenti ozračeni većom dozom radijacije od dozvoljene (od 20 do 100% veća doza)
- Rezultat: 28 potvrđenih pacijenata je ozračeno, od čega je devet umrlo (za pet se veruje od posledica zračenja), dva lekara osuđena na zatvor, Multidata Systems zabranjeni
- *The software was designed to allow a therapist to draw the placement of metal shielding blocks onscreen. Here's where the problem comes in. The doctors, for some reason, wanted five people to be present, rather than four. Unfortunately, the default shielding blocks for the program were set to four. They discovered they could jimmy it, however, by drawing all five blocks as a single block with a hollow middle. Of course, they neglected to consider a tiny little, life-ending glitch: drawing the hole one way gives the correct dosage, while drawing it another way prescribes double the correct dosage. They probably could have avoided a crisis if they'd run the calculations by hand, but why should they bother?*

POJMOVI

USABILITY



- Upotrebljivost: (mera) koliko dobro korisnici mogu da koriste funkcije sistema
- Usability se sastoji od nekoliko dimenzija (nekoliko aspekta upotrebljivosti korisničkog interfejsa:
  - Learnability, intuitiveness, memorability, (koliko se sistem lako uči i pamti, koliko je intuitivan)
  - Efikasnost
  - Podložnost greškama
  - Subjektivno zadovoljstvo
- Postoje i drugi aspekti (npr. koliko interfejs zamara korisnika), ali ovi su najvažniji



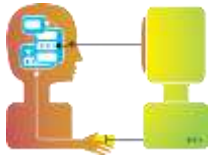
USABILITY DIMENZIJE

- Različite usability dimenzije nemaju isti značaj za različite korisnike ili u različitim tipovima aplikacija
  - Početnici treba softver koji se lako i brzo uči
  - Povremenim korisnicima treba intuitivnost, lako pamćenje i podsećanje pojedinih opcija
  - Ekspertima treba efikasnost



- Ali korisnik nije uniformno početnik ili ekspert
  - Domensko znanje (doktor je ekspert u svom zanimanju, ali početnik u korišćenju softvera)
  - Aplikativno znanje (korisnik je ekspert u jednom delu aplikacije, ali početnik u korišćenju nekih drugih funkcionalnosti tog istog softvera)
  - Učenje korisničkog interfejsa (vremenom postaje ekspert)
- Važno je znati korisnike kako bi se optimizovao korisnički interfejs za njih

USABILITY WIMP PROBLEMI (IBM 1994.)



- Dvosmisleni meniji i ikone (njihovo značenje nije jasno)
- Jezik koji omogućava navigaciju samo u jednom smeru
- Ograničenja u ulazu i direktnoj manipulaciji
- Ograničenja u selekciji i označavanju
- Nejjasne posledice nekih akcija

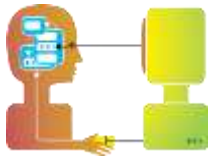


- Više koraka u interfejsu, nego što postoji akcija potrebnih da se uradi zadatak
- Kompleksne veze unutar i između aplikacije
- Neadekvatne poruke za feedback i konfirmaciju
- Nedostatak inteligencije i predviđanja (sledeće akcije korisnika)



- Neadekvatne poruke o grešci, help, dokumentacija i tutorijali

USABILITY WEB PROBLEMI (2004.)





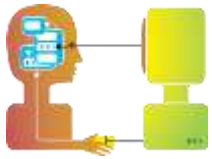
- Visual clutter (nedostatak praznog prostora, besmislene grafike, ...)
- Impaired information readability (loš izbor fonta, boja, grafike, ...)
- Incomprehensible components (čemu elementi inter. uopšte služe)
- Annoying distractions (elementi se pomeraju, blinking tekst)
- Confusing navigation (lavirint stranica u kojima se korisnik gubi)
- Inefficient navigation (preko stranice bez korisnog sadr. do cilja)
- Inefficient operations (vreme se gubi na previše operacija, download)
- Excessive or inefficient page scrolling (previše skrolovanja)
- Information overload (loša organizacija ili previše sadržaja)
- Design inconsistency (uvećana nekonzistentost sadržaja)
- Outdated or undated information (neažuriran sadržaj)
- Stale design caused by emulation of printed documents and past systems (šta reći?)

PRAKTIČNA OCENA USABILITY-A

- Može se saznati jednostavnim posmatranjem ili pričom sa korisnicima
- Nije naučni pristup, ali je dobar indikator
  - Da li korisnici često postavljaju pitanja ili posežu za dokumentacijom?
  - Da li se često čuju komentari tipa „dođavola“?
  - Da li se prečesto obavljaju nevažne akcije?
  - Da li postoje puno stvari koje treba ignorisati?
  - Da li ljudi žele da rade sa softverom?



OBJEKTIVNA OCENA USABILITY-A



- Naučni pristup
- Mogući načini da se izmeri usability

1. Time to complete a task.
2. Percentage of task completed.
3. Percentage of task completed per unit time (speed metric).
4. Ratio of successes to failures.
5. Time spent on errors.
6. Percentage of number of errors.
7. Percentage of number of competitors that do this better than the current product.
8. Number of commands used.
9. Frequency of help or documentation use.
10. Time spent using help or documentation.
11. Percentage of favorable to unfavorable user commands.
12. Number of repetitions of failed commands.
13. Number of runs of success and of failures.
14. Number of times the interface misleads the user.
15. Number of good and bad features recalled by users.
16. Number of available commands not invoked.
17. Number of regressive behaviors.
18. Number of users preferring your system.
19. Number of times users need to work around a problem.
20. Number of times the user is distracted from a work task.
21. Number of times the user loses control of a system.
22. Number of times the user expresses frustration or satisfaction.

From Tyldesley (1988).

USABILITY SPRAM CELOG SISTEMA

- Usability je samo jedna od karakteristika celog sistema
  - Funkcionlanost
  - Performanse
  - Cena
  - Sigurnost
  - **Usability**
  - Veličina
  - Pouzdanost
  - Standardi



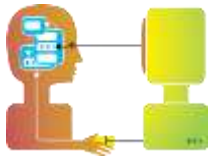
- Ne može se posmatrati nezavisno od ostatka
- Značaj usability-a zavisi od vrste i namene sistema





- Potrebno je ostvariti balans između svih aspekata sistema i često su potrebni kompromisi

USER EXPERIENCE (UX) ...



- Odnos između korisnika se ne posmatra samo kao odnos između korisnika i interfejsa, već se proširuje na proizvode, sisteme i procese čiji su interfejsi deo
- Kako korisnik doživljava korisnički interfejs, softver, ali i u svetlu svega ostalog što okružuje UI
- Dobar primer je iPhone jer UX korisnika ne dolazi samo od interfejsa već i od svega ostalog što dolazi uz iPhone







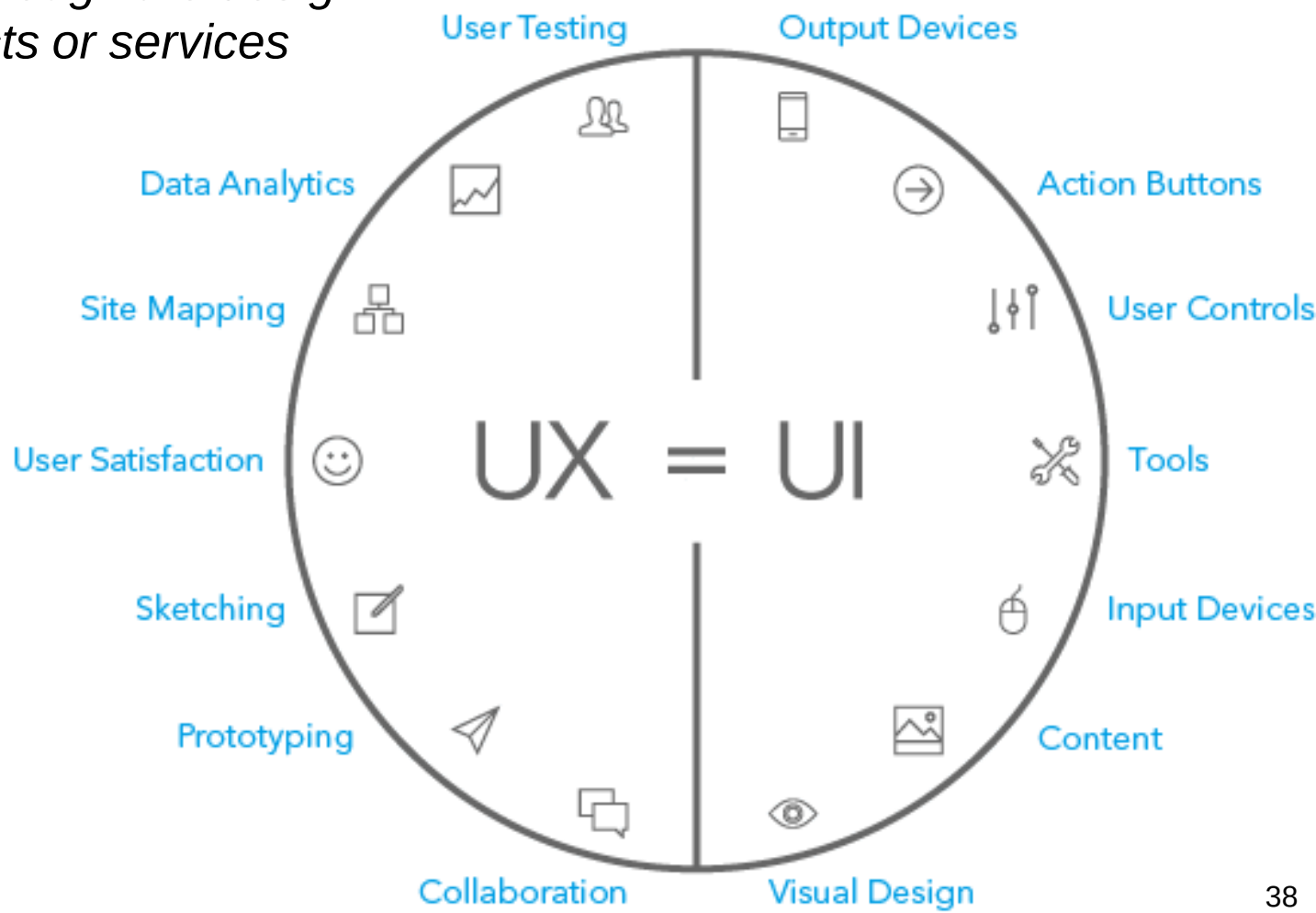




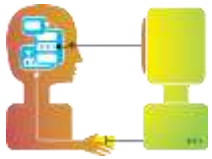
... USER EXPERIENCE (UX)

- UX = *The motivations, behaviours, and satisfaction we want to encourage through the design of our products or services*

- UI == UX ?



... USER EXPERIENCE (UX) …



- UI – brine o layout-u proizvoda i izgledom generalno
- UX – o osećaju koji korisnik ima dok koristi proizvod
- Odnosi se na
  - Lakoću korišćenja proizvoda
  - Brzinu korišćenja proizvoda
  - Privlačnost rada sa proizvodom
- Sve to utiče na korisnikovo zadovoljstvo ili nezadovoljstvo u radu sa proizvodom

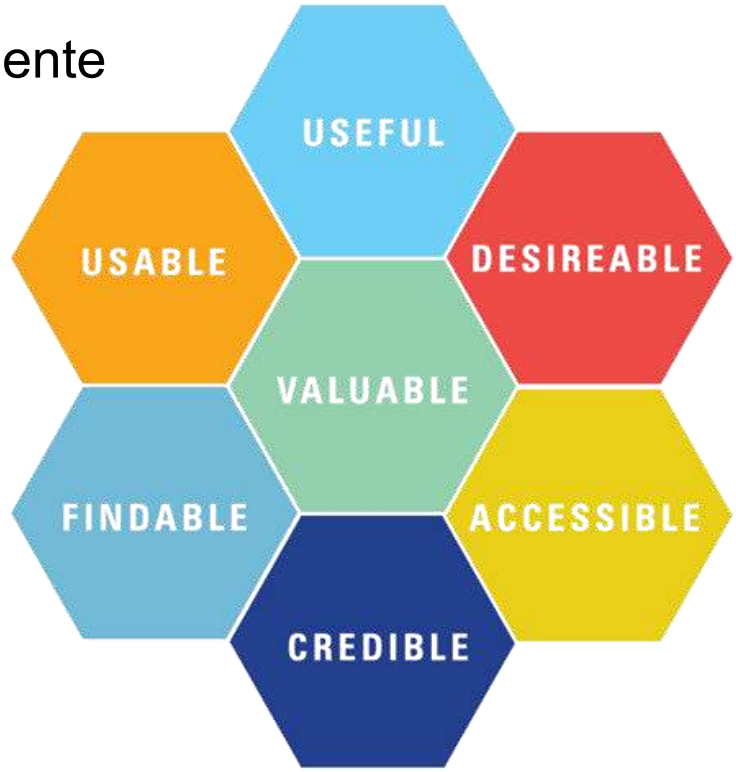


Primer sajta koji izgleda fenomenalno, ali na kome ni za ljubav Božiju, ne može da se pronađe željena stvar



... USER EXPERIENCE (UX) …

- Peter Morville User Experience Honeycomb
  - Ideja da se preko njega edukuju klijenti i da im se objasni šta je UX
- Obraćanjem pažnje na pojedinačne elemente dizajneri bi trebali da uoče nedostatke proizvoda iz UX ugla i da poboljšaju pojedinačne elemente
- Kako klijent intereaguje sa proizvodom
  - Proizvod ili zadovoljava korisnikova očekivanja ili ne
- Što je korisnik zadovoljniji, veća je verovatnoća da će se vratiti proizvodu i da će koristiti proizvod, što generalno znači veću zaradu



... USER EXPERIENCE (UX) …

- **Kako do dobrog UX?**
- Istražiti tržište – kako bi se odredili potencijalni korisnici i njihove potrebe, ciljevi, kao i načini ponašanja
  - na osnovu čega se grade persone, vrste korisnika koje će koristiti proizvod

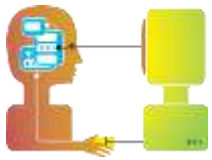


- Istražiti konkurenciju – kako bi se odredili kvaliteti/faktori najvažniji potencijalnim korisnicima

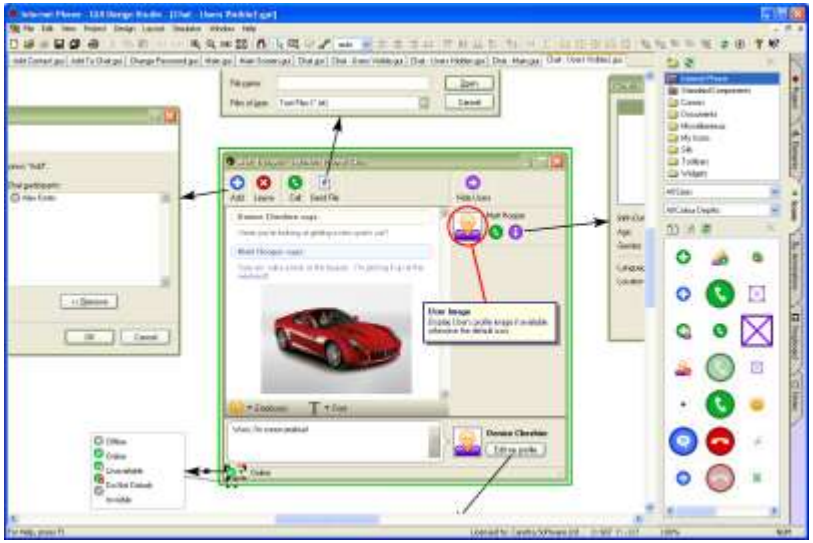


- Istražiti trendove – kako bi se odredila očekivanja potencijalnih korisnika
- Sve to omogućiće donošenje pravilnih odlika (i u dizajnu proizvoda i u usmeravanju razvoja proizvoda)

... USER EXPERIENCE (UX)



- **Kako do dobrog UX?**
- Napraviti prototip niske verodostojnosti (wireframe)
- Napraviti prototip visoke verodostojnosti





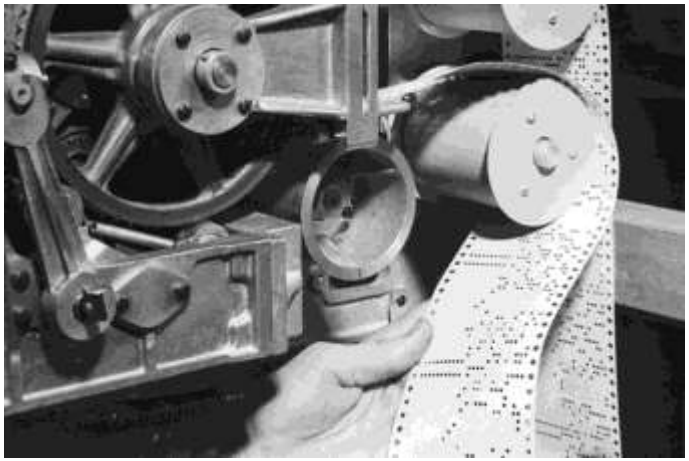
- Testirati proizvod sa korisnicima (usability testing)
- UX se non-stop razvija, i kada se proizvod pusti u produkciju, mora se raditi na njegovom unapređenju



ISTORIJA

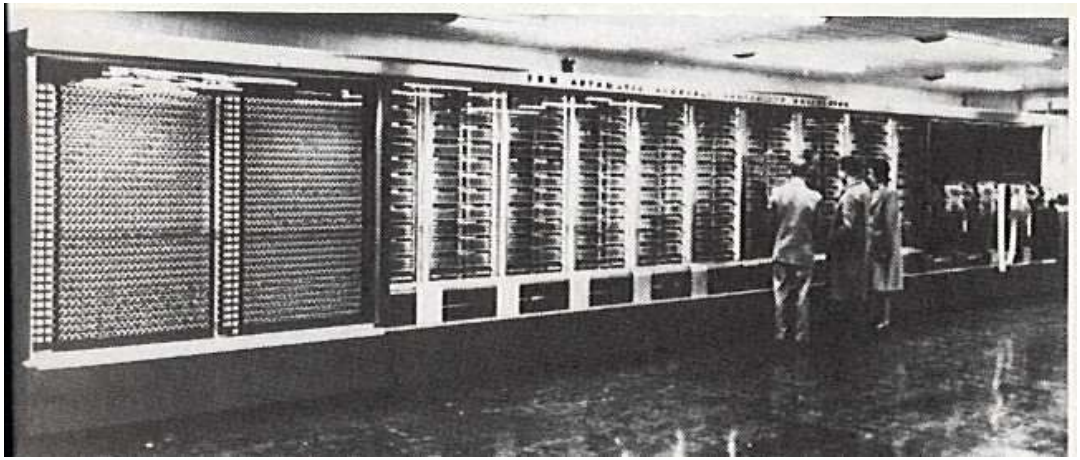
PRVA GENARACIJA RAČUNARA

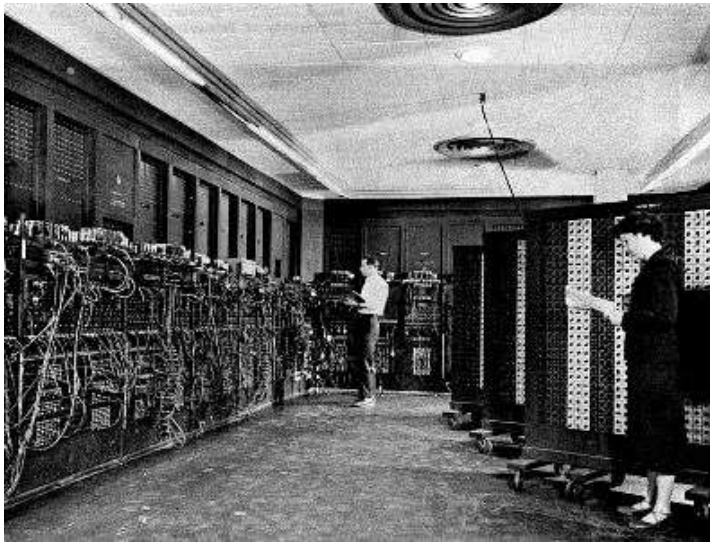
Mark I čitač papirnih traka



Programiranje Eniac-a







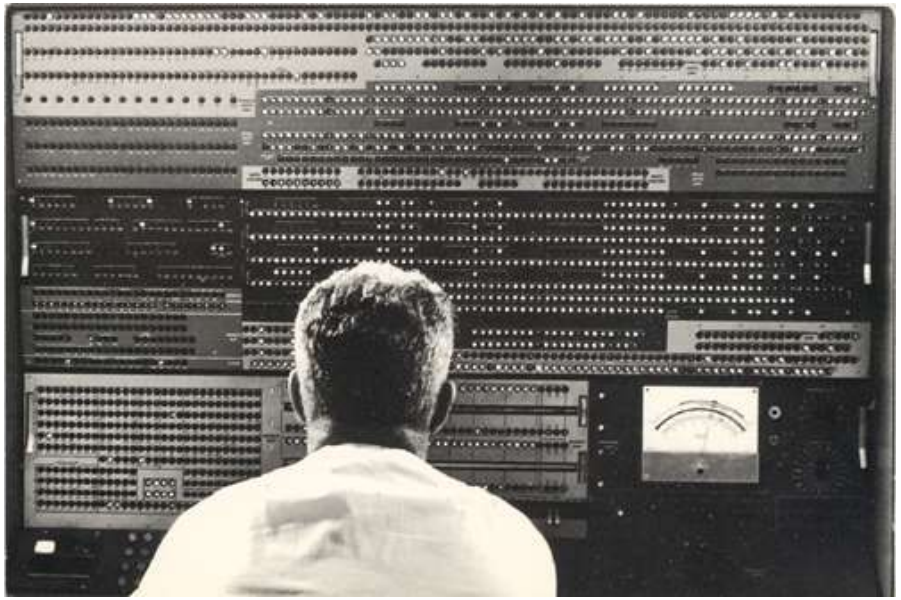
- Prekidači, bušene kartice

MEJN FREJM RAČUNARI

IBM SSEC (1948)



Stretch (1961)



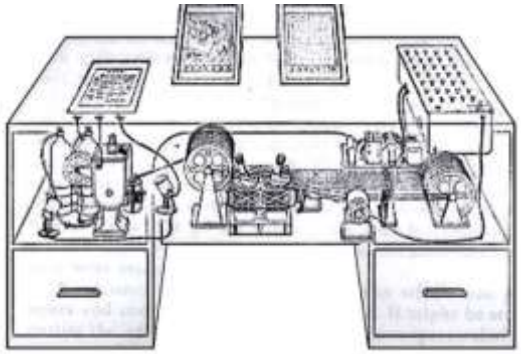
- Batch kompjuting



AS WE MAY THINK

- Vannevar Bush's "as we may think" (1945)
- „...publication has been extended far beyond our present ability to make real use of the record."
- Leading some 6,000 American scientists in the application of science to warfare







- Navigating the knowledge maze with a device he called memex
- "When the user is building a trail, he names it, inserts the name in his code book, and taps it out on his keyboard" (hyperlinks & bookmarks)
- Although today it is easy to equate memex with hypertext and the World Wide Web, Bush's inspiration for this idea came from the contemporary telephone exchange, which he described as a "spider web of metal, sealed in a thin glass container"

SISTEMI SA KOMANDNOM LINIJOM

- Request-response transakcija
- Nastao iz batch monitora povezanih na sistemsku konzolu
- Video-displej terminali (70tih)

VT100 terminal



ASR-33 Teletype



DEC PDP 1 (1961)

- Prvi komercijalni interaktivni računar
- Timesharing sistemi (mala preduzeća, laboratorije)



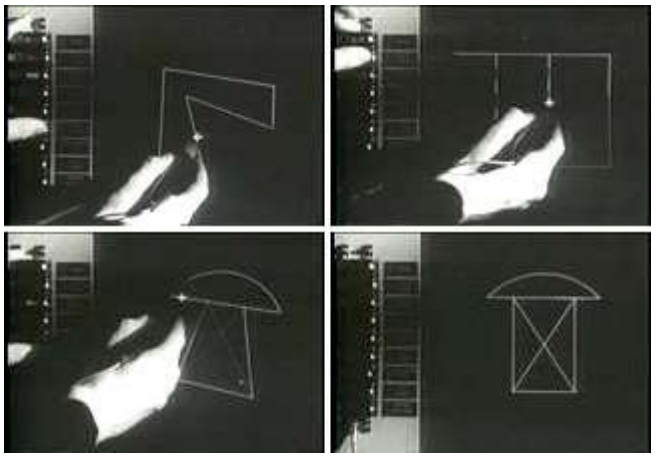




IVAN SUTHERLAND'S SKETCHPAD

- PhD na MITu 1962.
- *Heretofore, most interaction between man and computers has been slowed by the need to reduce all communication to written statements that can be typed. In the past we have been writing letters to, rather than conferring with, our computers (Sutherland, 1963, p. 329).*



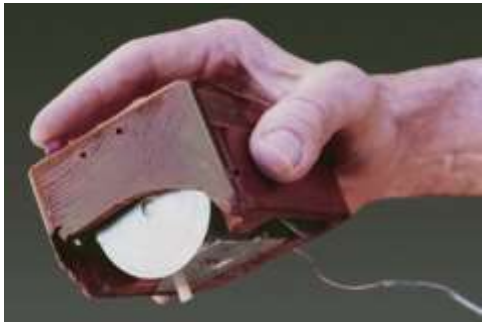


- Direktna manipulacija svetlostnim perom (prvi uređaj tog tipa):
  objekti su iscrtavani, risajzovani, hvatani, pomerani, proširivani, brisani
- Prve ikonice, koordinate na ekranu, orijentacija objekata, rekurzivne operacije

THE MOTHER OF ALL DEMOS



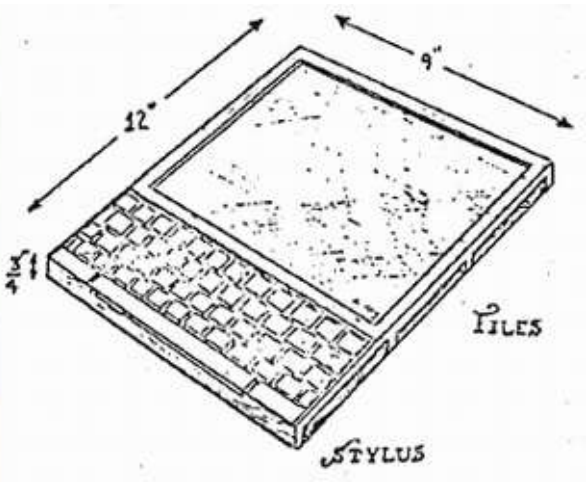
- Douglas C. Engelbart
- Pronalazak miša (1963.) – simboliše stvaranje HCIa, izmenilo način interakcije sa računarom
- Computer demonstration (1968.) u ACM/IEEE San Francisko, hw/sw kompletan sistem – oN-Line System, definisani skoro svi fundamentalni elementi savremenog računarstva:
  - Prozori (Windows)
  - Hipermedija
  - Grafika
  - Efikasna navigacija i ulaz/izlaz
  - Video konferencije
  - Računarski miš
  - Word procesing
  - Dinamičko linkovanje fajlova
  - Testiranje i obuka korisnika
  - Personalna anotacija
  - Revision control
  - Real time kolaboracija

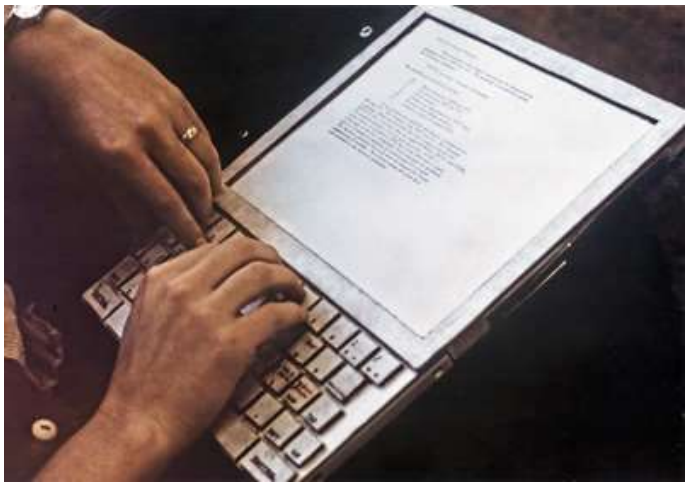


THE KIDDICOMP CONCEPT

- Alan Kay (1968.) – koncept tokom PhD rada
- *A personal computer for children of all ages* –računara veličine knjige, Dynabook računar 1972. (Xerox PARC)
- Definiše zahteve i koncepte portabilnog edukativnog računara, ali namenjenog pre svega deci
- Uključivao je GUI koncepte, Smalltalk kao sw komponentu, koncept računara koji funkcioniše kao živi organizam

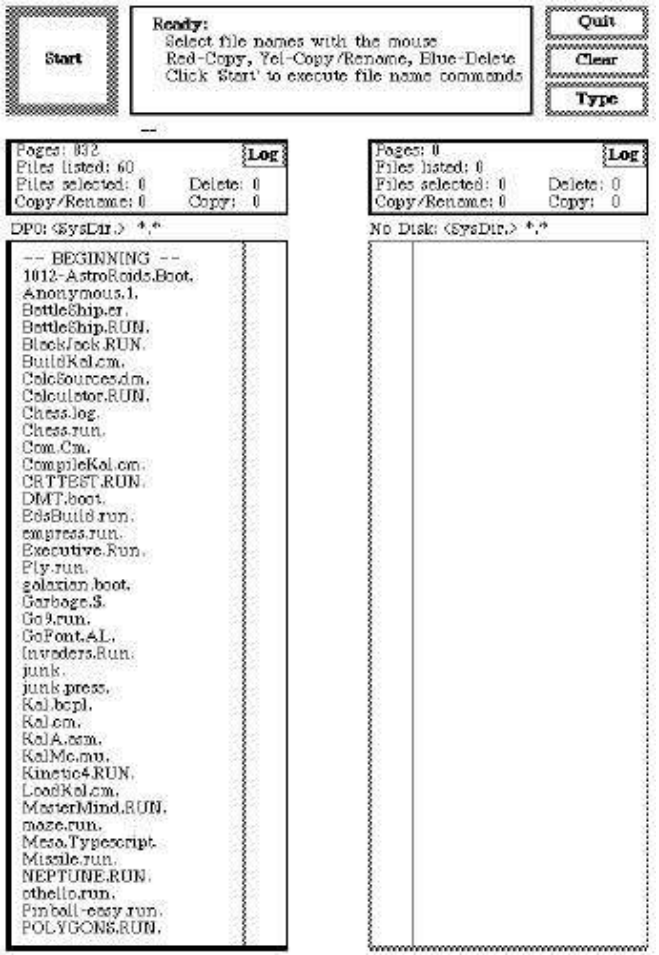






ALTO KOMPJUTER

- Sredina 70.tih, Xerox PARC, personalna radna stanica



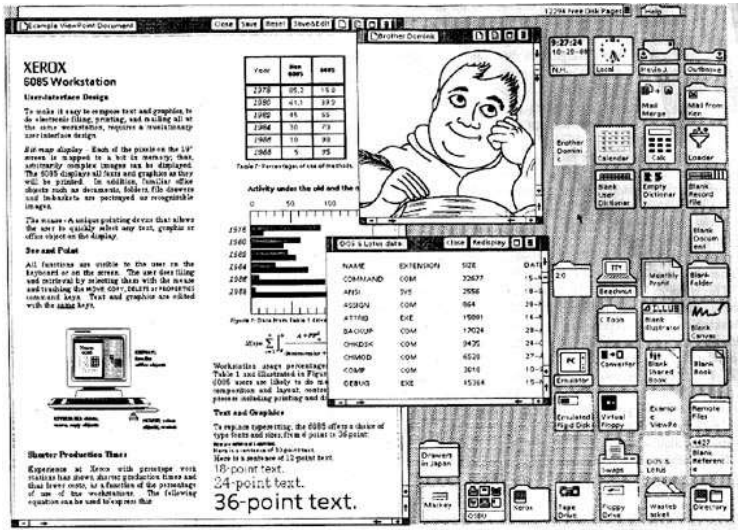


- Lokalni procesor, bitmapirani displej (monohromatski) i miš
- Napredan GUI (za to doba)
- Prozori, skrolovanje, konice, slajderi (nedostaju meniji)
- Kontrole su sve bile flat, bez 3D efekata

THE XEROX STAR 8100 INFORMATION SYSTEM

- Prva komercijalna verzija Alto računara (1981.), doživela finansiski neuspeh (spora i skupa)
- Prvi komercijalni rač. sistem sa GUIem – WINP sistem (windows, icons, menus, and a pointing device)
- Prvi rač. sistem baziran na usability engineering
- Podržavao je direktnu manipulaciju i WYSIWYG (what-you-see-is-what-you-get)



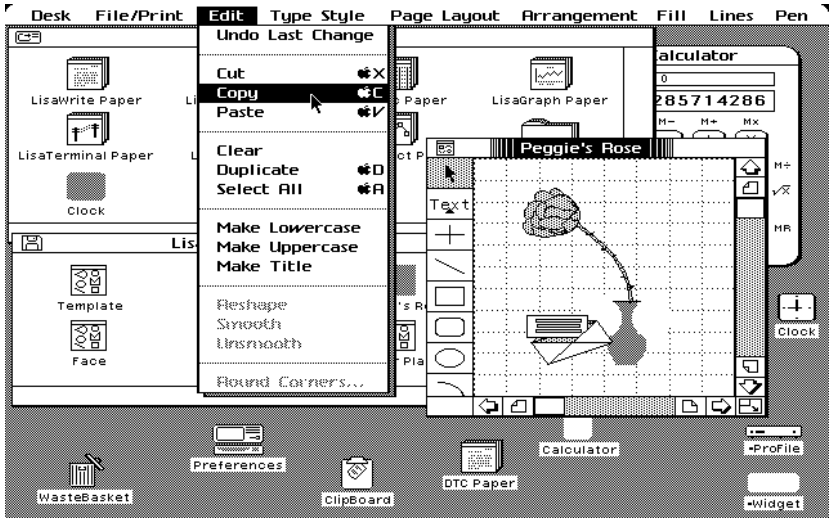


- Zamišljen kao *office automation system*
- *Ethernet* i email
- Desktop metafore: *Interactions that deal with files, not programs. So users "open a document," rather than "invoke an editor."*

APPLE LISA (1982.)



- Bazirano na idejema predstavljenim u Star-u
- Personalniji (bliže korisniku), manje office alat
- Jeftiniji od Star-a, ali i dalje preskup
- Konceptualni uspeh, ali komercijalni neuspeh
- Koristio GUI (WIMP)
- Namenjen velikom preduzećima, trebao da poveća produktivnost tako što bi učinio računare lakšim za korišćenje
- Aplikacije koje je sadržao:
  - LisaWrite: word procesor
  - LisaCalc: spread sheet
  - LisaGraph: charts
  - LisaList: outline builder, idea man.
  - LisaProject: project scheduler
  - LisaDraw: drawing program
  - LisaTerminal: modem comm. software



BIRTH OF HCI (1983.)

- Prva ACM SIGCHI konferencija (1983.)
- Publikova rad: Card, Moran, and Newell, "The Psychology of Human-Computer Interaction" (1983.),
- Pojava Apple Macintosh (1984.)
- Ben Shneiderman (1983.)
  - Direktna manipulacija (pojam za desktop metaforu)
  - Vizuelna manipulacija objektima koji se manipulišu
  - Fizička akcija umesto unosa teksta
  - Rezultat akcije je odmah vidljiv

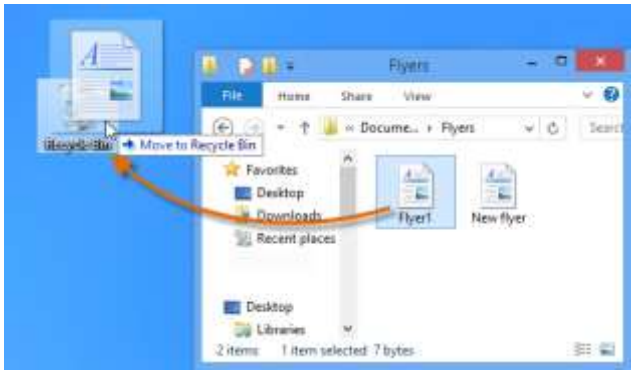








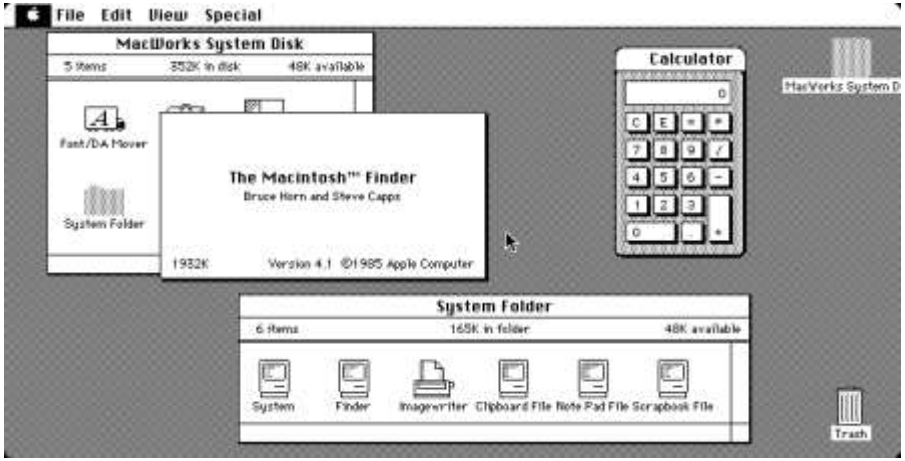




APPLE MACINTOSH (1984.)

- Prvi PC sa GUI-em pristupačan većem broju korisinika (~$2500)
- Dobri intarface guidelines
- Nisu bili pioniri u tehnika, ali su pametno kopirali
- Podržali 3rd party aplikacije
- Kvalitetna grafika i laserski štampač
- Imao samo 128k memorije
- MacWrite, MacPaint







MS WINDOWS (1985.)

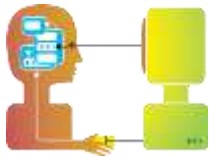
- Planiran za 1983.
- Windows 1.01 razočarenje
- Graphical, 16-bit multi-tasking shell on top of an existing MS-DOS installation
- Omogućavao korišćenje grafičkih programa dizajniranih baš za Windows, ali i MS-DOS aplikacije
- "turn the computer into a graphics-rich environment while using less memory"





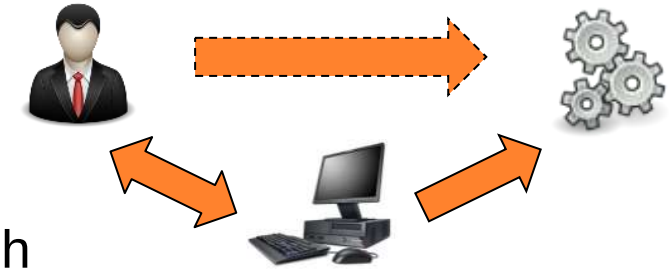
INTERAKCIJA

MODELI INTERAKCIJE



- Interakcija – komunikacija između čoveka i računara
- Kako su oba učesnika kompleksna, interfejs mora da efektivno prevodi između njih
- Modeli interakcije omogućuju sagledavanje interakcije, njihovo bolje razumevanje i predstavljaju način da se odrede potencijalni problemi, kao i njihovi izvori
- Modeli interakcije predstavljaju i efikasan alat za poređenje različitih tehnike interakcije

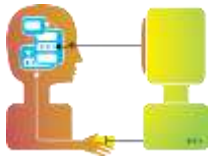
- Jedan od najuticajnih modela interakcije je Normanov model tokova izvršavanja i evaluacije
  - Opisuje interakciju na osnovu korisnikovih ciljeva i akcija



NORMANOV MODEL INTERAKCIJE ...



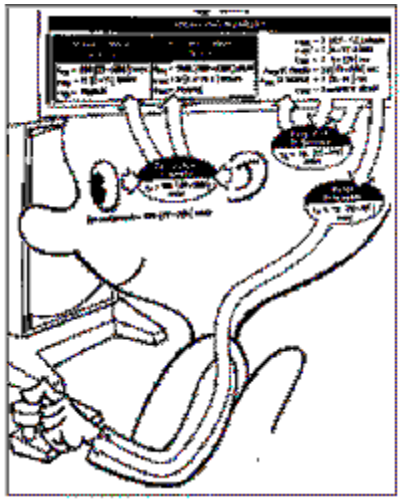
... NORMANOV MODEL INTERAKCIJE



- Svrha interaktivnog sistema je da pomogne korisniku da ostvari svoj cilje (goal) iz određenog domena primene (application domain)
- Cilj je željeni rezultat, izlaz iz sistema, akcije su dozvoljene operacije sistema (koncepti domena primene)
- Primer: Brightness u PhotoShop-u
  - Namera da se poveća brightness slike
  - Planiranje akcija (za početnika to znači pretraživanje menija dok se ne pronađe brightness opcija)
  - Korisnik izvršava akciju(e)
  - Korisnik osmatra, interpretira i evaluira reakciju sistema (izlaz iz sistema) spram originalne namere („Da li je to brightness opcija? Da li kada pokrenem opciju, radi to što želim?")
  - U zavisnosti od reakcije sistema i evaluacije, plan se modifikuje ("Da li da probam sledeći meni?") ili se menja cilj ("Umesto promene osvetljenja, mogu da docrtam brkove.")
- Dobri interfejsi imaju manji broj ponavaljanja (lako savladati)



GOMS ...



- GOMS (Goals Operators Methods Selection rules)
- Predstavlja pristup opservaciji interakcije između čoveka i računara
- Razvijen je od strane Stuart Card-a, Thomas P. Moran-a i Allen Newell -a, 1983. godine i publikovana u njihovoj knjizi The Psychology of Human Computer Interaction
- GOMS metodom korisnička interakcija sa računarom se razlaže na svoje elementarne akcije (psihološke, kognitivne ili perceptivne)
- Koristeći ove elementarne akcije kao osnovu, može se analizirati korisnički interfejs (ili modelovati interakcija)

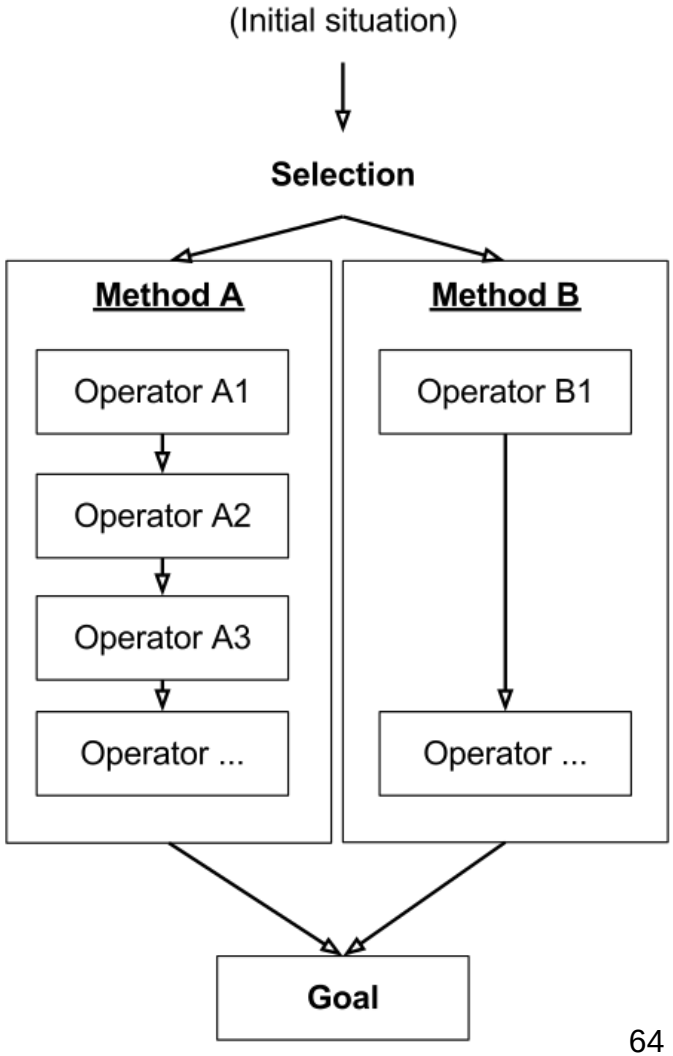
... GOMS ...

- Postoje mnoge varijacije GOMS metode, ali za sve varijacije definicije osnovnih koncepata su iste:

  1) Goals (korisnički cilj) predstavlja ono šta korisnik namerava da uradi

  2) Operators (operatori) predstavljaju osnovne akcije koje korisnik mora da sprovede kako bi koristio sistem

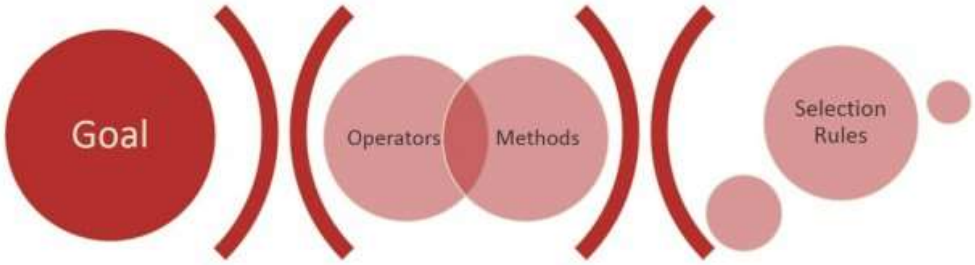
  3) Methods (metode) su sekvence operatora kojima se postiže cilj, odnosno podzadaci koje korisnik može rešavati po različitim putanjama radi postizanja glavnog cilja,



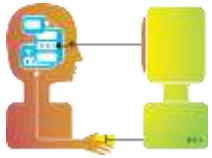
... GOMS ...

4) Selection rules (selekciona pravila) predstavljaju mehanizam za izbor putanje završetka zadatka, a i za definisanje pravila (rules) različitim korisnicima (budući da može postojati više metoda koje su na raspolaganju za postizanje jednog cilja, selekciona pravila se koriste u tačkama gde korisnik odlučuje koju će metodu dalje koristiti). U tipičnim GOMS analizama ova selekciona pravila su često ignorisana

- Postoj određena fleksibilnost za dizajnere, odnosno alnalitičare, prilikom definisanja ovih entiteta
- Recimo, ono što bi jednom analitičaru predstavljalo operator, drugom može da predstavlja cilj
- Nivo granularnosti je podešen tako da uhvati ono što u datoj situaciji evaluator želi da ispita



... GOMS ...



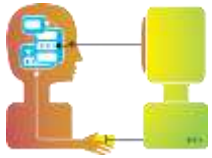
GOAL: TAKE PICTURE ON IPHONE

- GOAL: START-APPLICATION
  - REACH -FOR-IPHONE
  - RETRIEVE-HOME-SCREEN
  - TAP-APPLICATION-ICON
  - WAIT-FOR-APPLICATION-LOAD
  - VERIFY-APPLICATION-LOAD
- GOAL: CAPTURE-SUBJECT-PHOTOGRAPH
  - CHANGE-GAZE-TO-SUBJECT
  - POINT-CAMERA-TOWARD-SUBJECT
  - VERIFY-FACE-DETECTED
  - TAP-SHUTTER-LCD-BUTTON
  - VERIFY-APPLICATION-LOADING
- GOAL: VERIFY-APPLICATION-RESPONSE
  - WAIT-FOR-APPLICATION-RESPONSE
  - VERIFY-IMAGE-RESPONSE

GOAL: Metoda_dodavanja_teksta_na_spoljnu_ivicu_objekta selekcijom_objekta_i_korišćenjem_Fit_Text_ To_Path_opcije_Text_padajućeg_menija

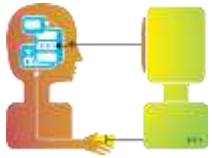
Pomeri_kursor_miša_iznad_objekta
Levi_klik_mišem_na_objekat
Pomeri_kursor_miša_iznad_Text_opcije_glavnog_menija
Levi_klik_na_Tekst_opciju_glavnog_menija
Pozicioniraj_miša_iznad_Fit_Text_To_Path_opcije_ padajućeg_menija
Levi_klik_mišem_na_Fit_Text_To_Path_opciju_glavnog_menija
Unesi_željeni_tekst]

... GOMS ...



- Uspeh GOMS metode posebno se pokazao u projektu *Ernestine* Engleske telefonije
  - Ergonomski dizajnirane stanice za rad su poređene sa starim u cilju utvrđivanja pobojšanja produktivnosti telefonskih operatera
  - GOMS metodom došlo se do procene da će produktivnost opasti za 3%
  - Kako su ove radne stanice zahtevale manje pritiskanja tastera od originalnih nije bilo jasno zbog čega se produktivnost smanjila
  - GOMS metodom utvrđenoje da nove radne stanice nisu iskorišćavale besposleno vreme operatera
  - GOMS metoda ne samo da je dala tačnu procenu (analizom razgovora utvrđeno je da je produktivnost zaista opala 4%), već je dala i objašnjenje nastale situacije

... GOMS

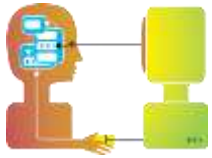


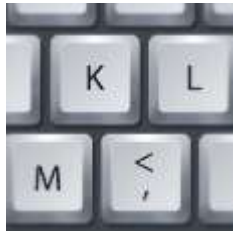
- Slabost GOMS metode leži u tome da ne uzima u obzir nepredvidivost korisnika (kao što su ponašanje zavisno od napora, socijalnog okruženja ili organizacionih faktora)

- Tehnologija je veoma eksplicitna kada se radi o bazičnim operacijama pokreta, ali slabija u pogledu kognitivnih akcija
  - Evidetno je da ovakve greške ne mogu biti sprečene, ali ni jedan od GOMS modela ne pokriva ovakav tip greške
  - Sve tehnike rade pod pretpostavkom da korisnik u svakom trenutku zna šta da uradi
  - Ovo je jedino prihvatljivo u slučaju ekspertskih korisnika, ne i u slučaju novajlija

- Kada stati, tj. do kog nivoa detalja ići?

KLM ...





- KLM (Keystroke Level Model)
- Predstavlja naučni pristup interakciji čoveka i računara baziran na CMN-GOMS metodi
- Razvijen je od strane Card-a, Moran-a i Newell-a i opisan u njihovoj knjizi The Psychology of Human Computer Interaction, 1983 godine
- KLM je metoda kojom pojedinci ili kompanije mogu proceniti vreme nepohodno za izvršavanje jednostavnih zadataka unosa podataka korišćenjem računara i miša
- Korišćenjem ove metode pojedinci često pronalaze efikasniji i pojednostavljen način izvršavanja zadataka
- KLM metoda predstavlja jednostavnu analizu neophodnih koraka u procesu, zatim reorganizaciju koraka, kao i izbacivanje nepotrebnih koraka

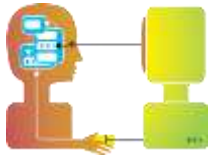
... KLM ...

- Dizajnirana je tako da bude jednostavnija za primenu od ostalih GOMS metoda čija primena obično podrazumeva angažovanje stručnjaka sa dubokim razumevanjem date metode
- KLM-GOMS se obično koristi u situacijama gde je potrebna minimalna količina posla i interakcije sa interfejsom



- Kalkulacije vremena, kao i broj koraka neophodnih za tačno izračunavanje ukupnog vremena postizanja cilja rastu brzo sa povećanjem broja zadataka
- Zbog toga je KLM-GOMS metoda najpodesnija za procenu zadataka čije izvršenje u proseku vremenski zahteva manje od 5 minuta za izvršenje

... KLM ...

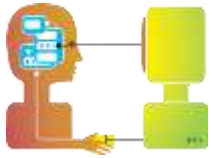


- Korisnički ciljevi se u KLM metodi modeluju kao sekvence operacija
- Svakoj operaciji dodeljeno je trajanje sa namerom da modeluje vreme koje bi bilo potrebno iskusnom korisniku da izvrši operaciju
- Operatore možemo podeliti na motorne, mentalne i operatore odgovora sistema

Standardne vrednosti KLM operatora:

K- pritisak na dati taster tastature
B – pritisak na taster miša
P – pointiranje, pomeranje pointer uređaja na cilj
H – homing, premeštanje ruke sa/na tastature sa/na miša
D – crtanje linije pomoću pointer uređaja
M – mentalna priprema za fizičku akciju
R – Reakcija sistema, može se ignorisati ako korisnik ne čeka na nju (eho tokom unosa).

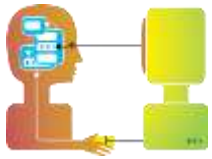
... KLM ...



Operator	Napomena	Vreme(s)
K	Dobar (90 r/m) Srednji(40 r/m) Početni	0.12 0.28 1.20
B	Dole ili gore klik	0.10 0.20
P	Fitt-ov zakon prosečno	1.10
H	na/sa tastature	0.40
D	crtanje	-
M	Mentalna priprema	1.35
R	Reakcija sistema	-

Tabela standardnih vrednosti KLM operatora

... KLM



Tab. 3.6. KLM: Metoda_dodavanja_teksa_na_spoljnu_ivicu_objekta_selekcijom_objekta_i_korišćenjem_ Fit_Text_To_Path_opcije_Text_padajućeg_Menija

Pomeri_kursor_miša_iznad_objekta	**H<miš>MP**
Levi_klik_mišem_na_objekat	**B<L>**
Pomeri_kursor_miša_iznad_Text_opcije_glavnog_menija	**MP**
Levi_klik_na_Tekst_opciju_glavnog_menija	**B<L>**
Pozicioniraj_miša_iznad_Fit_Text_To_Path_opcije_ padajućeg_menija	**MP**
Levi_klik_mišem_na_Fit_Text_To_Path_opciju_glavnog_menija	**B<L>**
nesi_željeni_tekst	**H<tastatura>10*(MK)**

H+M+P+B+M+P+B+H+10*(M+K) = 2H+12M+2P+10K = 31,2

- Voditi računa da je KLM samo vodič, ne i prorok
- Početne pretpostavke utiču na krajni rezultat
- Takođe, najbrže ne mora u svakoj situaciji biti i najbolje

STILOVI INTERAKCIJE

- Stilovi interakcije – metodologija preko koje se realizuje komunikacija između korisnika i sistema

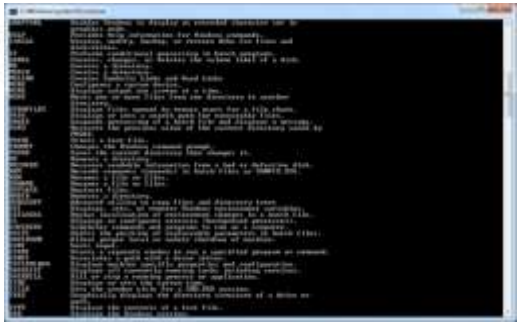
- Komandna linija
- WIMP
- Meniji
- Filin forma
- Direktna manipulacija
- Antropomorfni





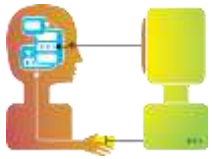


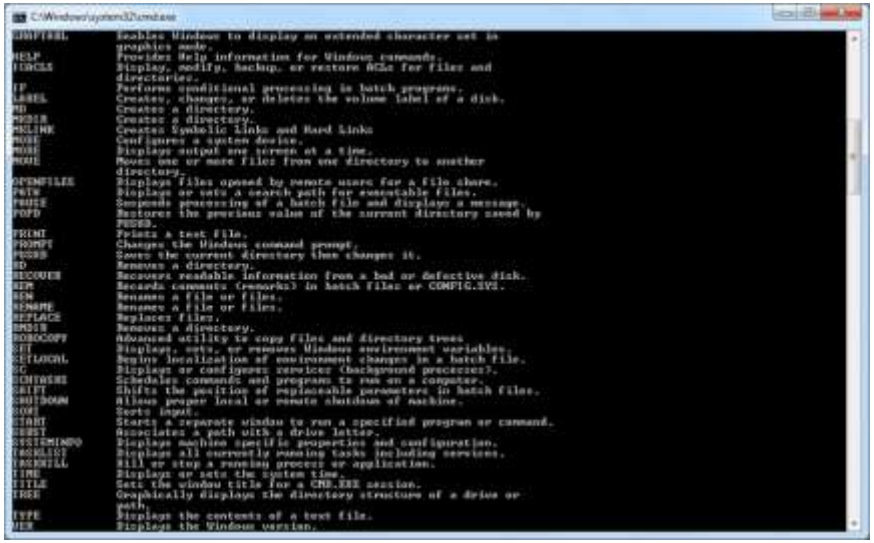






KOMANDNA LINIJA





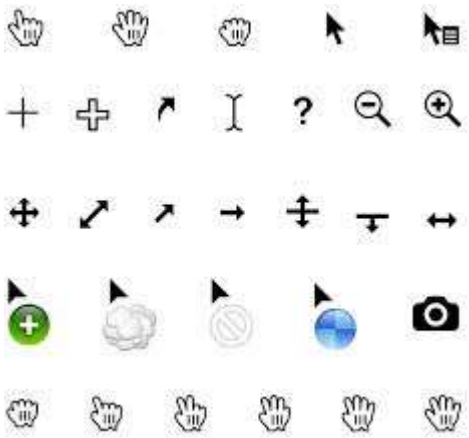
- Najstariji oblik interakcije
- Vrlo linearan oblik interakcije
- Korisnik unese komandu, sistem je izvrši
- Nakon što sistem izvrši komandu, korisnik može da unese novu komandu
- Komanda može biti karakter, skraćenica, reč, više reči, kod
- Prednost:
  - Moćan zato što pruža brz direktan pristup svim funkcijama sistema
  - Fleksibilan, jer se lako proširuje, može da pordži veliki broj varijacija kroz opcije i parametre
- Mane:
  - Naredbe moraju da se pamte
  - Pregled svih komandi nije dostupan na ekranu
  - Obskurni sa kompleksnom sintaksom
  - Vrlo podložni greškama i netrpeljivi prema greškama

WIMP

- Windows
- Icons
- Menus
- Pointers



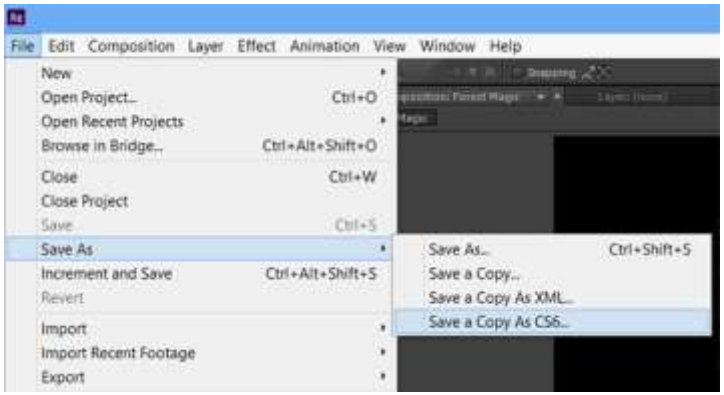




MENIJI

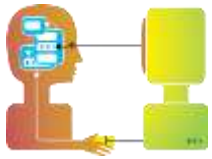
- Skup opcija ili izbora od kojih korisnik bira jednu pomoću miša ili tastature
- Vizuelni feedback kad je opcija selektovana
- Prednost:
  - Prepoznavanje a ne pamćenje
  - Kompleksne akcije se dele na više jednostavnih koraka (interakcija) – dobro za početnike, loše za eksperte (dodatne tehnike da se ovo prevaziđe – prečice?)



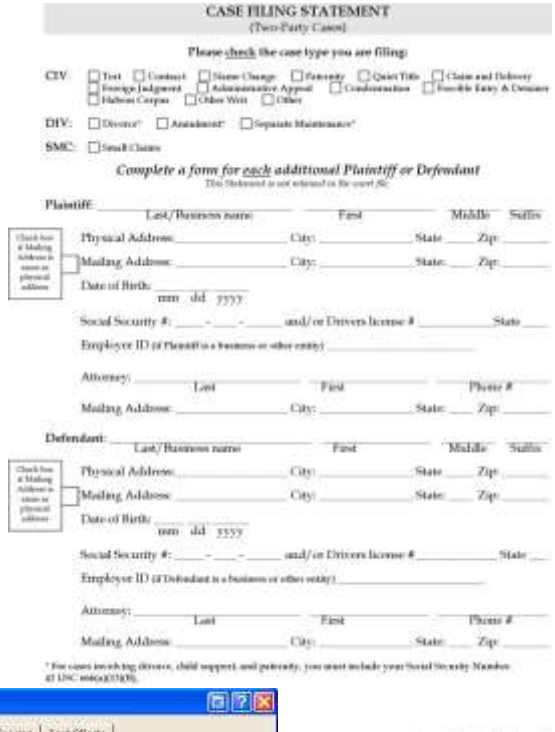


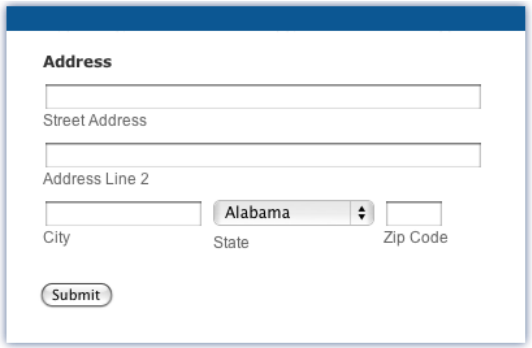
- Mana:
  - Upotrebljivost zavisi od kvaliteta labela/opisa ponuđenih opcija (njihovo značenje i razumevanje) – ako su labele loše, opašće produktivnost a porasti broj grešaka

FORME



- Koristan za prikupljanje informacija
- Podaci se ili unose ili se biraju ponuđene opcije preko većeg broja kontrola
- Familijarnost sa papirnatim formama
- Ako je dobro dizajniran, pomoćiće korisniku da razume svrhu forme i omogućiti mu da brzo i lako unosi podatke
- Loše dizajnirana forma biće neefikasna i produžiti unos podataka

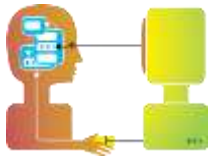








DIREKTNA MANIPULACIJA



- Korisnik manipuliše vizuelnim reprezentacijama objekata
  - Kontinualna vizuelna reprezentacija
  - Fizička (konkretna) akcija (m. down/up) vizuelne reprezentacije 3D dugmeta
  - Brze akcije (rezultat vidljiv skoro trenutno)
  - Inkretementalne akcije (skrol bar se pomera u malim ili velikim koracima)
  - Reverzibilne akcije (undo)
  - Akcije sa trenutno vidimljim efektima (bez dodatnog napora korisnika)





  - Oslanje se na perceptualne i motorne veštine čoveka
  - Prirodan (ne zavisi od jezičkih sposobnosti
  - Ne može se svaka funkcija interfejsa (korisnička operacija) direktno (lako) mapirati na fizičku akciju

ANTROPOMORFNI INTERFEJSI

- Interfejs komunicira (intereaguje) sa korisnikom na isti način kako bi to činila dva čoveka
- Uključuje govor (prirodni jezik, a ne jezik namenjen za upravljanje računarom), gestovi (pokreti ruku), facijalnu ekspresiju i pokret oka
- Razvoj ovog tipa interfejsa, oslanja se poznavanje ljudskog ponašanja (šta znače reči, gestovi, pogledi, facijalne eskpresije)



healthcare avatars





- Još uvek eksperimentalno, ali sve više u upotrebi (kontrola putem glasa, gesta, i sl.)

POST-WIMP INTERFEJSI

- Za kompleksne zadatke (Computer Aided Design)
- Za zadatke koji trebaju da se izvršavaju u paraleli (grupni rad)
- Za interaktivne računarske igre
- Za VR/AR sisteme







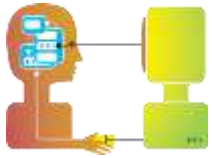






- Paralelna interakcija
- Kontinualna interakcija
- Multi-mode

POREĐENJE STILOVA INTERAKCIJE

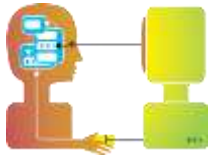


DISKUSIJA

- Lakoća učenja (knowledge in the head vs. world)
- Poruke o grešci
- Efikasnost
- User experience
- Sinhronizacija
- Programiranje
- Pristupačnost (accessibility)

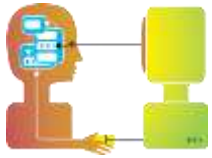
ŽIVOTNI CIKLUS RAZVOJA KORISNIČKOG INTERFEJSA

DIZAJNIRANJE ZA KORISNIKA ...



- Računari su danas dostupni gotovo svima, te korisnici računara mogu biti vrlo raznovrsni (različito znanje i računarske veštine)
- Zato korisnički interfejs mora biti prilagođen korisniku i njegovim potrebama
- Korisnički interfejs koji nije prilagođen korisniku može da omogući da se nešto uradi, ali to ne mora biti ono što korisnik želi i na način koji odgovara korisnikovim potrebama da reši zadatak
- Koriniskički interfejs može biti upotrebljiv ali ne i koristan
- Naravno, ne može interfejs biti takav da bude prilagođen baš svakom pojedinačnom softveru
- Ali mora biti prilagođen korisniku kojem je namenjen
- Korisnik ne bi trebalo da brine o tome kako sistem funckioniše (sem ako mu to, naravno, nije zadatak)

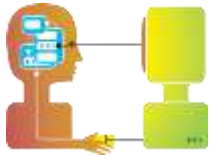
... DIZAJNIRANJE ZA KORISNIKA



- Zbog toga je neophodno dizajnirati korisnički interfejs za korisnika, a to je moguće samo ako se korisnik i njegove potrebe dobro upoznaju

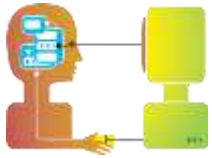
- Da bi se napravio stvarno upotrebljiv korisnički interfejs, mora se voditi računa:
  - kako ljudi koriste računar (na koji način intereaguju sa njim)
  - o korisnikovim karakteristikama (osobinama) koji direktno utiču na interakciju
  - o korisnikovom znanju, iskustvu i veštinama
  - o korisnikovim potrebama, zadacima i poslu
  - o korisnikovim psihološkim karakteristikama
  - o korisnikovim fizičkim karakteristikama
  - o tome da se primene predložene metodologije i tehnologije za upoznavanje svih prethodno navedenih aspekta

USER CENTER DESIGN ...



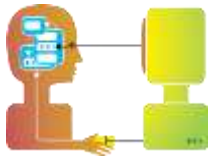
- Proces projektovanja/razvoja korisničkog interfajsa koji uključuje korisnika u sve faze projektovanja i razvoja
- Fokus proučavanja je korisnik, ali i **razumevanje njegovog zadatka**, kao i **razumevanje (upoznavanje) okruženja** u kojem će korisnik obavljati zadatak (organizacioni, sociološki i fizički aspekti okruženja)
- Gotovi svi pristupi basirani na UCD dele tri karakteristike
  - Iterativni dizajn beziran na brzom razvoju prototipa
  - Rani (od samog početka) na korisnike i njihove zadatke
    - Analiza korisnika: ko su korisnici
    - Analiza zadataka: šta treba korisnici da urade
    - Uključivanje korisnika evaluatora, konsultanata i, ponekad, uključivanje korisnika kao dizajnera
  - Stalna (re-)evaluacija tokom čitavog i iteraktivnog postupka
    - Korisnici učestvuju u svakoj iteraciji
    - Svaki prototip je na neki način evoluiran

... USER CENTER DESIGN



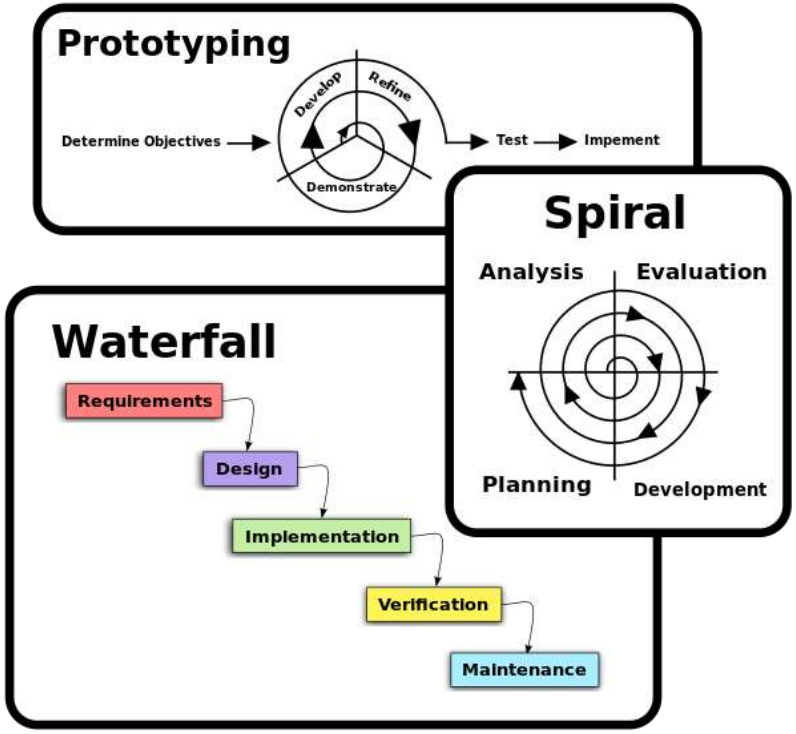
- *Human-Centered Design Processes for Interactive Systems* (ISO 13407, 1997) – osnovi principi i aktivnosti koje treba sprovoditi u UCD dizajnu da se ostvario što veći usability sistema
- Četiri osnovan principa UCD dizajna (ISO 13407, 1997 str. 7)
  1. Aktivno uključivanje korisnika
  2. Adekvatna alokacija funkcija između korisnika i sistema
  3. Iterativnost čitavog procesa dizajna
  4. Multidisciplinarni timovi
- Četiri osnovne aktivnosti UCD dizajna (ISO 13407, 1997 str. 10)
  1. Razumevanje i specifikacija konteksta upotrebe
  2. Specifikacija korisničkih i organizacionih zahteva
  3. Razvoj prototipova
  4. Evaluacija rešenja (prototipa) spram kor. zahteva zajedno sa korisnicima

USABILITY ENGINEERING

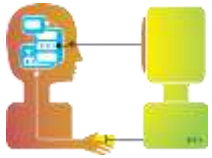


- Nije lako izgraditi dobar korisnički interfejs i ostvariti usability sistema
- Prateći strogo definisane procese i vodeći računa o detaljima
- Koriskički interfejs, kao i čitav softver, prati određeni životni ciklus razvoja softvera
  - Kaskadni model
  - Iterativni životni ciklus baziran na brzom razvoju prototipa

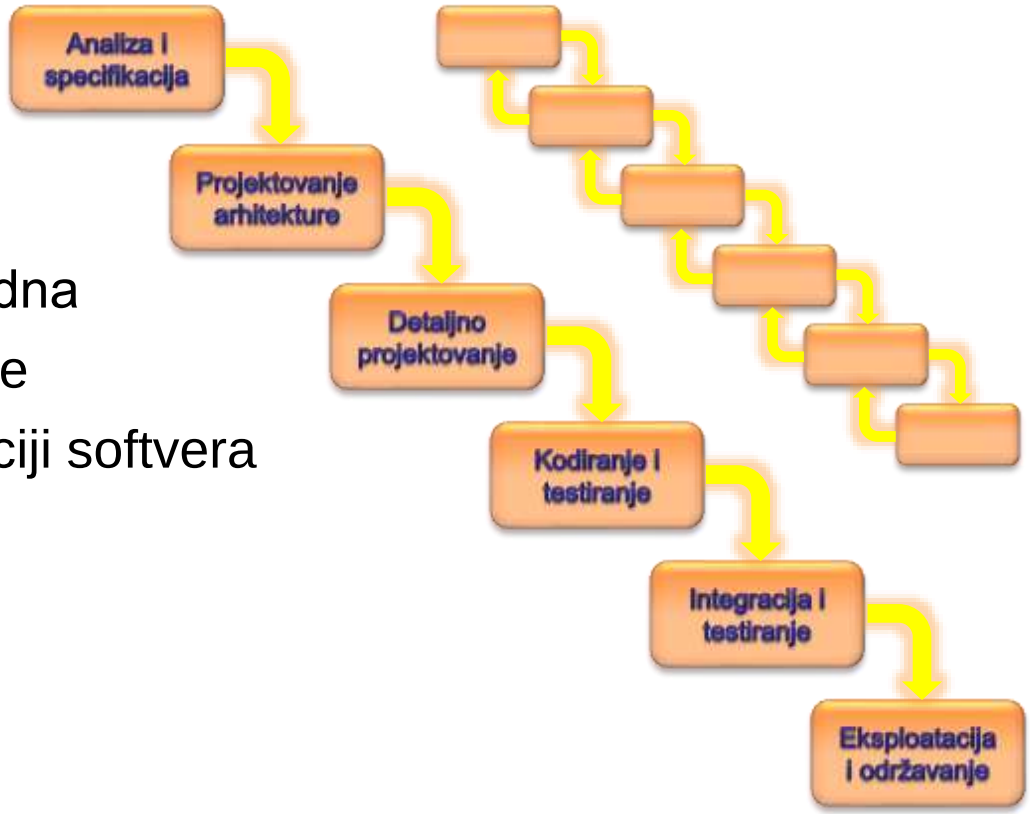




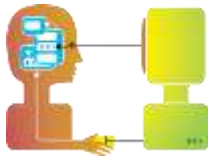
KASKADNI MODEL ŽIVOTNOG CIKLUSA ...



- UCD nikako ne ide ruku pod ruku sa kaskadnim modelom
- Niz strogo definisanih koraka
- Aktivnost ne može da počne ukoliko se nije završila prethodna
- Dobar za izradu dokumentacije
- Formalne procedure u realizaciji softvera
- Sekvencijalan
- Striktna podela uloga
- Analitika, ljudski pristup
- Postoji i iterativna
- Propagacija greške
- *Think first, code second*



... KASKADNI MODEL ŽIVOTNOG CIKLUSA

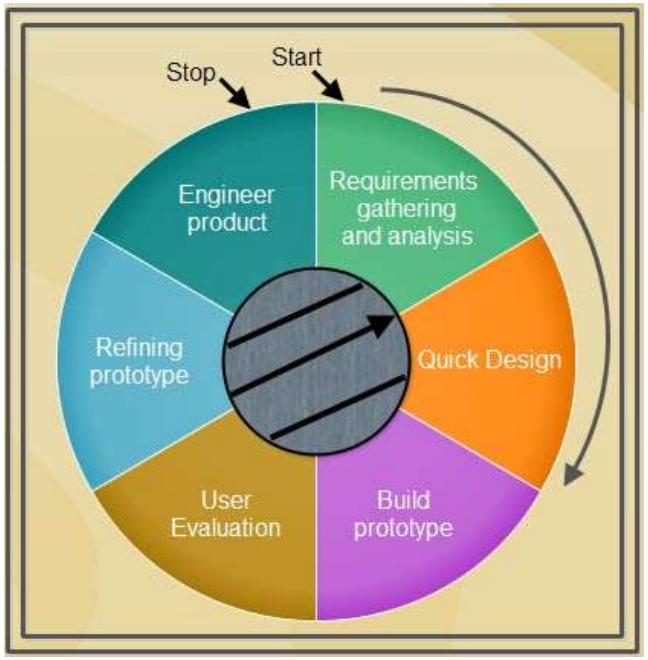


- Kaskadni životni ciklus je loš za razvoj korisničkog interfejsa
  - Razvoj korisničkog interfejsa je rizično i postoji velika verovatnoće da će se iz prve pogrešiti
  - Korisnici učestvuju samo u prvoj fazi i na kraju predzadnje, tako da će se saznati da li je interfejs odgovarajući tek na kraju kada je sve gotovo
  - Greške u korisničkom interfejsu su često dramatični i dovode do promena u zahtevima i dizajnu, pa će to dovesti do odbacivanja dobro napisanog i istestiranog koda



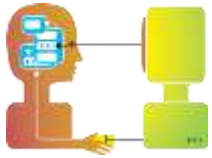
ITERATIVNI MODEL ŽIVOTNOG CIKLUSA ...

- Usability engineering se zapravo oslanja na iterativni dizajn
- Najverovatnije se neće pogoditi i napraviti sve kako treba iz prve
- Niz aktivnosti tokom razvoja softvera koja dovodi razvoja ranije nepotpune verzije softvera
- Omogućuje:
  - uključivanje korisnika odmah i tokom svih faza,
  - vidi i dopuni pristup
  - podržan je alatima
  - lako se upravlja njime
  - može da se odbaci, može da evoluira u kompletno rešenje
- Eventualno će se doći do upotrebljivog rešanja (nadamo se)

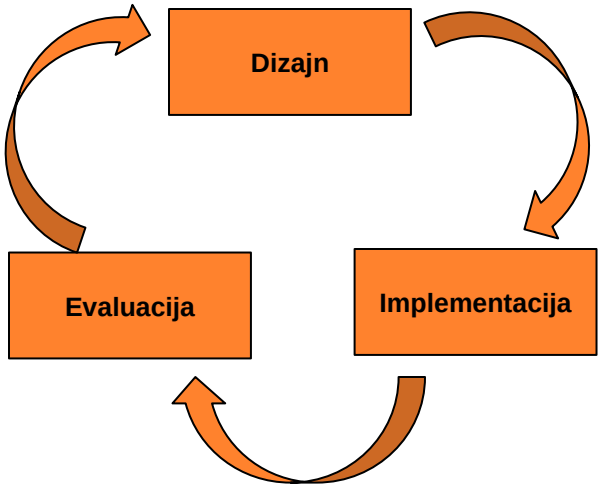


- Smernice i praksa mogu da ubrzaju proces (smanje broj iteracija)

... ITERATIVNI MODEL ŽIVOTNOG CIKLUSA ...

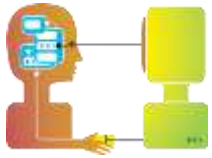


- Omogućuje umanjenje rizika koje dizajniranje korisničkog interfejsa nosi samo po sebi
- U pojednostavljenom obliku ima samo tri koraka
  - Dizajn (zamisli kako će da izgleda)
  - Implementacija (fizička realizacija izgleda)
  - Evaluacija (testiranje + korisnikovo mišljenje)
  - 3 koraka koje treba primeniti i u vašim zadacima

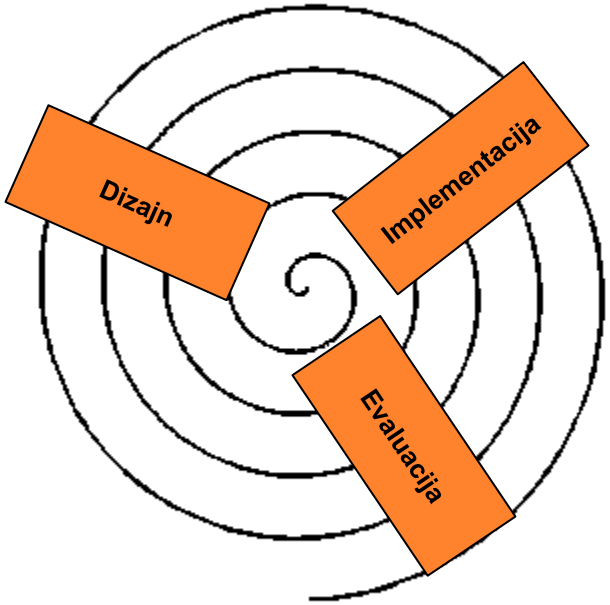


- Pogrešan način korišćenje iterativnog dizajna:
  - Svaka iteracija odgovara jednoj verziji softvera (jednom release-u)
    - Komentari (žalbe) iz evaluacije prenose se u sledeću verziju softvera (sledeći release)
  - Korisnici koji plaćaju za softver postaju vaši testeri
    - Neće biti srećni
    - Neće kupiti verziju broj 2
    - Early access – malo bolje (ne puno), samo se igra na psihu korisnika

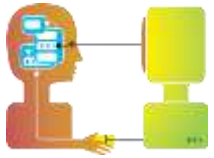
... ITERATIVNI MODEL ŽIVOTNOG CIKLUSA ...



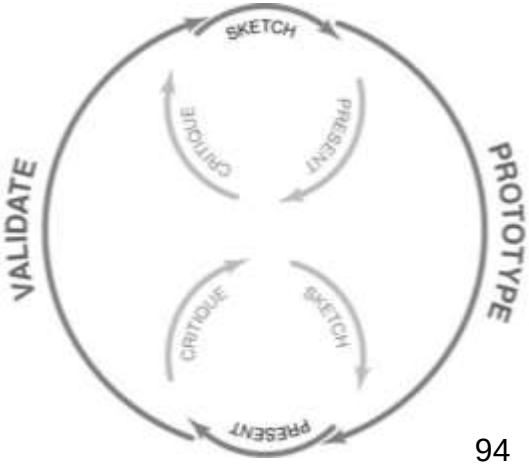
- Iterativni dizajn pod jedan podrazumeva upotrebu spiralnog modela
  - U startu se planira nekoliko iteracija (pre izlaska komercijalne verzije)
  - Svaki korak iteracije se pravi što je moguće jeftinijim
  - Radialna dimenzija spiralnog modela odgovara ceni koraka iteracije
  - Pod cenom se podrazumeve i koliko je verzija bliska krajnjem izgledu korisničkog interfejsa
    - U prvim koracima to može biti skeč na papiru ili neka vrsta jednostavnog HTML mokapa
    - Samo blaga senka krajnjeg proizvoda i njegove interakcije
    - Izrazito jeftino
    - Korisno da se sa korisnikom komuniciraju ideje i da se na osnovu njih sazna korisnik mišljenje



... ITERATIVNI MODEL ŽIVOTNOG CIKLUSA

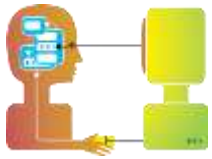


- Spiralni model je dobar za dizajniranje interakcije i korisn. interfejsa?
- Najveći rizik je u početku
  - Upotrebom jeftinih tehnika (brz razvoj prototipa) rizik se značajno smanjuje
  - Moguće je razvijati prototipe paralelno i brzo kako bi se istražile ideje i mogući pravci razvoja (odbacivanje početnih ideja gotovo da ništa ne košta)
- Kasnije faze koriste prototipe bogatijeg sadržaja – to su prototipovi koje planiramo da zadržimo i upotrebimo kao krajnje rešenje
  - Pretpostavka je da se nakon nekoliko koraka (cilkusa) naučilo dosta o korisniku i njegovom zadatku da ne dođe do nekih većih promašaja u koris. interfejsu
- Više iteracija uopšteno dovode do boljeg interfejsa
  - Postepeno, rafinisanjem koraka pronalazi se najbolji dizajn (ništa slučajno, već sa određenim ciljem)
  - Zadržava se ono što valja, a redizajniraju loši delovi
- Samo krajnji koraci se prikazuju svetu

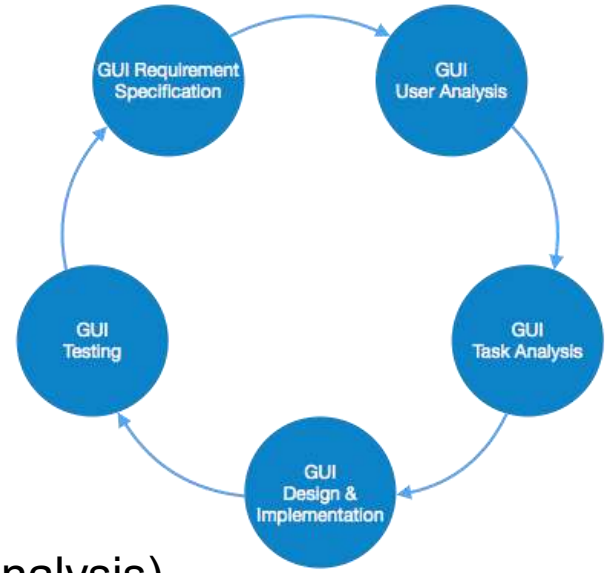


ANALIZA ZAHTEVA KORISNIČKOG INTERFEJSA

ANALIZA PROBLEMA



- Prvi korak u razvoju korisničkog interfejsa je sakupljanje svih podataka neophodnih da se pristupi razvoju interfejsa
  - Nikako bacanje na tastaturu kao na granatu, već postepeno, sakupljanjem svih neophodnih informacija i dobrim planiranjem, modelovanjem
- Moguće je postavljati svakakva pitanja vezana za sistem koji se razvija, ali najčešća (kada se radi o korisničkom interfejsu)



1. Ko su korisnici? (User analysis)
2. Koji je domen primene sistema koji se razvija (kontekst njegove upotrebe i mesto gde će se koristiti)? (Domain analysis)
3. Šta korisnici žele da urade sa sistemom? (Task analysis)
4. Koje zahteve prethodna tri koraka nameću pred dizajn korisničkog interfejsa? (Requirement analysis)

PIKUPLJANJE ZAHTEVA – ŠTA JE U FOKUSU?

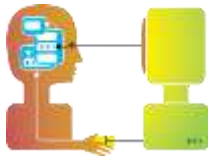
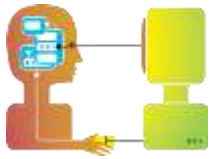


Table 3.1 Areas of investigation and information gathered for UI design

Focus of investigation	Information gathered
The domain	Wider specialist knowledge Specific knowledge for a computer system
The users	Who they are; focuses on the real (primary) users, but also considers other stakeholders (secondary users)
Characteristics of the users	Age, sex, culture, physical abilities and physical disabilities, educational background, computer/IT experience, motivation, attitude, enjoyment, satisfaction
Characteristics of the tasks	Are the tasks easy, complex, novel, variable, repetitive, frequent or infrequent, single tasks or multitasking, time critical, requiring individual or collaborative working? Are there safety issues in relation to the work?
Physical environment	Noise, stress, comfort, dirt, dust, heating, lighting, ventilation, furniture, working space, individual offices, open-plan areas, equipment layout, hazards in the workplace
Social environment	Pressure of work, individual or collaborative working, individual offices or open-plan areas
Organizational environment	Organizational mission and aims, organizational attitude to IT, organizational policies, job design, and roles
User support environment	Availability of training, availability of colleagues/experts, availability of manuals or online help
Qualitative usability aspects	General, often unquantifiable goals, such as easy to learn, UI intuitiveness
Quantitative usability goals	Measurable goals, such as usability metrics
Constraints	Costs, timescales, budgets, technology hardware and software
Trade-offs	Conflicting/contradictory requirements

KAKO PRIKUPITI ZAHTEVE



- Postoje različite tehnike koje se mogu upotrebiti za prikupljanje informacija o korisnicima, domenu i zadacima koje oni obavljaju
  - Observacija korisnika
  - Intervjuisanje korisnika
  - Grupni intervjui
  - Prikupljanje informacija upotrebom upitnika i anketa
  - Proučavanje dokumentacije
  - Proučavanje sličnih proizvoda
- Ove tehnike se koriste za prikupljanje zahteva, ali i u kasnijim fazama (pre svega u evaluaciji korisn. interfejsa)

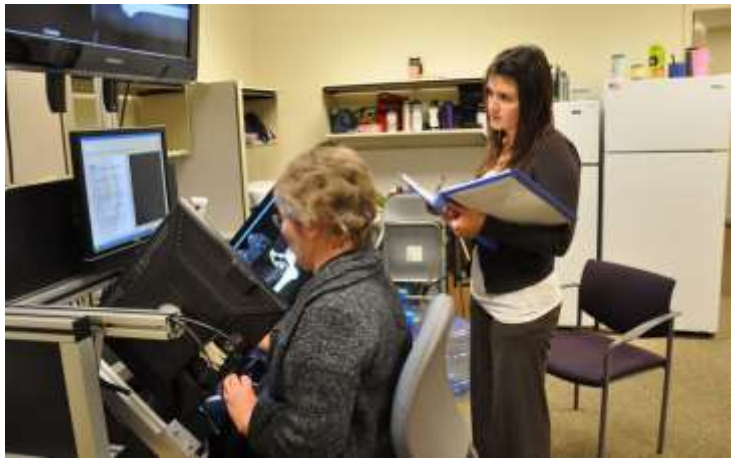


"We're out of crayons..."

- Prepreke:
  - Teško da programeri i korisnici ostvare kontakt usled barijera koje nameću kompanije (tehnička podrška štiti programere, marketing štiti korisnike)
  - Neke korisnike nije lako dobiti, ili su suviše skupi (lekari, direktori, ...)

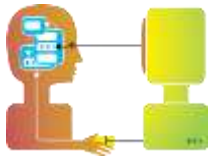
OBSERVACIJA KORISNIKA ...

- Najbolji način da se sazna sve o korisniku je posmatranjem korisnika u njegovom prirodnom okruženju, dok realizuje svoje zadatke (lako je saznati šta u sistemu koji koriste vole/ne vole)
- Direktna observacija od strane dizajnera korisn. interfejsa
  - Korisnik se direktno posmatra dok rešava zadak ili u kontrolisanim uslovima/ laboratoriji (controlled study) ili u prirodnom okruženju (field study)
  - Iako vrlo korisna, ima svojih ograničenja
    - samo jedan prolaz (nema ponavljanja) te se nešto može i propustit
    - vrlo nametljivo za korisnika
    - može dovesti do toga da se korisnik ne ponaša prirodno zato što zna da ga posmatraju



  - Vrlo značajna za inicijalnu fazu dizajniranja interfejsa, ali je uvek dobro imati nekakav zapis opservacije za kasnije referenciranje

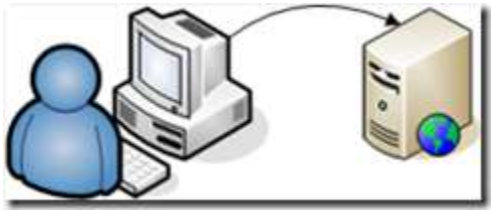
... OBSERVACIJA KORISNIKA





- Indirektna observacija
  - Ponašanje korisnika se indirektno loguje i snima (video/audio)
  - Prati se više stvari/akcija od jednom, te može biti više podataka za analizu i može oduzeti više vremena
  - Treba dobro isplanirati šta se želi snimati (koje se informacije žele saznati i kakvi se podaci žele uzeti)
  - Problem podašavanja i postavljanja opreme (ne sme da ometa sam rad i prirodnost okruženja a da se opet preuzmu svi želeni podaci)





- Obe se tehnike mogu kombinovati
- Zbog količine podataka mora se dobro izbalansirati

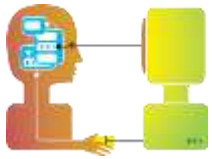
INTERVJUISANJE KORISNIKA ...

- Intervjuisanje podrazumeva razgovor sa korisnikom i/ili njegovo ispitivanje
- Omogućuje prikupljanje informacija brzo i na prijateljski način
- Moraju se planirati unapred i vrlo pažljivo
  - ko će sve biti intervjuisan
  - koja će se pitanja postavljati da se dobiju relevantni podaci
  - koliko će intervju trajati





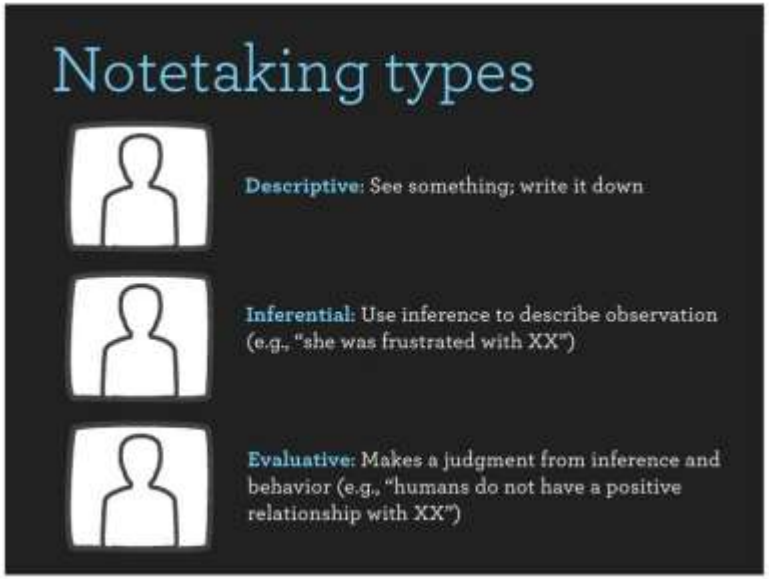
... INTERVJUISANJE KORISNIKA ...



- Postoje u osnovi dve vrste intervjua

  Strukturirani – unapred definisana pitanja koja se postavljaju bez odstupanja na unapred određen način

  Fleksibilni – postavljaju se teme koje se ispituju, ali nema unapred definisanih pitanja, ispitivač prati odgovore ispitanika u skladu sa njima određuje dalji pravac ispitivanja (traži pojašnjenja odgovora i sl.)

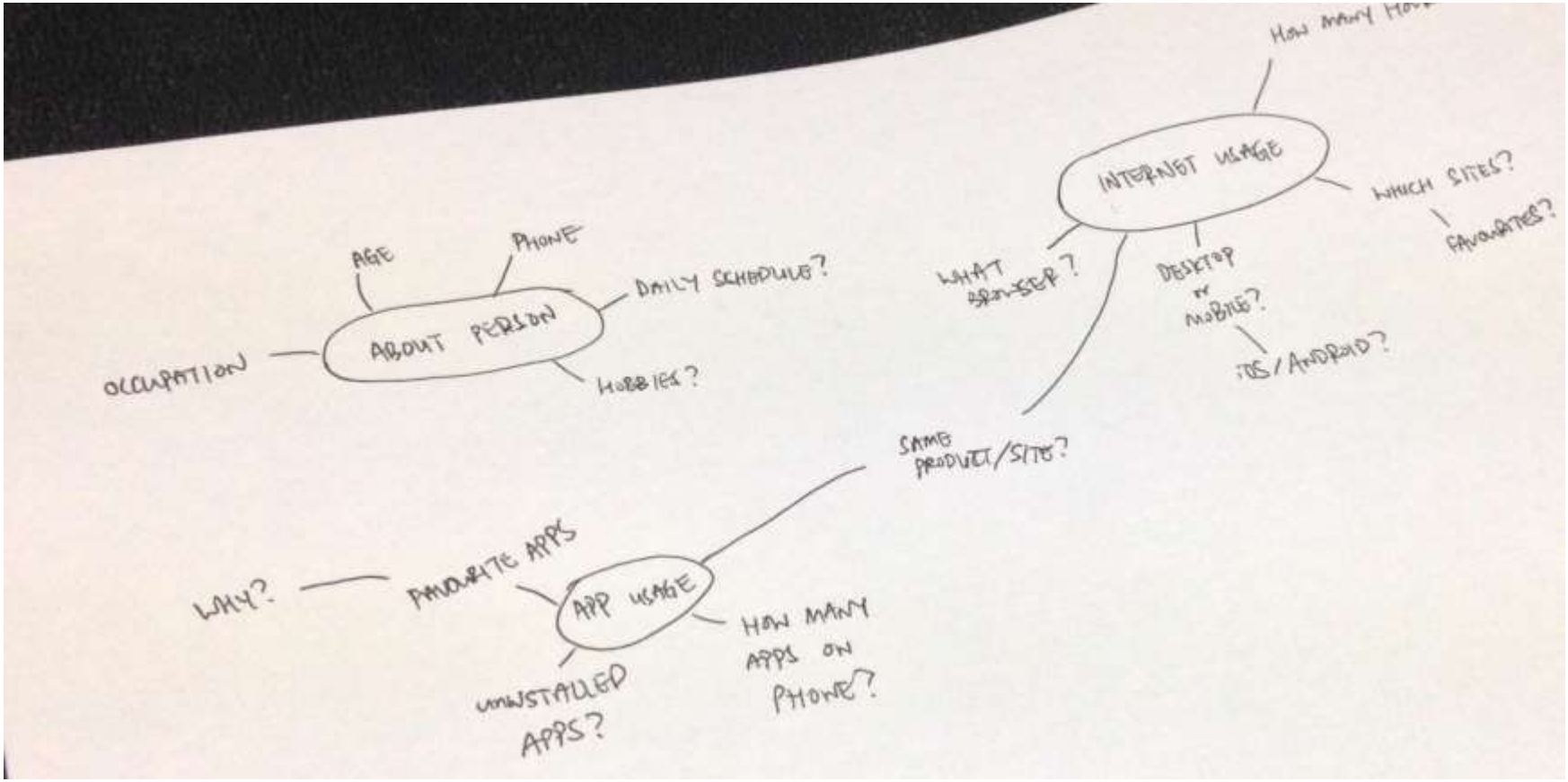


  - Fleksibilni intervju je manje formalan
  - Ali je dobro uvek imati spreman grub plan intervjuisanja (teme o kojima se želi razgovarati)

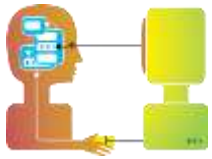
- Vrlo važno voditi računa o osećanjima onog koji se intervjuiše (prijateljsko okruženje, ispitanik ne sme da se oseća posramljeno ili ugroženo)

... INTERVJUISANJE KORISNIKA …

- Tematske mape mogu pomoći oko usmeravanja pitanja i samog flaksibilnog interrvjua, olakšavaju ispitivaču da se drži teme i plana



... INTERVJUISANJE KORISNIKA



- Što je intervju strukturiraniji, to je lakše za ispitivača, lakše je obraditi podatke koji se ovim putem dobiju
- Fleksibilniji intervjui omogućuju da se ispita više tema i da se otkriju relevantne stvari o sistemu kojih se ispitivač nije mogao setiti ili nije ni znao da postoje, ali su teži za ispitivača
- Prilikom osmišljavanja intervjua mora se voditi računa da pitanja ne usmeravaju odgovore korisnika (sugerišu korisniku šta da odgovori)
- Podatke dobijene fleksibilnim intervjuom je teže obraditi
- Dobra praksa je snimiti odgovore za kasniju analizu, ali voditi računa jer mogu uticati na ponašanje korisnika i odgovore koje budu davali (ljudi su obazriviji kada se zna da se njihovi odgovori snimaju)
  - Naravno, nikako ne snimati bez dozvole ispitanog!



GRUPNI INTERVJUI KORISNIKA

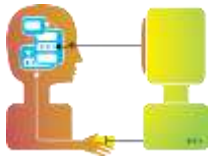
- Slično intervjuu jedan na jedan ali se sada intervjuiše grupa
- To su korisnici koji čine fokus softvera koji se razvija – focus group (workshops)
- Široki spektar mišljenja, ako se odradi kako treba
- Mora da prati dnevni red
- Ne sme da dominira jedna ličnost
- Treba podsticati sve učesnike





- Dobar da se dostigne koncenzus, ali i da se istakne ono oko čega dolazi do konflikta ili neslaganja

UPITNICI I ANKETE ...

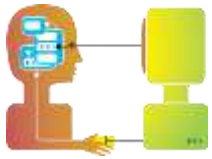


- Imaju potencijal da se u kratkom roku ispita veliki broj ljudi, pri čemu anketirani ne moraju sve vreme biti nadgledani
- Pošto ne moraju biti nadgledani, upitnici moraju biti dobro osmišljeni
  - Ne smeju biti dosadni
  - Pitanja moraju biti jednostavna i konkretna
  - Broj pitanja bi trebao da bude ograničen (da anketa ne traje predugo)
  - Pitanja ne smeju biti nejasna i dvosmislena
  - Pitanja treba da obezbede da se preuzme željena informacija
  - Omogućiti korisnicima da daju odgovore koji nisu predviđeni upitnikom



- Početi pilot studijom, pa nakon toga korigovari, pre nego što se ide na veliki broj ispitanika
- Ako je planirana statistička analiza (velikog broja) podataka, konsultovati statističara pre pravljenja upitnika, tj. uključiti ga u sve faze

... UPITNICI I ANKETE



- Drugačiji pristup u odnosu na intervju – sa fleksibilnog i prijateljskog se prelazi na formalne nedvosmislene upite za sakupljanje preciznijih informacija
- Gruba se mogu podeliti na dve vrste:
  - Upitnici otvorenog tipa
  - Upitnici zatvorenog tipa

BACKGROUND

Did you work with digiCamControl? no ○ ○ yes

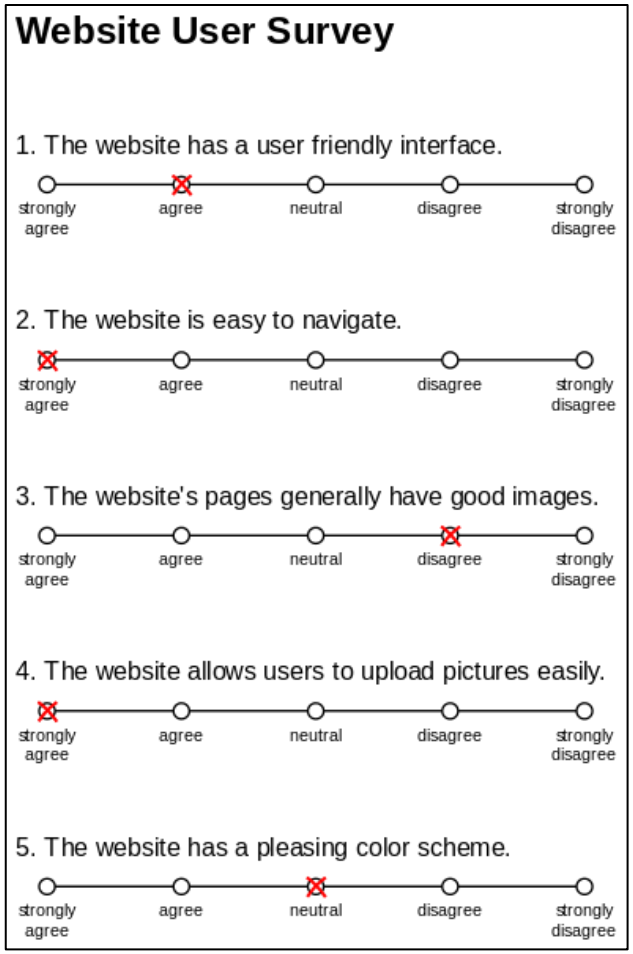
If the previous answer is yes, for how long did you work with it? ☐ months.

Optionally provide comments regarding your work with digiCamControl:

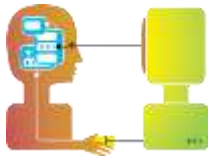
Criteria / Rating	Very satisfied	Satisfied	Neither satisfied nor dissatisfied	Dissatisfied	Very Dissatisfied
1-Accessibility					
The page capable of being understood and navigated even if users do not have the ability to identify specific colors or differentiate between colors.	☐	☐	☐	☐	☐
Access to information is quick and efficient.	☐	☐	☐	☐	☐
The time it takes for each Web page to load is acceptable. (the overall site speed is acceptable)	☐	☐	☐	☐	☐
The website is free from technical problems (hyperlink errors, programming errors etc.) (i.e. you did not face any technical problems while browsing this website)	☐	☐	☐	☐	☐
How satisfied are you from the accessibility of this web site?	☐	☐	☐	☐	☐
2-Interactivity					
This website provides popular forms of computer-mediated communication such as e-mail, video, audio or text chat with customers.	☐	☐	☐	☐	☐
You feel the this website is interactive (clear animations, hints, examples and messages)	☐	☐	☐	☐	☐
You have the sense of control while using this website.	☐	☐	☐	☐	☐
It is easy to find links on each webpage such as links back to the homepage, hyperlinks on key words, and hyperlinks suggesting additional information.	☐	☐	☐	☐	☐
You found customer support activities while using the website (such as status tracking, feedback options), market research activities (such as taking product surveys), personal choice helpers (such as key word searches, dealer locators) (i.e. it was easy for you to complete virtual buying process)	☐	☐	☐	☐	☐

UPITNICI ZATVORENOG TIPA ...

- Zahtevaju od korisnika da izabere
  - jedan od predefinisanih odgovora na postavljeno pitanje, ili
  - neku od ponuđenih vrednosti na nekoj skali vrednosti
  - šta od ta dva zavisi on nivoa detalje do kojih se želi doći upitom
- Multipoint rating scale (semantic differential)
  - Preporučuje se oko 7 vrednosti, mada se često koristi i skala sa 3, 5 ili 10 vrednosti
  - Početna i krajnja vrednost moraju zaista biti alternativne (suprotne) čime se pomaže korisniku da lakše odredi na kojoj poziciji se nalazi njegov/njen odgovor
  - Par dobro osmišljenih prideva je bolje od "očajno" do "odlično"



... UPITNICI ZATVORENOG TIPA

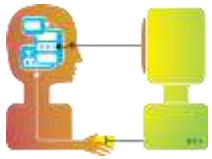


- Likertova skala (Likert scale)
  - Psihometrijska skala za skaliranje odgovora
  - Odgovori su dizajnirani tako da odogovaraju statistici grupe ljudi koji će biti anketirani
  - I tako da krajnji elementi imaju jaku pozitivnu, odnosno negativnu konotaciju u odnosu na postavljeno pitanje

EASE OF LEARNING

		-3	-2	-1	0	1	2	3		NA
I learned to use it quickly.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
I easily remember how to use it.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
It is easy to learn to use it.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
I quickly became skillful with it.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
Exploring new features by trial and error is easy.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
Help messages on the screen are helpful.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
Help content is clear.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
Tutorial content is clear.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
Help content is helpful.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O
Tutorial content is helpful.	strongly disagree	O	O	O	O	O	O	O	strongly agree	O

UPITNICI OTVORENOG TIPA



- Omogućuju korisniku da napiše šta god želi u svom odgovoru
- Obično počinju sa:
  - "Šta?", "Kako?", ...
- Ograničavanjem prostora za odgovor ili broja reči može uticati na korisnika da bude koncizan i precizan, što olakšava kasniju analizu
- Ovi upitnici su bogatiji u količini informacija koje nose, ali su teži za analizu
- Mogu se kombinovati sa upitnicima zatvorenog tipa

Optionally provide comments regarding ease of learning:

PROUČAVANJE DOKUMENTACIJE/SOFTVERA

- Proučavanje dokumentacije
  - Relevantna literatura, poslovnici, zakoni, uputstva
  - Dobro za shvatanje osnovnog o zadatku koji treba da se podrži
  - Dobro za prikupljanje podataka o zadacima i akcijama
  - Ne zahteva vreme korisnika
  - Uvek mora da se proučava u svetlu korisnika, jer korisnici možda to ne rade tako kako bi trebalo po dokumentaciji, pravilniku ili zakonu...



- Proučavanje softvera sličnog tipa
  - Dobro da se brzo dođe do nekih ideja
  - Dobro da se brzo dođe do inicijalnih zahteva
  - Dobro da se brzo dođe do alternativnog dizajna



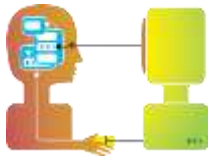
UPOZNAVANJE KORISNIKA ...

- Kako programeri korisničkog interfejsa nisu i korisnici sistema, potrebno je saznati ko su zapravo korisnici sistema
- Toliko očigledno, da se ponekad na to zaboravi, ali zaboraviti korisnika je direktan put ka lošem interfejsu
- Odrediti primarne korisnike (one koji će zaista koristiti softver, ali i tu ima nekoliko nivoa – onaj koji koristi softver, onaj koji plaća/naručuje softver, softverska podrška) ali i sekundarne (koji su pogođeni akcijama softvera)



  - Operator na šalteru Informatike je direktni primarni korisnik (koristi softver), njegov direktor je indirektni primarni korisnik (naručilac softvera), građanin koji plaća račun je sekundarni korisnik
- Fokus je uvek na primarnim (direktnim) korisnicima, pa tek posle na sekundarnim
- Zajedno se zovu stakeholders (zainteresovane strane)

... UPOZNAVANJE KORISNIKA



- Postoje različiti aspekti korisnika koje je potrebno saznati da bi se korisnik sistema u potpunosti analizirao/upoznao
  - Karakteristike korisnika
  - Profili korisnika
  - Šta korisnik želi
    - Želje koje iskazuju u samom razgovoru (expressed needs)
    - Želje koje se identifikuju tokom razgovora a za koje korisnik možda ni ne zna da ih ima (felt needs)
  - Šta korisnik očekuje



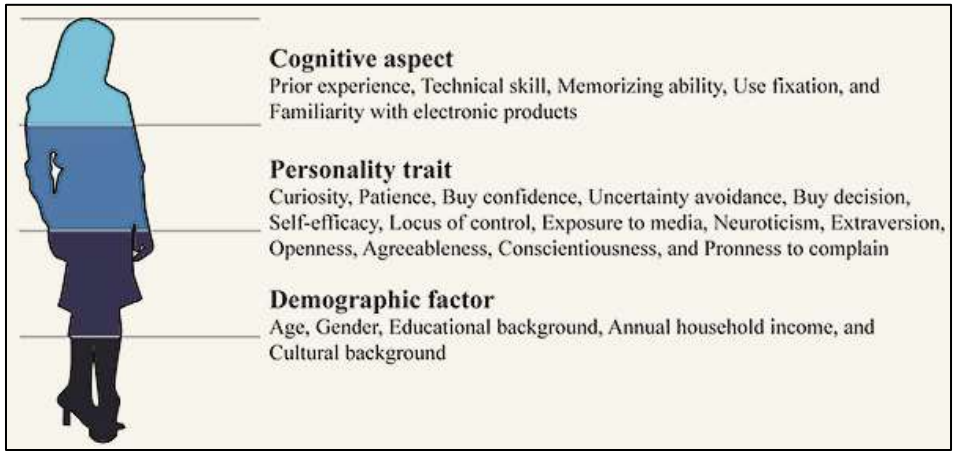






KARAKTERISTIKE KORISNIKA ...

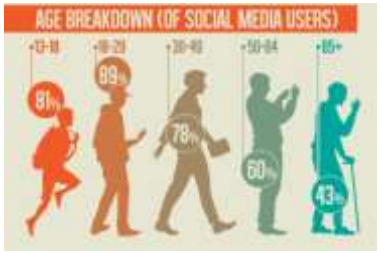
- Važan aspekt dizajniranja korisničkog interfejsa je da se obezbedi da kor. interfejs odgovara atributima korisnika, tj. njegovim karateristikama
- Tipične karakteristike:
  - Pol, godište, kulturološka grupa (prepoznavanje ikonica), jezik koji govore
  - Obrazovanje (pismenost, računarska pismenost, snalaženje sa brojevima)
  - Fizičko ograničenje
  - IT znanje, veštine (da li zna da kuca?) i iskustvo
  - Psihološke karakteristike korisnika – motivacija i stav preme upotrebi računara i digitalizaciji u njihovom poslu (koliko god da je dobar ili loš softver uspeh prečesto zavisi od stava korisnika prema upotrebi računara u njihovom poslu – npr. PACS, doobia)
  - Iskustvo u upotrebi domenskih aplikacija



... KARAKTERISTIKE KORISNIKA ...

- Mnoge aplikacije imaju različite vrste korisnika
  - korisnici imaju različite uloge (student, nastavnik, administrator)



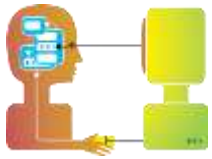


  - različiti korisnici prema njihovim karakteristikama (godište, motivacija)
  - persone (svaka korisnička klasa dobija fiktivnog predstavnika





... KARAKTERISTIKE KORISNIKA ...

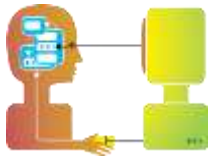


- Profilisanje korisnika je lakše kada je korisnik poznat
  - Pitate konkretne korisnike
  - Kada se ne zna kosu tačno korisnici, to se mora istražiti intervjuisanjem adekvatnih ljudi u organizaciji (domenski eksperti, menadžeri)

able 3.3 User Profile of ATM Customers (from Chapanis, 1996)

User characteristics	ATM customer characteristics
Age	Will range in age from about 12 to 80+
Sex	Both male and female
Physical limitations	May be fully able-bodied or may have some physical limitations in relation to hearing, sight, mobility, use of hands, or wheelchair use Will be of varying heights
Educational background	May have only minimal education qualifications and possess limited literacy and numeracy skills
Computer/IT use	May have little or no prior experience of computer or IT use
Motivation	May be very motivated to use the ATM, particularly if they can do their banking quickly and avoid waiting in long lines at the bank
Attitude	Attitudes to use may vary, depending on the services the ATM offers, the reliability of the technology itself, and the attitude of users toward computers

... KARAKTERISTIKE KORISNIKA ...

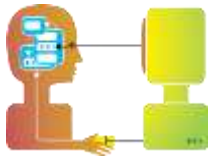


- Identifikovati grupe korisnika unutar opšte populacije korisnika
  - Identifikovanjem grupa, mogu se identifikovati i načini ponašanja

Table 3.4 ATM User Groups (adapted from Stone, 2001)

User characteristic	ATM customer characteristics, by group		
	Teens/young adults	Young adults to middle age	Middle age to senior citizens
Age	12 to 25.	25 to 50.	50 to 80+.
Sex	Both male and female.	Both male and female.	Both male and female.
Physical limitations	May be fully able-bodied or may have some physical limitations in relation to, for example, hearing or sight. Will be of varying heights.	May be fully able-bodied or may have some physical limitations in relation to, for example, hearing or sight. Will be of varying heights.	May be fully able-bodied or may have some physical limitations in relation to, for example, hearing or sight, mobility, or use of hands. Will be of varying heights.
Educational background	May have minimal or no educational qualifications.	May have only minimal educational qualifications.	May have only minimal educational qualifications.
Computer/IT use	Probably have some prior experience of computer or IT use.	May have little or no prior experience of computer or IT use.	May have little or no prior experience of computer or IT use.
Motivation	Probably very motivated to use the ATM, especially in relation to their banking habits.	Could be very motivated to use the ATM, especially if they can do their banking quickly and avoid standing in line at the bank.	Could be very motivated to use the ATM, but would probably prefer to stand in a line in the bank.
Attitude	Attitudes to use may vary, depending on the services the ATM offers and the reliability of the technology itself.	Attitudes to use may vary, depending on the services the ATM offers and the reliability of the technology itself.	Attitudes to use may vary, depending on the services the ATM offers and the reliability of the technology itself.

... KARAKTERISTIKE KORISNIKA

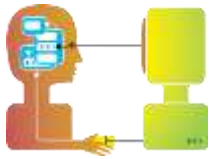


- Uvek proveriti pretpostavke
- Nakon čega sledi analiza i pretvaranje zaključaka u zahteve

Table 3.5 Translating User Characteristics into UI Design Requirements (middle age to senior citizen group)

User characteristics	ATM UI requirements
Age range from 12 to 80+	ATM screen height needs to accommodate users of varying height.
May be fully able-bodied or may have some physical limitations	ATM screen height needs to accommodate able-bodied users as well as users with walking sticks or those who use wheelchairs. Arthritis of the hands could be a problem, so any controls used should accommodate this.
May have some physical limitations in relation to hearing	All user inputs should have both visual and auditory feedback.
May have some physical limitations in relation to sight	Screen text should be of a reasonably large font, in order to be read by both the visually impaired and unimpaired.
May have some physical limitations in relation to use of hands	Touchscreens, if used, should have target areas that are large enough to locate with limited manual dexterity. Touchscreens, if used, should be sensitive enough to respond to users with decreased strength in fingers or hands.
Little or no experience of computer/IT use	The application should be easy to use (i.e., the tasks users want to undertake should be simple to perform). The application should be easy to learn (i.e., the user should be able to use the system without help, training, or instruction).

GREŠKE U ANALIZI KORISNIKA



- Opis idealnog korisnika, umesto onog što zaista jeste
  - Npr. idealni student, umesto onakvog kakav je
  - Npr. svi naši korisnici imaju Android telefon
  - Npr. korisnik zna da koristi računar (JKP Parking servis primer)
- Opis kakav bi korisnik trebao da bude a ne kakav jeste je zapravo iznošenje zahteva a ne karkteristika postojećeg korisnika
  - To nije ono zbog čega se radi analiza korisnika, već da se vidi da li nešto nedostaje i šta treba da se doradi da se ispune zahtevi



"Yes, I'm a real Genie... but you're asking me to understand your client's requirements and even I can't do that!"

ANALIZA DOMENA ...

- Odnosi se na ekspertsko i specijalističko znanje, domen za koji se aplikacija razvija
- To potrebnih informacija se dolazi ispitivanjem eksperata iz datog domena, proučavanjem dokumentacije i literature iz date oblasi (za razvoj finansijskog softvera moraju se konsultovati stručnjaci iz oblasti finansija, ali i odgovarajuća literatura, ako i zakoni)
- Pored eksperata uvek uključiti i primarne korisnike

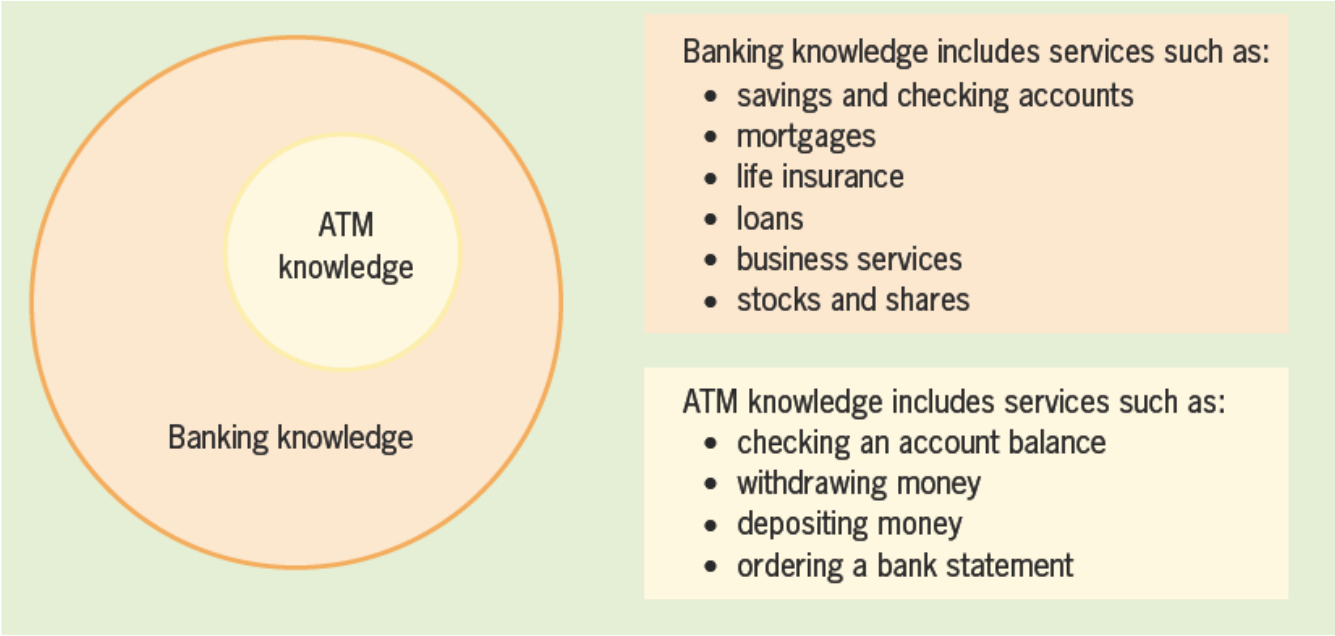
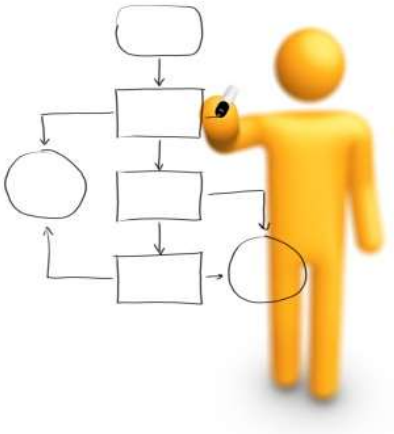
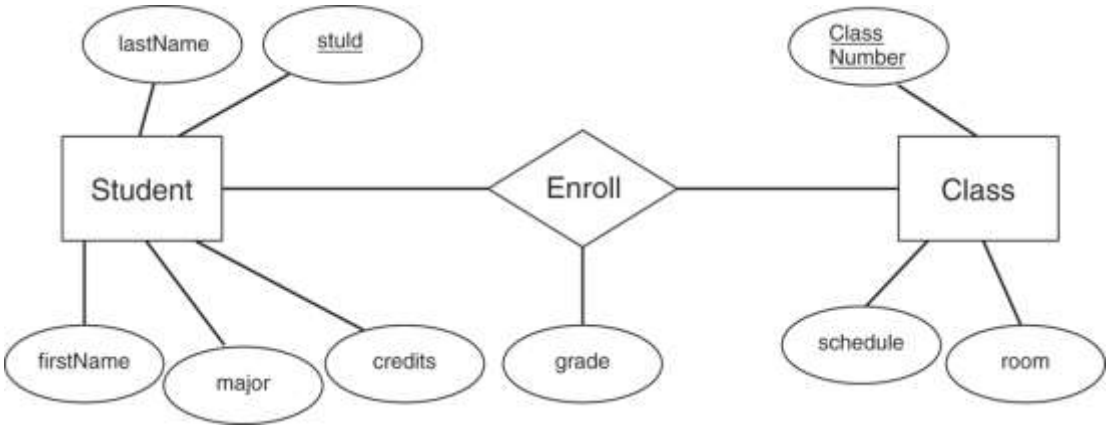


Figure 3.3 ATM Domain Knowledge Is a Subset of Total Banking Domain Knowledge. (From Stone, 2001.)

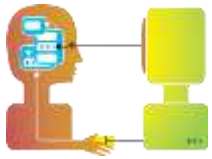
... ANALIZA DOMENA

- Osnovni zadatak ove analize je da identifikuje entite domena, tj. sve stvari koje su uključene u proces
- Entiteti domena:
  - Realni objekat (osoba, kuća, dokument, automobil, ...)
  - Apstraktni koncept (veličina, boja, radno mesto, ...)
  - Događaj (rođenje, upis, isplata, ispit, ...)
  - Odnos (nastavnik-predmet, proizvod-prodacnica, ...)
- Modelovanje domena
  - Dijagram toka podataka
  - ER dijagram
  - Dijagrama stanja





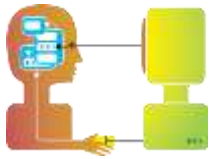
ANALIZA ZADATKA ...



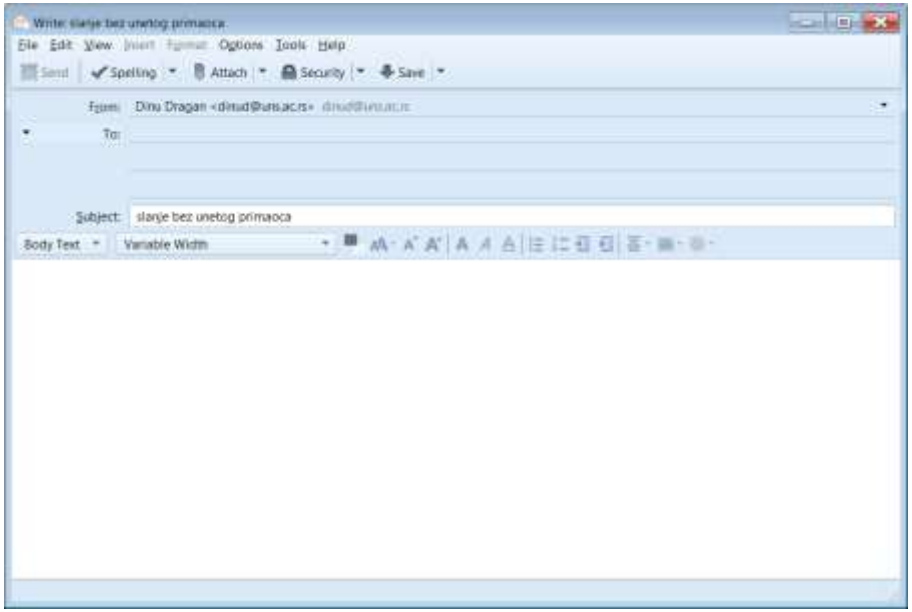
- Da bi se napravio dobar sistem sa dobrim korisničkim interfejsom, moraju se odrediti ciljevi koje korisnik namerava da ostvarim pomoću tog sistema
- Ovaj korak analize ima za zadatak da odredi koja je namena sistema
- Korisnički posao može se opisati kroz
  - Ciljeve (goals) – krajnji rezultat koji se želi ostvariti, opisano na višem nivou apstrakcije (Šta se želi ostvariti)
  - Zadatke (tasks) – strukturirani skup povezanih aktivnosti koji se primenjuju u sekveni kako bi se ostvario željeni cilj, neki ciljevi se mogu ostvariti na različite način (različiti zadaci)
  - Akcije (actions) – individualna operacija ili korak koji se moraju izvesti kako bi se realizovao zadatak



... ANALIZA ZADATKA ...

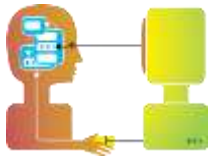


- Neki zadaci se mogu podeliti u podzadatke
- Da bi se neki ciljevi mogli ostvariti, potrebno je znati i preduslove (preconditions)
  - Primer: pre nego što počne da koristi facebook, korisnik mora da se loguje na sistem
  - Preduslovi su značajni za dobar dizajn kor. interfejsa
  - Omogućuju da se identifikuju situacije koje će dovesti do greške, jer korisnici često ne ispune preduslove
  - Samim tim omogućuju da se greške izbegne ili da se učine bezazlenim



  - Primer: da bi se poslao mejl, mora se znati adrese primaoca, mejl klijent ne dozvoljava klik na send dugme ako nije uneta adresa primaoca

... ANALIZA ZADATKA ...

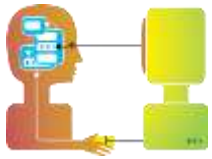


- Analiza korisničkih zadataka dovode do toga da se identifikuje šta treba da se uradi, ali ne i kako
- Hijerarhijska dekompozicija je dobar pristup u analizi zadatka; krene se od osnovnog (glavnog) zadatka sistema, te se on dekomponuje na svoje podzadatke
- Kao i korisnici i zadaci imaju svoje karakteristike koje utiču na odluke u izboru i dizajniranju kor. interfejsa, kao i na izbor interakcionih uređaja

Table 4.1 Characteristics of Tasks

The extent to which tasks vary from one occasion to another
Whether tasks will be carried out regularly, infrequently, or only once
The knowledge and kinds of skill required to perform tasks
How much the work is affected by changes in the environment
Whether time is critical for the work
Whether there are safety hazards
Whether the user will do the work alone or with others
Whether the user will normally be switching between several tasks

... ANALIZA ZADATKA ...

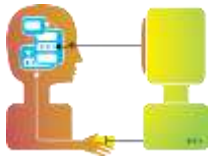


- Primer: karakteristike zadatka u podizanju novca sa bankomata

Table 4.2 Task Characteristics for Withdrawing Money from an ATM

Does the task vary from one occasion to the next?	No.
How frequently is the task carried out?	May be daily, weekly, or less frequently.
What kinds of skills or knowledge are needed?	Must remember PIN to access machine.
Is the task affected by the environment?	Weather conditions could affect use of machine (e.g., the user may be wearing gloves in winter, it may be raining, bright sunlight may make reading the display difficult).
Is the task time critical?	Users may be in a hurry when using the ATM, since ATMs are often used for their speed and convenience.
Are there any safety or security hazards?	There are no safety hazards in the use of the ATM itself. However, the users' personal safety in relation to onlookers and the safeguarding of their PINs and the cash withdrawn are considerations.
Will the work be done alone or with others?	The work will be done alone.
Will the users normally be switching between several tasks?	Many users will check their balance before withdrawing money. The users will not switch between tasks when withdrawing money, but external factors (like children) may divert their attention.

... ANALIZA ZADATKA ...

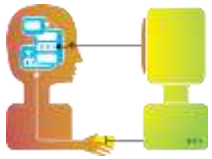


- Do jednog cilja može se doći na više načina

Table 4.3 Different Task Sequences for Sending a Letter

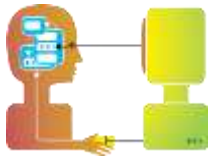
How Bill Smith sends a letter	How Brenda Jones sends a letter
Write the letter.	Get an envelope.
Get an envelope.	Address the envelope.
Address the envelope.	Write the letter.
Put a stamp on the envelope.	Put the finished letter in the envelope.
Put the finished letter in the envelope.	Put a stamp on the envelope.

... ANALIZA ZADATKA



- Do podataka o zadacima dolazi se intervjuom i observacijom korisnika
- Pitanja o zadacima:
  - Gde se izvršava zadatak?
  - Kakvo je okruženje u kojem se izvršava zadatak? Prljavo, bučno, opasno?
  - Koliko se često izvodi zadatak?
  - Koji su vremenski okviri za realizaciju zadatka?
  - Koji su resursi na raspolaganju a potrebni da se izvede zadatak?
  - Na koji način se stiče znanje potrebno da se realizuje zadatak?
  - Šta može poći po zlu? (Exceptions, errors, emergencies)
  - Ko još učestvuje u zadatku?

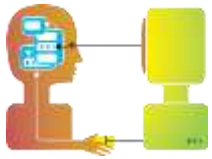
GREŠKE U ANALIZI ZADATKA



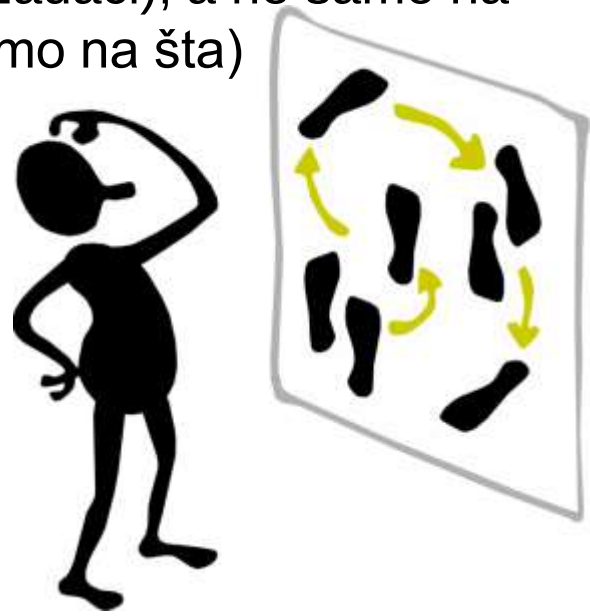
- Razmišljanje iz aspekta sistema a ne korisnika
  - "Obavestiti korisnika o sastanku," (ovo je zahtev)

  a treba:
  - "Obavesti me o sastanku" (zadatak)
  - Nekada je to samo stvar semantike, ali je često fokus na to šta sistem može, a ne šta korisnik želi
- Suviše rano se fokusira na dizajn interfejsa
  - "Sistemsko zvono će obavestiti korisnika ..." (to je pre detaljno i zavisi od implementacije)
- Beleži se šta korisnik zna (konkretno), a ne šta radi (esencijalno)
  - Čuvanje fajla na disk umesto postaram se da je moj rad sačuvan
- Dupliranje loše prakse iz drugog softvera
- Ignorisanje dobrih stvari iz prakse i njihovo ne prenošenje u sistem (npr. beleške asistenta na dokumentima)

KAKO DO BOLJE ANALIZE KORISNIKA/ZADATKA ...



- Koncetrisati se na pitanja **zašto** (cilj) i **kako** (podzadaci), a ne samo na šta (jer korisnici teže da se fokusiraju upravo i samo na šta)
  - To će da učini analizu apstraktniju, ali i detaljniju (precizniju)
- Tražiti slabosti i probleme u tome kako korisnici ostvaruju ciljeve i rade svoje zadatke (u trenutnoj situaciji)
  - Koji zadaci često rezultuju neuspehom? Zašto?





  - Koji su zadaci nevažni a oduzimaju previše vremena?
  - Koji zadaci nerviraju korisnike? Kako bi korisnici to popravili?

... KAKO DO BOLJE ANALIZE KORISNIKA/ZADATKA

- Istraživanje (ispitivanje) konteksta korisnikovog rada
  - Observacija rada korisnika u realnom okruženju
  - Konkretna pitanja
  - Uspostavljanje "master-apprentice"
    - Korisnik pokazuje "Kako" i priča o tome
    - Intervjuista gleda, hvata beleške i postavlja pitanja
  - Dovođenje u pitanju pretpostavki i stalno preispitivanje





- Participatory Design (uključivanje svih učesnika u dizajn kor. interfejsa)
  - Uključiti reprezantivne korisnike u sam dizajn (uvek! ali i u sve faze, od prikupljanja zahteva i njihove analize)
  - Npr. kako napraviti aplikaciju za knjigovodstvo bez konsultacija sa iskunim knjigovođom (ekspertsko znanje)

FAKTORI U FORMIRANJU ZAHTEVA

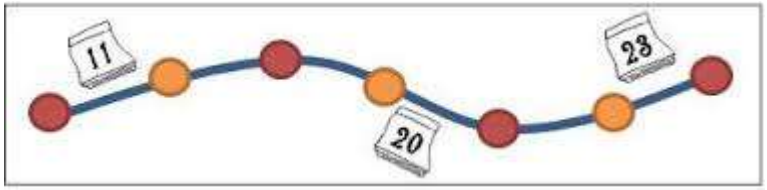
- Tokom razvoja rač. sistema uvek će postojati ograničenja i kompromisi
  - Cena razvoja/budžet/rokovi
  - Dostupna tehnologija i povezanost (interoperabilnost) sa hardverom i drugim softverom
  - Stavovi/motivi individualnih zainteresovanih strana (stakeholders)
  - Kontradiktorni zahtevi
  - Organizaciona politika



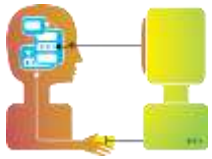




"The client kept changing the requirements on a daily basis, so we decided to freeze them until the next release."



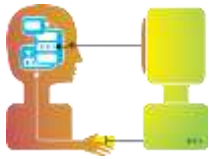
PROBLEMI TOKOM FORMIRANJA ZAHTEVA



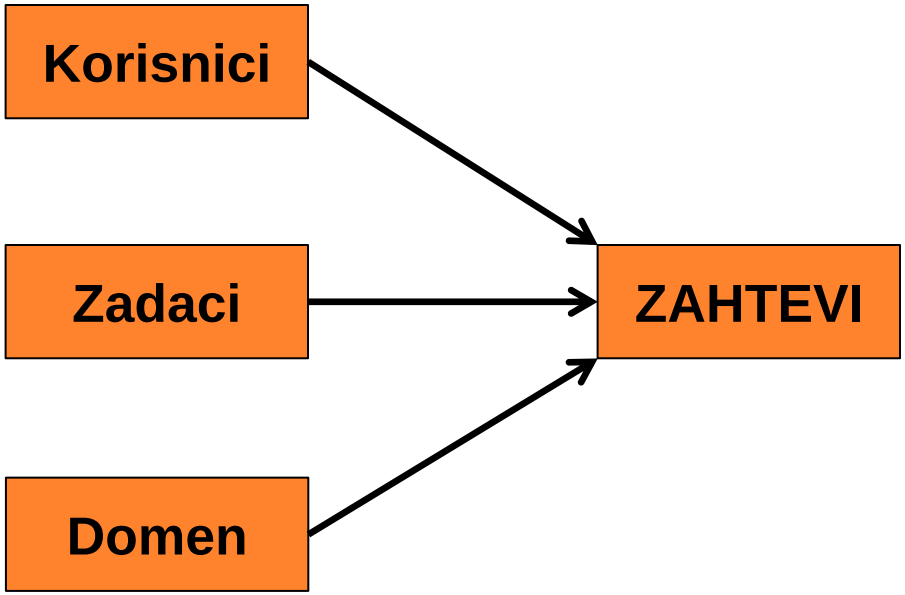
- Velika lista ograničenja i problema, na neke se ne može uticati
  - Nedovoljan broj korisnika/zainteresovanih strana, nekompletna lista zahteva
  - Loša evidencija zahteva i promena funkcija usled otkrivanja novih zahteva, može dovesti do ispuštanja nekog zahteva ili pogrešne interpretacije
  - Loša koordinacija faze prikupljanja zahteva
  - Loša komunikacija između korisnika na različitim nivoima odlučivanja, loša komunikacija sa program. timom
  - Određivanje domena nije uopšte lako i trivijalno, greške mogu rezultovati pogrešnim zahtevima
  - Ljudi koji razumeju problem možda nisu dostupni ili zainteresovani
  - Organizacioni i politički faktori
  - Zainteresovani učesnici ne znaju šta žele od sistema
  - Okruženje se stalno menja kao i ljudi na pozicijama, te zahtevi mogu zastareti usled tih promena
  - Prenagljeno zaključivanje (opis idealnog stud. a ne kakvi su stud. zaista)



FORMIRANJE ZAHTEVA ...



- Zahtevi vezani za interfejs su deo zahteva čitavog sistema
- Trebaju da obezbede opis onog šta sistem treba da uradi, ne i kako
- Analizom skupljenih podataka iznose se zaključci koji se beleže u dokument (specifikaciju) zahteva



Dokument zahteva:

1. Arhitektonski zahtevi (opisuje potrebe arhitekture sistema)
2. Funkcionalni zahtevi (funkcionalnost koja se mora izvršiti)
3. Nefunkcionalni zahtevi (karakteristike sistem kao zahtevi kvaliteta)
4. Zahtevi ograničenja (margina rada sistema)

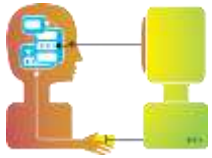
... FORMIRANJE ZAHTEVA ...

- Ne postoji standardni način da se napiše dokument zahteva
  - Zahtevi koji se odnose na karakteristike korisnika
  - Zahtevi koji se odnose na zadatke i njihove karakteristike
  - Usability zahtevi
  - Zahtevi koji se odnose na domen i radno okruženje
  - Limitirajući faktori, ograničenja i preduslovi
  - ...
- Pišu se prirodnim jezikom koji može svako razumeti, jer će ga čitati ljudi različitog profila
  - Govorni jezik može predstavljati i problem jer se može interpretirati na više načina
- Smernice za pisanje dokumenta zahteva



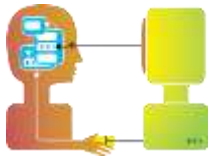


... FORMIRANJE ZAHTEVA



Guideline	Description
Define standard templates for describing requirements.	You should define a set of standard formats for different types of requirements and always express requirements using that format. This makes it less likely that important information will be missed out and makes it easier for the reader to understand the different parts of the requirement.
Use language simply, consistently, and concisely.	Do not write requirements using convoluted language, but follow good writing practice such as short sentences and paragraphs, using lists and tables and avoiding jargon wherever possible.
Use diagrams appropriately.	You should not develop complex diagrams but should use diagrams to present broad overviews and to show relationships between entities.
Supplement natural language with other descriptions of requirements.	Do not try to write everything in natural language. If readers of the requirements document are likely to be familiar with other types of notation (e.g., equations), you should not hesitate to use these notations.
Specify requirements quantitatively.	Wherever possible, you should specify your requirements quantitatively. This is often possible when you are specifying the properties of a system such as reliability, usability, or performance.

FORMIRANJE ZAHTEVA: POTREBNO ZNANJE



- Na kvalitet, sveopštost i tačnost dokumenta zahteva, pored informacija o korisniku, zadacima, domenu, potrebno je i znanje iz samog dizajna korisničkog interfejsa
- Ovo znanje dolazi iz teorije, kognitivne psihologije i delom iz iskustva koje dolazi proučavanjem dobrog korisničkog interfejsa



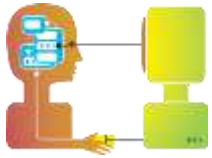
- Upotreba specifičnog i specijalizovanog znanja, formulisanog kroz:



  - smernice (abstrakna upustva visokog nivoa opštosti koje zahtevaju interpretaciju da bi se mogli primeniti) i
  - pravila (detaljna, specijalizovana uputstva niskog nivoa opštosti koja se mogu pratiti uz minimalno interpretacije)

- Teško definisati, nisu apsolutna i zavisi od izvora uputstva kao i konteksta upotrebe
- Treba uvek primenjivati obazrivo i u skladu sa kontekstom – uzimati u obzir korisnike, njihove zadatke i domen

ČETIRI PSIHOLOŠKA PRINCIPA



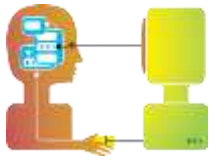
1. **Korisnici vide ono što očekuju da vide.**
   Ako su OK dugme i Cancel na jednom dijalogu ne jednoj poziciji, a na sledećem su na drugoj poziciji, korisnik na drugom dijalogu može videti da je OK na mestu sa prvog dijaloga

2. **Korisnici se fokusiraju samo na jednu stvar.**
   U bučnom okruženje (velika gužva) korisniku može da odluta pažnja, kor. interfejs treba da ga podseća na to šta treba da uradi sledeće.

3. **Lakše je opažati strukturirani razmeštaj kontrola (layout).**
   Teško je uočiti gde treba klilnuti, ako korisnički interfejs ne daje nagoveštaje gde se može kliknuti (linijama, bojom, senkom).

4. **Lakše je prepoznati nešto nego se podsećati.**
   Daleko je lakše prepoznati nešto sa menija ili dugmeta, nego se prisećati komande.

KORISNICI VIDE ONO ŠTO OČEKUJU DA VIDE



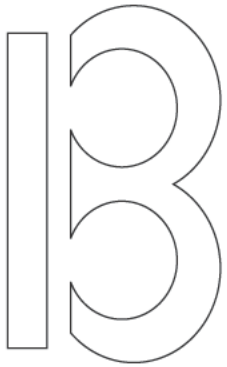


Figure 1.3 An ambiguous shape?





Figure 1.4 ABC

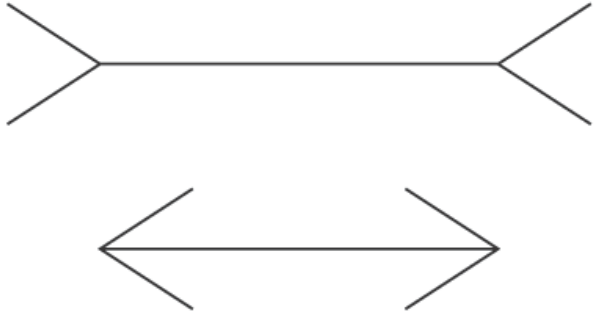
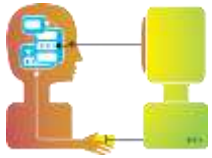


Figure 1.6 The Muller–Lyer illusion – which line is longer?

KOR. SE FOKUSIRAJU SAMO NA JEDNU STVAR

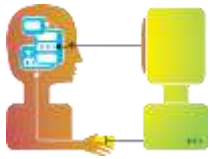


The Cocktail Party Effect

Do you remember the last time you were at a party or in a crowd and you were surrounded by a sea of faces and a babble of voices? How long was it before you found yourself turning your attention to one specific conversation? What happened to the other voices and faces? Did they simply become a blur? While being involved in the one conversation, did you find yourself overhearing anyone else's conversation? Perhaps it was a piece of gossip, which you could not resist listening to. What happened when you tuned in to this other conversation? Were you able to carry on with your conversation or did you become distracted?

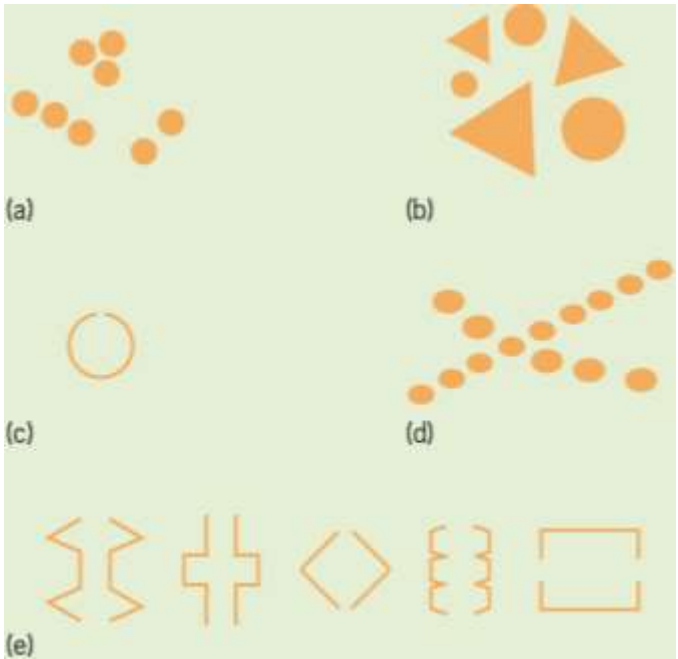
This everyday experience of focusing on one particular activity while switching between others has become known as the **cocktail party phenomenon** (Cherry, 1953). You probably found that, after the initial impression of chaos, you found yourself attracted to one group and one conversation, and the others faded. But, for example, if you heard your own or a familiar name mentioned elsewhere in the room, you may have found that your attention switched in this new direction and that you then lost the thread of the former conversation.

LAKŠE JE OPAŽATI STRUKTURIRANI LAYOUT



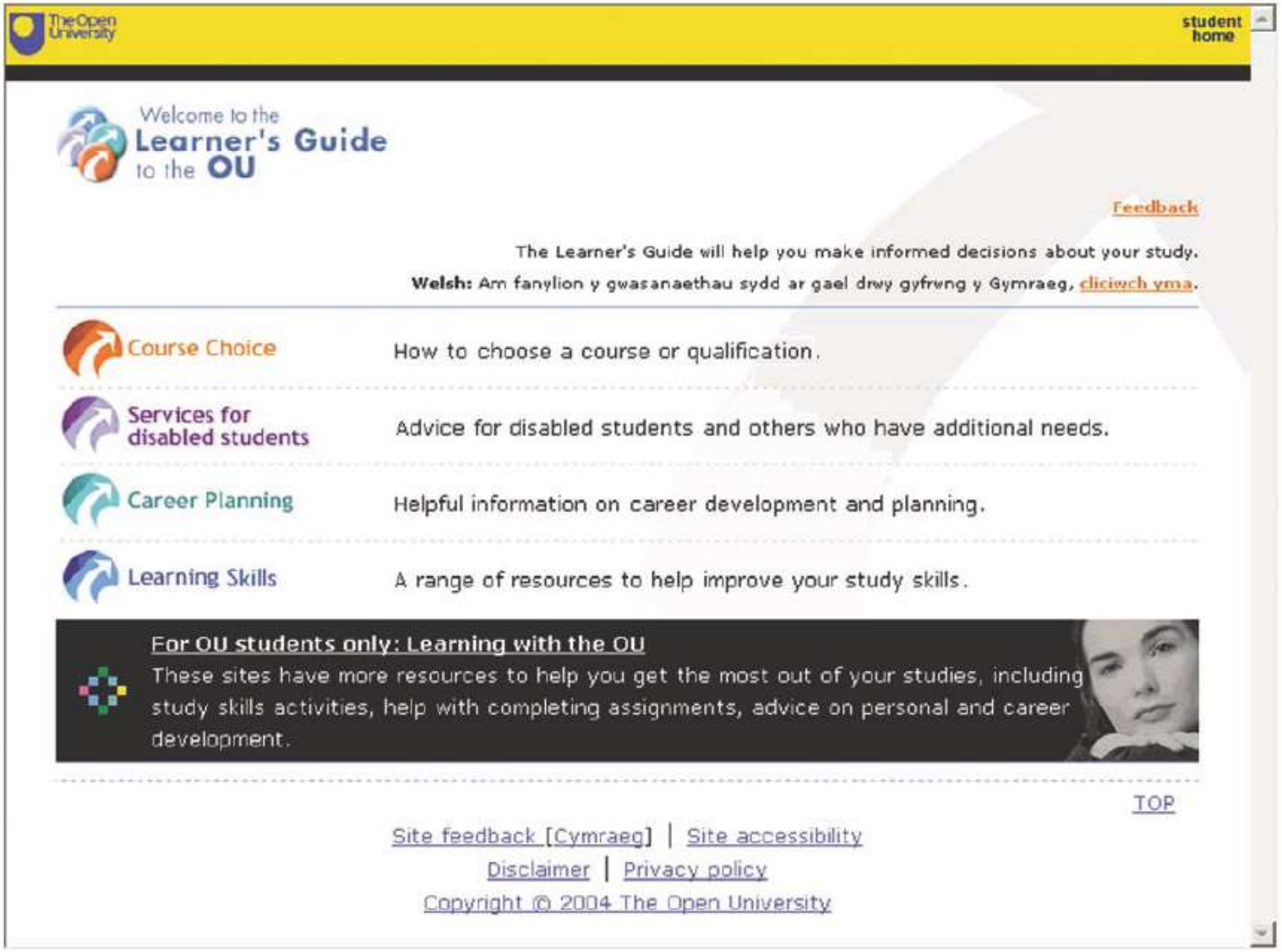
Destination	Flight	Carrier	Depart	Arrive	Rates Business	Standard
Aberdeen	4171	BA	0845	0945	£155	£102
Dublin	664	FR	1035	1135	£149	£100
Toulouse	8064	AF	1110	1410	£307	£182
Frankfurt	4618	LH	1115	1355	£222	£152
Amsterdam	2045	UK	1130	1335	£222	£152
Copenhagen	8363	BA	1145	1445	£315	£187
Paris-CDG	1803	BA	1150	1400	£248	£165
Exeter	446	JY	1205	1305	£155	£102
Glasgow	1903	BA	1210	1310	£155	£102
Munich	4526	LH	1225	1525	£301	£179
Geneva	8413	BA	1235	1420	£222	£152
Aberdeen	4172	BA	1245	1345	£155	£102

- Zakoni: blizine, sličnosti, zatvorenosti, kontinualnosti, simetrije

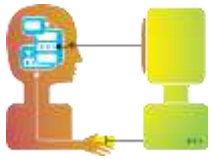


Aberdeen (BA4171) Dep: 0845; Arr: 0945
(B/S: £155/102)
Dublin (FR664) Dep: 1035; Arr: 1135
(B/S: £149/100)
Toulouse (AF8064) Dep: 1110; Arr: 1410
(B/S: £307/182)
Dest: Frankfurt (LH4618) Dep: 1115; Arr: 1355
(B/S: £222/152)
Dest: Amsterdam (UK2045) Dep: 1130; Arr: 1335
(B/S: £222/152)
Dest: Copenhagen (BA8363) Dep: 1145; Arr: 1445
(B/S: £315/187)
Dest: Paris-CDG (BA1803) Dep: 1150; Arr: 1400
(B/S: £248/165)
Dest: Exeter (JY446) Dep: 1205; Arr: 1305
(B/S: £155/102)
Dest: Glasgow (BA1903) Dep: 1210; Arr: 1310
(B/S: £155/102)
Dest: Munich (LH4526) Dep: 1225; Arr: 1525
(B/S: £301/179)
Dest: Geneva (BA8413) Dep: 1235; Arr: 1420
(B/S: £222/152)
Dest: Aberdeen (BA4172) Dep: 1245; Arr: 1345
(B/S: £155/102)

LAKŠE JE PREPOZNATI, NEGO SE PODSEĆATI



FORMIRANJE ZAHTEVA: PRINCIPI IZ ISKUSTVA



- Vidljivost (visibility):
  Da li su kontrole vidljive, jasne?
- Očiglednost, dostupnost (affordance):
  Da li je očigledno kako se kontrole koriste ?
- Obaveštavanje (feedback):
  Da li je jasno iskazana reakcija sistema na korisničku akciju



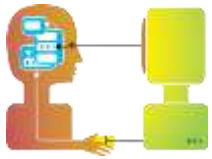
Figure 5.7 Door handles. The one on the left affords pulling. The one on the right affords levering. What about the one in the middle?

Visibility, Affordance, and Feedback

The controls on newer VCRs are generally more visible than on older models. There are good mappings between the controls and their effects — that is, the functions provided by the VCR interface map onto or are related to the user's goals and needs. Controls often have just one function; for example, there may be several push buttons that do only one thing, like an on/off button that simply turns the VCR on or off. There is affordance, as the buttons make it obvious how they are operated (i.e., they afford pressing). There is feedback on the display, and the system is generally understandable. In general, the relationships between the user's goals, the required actions, and the results are sensible, meaningful, and not arbitrary.

SMERNICE ZA RAZVOJ KORISNIČKOG INTERFEJSA

SMERNICE ...



- Razlikuje se u nivou detalja i tipu, od opštih do platformskih, od dizajnerskih do programerskih
- Smernice za razvoj na Microsoft platformi

  https://msdn.microsoft.com/en-us/library/windows/desktop/dn688964(v=vs.85).aspx



- Smernice za razvoj na Mac platformi

  https://developer.apple.com/go/?id=osx-hig

- Smernice za razvoj na Andriodu

  https://developer.android.com/guide/practices/ui_guidelines/index.html

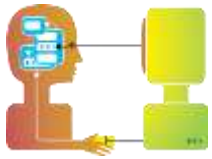
- Smernice za razvoj na iOSu

  https://developer.apple.com/ios/human-interface-guidelines/overview/design-principles/

- Smernice za razvoj na Linuxu

  https://developer.gnome.org/hig/stable/

... SMERNICE



- Opštih smernica ima jako puno, kao da svaki istraživač ima svoj skup smernica
- Dosta se preklapaju i razlikuju se u nivou detalja i broju stavki
  - Stručnjaci se slažu (manje-više) oko toga šta čini dobar kor. interfejs i nakojim principima se bazira dobar usability, već se ne slažu oko toga kako te principe organizovati u mali kompaktan skup upustava

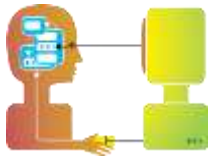


- Primeri smernica:
  - Ben Šnajdermanovih „osam zlatnih pravila“
  - Džob Nilsenova pravila
  - Normanova pravila
  - Tognazzini principi





OSAM ZLATNIH PRAVILA ...



Shneiderman's 8 Golden Rules of Interface Design

The principles	Questions to consider	Mark Complete
1. Strive for consistency	Is the style of this element maintained across your site/app? Is this content placed in the correct location according to the site hierarchy? Does this follow the conventions for your chosen platform? How can you make your designs more consistent?	☐
2. Enable frequent users to use shortcuts	Are there shortcuts available for your more experienced users? Who is this product designed for? Will there be a need to consider experienced users? How can you make it easier and quicker for experienced users?	☐
3. Offer informative feedback	Does the user know where they are at in the process? Does the user know what they have done after performing this action? How are you communicating this feedback to your user?	☐
4. Design dialogue to yield closure	Does the user have to do any guessing here? Is it clear and obvious enough for your intended audience? Are there any next steps for the user? How are you communicating the system status with the user?	☐
5. Offer simple error handling	Have you done everything imaginable to prevent this error from happening on your end? Is this error avoidable in the first place? If the user does make an error, how easy is it for them to fix it?	☐
6. Permit easy reversal of actions	How many steps does the user have to take to reverse their actions? Will the user quickly realize they need to reverse the action in the first place? How can you make your users detect the possibility of reversal?	☐
7. Support internal locus of control	Will the user feel in control at this specific touch point in your app? Will they be surprised in an unpleasant manner? Does the site feel easily navigable? Does the user feel safe and in control? How can you make the user feel more safe and in control?	☐
8. Reduce short-term memory load	Are there enough visual cues here for the user to find the functionality or item? Do they have to remember things to understand what's going on? How can you help the user recall?	☐

… OSAM ZLATNIH PRAVILA ...



(1. pravilo) Težiti konzistentnosti ...

- Identična terminologija treba biti korišćena u svim elementima korisničkog interfejsa (odzivima sistema, menijima, dokumentaciji i sličnom)
- Slične situacije treba da zahtevaju slične setove korisničkih akcija
- Korisnicima treba omogućiti korišćenje standarda na koje su naviknuti



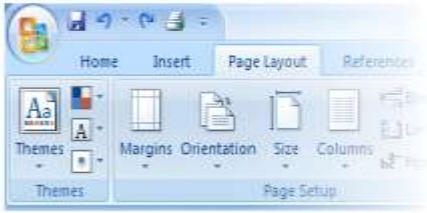
- Ne treba dozvoliti da se korisnik zapita da li različite reči, oznake, situacije ili akcije znače isto što i u drugim okruženjima
- Pridržavanje ovom pravilu najlakše se ostvaruje korišćenjem konvencija platforme koja je u opticaju

... OSAM ZLATNIH PRAVILA ...

... (1. pravilo) Težiti konzistentnosti ...



Microsoft Word

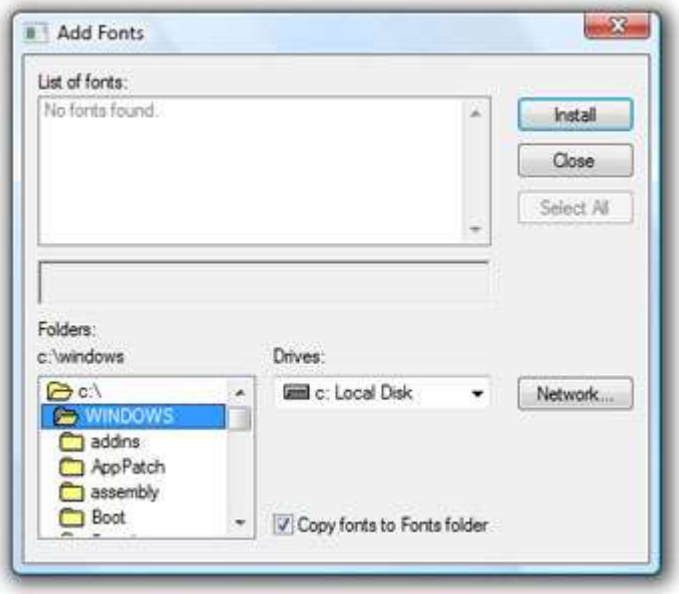


Microsoft Excel



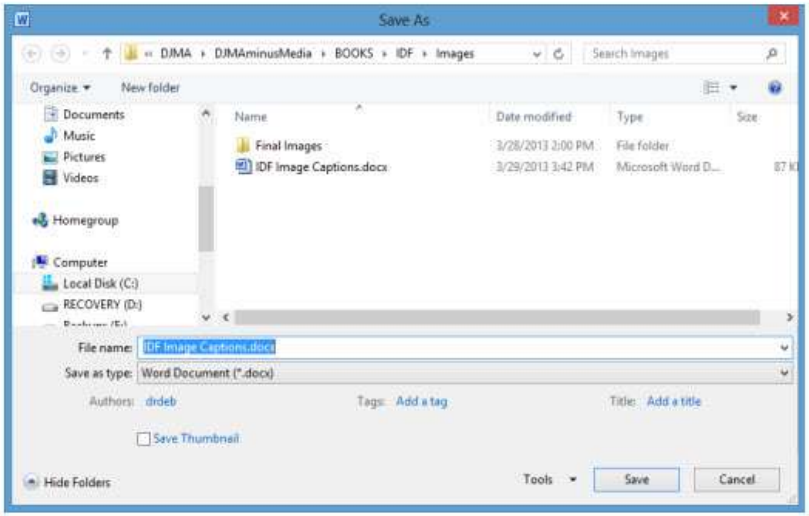
Microsoft Powerpoint

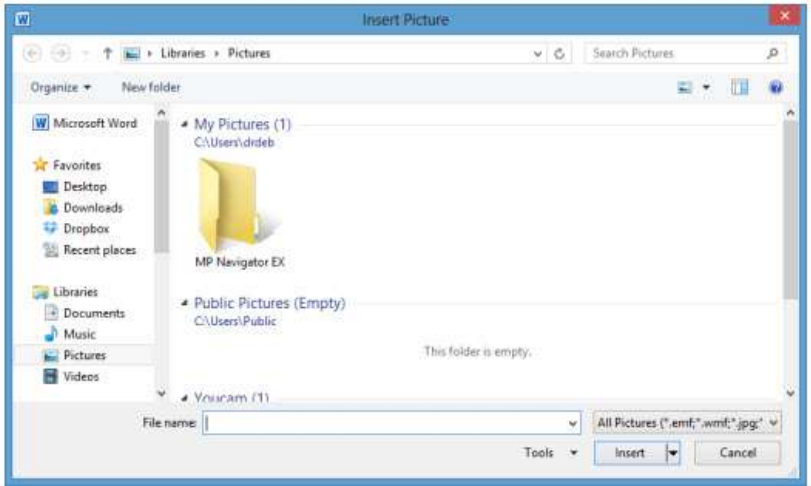


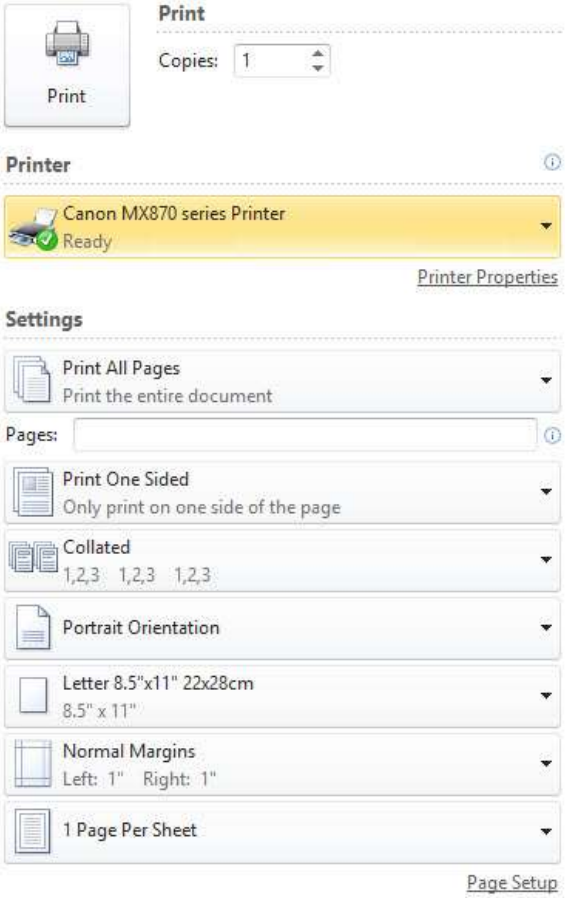


... OSAM ZLATNIH PRAVILA ...

... (1. pravilo) Težiti konzistentnosti







... OSAM ZLATNIH PRAVILA ...

(2. pravilo) Stremiti univerzalnoj upotrebljivosti ...

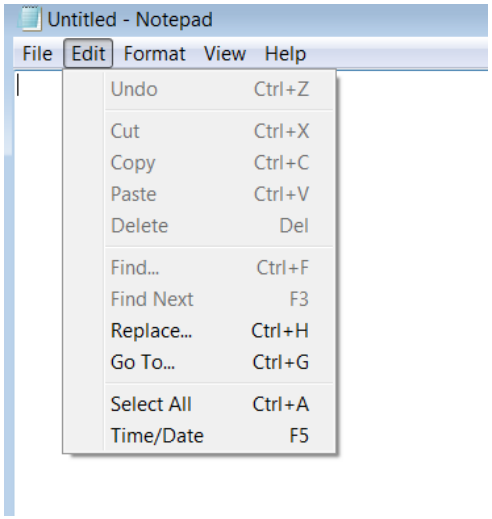
- Prepoznavanje potreba različitih korisnika
- Početnici naspram eksperta, ljudi sa invaliditetima, starost, tehnološka različitost
- Kako raste frekvencija upotrebe softvera, tako i korisnik sve više želi da redukuje broj akcija koje sprovodi i ubrzati tempo interakcije

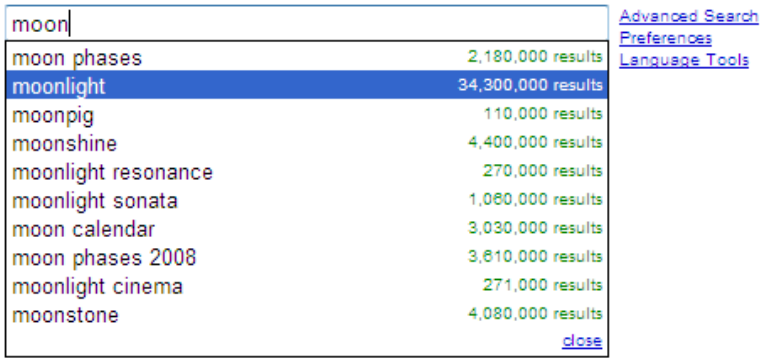


- Prečice, funkcijski tasteri, skrivene komande i makro funkcije mogu biti od velikog značaja naprednom korisniku
  - Novi korisnici najčešće ne primećuju akcije koje mogu ubrzati njihov rad, dok frekventniji korisnici ovakve akcije pametno iskorišćavaju
  - To je razlog zbog kog takve akcije treba omogućiti, kako bi sistem bio pogodan i za nove i za frekventne korisnike

... OSAM ZLATNIH PRAVILA ...

... (2. pravilo) Stremiti univerzalnoj upotrebljivosti ...



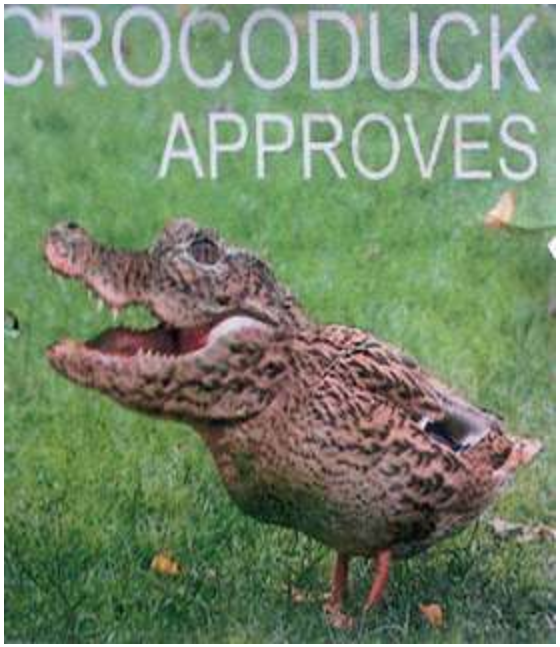




... OSAM ZLATNIH PRAVILA ...

(3. pravilo) Davati informativni feedback ...

- Za svaku korisničku akciju trebalo bi da dođe do nekog odziva sistema kroz odgovarajući feedback u razumnom vremenu
- Za frekventne akcije i akcije sa malim posledicama odziv može biti neupadljiv i umeren
- Za nefrekventne i akcije sa velikim posledicama odziv mora biti naglašeniji i jači
- Informativni feedback mora biti smislen i informativan



... OSAM ZLATNIH PRAVILA ...

... (3. pravilo) Davati informativni feedback ...

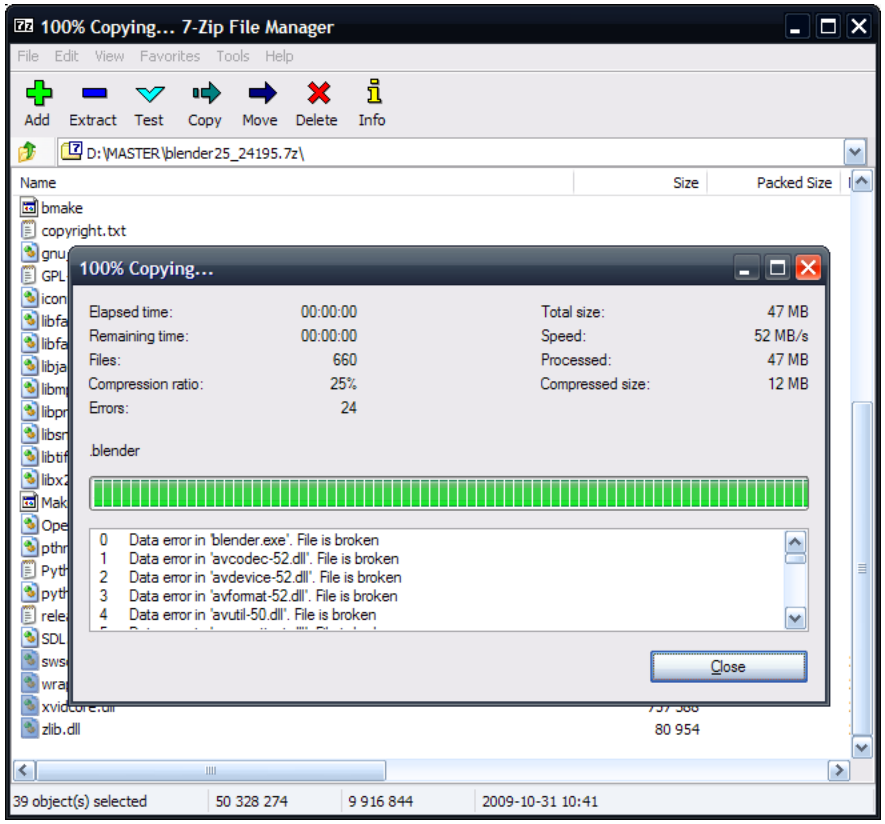






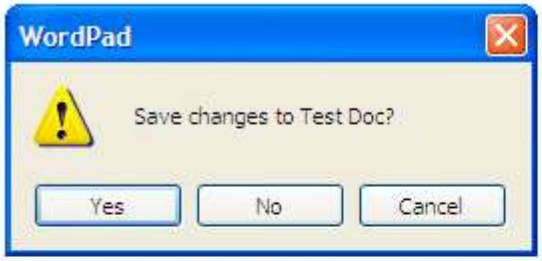
... OSAM ZLATNIH PRAVILA ...

... (3. pravilo) Davati informativni feedback

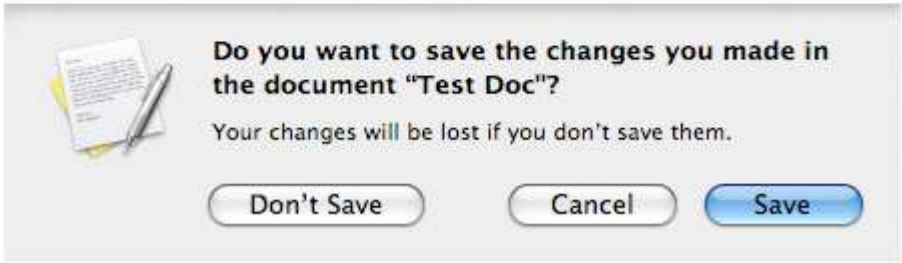




Windows XP WordPad save dialog:



OS X TextEdit save dialog:



... OSAM ZLATNIH PRAVILA ...

(4. pravilo) Projektovati dijaloge naglašene zatvorenosti ...

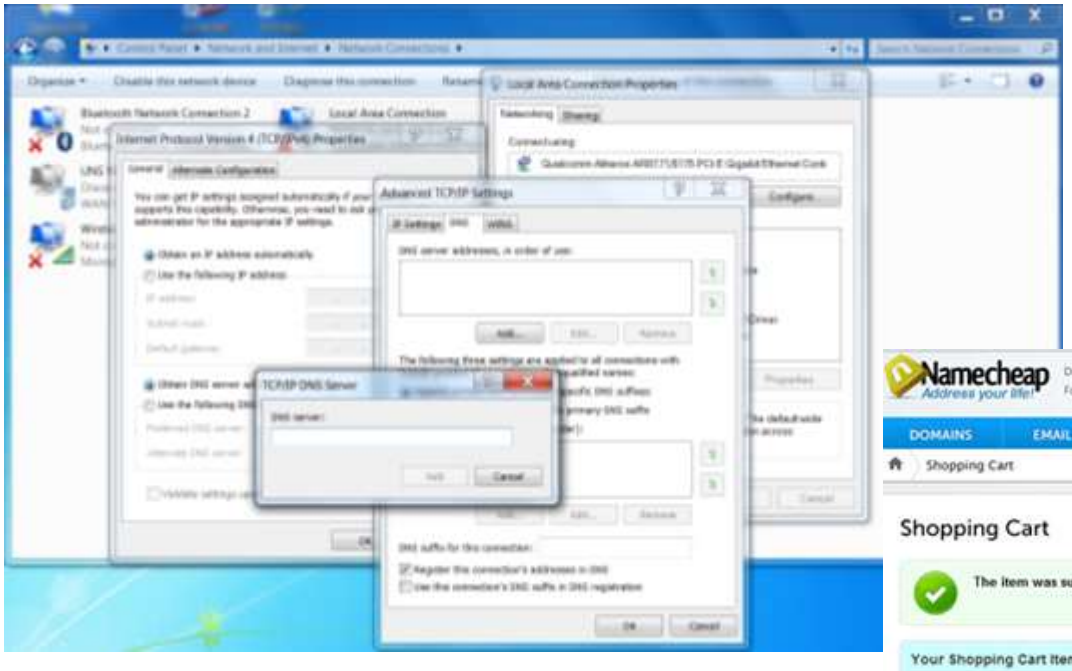
- Sekvence akcija trebalo bi da budu organizovane u grupe sa jasno definisanim početkom, sredinom i krajem
- Informativni odziv na kraju grupe akcija daje korisniku satisfakciju za dobro obavljen posao i priprema ga za sledeću grupu akcija
- Sprovoditi korisnika kroz sve korake
- Jasno signalizirati korisniku da je stigao do kraja i da je ispunio sve zahteve

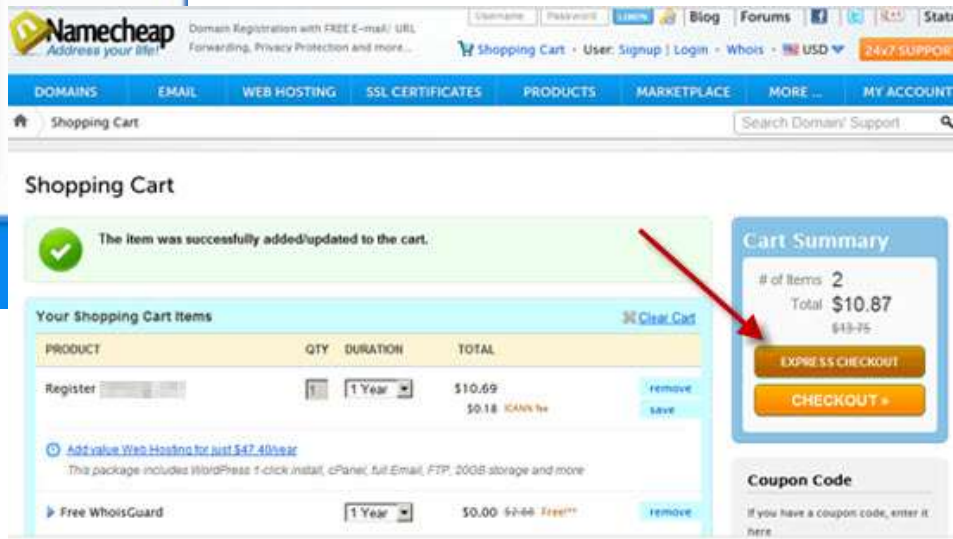


... OSAM ZLATNIH PRAVILA ...



... (4. pravilo) Projektovati dijaloge naglašene zatvorenosti





... OSAM ZLATNIH PRAVILA ...



(5. pravilo) Ponuditi prevenciju i rukovanje greškom ...

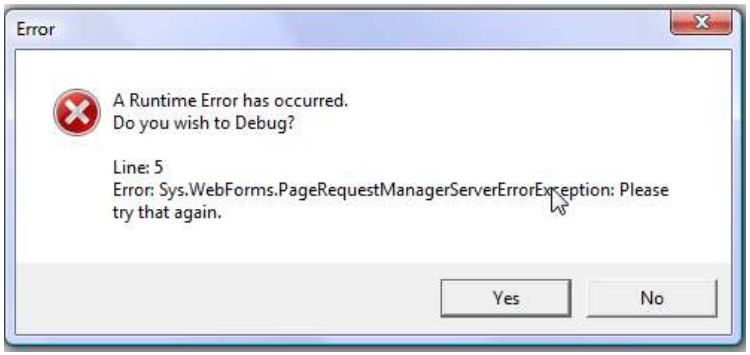
- Koliko god je to moguće, dizajnirati sistem tako da korisnik ne može napraviti ozbiljnu grešku
- Ukoliko je greška već napravljena, sistem bi trebalo da bude u mogućnosti da prepozna grešku i ponudi jednostavne mehanizme za obradu greške, odnosno oporavak od nje
- Poruke o grešci treba da budu napisane jasnim jezikom, koji ne uključuje kod

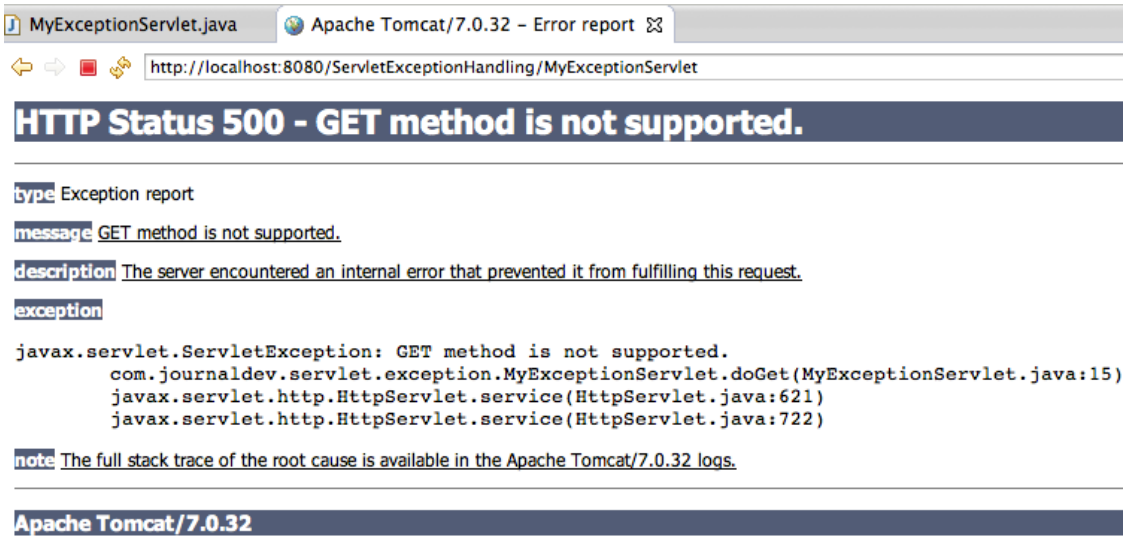


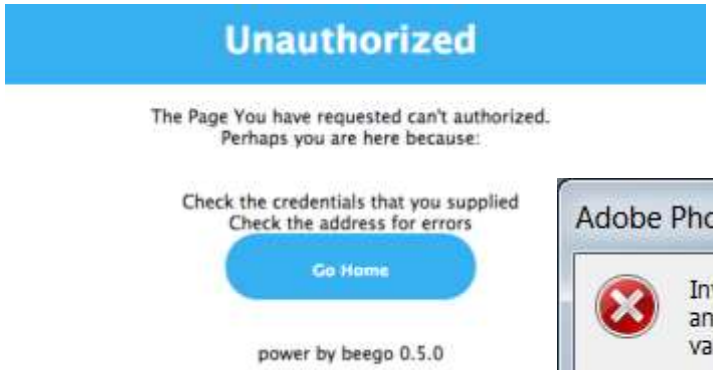
... OSAM ZLATNIH PRAVILA ...

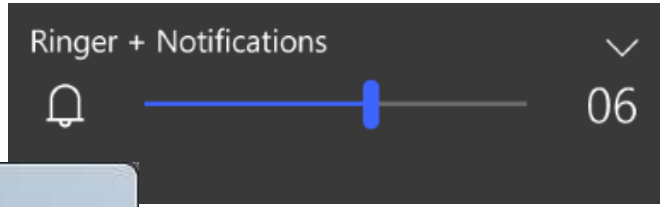


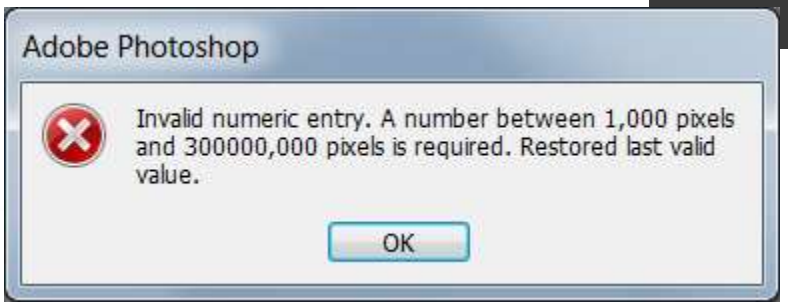
... (5. pravilo) Ponuditi prevenciju i rukovanje greškom

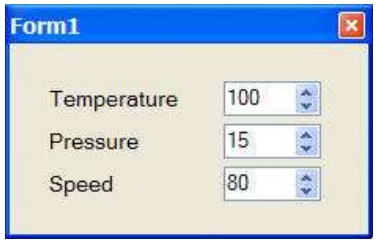












... OSAM ZLATNIH PRAVILA ...



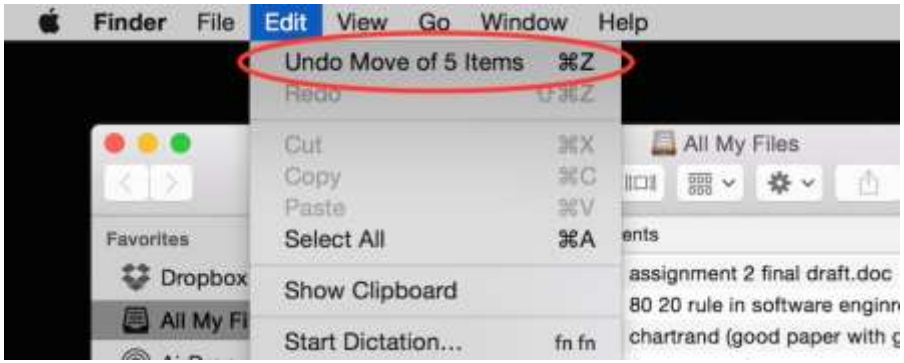
(6. pravilo) Dozvoliti poništavanje efekata ...



- Korisniku treba dati slobodu i mogućnost biranja, kao i donošenja sopstvenih odluka
- Ostvaruje se pružanjem mogućnosti korisniku da odabere koje akcije i zadaci će se izvršavati u kom trenutku (sistem ne bi trebalo da o tome sam odlučuje ili da na neki način nameće određene akcije korisniku)
- Undo i Redo su osnovni alati za ostvarivanje ovakvog cilja
- Ako je ovo pravilo ispoštovano, korisnik se oseća opuštenije u radu jer zna da se akcije koje su dovele do greške mogu poništiti
- Ovo dalje omogućava korisniku da nepoznate opcije programa uči kroz istraživanje - pomoću pokušaja i pogrešaka
- Jedinice u kojima se akcije mogu poništiti mogu biti različitog obima: može se poništavati pojedinačna akcija, ceo unos ili kompletna grupa akcija

... OSAM ZLATNIH PRAVILA ...

… (6. pravilo) Dozvoliti poništavanje efekata

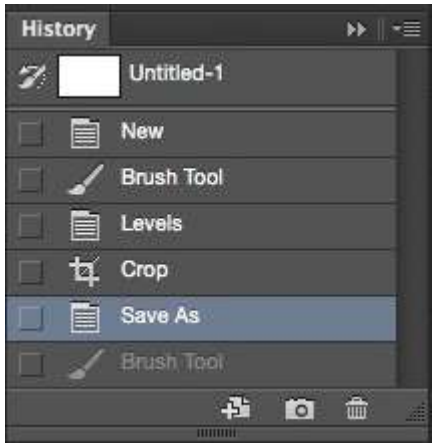












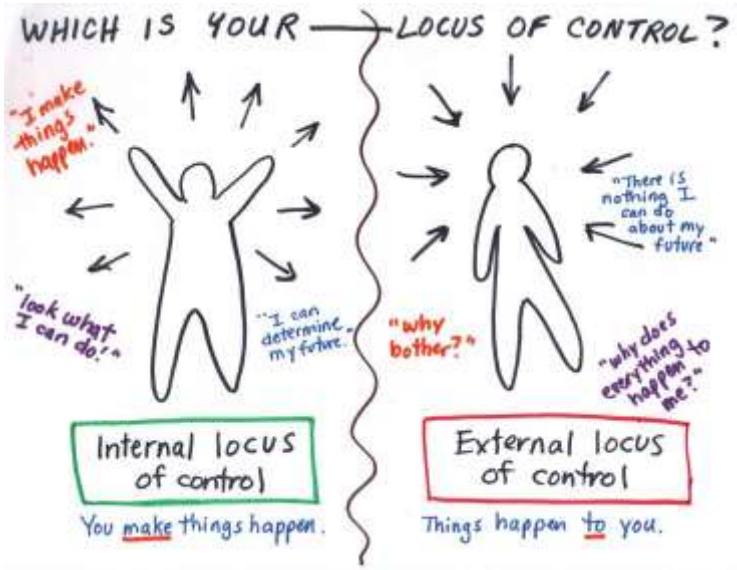


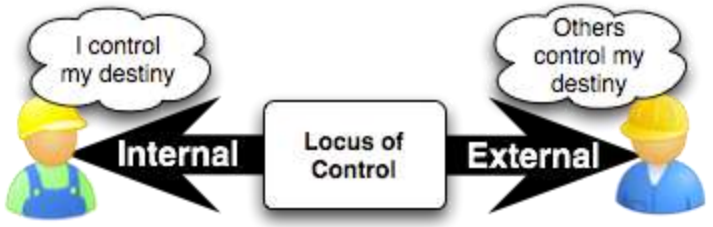
... OSAM ZLATNIH PRAVILA ...

(7. pravilo) Interno podržavati kontrolu ...

- Iskusni korisnici žele da imaju punu kontrolu nad sistemom i da sistem funkcioniše u skladu sa njihovim komandama
- Sistem treba dizajnirati tako da su korisnici inicijatori akcija, a ne samo posmatrači

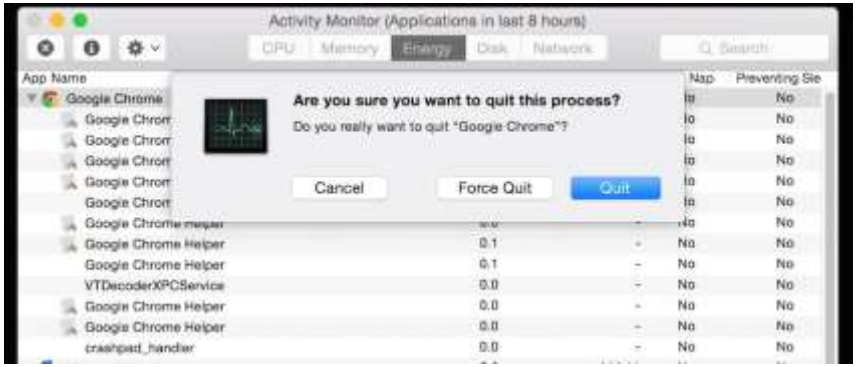


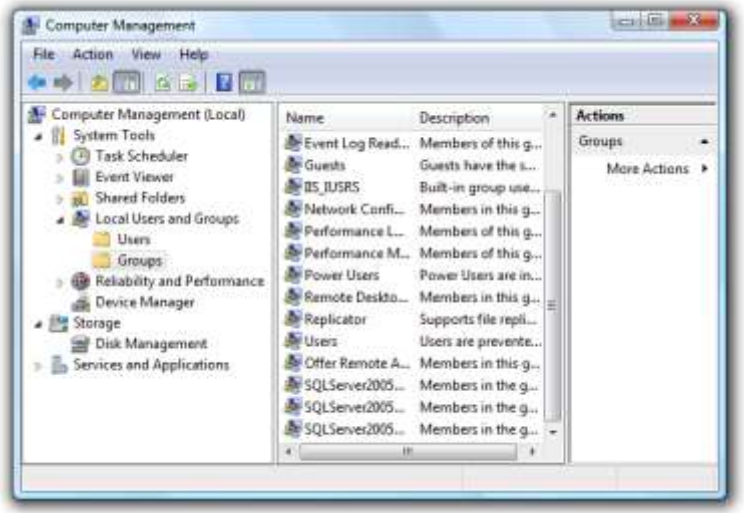


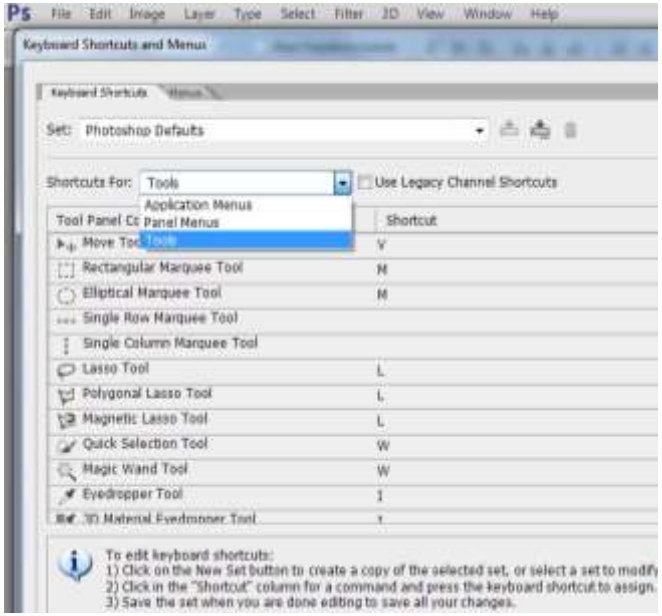


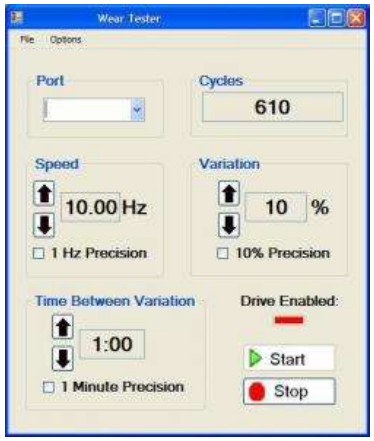
... OSAM ZLATNIH PRAVILA ...

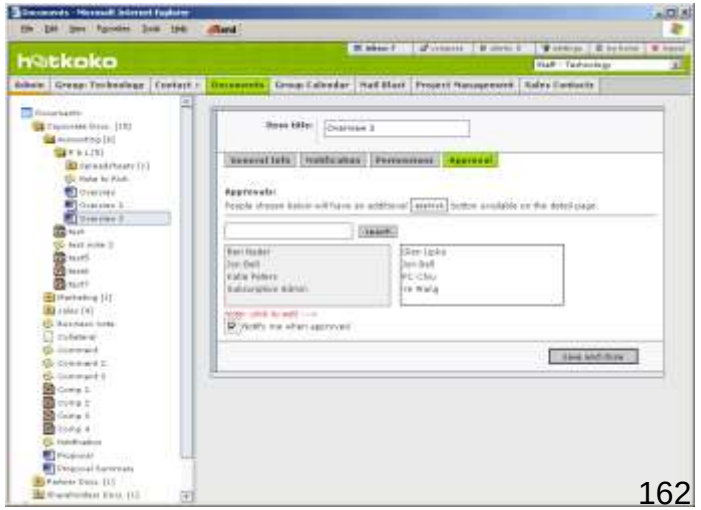
… (7. pravilo) Interno podržavati kontrolu







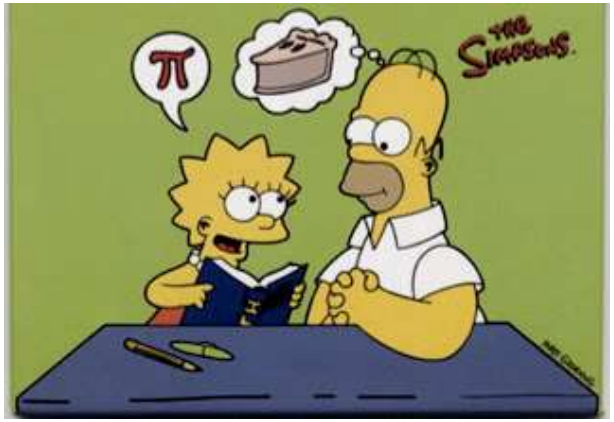




... OSAM ZLATNIH PRAVILA ...

(8. pravilo) Redukovati opterećenje radne memorije ...

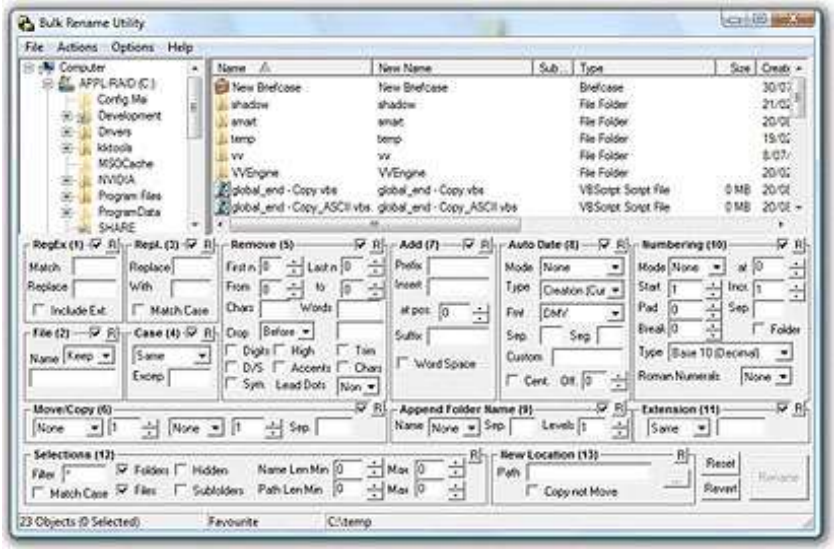
– Ograničenja kratkotrajne memorije kod ljudi zahtevaju da svi prikazi budu jednostavni, da višestrani prikazi budu konsolidovani, zahtevi za pomeranjem prozora da budu minimalni i da vreme učenja bude podeljeno između kodova, mnemonika i sekvenci akcija



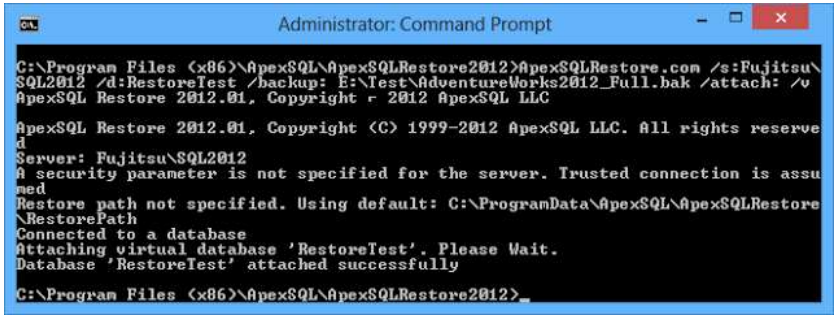


... OSAM ZLATNIH PRAVILA

… (8. pravilo) Redukovati opterećenje radne memorije

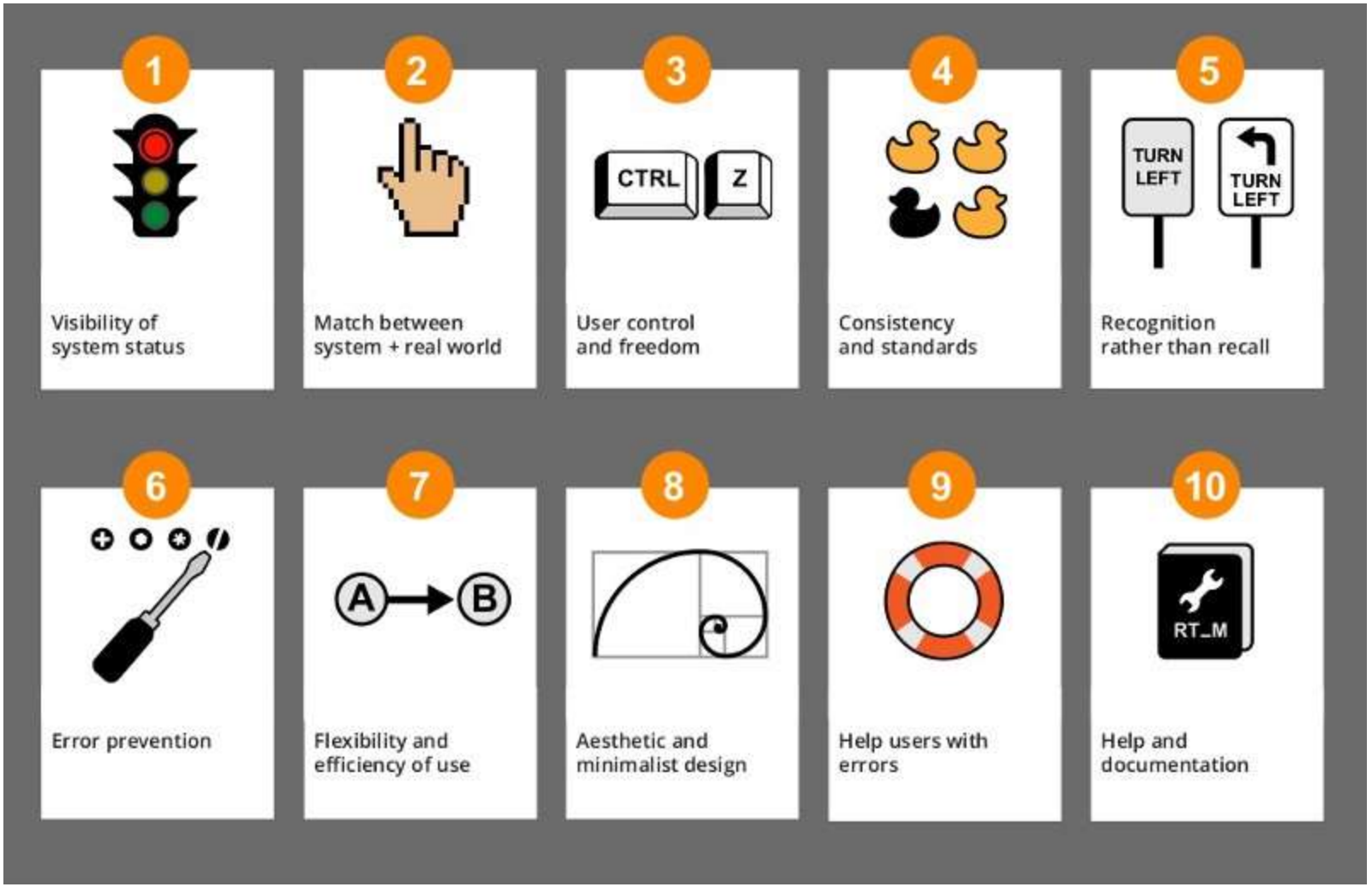








NIELSENOVA PRAVILA …



… NIELSENOVA PRAVILA …

Ten Usability Heuristics by Jakob Nielsen



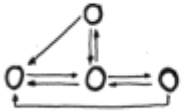
Visibility of system status

Give the users appropriate feedback about what is going on.



Match between system and the real world

Use real-world words, concepts and conventions familiar to the users in a natural and logical order.



User control and freedom

Support undo, redo and exit points to help users leave an unwanted state caused by mistakes.



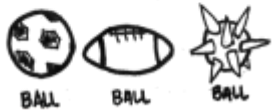
Error prevention

Prevent problems from occurring: eliminate error-prone conditions or check for them before users commit to the action.



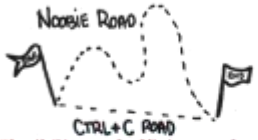
Aesthetic and minimalist design

Don't show irrelevant or rarely needed information since every extra elements diminishes the relavance of the others.



Consistency and standards

Follow platform conventions through consistent words, situations and actions.



Flexibility and efficiency of use

Make the system efficient for different experience levels through shortcuts, advanced tools and frequent actions.



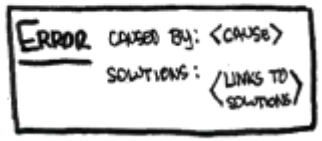
Recognition rather than recall

Make objects, actions, and options visible at the appropriate time to minimize users' memory load and facilitate decisions.



Help and documentation

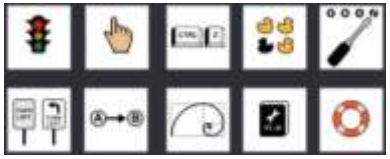
Make necessary help and documentation easy to find and search, focused



Help users recognize, diagnose, and recover from errors

Express error messages in plain language (no codes) to indicate the problem and suggest solutions.

… NIELSENOVA PRAVILA …



(1. pravilo) Vidljivost statusa sistema ...



– Sistem bi stalno trebalo da obaveštava korisnika o svojim aktivnostima kroz odgovarajuće poruke („feedback“) upotrebljene na pravom mestu, u pravo vreme i u odgovarajućem vremenskom roku

– Da li je korisnik u svakom trenutku svestan u kom statusu se sistem nalazi? I koja je sledeća faza?



– Primeri su dosta jednostavni: „download“ proces prilikom kog nas sistem obaveštava koji deo fajla je „skinut“ i koliko vremena je do kraja procesa

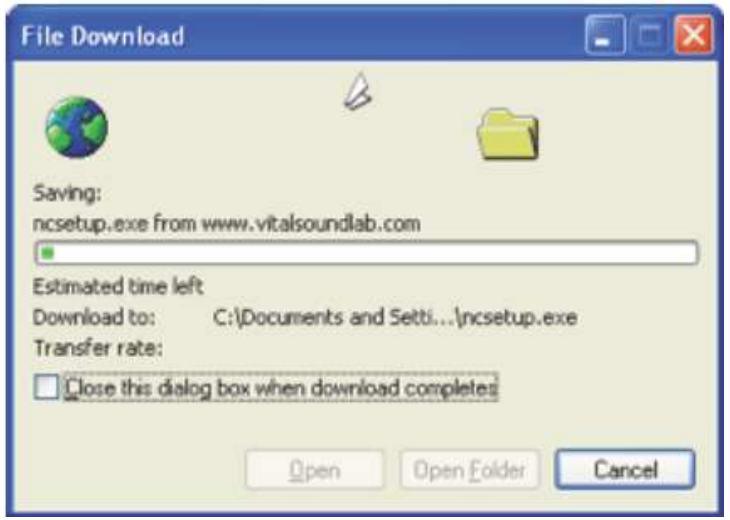
… NIELSENOVA PRAVILA …

… (1. pravilo) Vidljivost statusa sistema ...

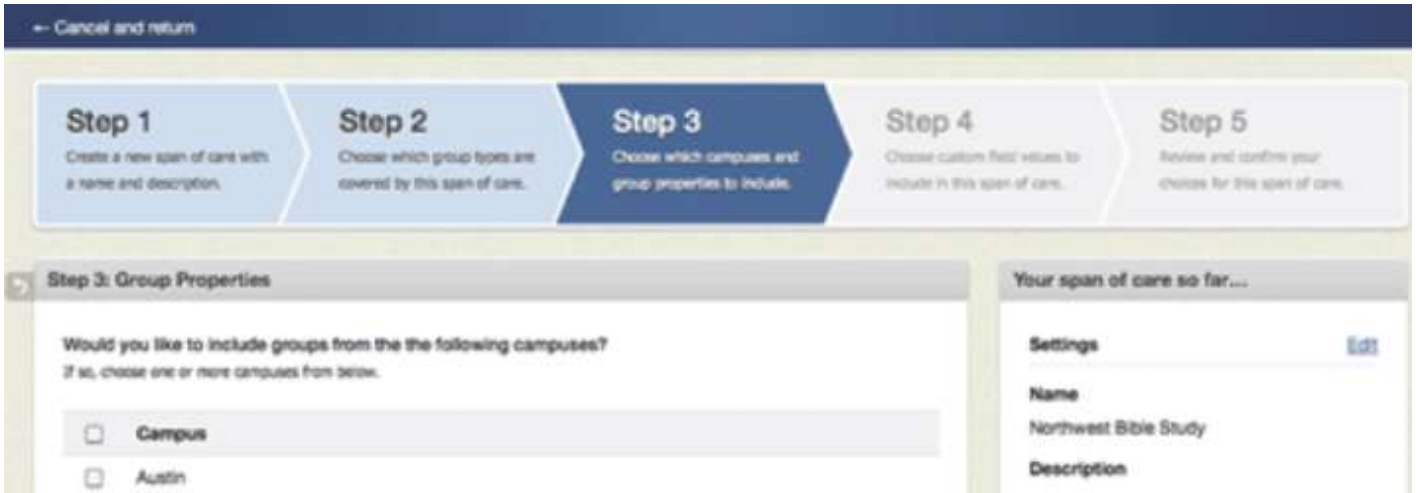
Hyperlink
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Buy Now Normal
Buy Now Hover
Buy Now On press
Buy Now Disabled

OFF
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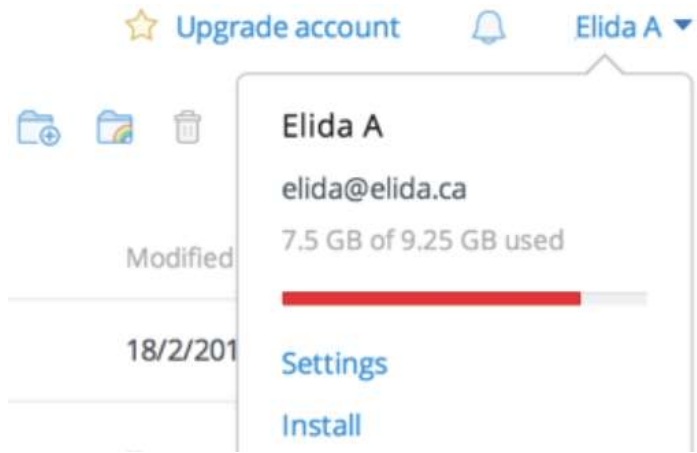


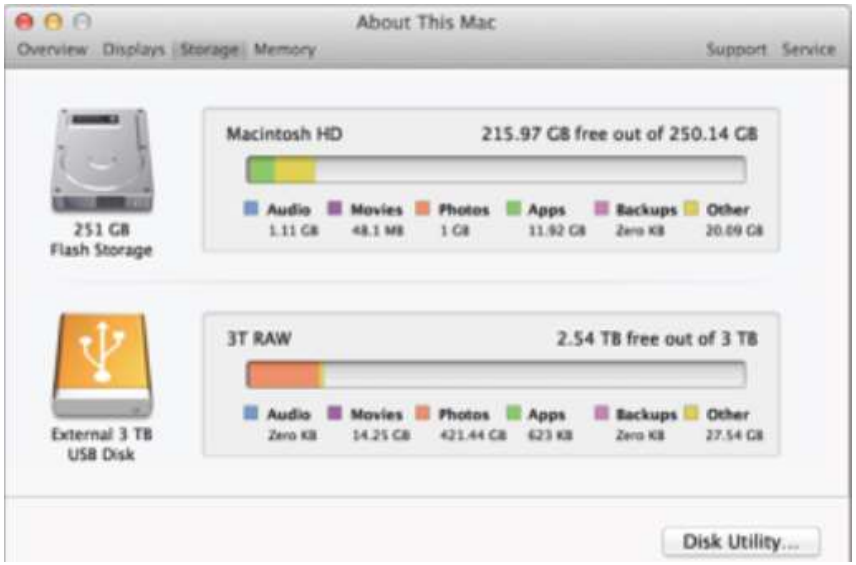
… NIELSENOVA PRAVILA …

… (1. pravilo) Vidljivost statusa sistema

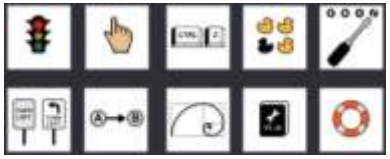




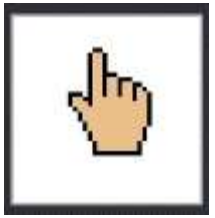




… NIELSENOVA PRAVILA …



(2. pravilo) Veza sa stvarnim svetom ...



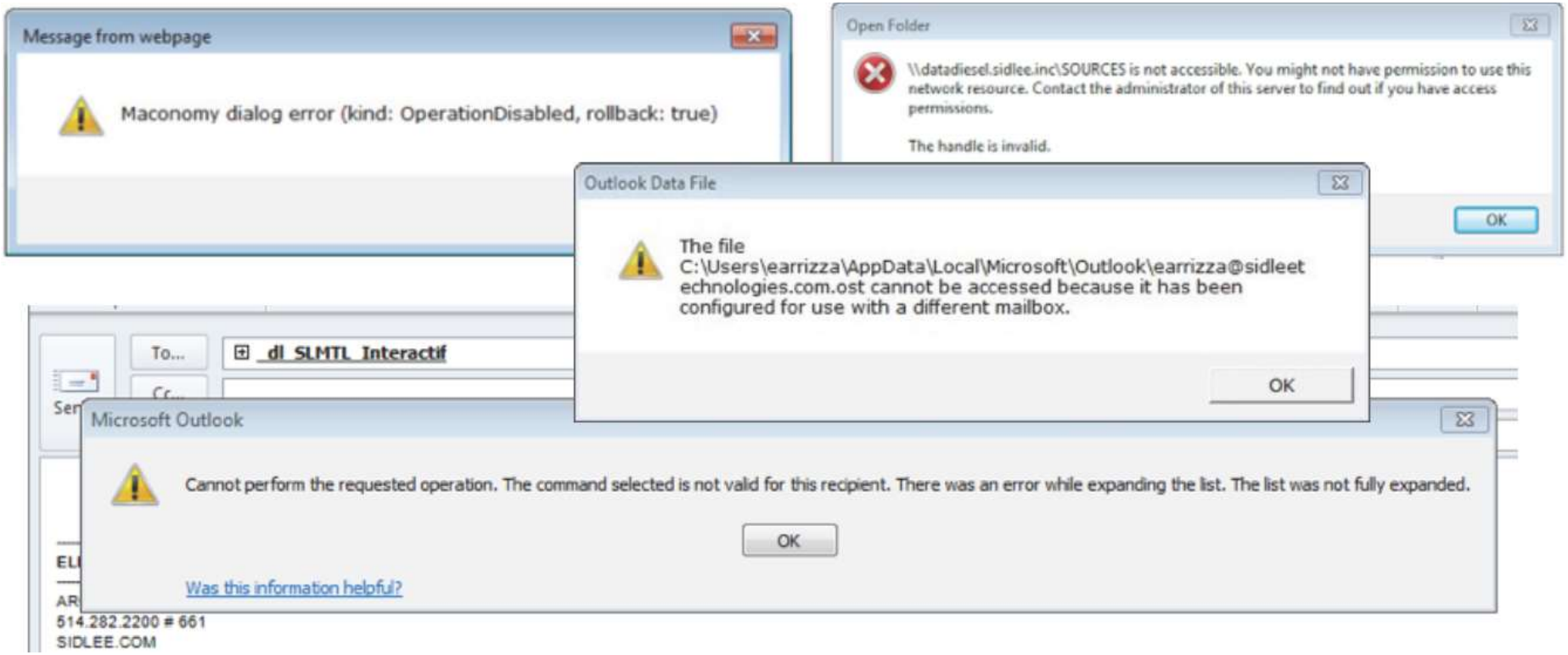
- Sistem bi trebalo da prati i poštuje konvencije i logiku iz stvanog sveta što je moguće više i da omogući korisniku da brzo razume kako sistem funkcioniše
- Trebalo bi da prati konvencije iz stvarnog sveta, tako da se informacije prikazuju u logičkom i prirodnom redosledu



- Na primer, sistem bi trebalo da „govori“ jezik korisnika – koliko ste puta dobili poruku „Error 157935842930589“??? Sistem bi trebalo da nam pruži informaciju tipa „Printer je u zastoju, nema papira. Molimo dodajte jos“
- Programeri i krajnji korisnici često (ako ne i uvek) imaju drugačije viđenje kako bi sistem trebao da funkcioniše i ignorišući mentalni model krajnjih korisnika, programeri stvaraju aplikacije koje su teške i neefikasne za korišćenje. („Korisnik je glup“ nije mentalni model!!!)

… NIELSENOVA PRAVILA …

… (2. pravilo) Veza sa stvarnim svetom ...



… (2. pravilo) Veza sa stvarnim svetom













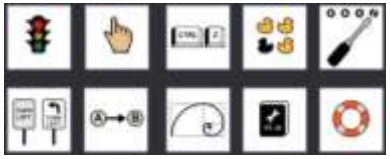








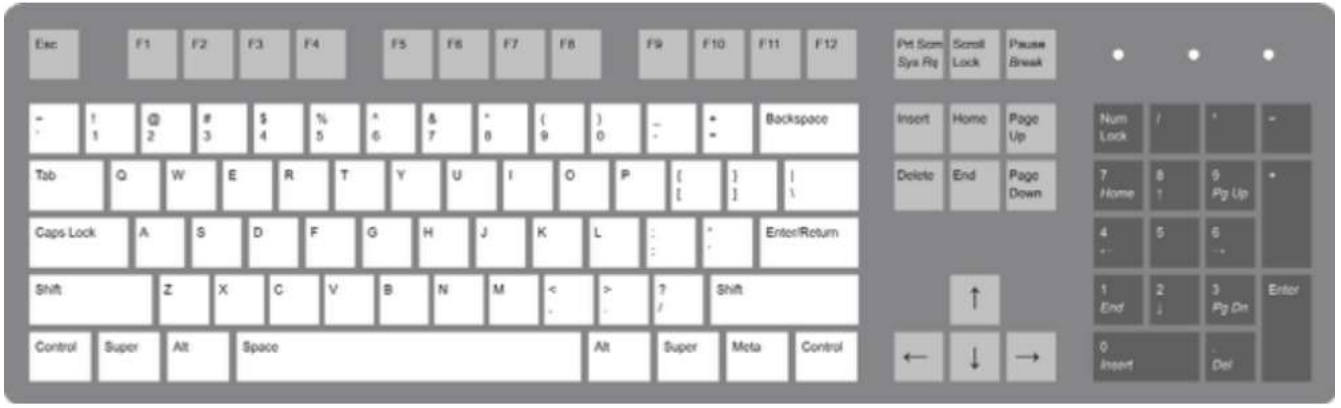
… NIELSENOVA PRAVILA …



(3. pravilo) Korisnikova kontrola i sloboda ...

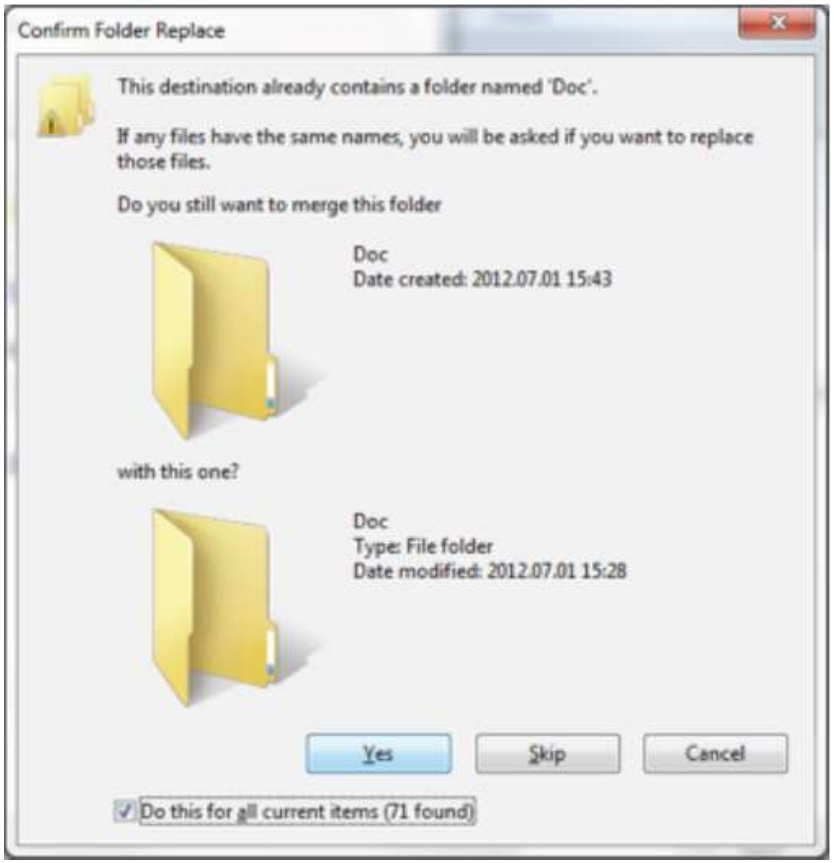


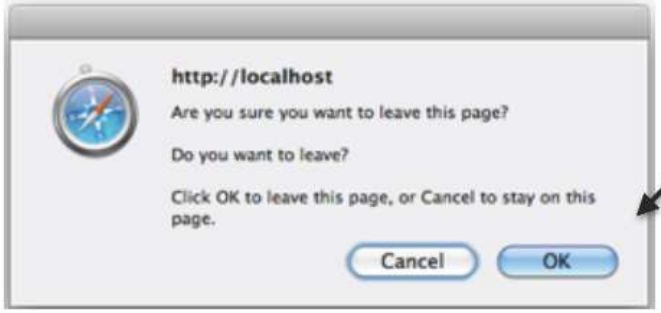
- U mnogim situacijama, korisnik odabere neke komande sasvim slučajno, greškom i sistem bi trebalo da obezbedi „bezbolni“ povratak na prethodno stanje
- Podržati Undo/Redo
- Primeri na vebu su mnogobrojni: logo koji uvek vodi na početnu stranicu, dugme nazad, izađi itd



… NIELSENOVA PRAVILA …

… (3. pravilo) Korisnikova kontrola i sloboda ...





unneeded encoding

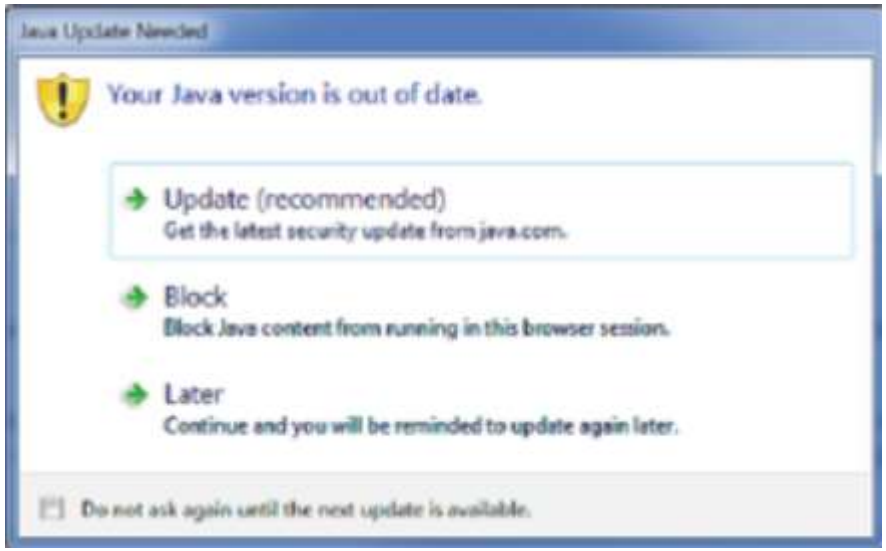




… NIELSENOVA PRAVILA …

… (3. pravilo) Korisnikova kontrola i sloboda ...







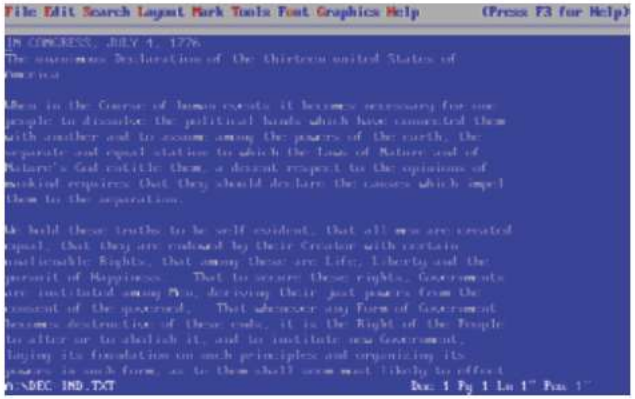
… (3. pravilo) Korisnikova kontrola i sloboda

WYSIWYG



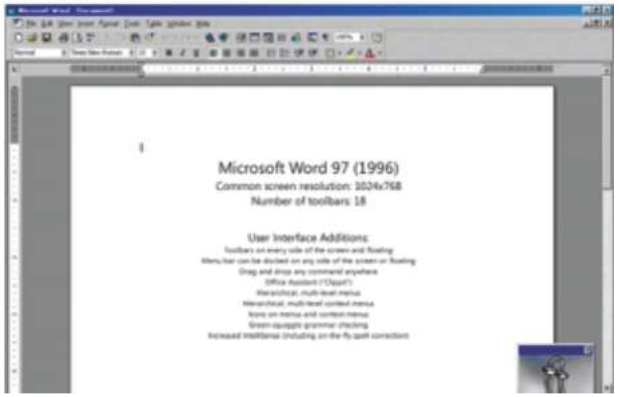
IBM Selectric Typewriter

NO WYSIWYG



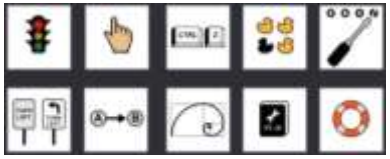
WordPerfect 2.1 for DOS

BACK TO WYSIWYG

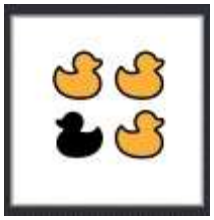


MS WORD 97

… NIELSENOVA PRAVILA …



(4. pravilo) Konzistentnost i standardi ...



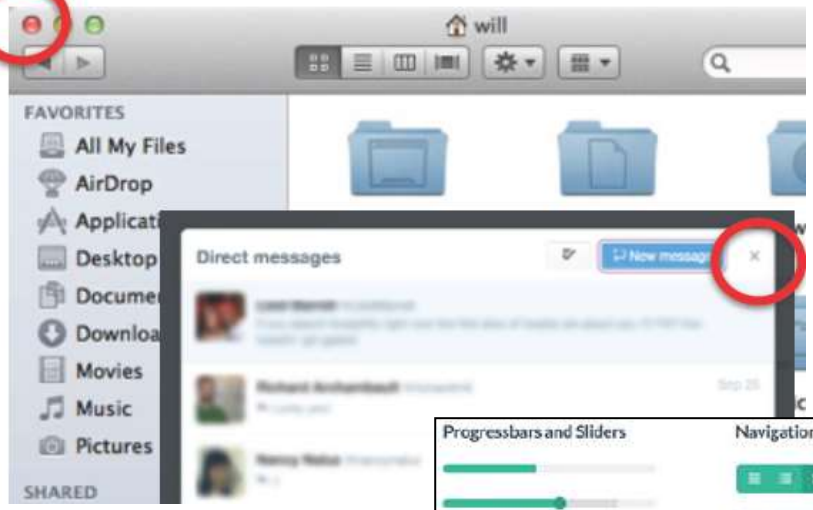
- Neki simboli i ikone uvek imaju isto značenje i što je više moguće trebalo bi ih korititi
- Korisnik ne bi trebalo da se zapita da li raličite reči, situacije ili naredbe imaju isto značenje
- Pratiti konvencije vezane za platformu
- Na primer, email se označava pismom, štampač sličicom štampača, zastavice koje označavaju promenu jezika su u desnom gornjem uglu stane
- Doslednost, tj. konzistentnost znači da jedan stil, set boja, font, logo, „layout“ (izuzetak može biti jedino početna strana), redosled ikona uvek treba da prati istu logiku, da ima isto značenje i poziciju
- Na primer, kada korisnik treba da odabere da nesto sačuva ili eliminiše, „save“ dugme bi trebalo da bude uvek sa iste strane, na svakom delu veb sajta na kom postoji „save“ opcija

… (4. pravilo) Konzistentnost i standardi

?







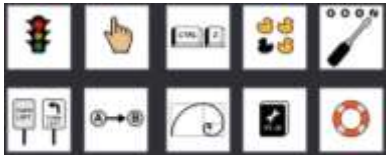


Primary button
(Encouraged Action)

Secondary



… NIELSENOVA PRAVILA …



(5. pravilo) Prevencija grešaka ...



– Kad god je to izvodljivo eliminisati situacije koje dovode do greške

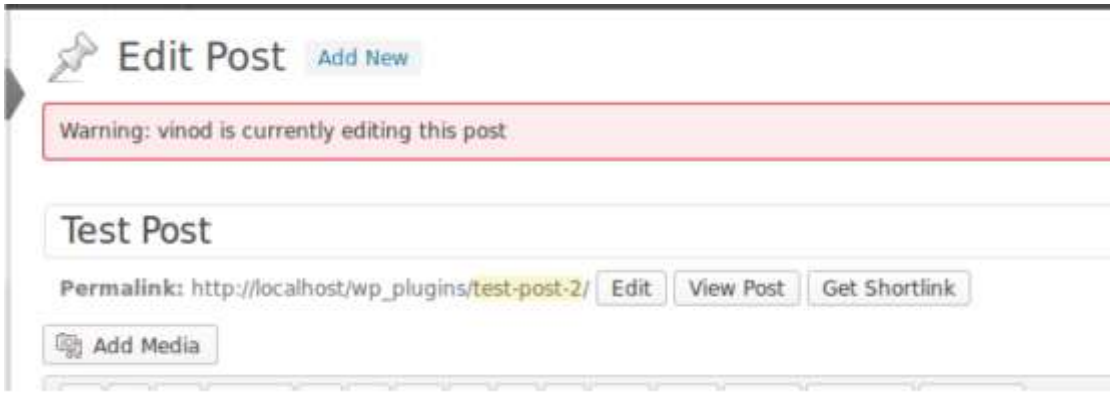
– Ukoliko u sistemu postoje neke akcije koje mogu prouzrokovati trajne i opasne posledice, korisniku treba pružiti opomenu

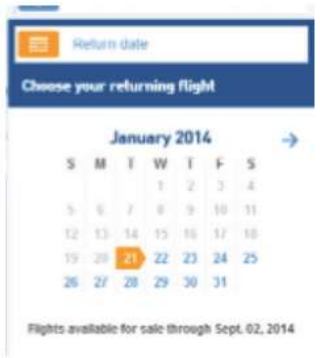


– Na primer, „Da li ste sigurni da želite ovo da izbrišete?“

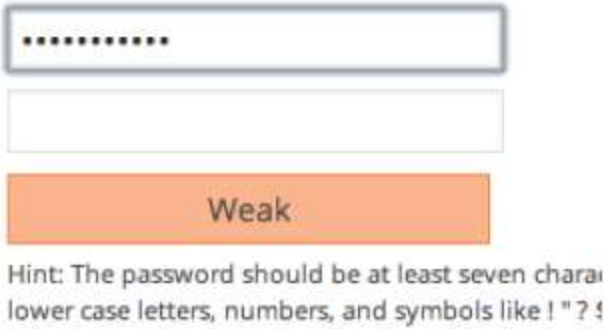


… (5. pravilo) Prevencija grešaka …



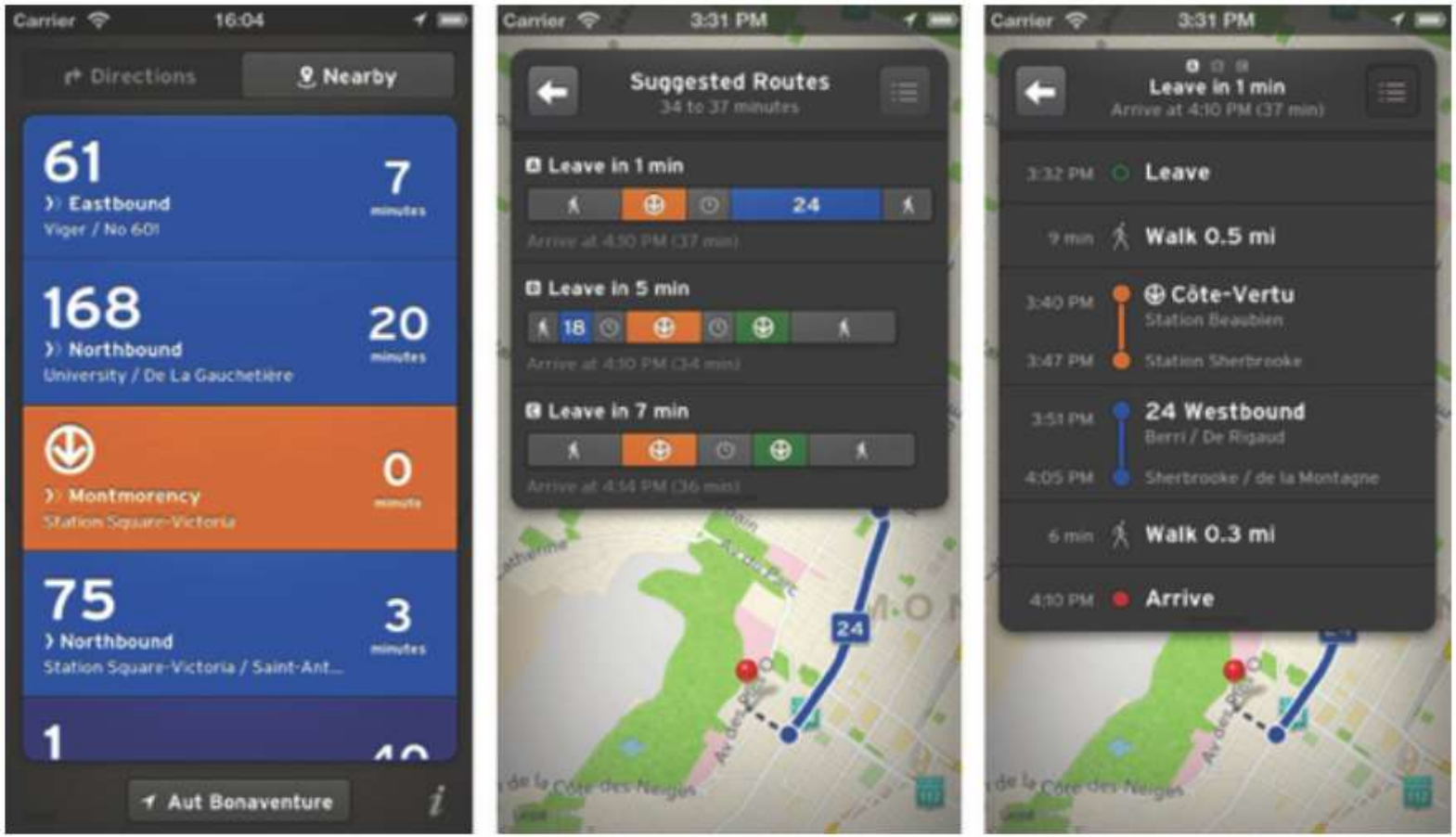


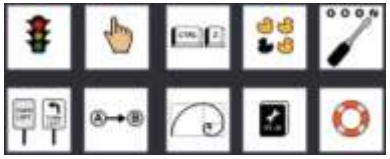




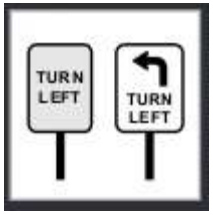
… NIELSENOVA PRAVILA …

… (5. pravilo) Prevencija grešaka





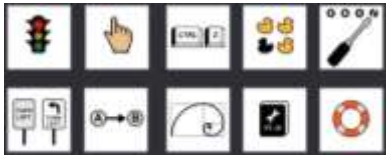
(6. pravilo) Prepoznavanje, pre pamćenja ...



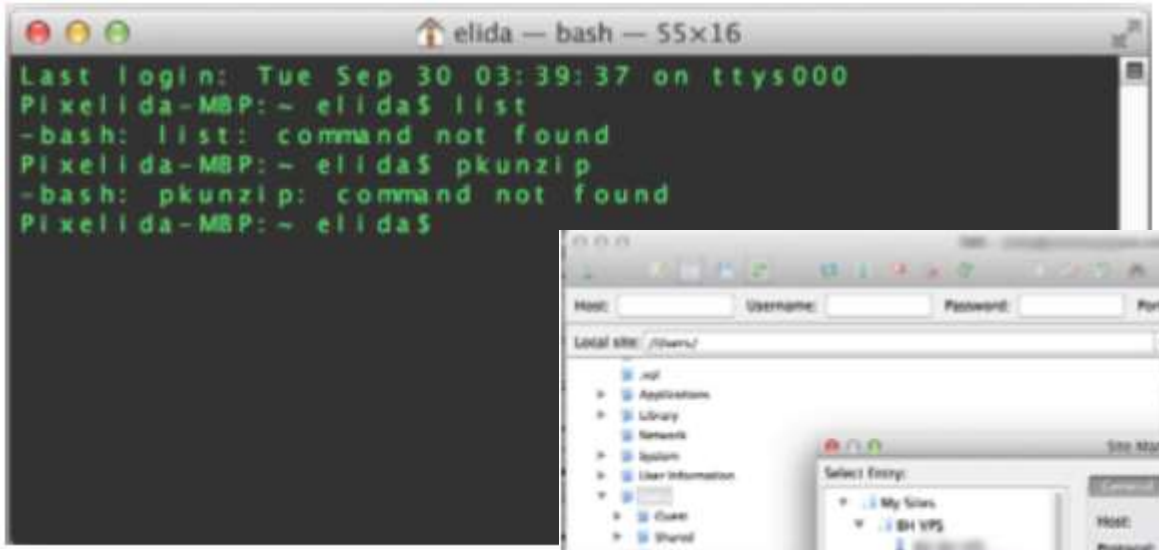
- Trebalo bi minimizovati opterećenje korisnikove memorije tako što su svi elementi interakcije vidljivi
- Korisnik ne treba da pamti kako da koristi sistem
- Dizajneri bi trebali da učine objekte, akcije i moguće opcije sistema vidljivim kako bi korisnik lakše prepoznavao kako da rukuje sa sistemom
- Korisnik ne bi trebalo da pamti informacije (istoriju) sa prethodnog dijaloga u radu sa trenutnim dijalogom
- Instrukcije za rad sa sistemom trebale bi da su stalno vidljive i dostupne
- „Kako ono beše? I šta sam posle toga pritisnuo?“ nisu najbolji pokazatelji performansi sistema

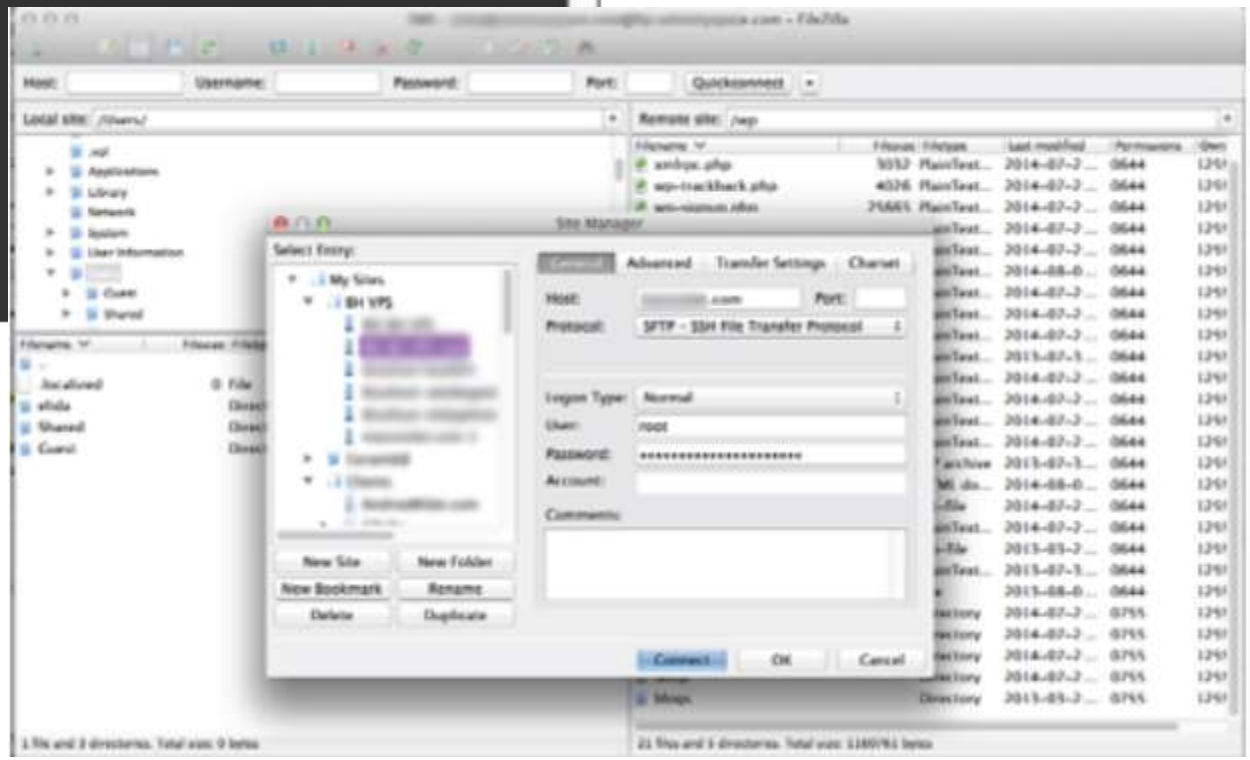


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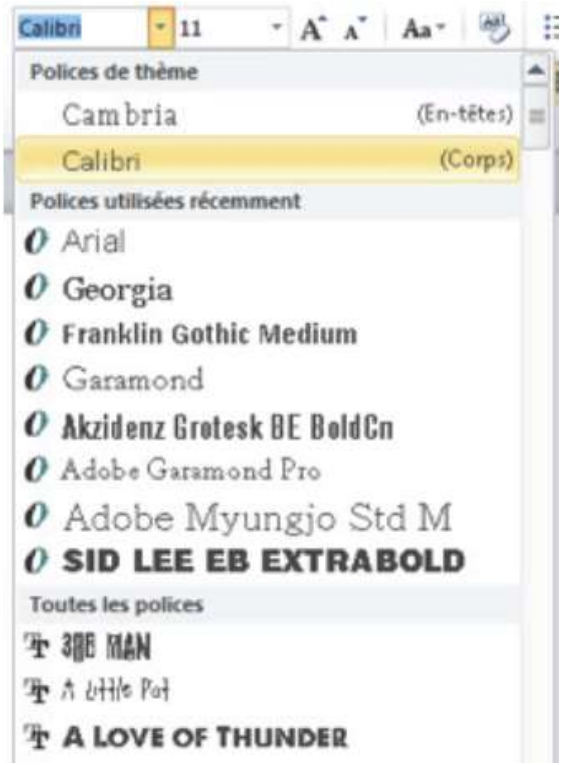
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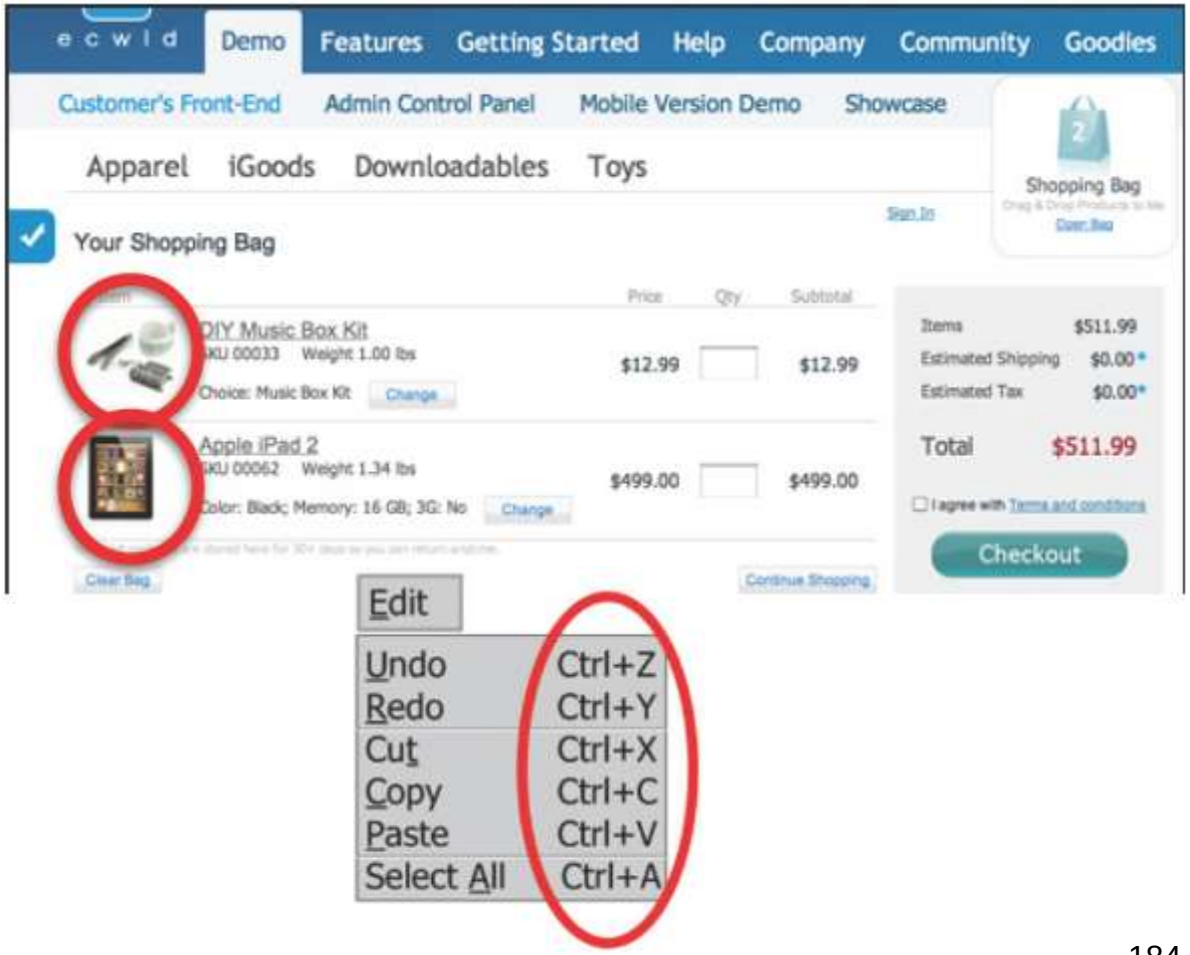




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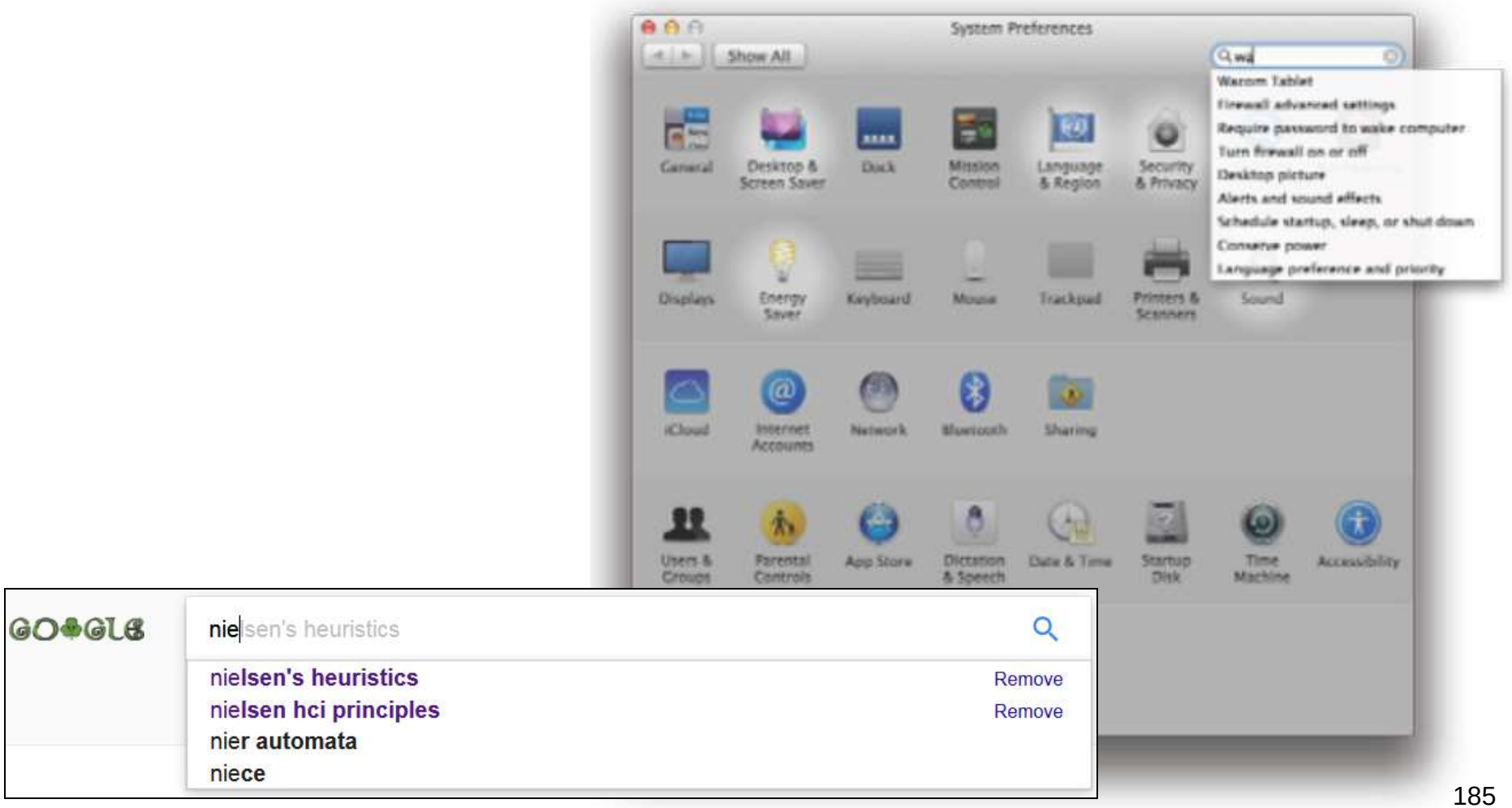
… (6. pravilo) Prepoznavanje, pre pamćenja ...



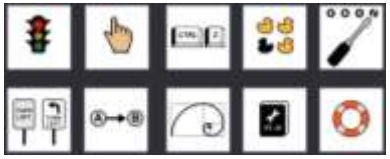


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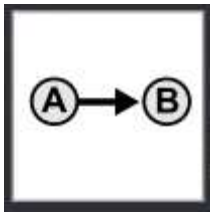
… (6. pravilo) Prepoznavanje, pre pamćenja



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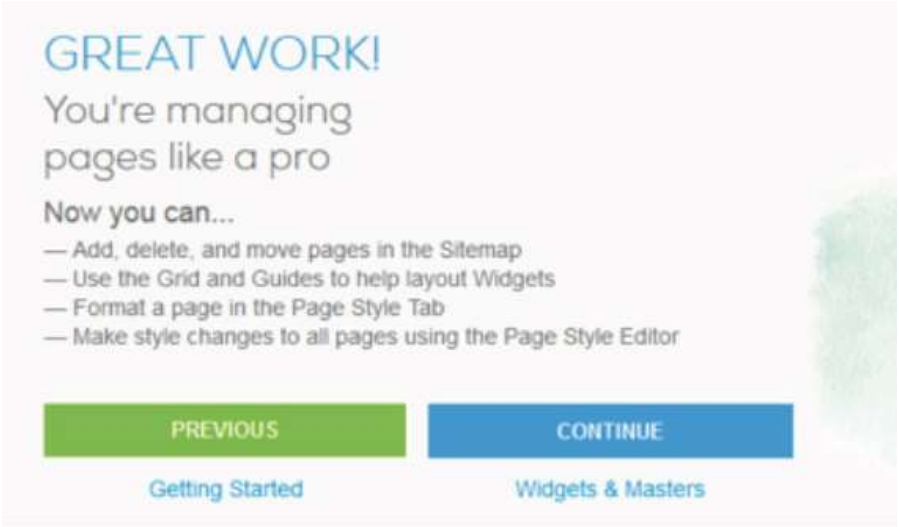
(7. pravilo) Fleksibilnost i efikasnost ...



– Omogućiti napradnim korisnicima da ubrzaju svoj rad putem prečica

– Omogućiti korisnicima da podešavanje okruženja





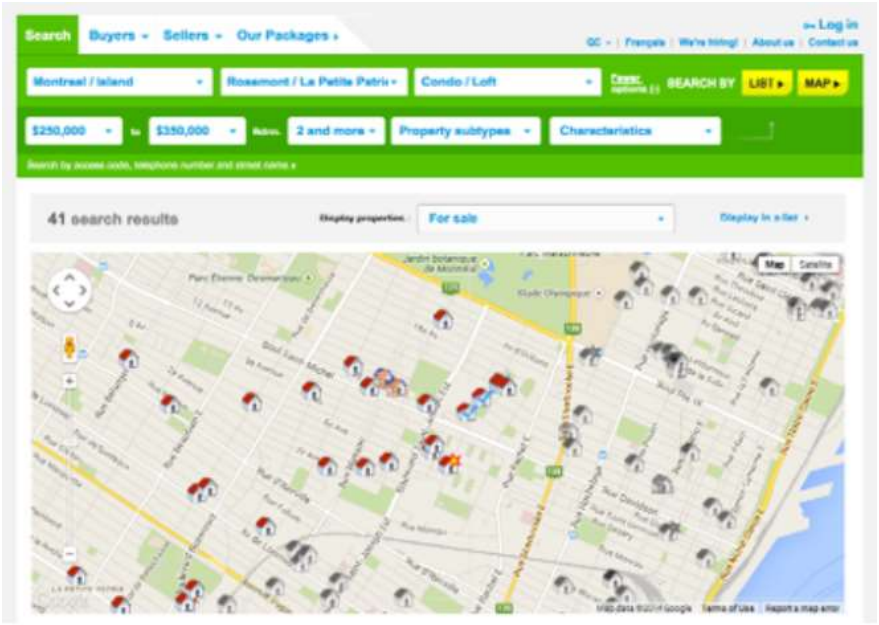
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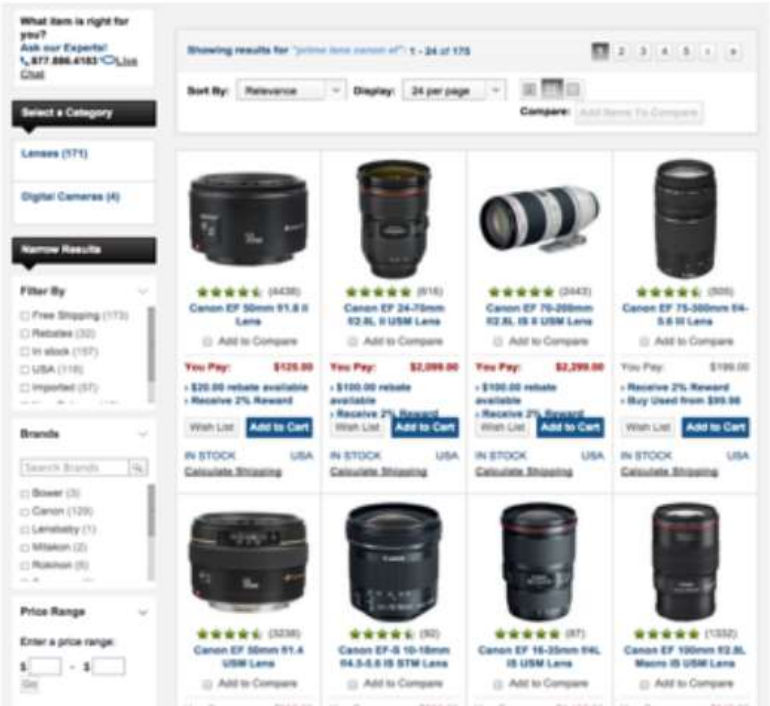
… (7. pravilo) Fleksibilnost i efikasnost





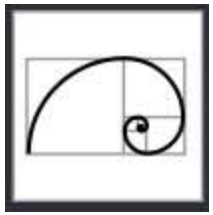






… NIELSENOVA PRAVILA …

(8. pravilo) Estetičan i minimalističan dizajn ...



– Izbegavati puno teksta i nepotrebnih informacija

– Dijalozi ne bi trebalo da sadrže informacije koje nisu relevantne za rešavanje korisnikovog zadatka



– Svaki dodatni (nepotrebni) podatak zauzima mesto bitnom podatku i smanjuje vidljivost bitnih podataka

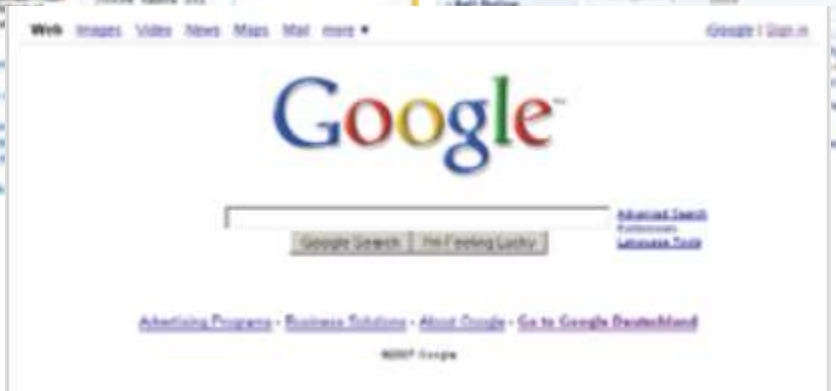


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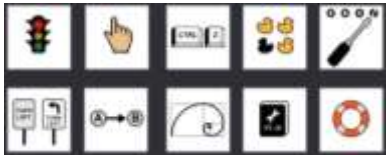
… (8. pravilo) Estetičan i minimalističan dizajn ...



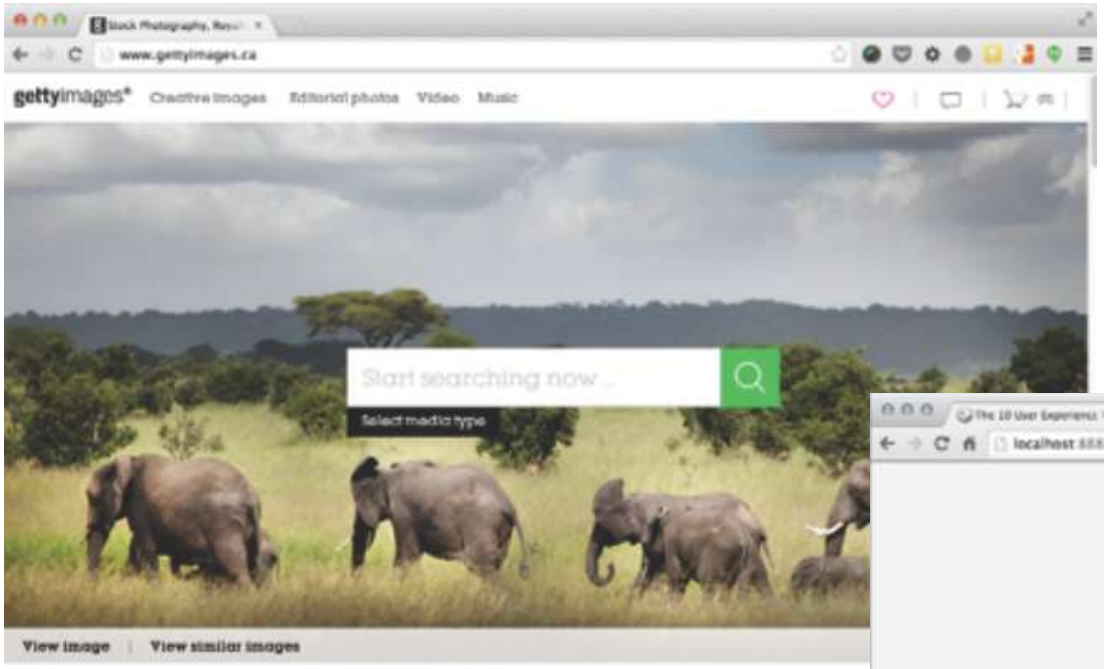


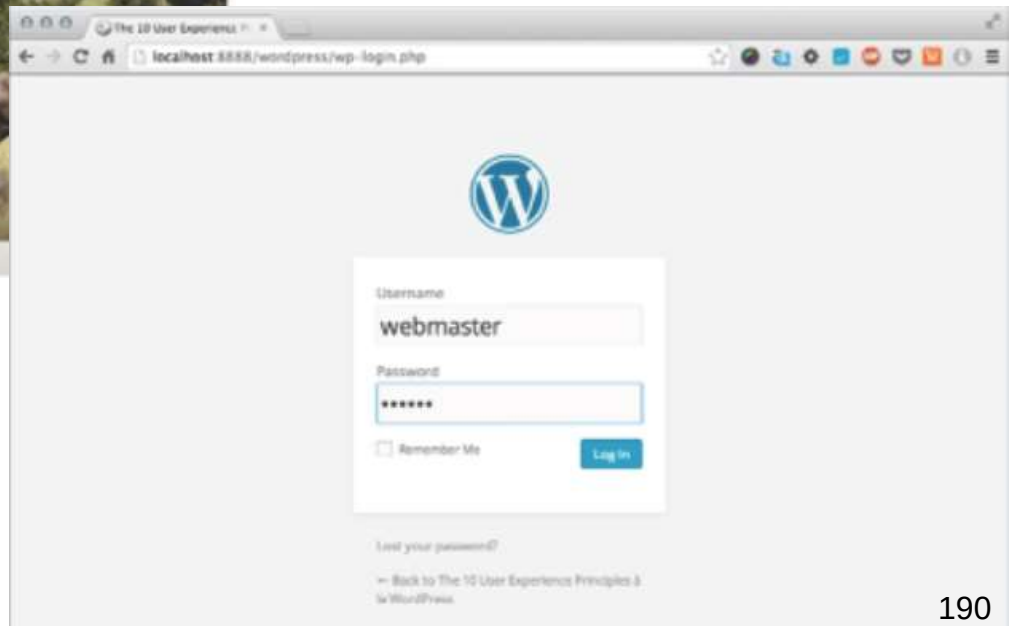


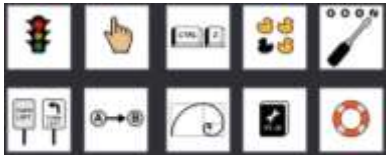
… NIELSENOVA PRAVILA …



… (8. pravilo) Estetičan i minimalističan dizajn







(9. pravilo) Pomoć korisniku da prepozna, dijagnostikuje i ispravi grešku ...

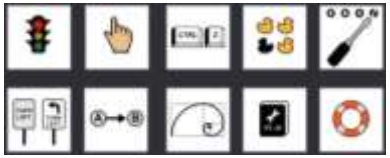


- Svakako ne isto kao i obezbediti prevenciju od grešaka
- Ako je greška načinjena, zanači da prevencija nije bila dovoljno dobra
- Ovaj princip se odnosi na poruke koje korisnik dobija od sistema („error messages“) – trebalo bi da informišu korisnika o nastaloj grški i da ga uputi kako da problem reši
- Poruke o grešci treba da budu napisane jasnim jezikom, da precizno opišu grešku i da sugerišu oporavak od greške na konstruktivan način
- Jedan od primera gde je ovaj princip lepo primenjen je istorija u Photoshop-u



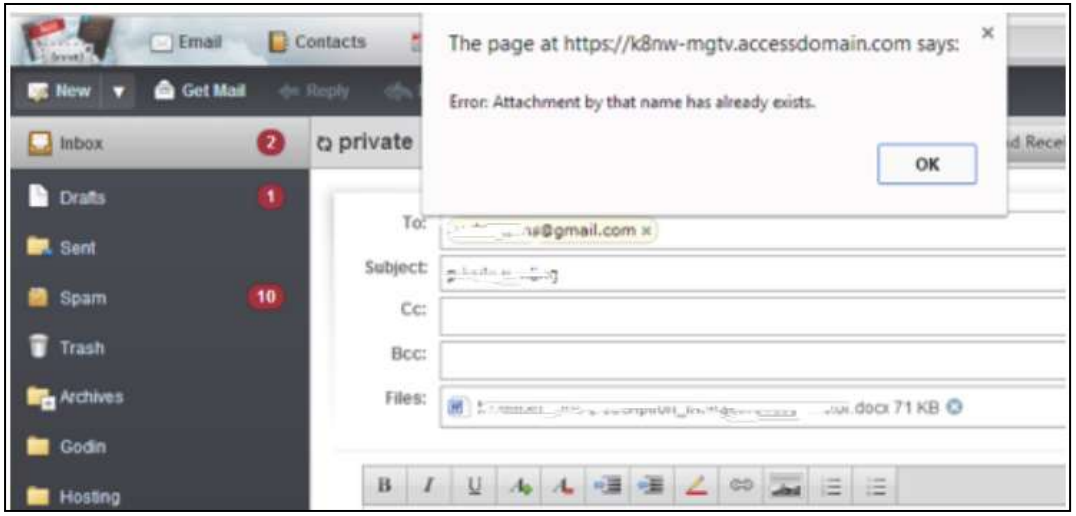


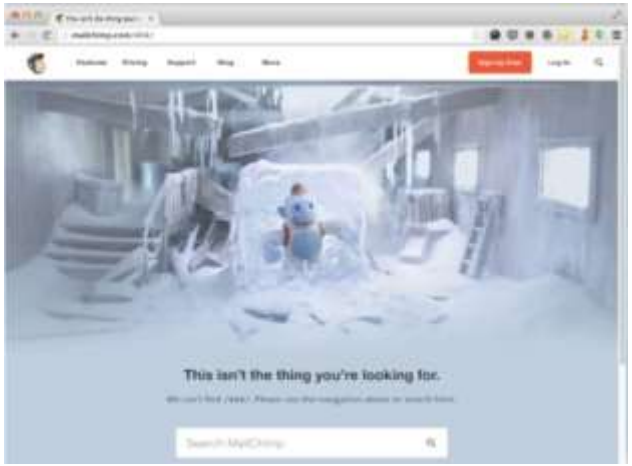
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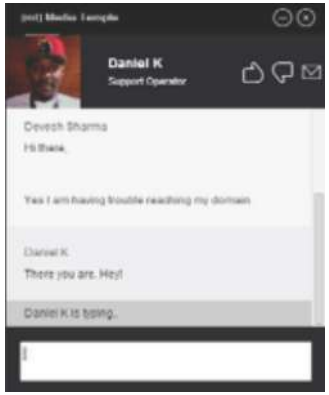
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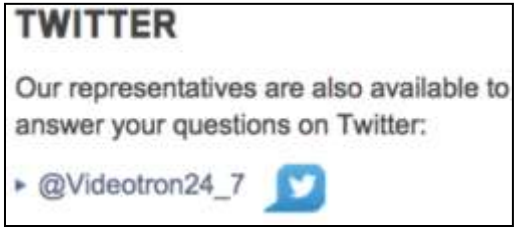


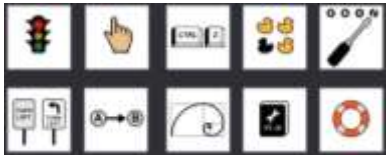












(10. pravilo) Pomoć i dokumentacija ...

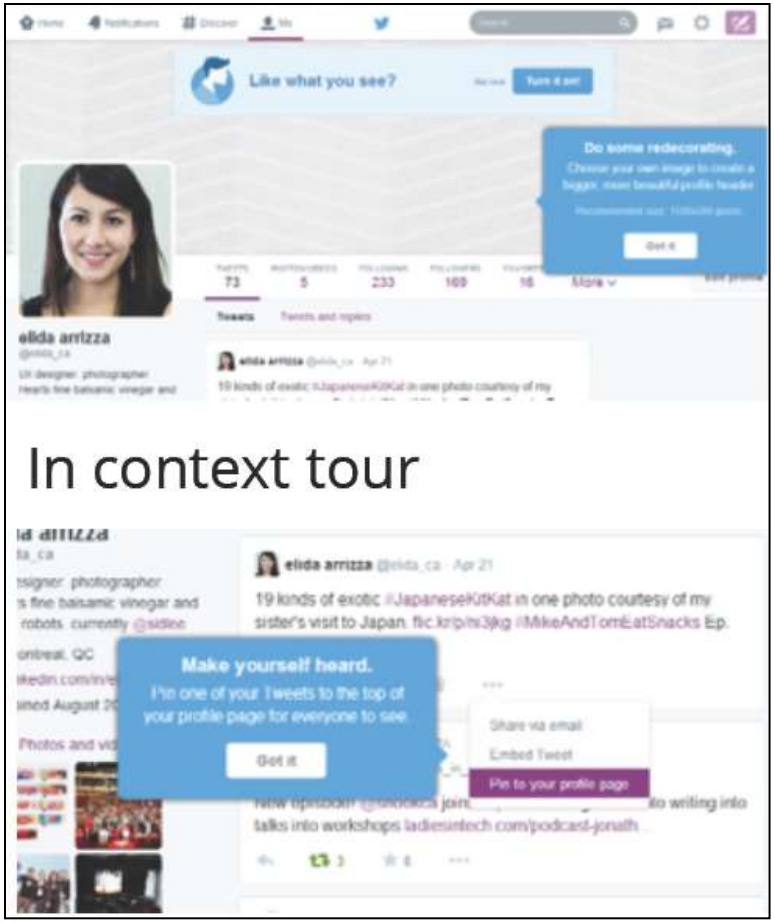


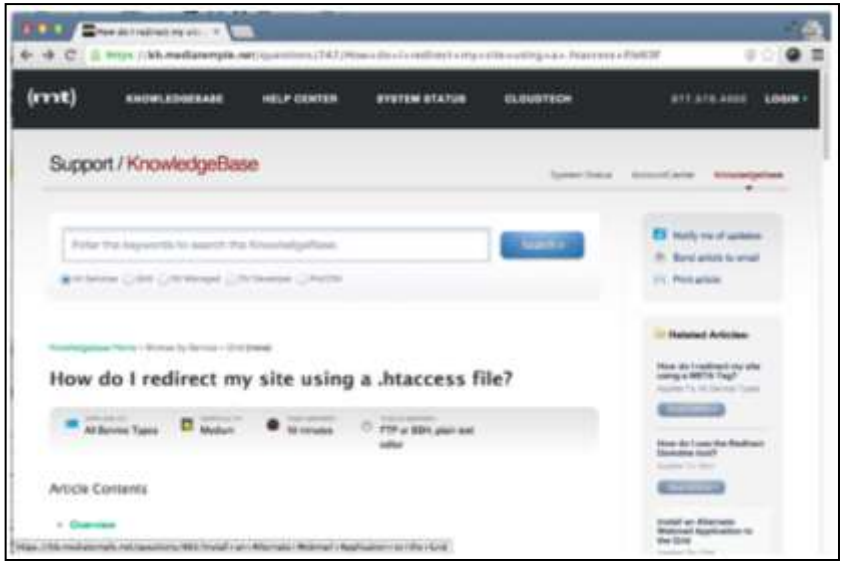
- U idealnom slučaju sistem može da se koristi bez ikakve dokumentacije ili pomoći
- Jednostavno, svaki sistem bi trebao da ima stavku pomoć sa instrukcijama o korišćenju istog
- Poželjno je da pomoć bude laka za čitanje, sa slikama i jednostavna za pretragu
- Pomoć bi trebala da bude fokusirana na korisničke zadatke
- Ne bi trebala da bude predugačka kako ne bi odvratila korisnika od korišćenja dokumentacije



… NIELSENOVA PRAVILA

… (10. pravilo) Pomoć i dokumentacija







TOG'S FIRST PRINCIPLE …



1. Aesthetics
2. Anticipation
3. Autonomy
4. Color
5. Consistency
6. Defaults
7. Discoverability
8. Efficiency of the User
9. Explorable Interfaces
10. Fitts's Law
11. Human Interface Objects
12. Latency Reduction
13. Learnability
14. Metaphors, Use of
15. Protect Users' Work
16. Readability
17. Simplicity
18. State: Track it
19. Visible Navigation

… TOG'S FIRST PRINCIPLE …



(1. pravilo) Aesthetics

- **Principle: Aesthetic design should be left to those schooled and skilled in its application: Graphic/visual designers**

- **Principle: Fashion should never trump usability**

  Generating artificial obsolescence through fashion is a time-honored and effective way to sell everything from clothing to cars. A new fashion should not and need not detract from user-performance: Enormous visual and even behavioral changes can be carried out that either do not hurt productivity or markedly increase it.

- **Principle: User test the visual design as thoroughly as the behavioral design**

  User test after aesthetic changes have been made, benchmarking, where applicable, the new design against the old. Ensure that learnability, satisfaction, and productivity have been improved or at least have stayed the same. If not, newly-added aesthetics that are causing a problem need to be rethought.

… TOG'S FIRST PRINCIPLE …



(2. pravilo) Anticipation

- **Principle: Bring to the user all the information and tools needed for each step of the process**

  Software and hardware systems should attempt to anticipate the user's wants and needs. Do not expect users to leave the current screen to search for and collect necessary information. Information must be in place and necessary tools present and visible.

  Anticipation requires that designers have a deep understanding of both the task domain and the users in order to predict what will be needed. It also requires sufficient usability testing to ensure the goal has been met: If a tool or source for information is there on the screen, but users can't find it, it may as well not even be present.

  The penalty for failing to anticipate is often swift and permanent, particularly if you do not have a captive user, as is the case with public websites and apps, for example. Those users will probably never return from their search. Even if you do have a captive user, you probably don't have a captive client, and if the client's employees are wasting time trying to find required resources, your competitors will have a good story to tell when it is time to make their next pitch.

… TOG'S FIRST PRINCIPLE …



(3. pravilo) Autonomy …

- **Principle: The computer, interface, and task environment all "belong" to the user, but user-autonomy doesn't mean we abandon rules**

  Give users some breathing room. Users learn quickly and gain a fast sense of mastery when they are placed "in charge." Paradoxically, however, people do not feel free in the absence of all boundaries (Yallum, 1980). A little child will cry equally when confined in too small a space or left to wander in a large and empty warehouse. Adults, too, feel most comfortable in an environment that is neither confining nor infinite, an environment explorable, but not hazardous.

- **Principle: Enable users to make their own decisions, even ones aesthetically poor or behaviorally less efficient**
  - Autonomy means users get to decide what keyboard they want to use, how they want their desktops to look (even if they like clutter), and what kind of apps they want to run. When developers take that kind of control away, users can be left frustrated and angry.

… TOG'S FIRST PRINCIPLE …



… (3. pravilo) Autonomy …

- **Principle: Exercise responsible control**

  Allowing users latitude does not mean developers should abandon all control. On the contrary, developers must exercise necessary control. Users should not be given so much rope they hang themselves. However, some developers today are not only taking excessive control, but making huge HCI errors in the process, like restricting text to fonts and sizes that people with ordinary eyesight can't read. They offer editing schemes that require the user to use their fat finger to place the text cursor with pixel-precision accuracy just to avoid adding the necessary arrow keys to their aesthetically perfect, but functionally crippled, keyboard.

  They also set an arbitrary timing and movement threshold for determining whether a user is or is not pressing a link on purpose, rather than her just pausing for an instant at the start of an upward swipe for scroll, for example. They then offer the user no way to alter that threshold, so many users find themselves triggering links to unwanted pages many, many times per day. That is an irresponsible application of control. We learned 30 years ago that users needed access to a slider for mouse double-clicking. Touch users need the same thing for link timing.

… TOG'S FIRST PRINCIPLE …



… (3. pravilo) Autonomy

- **Principle: Use status mechanisms to keep users aware and informed**

  No autonomy can exist in the absence of control, and control cannot be exerted in the absence of sufficient information. Status mechanisms are vital to supplying the information necessary for users to respond appropriately to changing conditions.

- **Principle: Keep status information up to date and within easy view**

  Users should not have to seek out status information. Rather, they should be able to glance at their work environment and be able to gather at least a first approximation of state and workload.

- **Principle: Ensure status information is accurate**

  Status information can be up to date, yet inaccurate. At the time of this writing, when a user updated an iPhone or iPad to a new generation of system software, a progress indicator would appear showing that it will take approximately five minutes to complete the task. Actually, it typically takes an hour or more. (The new system itself would update in five minutes, but then all the other tens or hundreds of megabytes of information on the phone had to be re-uploaded.) The user, having been lied to, was left with no way to predict when she might actually get her device back. Such a user is not feeling autonomous.

… TOG'S FIRST PRINCIPLE …



(4. pravilo) Color …

Color blindness

- **Principle: Any time you use color to convey information in the interface, you should also use clear, secondary cues to convey the information to those who cannot see the colors presented.**

  Most people have color displays nowadays. However, approximately 10% of human males, along with fewer than 1% of females, have some form of color blindness.

- **Principle: Test your site to see what color-blind individuals see**

  Search Google for simulation tools. For example, for websites, you might try http://enably.com/chrometric/. For images, http://www.colblindor.com/coblis-color-blindness-simulator/.

… TOG'S FIRST PRINCIPLE …



… (4. pravilo) Color

Color as a vital interface element

- **Principle: Do not avoid color in the interface just because not every user can see every color.**

  Color is a vital dimension of our limited communication abilities. Stripping away colors that a person who is color blind can't see does no more for that person than turning off the entire picture does for a person who is completely without sight. It's the presence of an alternate set of cues for that person that is important.

- **Principle: Do not strip away or overwhelm color cues in the interface because of a passing graphic-design fad.**

  Generating artificial obsolescence through fashion is a time-honored and effective way to move products from clothing to cars. A new fashion should not and need not, however, detract from user-performance. User test after making aesthetic changes, benchmarking the new design against the old. Ensure that learnability, satisfaction, and productivity have improved or at least stayed the same. If not, newly added aesthetics that are causing a problem need to be rethought.

… TOG'S FIRST PRINCIPLE …



(5. pravilo) Consistency …

Levels of Consistency

- **Principle: The importance of maintaining strict consistency varies by level.**

  The following list is ordered from those interface elements demanding the *least* faithful consistency effort to those demanding the *most*. (Many people assume that the order of items one through six should be exactly the reverse. This can lead to real confusion as users confront pages that look familiar, but act completely different.)

  1. **Top level consistency**
     - **Platform consistency:** Be generally consistent with *de jure* & *de facto* standards
     - **In-house consistency:** Maintain a general look & feel across your products/services

     Communicates brand and makes adoption of your other products and services easier and faster

  2. **Consistency across a suite of products, e. g., Microsoft Office**
     - General look & feel communicates family

… TOG'S FIRST PRINCIPLE …



… (5. pravilo) Consistency …

Levels of Consistency

- **Principle: The importance of maintaining strict consistency varies by level.**
  3. **The overall look & feel of a single app, application or service–splash screens, design elements, etc.**
     - A visual designer should establish a purposeful & well thought-through visual language, shaped by usability testing. User behaviors should be fully transferable throughout the product.
  4. **Small visible structures, such as icons, symbols, buttons, scroll bars, etc.**
     - The appearance of such objects needs to be strictly controlled if people are not to spend half their time trying to figure out how to scroll or print. Their location is only just slightly less important than their appearance. Where it makes sense to standardize their location, do so.

… TOG'S FIRST PRINCIPLE …



… (5. pravilo) Consistency …

Levels of Consistency

- **Principle: The importance of maintaining strict consistency varies by level.**
  5. **Invisible structures**
     - Invisible structures refers to such invisible objects as Microsoft Word's clever little left border that has all kinds of magical properties, if you ever discover it is there. It may or may not appear in your version of Word. And if it doesn't, you'll never know for sure that it isn't really there, on account of it's invisible. That is exactly what is wrong with invisible objects and why, if you insist on using them, rigid consistency becomes so important.
     - Some objects while, strictly speaking, visible, do not appear to be controls, so users, left to their own devices, might never discover their ability to be manipulated. If you absolutely insist on disguising a control, the secret rule should be crisp and clean, for example, "you can click and drag the edges of current Macintosh windows to resize them," not, "You can click and drag various things sometimes, but not other things other times, so just try a lot of stuff and see what happens."
     - Objects that convey information, rather than being used to generate information, should rarely, if ever, be made invisible. Apple has violated this in making the scroll bars on the Macintosh invisible until a user passes over them.

… TOG'S FIRST PRINCIPLE …



… (5. pravilo) Consistency …

Levels of Consistency

- **Principle: The importance of maintaining strict consistency varies by level.**
  6. **Interpretation of user behavior**
     - Changing your interpretation of a user's habitual action is one of the the worst things you can do to a user. Shortcut keys must maintain their meanings. A learned gesture must be interpreted in the standard way. If the button that carries the user to the next page or screen has been located at the bottom right for the last 30 years, don't move it to the top right. Changes that require a user to unlearn a subconscious action and learn a new one are extremely frustrating to users. Users may not even realize what has happened and assume that something has failed in their hardware or software.
     - If you want to attract existing users of someone else's product to your product, you should try to interpret your new user's commands in the same way by, for example, allowing them to reuse the same shortcut keys they've grown used to.

… TOG'S FIRST PRINCIPLE …



… (5. pravilo) Consistency …

Induced Inconsistency

- **Principle: It is just important to be visually inconsistent when things act differently as it is to be visually consistent when things act the same**

  Make objects that act differently look different. For example, a trash can is an object into which a user may place trash and later pull it back out. If you want to skip the "and pull it back out" functionality, that's fine. Just make it look like an incinerator or shredder or anything other than a trash can.

  Make pages that have changed look changed. If someone encounters an unfamiliar page on an updated website or in a revised app, they know to look around and figure out what's different. In the absence of such a cue, they will attempt to use the page exactly as they have always done, and it won't work.

… TOG'S FIRST PRINCIPLE …



… (5. pravilo) Consistency …

Continuity

- **Principle: Over time, strive for continuity, not consistency**

  If you come out with a completely re-worked area of your product or even a completely new product, it is important that people instantly recognize that something big has changed. Otherwise, they will jump into trying to use it exactly the way they always have and it just isn't going to work. "Uniformity" would mean that your next product would be identical to your last, clearly wrong, but "consistency" is little better in a field where so much growth will continue to take place. Our goal is continuity, where there is a thread that weaves through our various products and releases, guiding our users, but not tying us to the past.

… TOG'S FIRST PRINCIPLE …



… (5. pravilo) Consistency

Consistency with User Expectation

- **Principle: "The most important consistency is consistency with user expectations"—William Buxton**

  It doesn't matter how fine a logical argument you can put together for how something should work. If users expect it to work a different way, you will be facing an uphill and often unwinnable battle to change those expectations. If your way offers no clear advantage, go with what your users expect.

… TOG'S FIRST PRINCIPLE …



(6. pravilo) Defaults …

- **Principle: Defaults within fields should be easy to "blow away"**

  When a user activates a field, the current entry should be auto-selected so that pressing Backspace/Delete or starting to type will eliminate the current entry. Users can click within the field to deselect the whole, dropping the text pointer exactly where the user has clicked. The select-on-entry rule is generally followed today. (Sloppy coding, however, has resulted in the text cursor dropping at various unpredictable locations.)

- **Principle: Defaults should be "intelligent" and responsive**

  Not everything should have a default. If there isn't a predictable winner, consider not offering any default. It takes precious cognitive cycles to look at a default that covers maybe 25% of the cases and make the decision not use it. That same time could be spent entering the choice actually desired.

… TOG'S FIRST PRINCIPLE …



… (6. pravilo) Defaults …

- **Principle: Replace the word "default" with a more meaningful and responsive term**

  Users rarely have any idea of what "Default," in a given situation means. (They do know its literal meaning, of course, which is that the bank is going to take away the user's house. That always cheers them up.) Replace "Default" with "Revert to Standard Settings," "Use Customary Settings," "Restore Initial Settings," or some other more specific terminology describing what will actually happen. User test to find out what terms enable your users to accurately predict what your software will actually do.

… TOG'S FIRST PRINCIPLE …



… (6. pravilo) Defaults

- **Principle: Both your vocabulary and visual design must communicate the scope of a reversion**

  Make sure, through user testing, that users understand the extent of the restoration: Are they signing up to a benign restoration of just a few recent and localized items, or are they about to spend the next four days re-entering usernames and passwords in every app they own?

  User-test your restoration options to find out what users think the result of pressing the button will be. If you are going to do something benign, but they interpret it as potentially destructive, they won't use the option, leaving them with the same broken or partially broken system that made them consider using it in the first place. Likewise, if you wipe out hours of careful customization without properly preparing them, they may not be nearly as grateful as you might expect. (I once had a young chap in India help me out with a minor problem on my DVR. When he was finished, he had led me to reinitialize the hard disk, erasing every single program on the machine. That was a bit more restoration than I was looking for. After that, I was able to carry out the rest of my conversation with him without even using the phone. I had no idea I could yell that loud.)

  When designing tabbed objects, such as properties and preference windows, ensure that the visual design makes the scope of a restoration button clear. Individual tabbed "cards" should be visually separated from the surrounding window so that buttons may be placed either within the individual card or in the surrounding area, indicating whether the button action will apply only to the current tab or all tabs. There is never an excuse for leaving such a scope ambiguous. This is not a fashion decision.

… TOG'S FIRST PRINCIPLE …



(7. pravilo) Discoverability…

- **Principle: Any attempt to hide complexity will serve to increase it**

  Functional software does not have to look like a tractor; it can look like a Porsche. It cannot, however, look like a Porsche that's missing its steering wheel, brake, and accelerator pedal. Yet many tech companies in the late 1990s began purposely hiding their most basic controls, often to the serious detriment of their users. Why? Because they found it more important to generate the Illusion of Simplicity for potential buyers than to reveal the extent of complexity to their actual users.

  Businesses are driven to hide complexity from buyers because it can pay off in the short term: Most consumers who are potential buyers make judgements as to whether they can conquer a new machine not by sitting down and spending a day trying to learn it, but by gazing at the screen for ten minutes while the salesperson gives a demo. Stripping away scroll bars, hiding buttons, doing all the things that this section tells you not to do can all lead to increased profits, at least in the short term.

… TOG'S FIRST PRINCIPLE …



… (7. pravilo) Discoverability…

- **Principle: If you choose to hide complexity, do so in the showroom only**

  You need never decide whether to support potential buyers or eventual users. We are not working with fixed pieces of hardware. We work with either pure software or hardware driven by software. A designer can easily create a system that will fully support both buyer and user, switching appearance depending on current need. You can design the software for an operating system, for example, that will present itself in a very simple form in the store only to slowly open up like a flower, offering the user more and more accessibility and functionality as the user becomes progressively more skilled and more comfortable.

  Crippling an interface might help make the initial sale, but in the long run, it can lead to having your most important "sales force," your existing customer base, not only leave you, but tell your potential buyers to stay away as well.

… TOG'S FIRST PRINCIPLE …



… (7. pravilo) Discoverability…

- **Principle: If the user cannot find it, it does not exist**

  Not all buyers are naive. Even those that are don't stay that way long. Only the most persistent buyers/users will travel the web searching for a treasure map to features that you choose to hide from them. Most will simply turn to your competitors, taking you at your word that you just don't offer whatever they were after.

- **Use Active Discovery to guide people to more advanced features**

  With Active Discovery, you cease waiting for people to find something and, instead, offer it to them. In its ideal form, your system "realizes" they now need it and offers it to them. In most instances, we are far from being able to do that. A workable compromise:

  - **Mention to a user that a feature exists about the earliest he might need it**
  - **Repeat the message at intelligently spaced intervals. Not over and over again.**
  - **Stop mentioning it once either explored or adopted**

  The messaging might take the form of a "Did you know…" hint that you show during startup. (If you see a large percentage of your users are turning off these hints, it reflects that you are prematurely mentioning features, are giving them too many hints too often, or continue to tell them about features they have adopted. It is not necessary to give a helpful hint each and every time the user starts up your app. They are more likely to be read and appreciated if they occur on occasion.

… TOG'S FIRST PRINCIPLE …



… (7. pravilo) Discoverability…

- **Principle: Controls and other objects necessary for the successful use of software should be visibly accessible at all times**

  The object itself should either be view or enclosed by an object or series of objects (documents within folders, menu items within a menu, for example) that, in turn, are visibly accessible at all times.

  Exceptions can be made for systems that are used habitually, such as a mobile browser or reader, where:

  Screen size is so limited that it is impractical to display items not currently needed, and

  It would be difficult or impossible for the user to fail to trigger the appearance of the controls by accident, thus ensuring the user will discover their existence.

  These exceptions cover standard and widely-used operating system objects and behaviors on mobile devices, as long as users are given simple and obvious access to a help guide for using them.

… TOG'S FIRST PRINCIPLE …



… (7. pravilo) Discoverability…

- **Principle: There is no "elegance" exception to discoverability**

  A few designers, having fallen in love with the clean lines of smartphone apps, thought it would be great to visit those same clean lines on giant-screen computers. Wrong! Hiding functionality to create the Illusion of Simplicity is an approach that saps user-efficiency and makes products an easy target for competitors.

- **Principle: With the exception of small mobile devices, controls do not belong in the middle of the content area**

  Smartphone and tablet controls are sometimes forced into the content area because the screens are so small, that's the only area there is. Even there, you need to provide a standard trigger, such as a tap in the middle of the content area, that will simultaneously expose all the icons and buttons representing all the hidden controls so that users don't have to carry out a treasure hunt.

  On laptop and desktop computers, controls, and, in particular, hidden controls have no place inside the content area.

… TOG'S FIRST PRINCIPLE …



… (7. pravilo) Discoverability…

- **Principle: Communicate your gestural vocabulary with visual diagrams**

  Include a help page that shows the gestures your app can understand. Present the page when the user first opens the app and make it clear where the user will find this help page after that. In a mobile app, make the icon representing the page constantly visible or have it form part of the standard set that appears around the periphery when the user touches the triggering area of the mobile screen. For a magazines and similar media objects, make it the first open page (after the cover) of each and every issue, always there and available.

… TOG'S FIRST PRINCIPLE …



… (7. pravilo) Discoverability…

- **Principle: Strive for Balance**

  It is not 1980, when most people had never seen a computer, and we necessarily made everything highly visible. You can use subtly in design: Don't put an info icon next to every single item on the page. Instead, use overlays like this one from Google+ Snapseed that explain every symbol and every gesture at once:



  It is difficult to see with the overlay in place, but the developers have reinforced their Cancel and Apply arrows with permanent, written labels. How do you find out if such reinforcement is necessary? How do you find out whether the user can figure out what to press (in this case, an always-visible question mark at the extreme top right) to get help to begin with? Follow the next principle and apply the results.

… TOG'S FIRST PRINCIPLE …



… (7. pravilo) Discoverability

- **Principle: User-test for discoverability**

  To discover what information you need to communicate and to ensure you're successfully communicating it, you must do routine usability tests throughout a project. Test using a population that has your expected level of experience with the system and task domain. See if they can locate, identify, learn, and use the tools they need to perform the tasks you expect your users to perform. If they cannot, iterate the design until they can. Make use of Active Discovery, Dealer Modes, whatever you need to to ensure your users can discover and learn the features of your product.

  Not one of the errors I've discussed above would ever make it into production were user-experience groups conducting usability studies and altering the design based on those results.

… TOG'S FIRST PRINCIPLE …



(8. pravilo) Efficiency of the User …

- **Principle: Look at the user's productivity, not the computer's**

  In judging the efficiency of a system, look beyond just the efficiency of the machine. People cost a lot more money than machines, and while it might appear that increasing machine productivity must result in increasing human productivity, the opposite is often true. As a single example, forcing customers to enter telephone numbers without normal spacing or punctuation saves a single line of code and a handful of machine cycles. It also results in a lot of incorrectly captured phone numbers as people cannot scan clusters of ten or more digits to discover errors. (That's exactly why phone numbers are broken up into smaller pieces.) The amount of time wasted by just one person in your company trying to track down the correct version of an incorrectly entered phone number would sweep away the few minutes it would have taken to code the entry form so users could scan and correct their errors. Wrong numbers also can and do result in a lost sales. How many trillions of machine cycles would the profit from a single lost sale have covered?

… TOG'S FIRST PRINCIPLE …



… (8. pravilo) Efficiency of the User …

- **Principle: Keep the user occupied**

  Typically, the highest expense by far in a business is labor cost. Any time an employee must wait for the system to respond before they can proceed, money is being lost.

- **Principle: To maximize the efficiency of a business or other organization you must maximize everyone's efficiency, not just the efficiency of the IT department or a similar group**

  Large organizations tend to be compartmentalized, with each group looking out for its own interests, sometimes to the detriment of the organization as a whole. Information technology departments often fall into the trap of creating or adopting systems that result in increased efficiency and lowered costs for the information resources department, but at the cost of lowered productivity for the company as a whole. It is your job to run the studies that prove out whether new designs based on new, money-saving technologies will increase or decrease overall productivity among the affected workers and, if so, by how much and with what result to the corporation's bottom line.

  Work with HR or department heads to find out the average cost per hour of affected employees. (An honored rule of thumb is to take their hourly wage and multiply it by three to include all the other overhead associated with an employee, from rent to heat, lights, computer support, etc.) Multiply the overhead cost x the number of affected employees x the time an activity takes x the difference in productivity, positive or negative, in carrying out the activity to find the actual cost of a change. A positive number will help you sell your group and your design. A negative number will help your company from making a costly mistake.

… TOG'S FIRST PRINCIPLE …



… (8. pravilo) Efficiency of the User …

- **Principle: The great efficiency breakthroughs in software are to be found in the fundamental architecture of the system, not in the surface design of the interface**

  This simple truth is why it is so important for everyone involved in a software project to appreciate the importance of making user productivity goal number one and to understand the vital difference between building an efficient system and empowering an efficient user. This truth is also key to the need for close and constant cooperation, communication, and conspiracy between engineers and human interface designers if this goal is to be achieved.

  Look at the difference between the iPad and the netbook computers it crushed. The differences had nothing to do with what key you pressed to open an email. They had to do with things like not having to press any key at all.

… TOG'S FIRST PRINCIPLE …



… (8. pravilo) Efficiency of the User

- **Principle: Error messages should actually help**

  Error messages must be written by a skilled writer to:

  - Explain what's wrong
  - Tell the user specifically what to do about it
  - Leave open the possibility the message is improperly being generated by a deeper system malfunction

  "Error -1264" doesn't do any of these. Rare is the error message that covers even Point One well. Yours should cover all three. Your Quality Assurance group should be charged with the responsibility for reporting back to you any message that does not fulfill the criteria.

  Many other principles in this list affect efficiency, Latency Reduction and Readability in particular.

… TOG'S FIRST PRINCIPLE …



(9. pravilo) Explorable Interfaces …

- **Principle: Give users well-marked roads and landmarks, then let them shift into four-wheel drive**

  Mimic the safety, consistency, visibility, and predictability of the natural landscape we've evolved to navigate successfully. Don't trap users into a single path through a service, but do offer them a line of least resistance. This lets the new user and the user who just wants to get the job done in the quickest way possible a "no-brainer" way through, while still enabling those who want to explore and play what-if a means to wander farther afield.

- **Principle: Sometimes you do have to provide deep ruts**

  The earlier your users are on the experience curve, the more you need to guide them. A single-use application for accomplishing an unknown task requires a far more directive interface than a habitual-use interface for experts. The deepest ruts are wizards.

… TOG'S FIRST PRINCIPLE …



… (9. pravilo) Explorable Interfaces …

- **Principle: Offer users stable perceptual cues for a sense of "home"**

  Stable visual elements not only enable people to navigate fast, they act as dependable landmarks, giving people a sense of "home." A company logo on every page of a website, including every page of checkout, all enabling the user to escape back to the home page, makes users feel safe and secure. Paradoxically, such cues make it more likely that people will not escape back to the home page, secure in the knowledge that they easily can.

- **Principle: Make Actions reversible**

  People explore in ways beyond navigation. Sometimes they want to find out what would happen if they carried out some potentially dangerous action. Sometimes they don't intend to find out, but they do anyway by accident. By making actions reversible, users can both explore and can "get sloppy" with their work. A perfect user is a slow user.

… TOG'S FIRST PRINCIPLE …



… (9. pravilo) Explorable Interfaces …

- **Principle: Always allow "Undo"**

  The unavoidable result of not supporting undo is that you must then support a bunch of confirmation dialogs that say the equivalent of, "Are you really, really sure?" Needless to say, this slows people down.

  If you fail to provide such dialogs, in the absence of undo, people slow down even further. A study a few years back showed that people in a hazardous environment make no more mistakes than people in a supportive and more visually obvious environment, but they work a lot slower, taking great care to avoid making errors. The result was a huge hit in productivity.

  We usually think of the absence of Undo as being the sign of lazy programming, but sometimes people do it on purpose. For example, some ecommerce sites want to make it hard for you to take things back out of your shopping cart once you've put them in there. This turns out to be a backwards strategy: An ecommerce study we did at the Nielsen Norman Group looked at what happens when merchants make it really easy to take things out of shopping carts. As might be expected, people visiting these merchants were much more willing to throw things in, figuring, "I can always take it back out later." Except they didn't take them back out, because the deletion rate was no different. These user just bought more stuff.

… TOG'S FIRST PRINCIPLE …



… (9. pravilo) Explorable Interfaces

- **Principle: Always allow a way out**

  Users should never feel trapped inside a maze. They should have a clear path out.

- **Principle: Make it easy and attractive to stay in**

  A clear, visible workflow that enables people to understand where they are and move either backward or forward in a process will encourage people to stick with a task. Consider, as an example, a multi-step checkout procedure. Making the navigation visible by putting each step on a clearly-labelled tab will let people know where they are in the process. Clicking an earlier tab should allow people to jump back to correct an error or just change their mind by, for example, selecting a different delivery address. They should then be able to click on the tab they were originally on and resume their forward movement. When you either forbid people to move back or destroy all subsequent data if they do, they will not be happy with you. Even if they decide to grit their teeth and continue with the current sale, they are unlikely to ever return.

… TOG'S FIRST PRINCIPLE …



(10. pravilo) Fitts's Law …

- **Principle: The time to acquire a target is a function of the distance to and size of the target**

  Use large objects for important functions (Big buttons are faster). Use small objects for functions you would prefer users not perform.

  Use the pinning actions of the sides, bottom, top, and corners of your display: A single-row toolbar with tool icons that "bleed" into the edges of the display will be significantly faster than a double row of icons with a carefully-applied one-pixel non-clickable edge between the closer tools and the side of the display. (Even a one-pixel boundary can result in a 20% - 30% slow-down for tools along the edge.)

  While at first glance, this law might seem patently obvious, it is one of the most ignored principles in design. Fitts's law (often improperly spelled "Fitts' Law") dictates the Macintosh pull-down menu acquisition should be approximately five times faster than Windows menu acquisition, and this is proven out.

  Fitts's law predicted that the Windows Start menu was built upside down, with the most used applications farthest from the entry point, and tests proved that out. Fitts's law indicated that the most quickly accessed targets on any computer display are the four corners of the screen, because of their pinning action, and yet, for years, they seemed to be avoided at all costs by designers.

… TOG'S FIRST PRINCIPLE …



… (10. pravilo) Fitts's Law …

- **Multiple Fitts: The time to acquire multiple targets is the sum of the time to acquire each**

  In attempting to "Fittsize" a design, look to not only reduce distances and increase target sizes, but to reduce the total number of targets that must be acquired to carry out a given task. Remember that there are two classes of targets: Those found in the virtual world—buttons, slides, menus, drag drop-off points, etc., and those in the physical world—keyboards and the keys upon them, mice, physical locations on touch screens. All of these are targets.

- **Principle: Fitts's Law is in effect regardless of the kind of pointing device or the nature of the target**

  Fitts's Law was not repealed with the advent of smartphone or tablets. Paul Fitts, who first postulated the law in the 1940s, was working on aircraft cockpit design with physical controls, something much more akin to a touch interface rather than the indirect manipulation of a mouse. The pinning action of the sides and corners will be absent unless the screen itself is inset, but the distance to and size of the target continue to dictate acquisition times per Fitts's law just as always.

… TOG'S FIRST PRINCIPLE …



… (10. pravilo) Fitts's Law

- **Principle: Fitts's Law requires a stop watch test**

  Like so much in the field of human-computer interaction, you must do a timed usability study to test for Fitts's Law efficiency.

… TOG'S FIRST PRINCIPLE …



(11. pravilo) Human Interface Objects …

Human-interface objects are separate and distinct from the objects found in object-oriented systems. Our objects include folders, documents, buttons, menus, and the trashcan. They appear within the user's environment and may or may not map directly to an object-oriented program's object. In fact, many early GUI's were built entirely in non-object-oriented environments.

– **Principle: Human-interface objects can be seen, heard, felt, or otherwise perceived**

Human interface objects that can be seen are quite familiar in graphic user interfaces. Objects that are perceived by another sense such as hearing or touch are less familiar or are not necessarily recognized by us as being objects. Ring tones are auditory objects, for example, but we tend to just think of them as ring tones, without assigning any higher-level category to them.

- **Principle: Human-interface objects have a standard way of being manipulated**

- Buttons are pressed, sliders are dragged, etc.

… TOG'S FIRST PRINCIPLE …



… (11. pravilo) Human Interface Objects …

- **Principle: Human-interface objects have a standard way of being manipulated**

  Buttons are pressed, sliders are dragged, etc.

- **Principle: Human-interface objects have standard resulting behaviors**

  Dropping a document on a trash can does not delete it, it stores it in the trash can. Selecting "Empty Trash" is necessary to actually delete it.

- **Principle: Human-interface objects should be understandable, self-consistent, and stable**

… TOG'S FIRST PRINCIPLE …



… (11. pravilo) Human Interface Objects

- **Principle: Use a new object when you want a user to interact with it in a different way or when it will result in different behavior**

  If dropping a document on your delete-document icon will destroy it instantly and permanently, do not make it look like a trash can. People come with expectations about previously encountered objects. It's important not to confuse or water down such expectations. For example, if you use a trash can icon, but instantly destroy documents dropped into it, it broadens the rule for trash cans. Instead of the rule remaining: "Dropping a document on a trash can does not delete it. Rather, it stores it in the trash can. Selecting 'Empty Trash' is necessary to actually delete it," it will shift to, "Dropping a document on a trash can will destroy it either right now or sometime in the next six months to a year." That is not only confusing for your users, it is damaging to every other developer that uses the trash can icon in the proper way.

… TOG'S FIRST PRINCIPLE …



(12. pravilo) Latency Reduction …

– Wherever possible, use multi-threading to push latency into the background

Latency can often be hidden from users through multi-tasking techniques, letting them continue with their work while transmission and computation take place in the background. Modern web browsers can pre-fetch data, reducing the dead time when the user reaches the end of a task and must wait for the next page to appear.

– **Principle: Reduce the user's experience of latency**

- Acknowledge all button clicks by visual or aural feedback within 50 milliseconds
- Trap multiple clicks of the same button or object.

Because the Internet is slow, people tend to press the same button repeatedly, causing things to be even slower.

… TOG'S FIRST PRINCIPLE …

… (12. pravilo) Latency Reduction …

– **Principle: Keep users informed when they face delay**

Expected Delay	Indication
1/2 to 2 seconds	Use animated mouse cursor or other "busy" indicator 
> 2 seconds	Tell them potential length of wait
> 5 seconds	Use an animated progress indicator  Process must end by the time indicator is full!
> 10 seconds	Keep users a) informed & b) entertained
> 15 seconds	Same as >10 plus add at end a noticeable sound & strong visual indication so users know to return

… TOG'S FIRST PRINCIPLE …



… (12. pravilo) Latency Reduction

- **Principle: Make it faster to begin with**

  Eliminate any element of the application that is not helping. Be ruthless.

  The sluggish speed of the early web set users's expectations extremely low. (It also burst the Internet bubble when people realized they could get in their car and drive round-trip to the shopping center in less time than it took to "trick" the website into selling them something.) They have become less forgiving as time has passed.

  Mobile, which has an architecture more in keeping with traditional GUI applications than web browsing, has been reminding people that computers can be fast, and they are even more impatient with slow-downs. Wearables will come with an even higher level of expectation: No one waits to see what time it is, and they will not wait to see who is calling, what the temperature is outside, or any other information to be displayed.

  Automotive applications today are oftentimes sluggish, suffering from a fatal cocktail of weak hardware, poor design/coding practices, and high latency. Consider the car hurtling down the road at 88 feet per second (27 meters per second) while the user, eyes fixed on the flat panel display, waits to learn which of ACDC's many fine works he's currently enjoying. Imagine the rich irony when the accident report reveals it was "Highway to Hell."

… TOG'S FIRST PRINCIPLE …



(13. pravilo) Learnability …

Ideally, products would have no learning curve: Users would walk up to them for the very first time and achieve instant mastery. In practice, however, all applications and services, no matter how simple, will display a learning curve.

– **Principle: Limit the Trade-Offs**

Learnability and usability are not mutually exclusive. First, decide which is the most important; then attack both with vigor. Ease of learning automatically coming at the expense of ease of use is a myth.

How do you decide whether learnability or usability is most important? The first thing you must do is identify **frequency of use**: Are you working on a product or service that will be used only once or infrequently, or is it one that will be used habitually? If it's single-use, the answer is clear: Learnability. If someone will use this every day, eight hours a day for the rest of his or her life, the answer is equally clear: Usability.

Next, **who is the buyer?** If the person who will use it habitually will also make the buying decision, a product's reputation for learnability may be a key factor in making the sale. That's why you want to identify the most important of the two, then attack both.

… TOG'S FIRST PRINCIPLE …



… (13. pravilo) Learnability

- **Principle: Avoid only testing for learnability**

  Much usability testing involves running a series of tests at regular intervals with your spending only 20 minutes to an hour with each subject you recruit. You end up knowing everything about the initial learning curve and nothing about either the long-term curve or the end-state level of productivity.

  If you are working on an application that will be used habitually, go about it entirely differently: Work with HR to hire temporary workers. Then, have them spend a week or two coming up to speed on the interface, monitoring them with time tests to see what the overall learning curve and the eventual efficiency of your interface actually prove out to be.

… TOG'S FIRST PRINCIPLE …



(14. pravilo) Metaphors, Use of …

- **Principle: Choose metaphors that will enable users to instantly grasp the finest details of the conceptual model**

  Good metaphors generate in the users' minds a strong series of connections to past experiences from the real world or from a previous cyberspace encounter, enabling users to form a fast and accurate sense of your system's capabilities and limitations.

- **Principle: Bring metaphors "alive" by appealing to people's perceptions–sight, sound, touch, and proprioception/kinesthesia–as well as triggering their memories**

  Try making your concepts visually apparent in the software itself. If that proves impractical, make it visual apparent through an illustration. The illustration should be compact and meaningful. Test it to see if it works, then embed it in such a way that every user that needs to see it will see it.

… TOG'S FIRST PRINCIPLE …



… (14. pravilo) Metaphors, Use of …

- **Principle: Expand beyond literal interpretation of real-world counterparts**

  Most metaphors evoke the familiar, but can and usually should add a new twist. For example, an electronic newspaper might bear a strong resemblance to a traditional paper, but with hyperlinks than enable users to quickly dive as far into articles as their interest drives them, something quite impossible with their paper counterparts. Not only is there no need to slavishly copy a real-world object (skeuomorphism), but unnecessarily limiting the functionality of a software counterpart just to "perfect" the imitation is most often bad design.

  The inverse of skeuomorphism is abstraction, a prominent feature in so-called flat design, a fashion that took hold in 2013, turning once well-understood icons and other elements into meaningless abstractions and even false symbols. (For example, the icon for the browser on the iPhone became a compass, only connected to the concept of the web through the vaguest of abstractions. The iPhone has an actual compass, so they turned its icon into… another compass! Two compass icons: One tells you which way is north and the other connects you to your bank account. The Settings icon had originally looked like the inner workings of a clock, clearly carrying the message that this is an app that will let you see and affect the inside workings of the iPhone. That was abstracted to the point that it looks exactly like a large industrial fan.)

… TOG'S FIRST PRINCIPLE …



… (14. pravilo) Metaphors, Use of

- **Principle: If a metaphor is holding you back, abandon it**

  A metaphor is intended to empower your user. However, there are times when it can also hold back your design.

… TOG'S FIRST PRINCIPLE …



(15. pravilo) Protect Users' Work …

- **Principle: Ensure that users never lose their work**

  This principle is all but absolute. Users should not lose their work as a result of error on their part, the vagaries of Internet transmission, or any other reason other than the completely unavoidable, such as sudden loss of power to a client computer without proper power protection. We've gotten so used to being the victim of data loss that we often don't even notice it. So consider if what happens routinely on the web happened in real life:

  You go into Harrod's Depart. Store in London. After making your selections, you are asked to fill out a four-page form. A gentleman looks the form over, then points to the bottom of Page 3 at your phone number. "Excuse me," he says, "Look there. See how you used spaces in your phone number?" When you nod, he continues, "We weren't expecting you to do that," at which point, he picks up the four-page form and rips it to shreds before handing you a new, blank form.

  Of course, never in a thousand years would such an event take place at Harrod's, but another venerated British institution did exactly that to me almost twenty years into the miracle of the world wide web when I was invited to give them my emergency contact information for an upcoming flight into London. Every time I would fill out all eight fields on the form, it would come back with an error message about at least one field, having destroyed the entire contents of all eight of them! I'm sure it was carrying out this wanton destruction for my own good, but I could not for the life of me figure out from the messages what it actually wanted. 20 minutes and two browsers later, I gave up. I'm sure other passengers are abandoning their effort far earlier.

… TOG'S FIRST PRINCIPLE …



… (15. pravilo) Protect Users' Work …

Travel sites, in general, think nothing of repeatedly tossing all the information the user has entered about cities, times, days of travel, frequent flyer numbers, anything that takes time and trouble to type in. The user may attempt nothing more radical than leaving a half hour later, but that is apparently grounds to destroy their choice of departure city and date as well as arrival city and date. If the user is impolite enough to go to the bathroom, well, that sort of activity poses a significant security risk, so of course their entire evening's work must be destroyed, with a message explaining its been done for their own good.

Travel sites may be the tip of the iceberg, but websites in general, are notorious for their cavalier attitude when it comes to their user's hard work, and it doesn't stop there: Traditional applications continue to crash and burn, and the excuses for entire computer systems crashing and burning are at an end. Small portables can survive a power outage. It's no longer acceptable that many of today's high-end desktop computers and operating systems still do not support and encourage continuous-save. That, coupled with a small amount of power-protected memory, could eliminate the embarrassment of $5000 machines offering less reliability than 10-cent toys.

… TOG'S FIRST PRINCIPLE …



(16. pravilo) Readability …

- **Principle: Text that must be read should have high contrast**

  Favor black text on white or pale yellow backgrounds. Avoid gray backgrounds.

- **Principle: Use font sizes that are large enough to be readable on standard displays**

  You must demand of your marketing people that they tell you what the expected range of your customer's standard displays will be. You then need to work with your graphic design and engineering people to ensure that your code will show up in appropriate sizes across that range of displays. It need not be one-size-fits-all. For example, CSS can mold itself to the system in which it finds itself.

… TOG'S FIRST PRINCIPLE …



… (16. pravilo) Readability …

- **Principle: Favor particularly large characters for the actual data you intend to display, as opposed to labels and instructions.**

  For example, the label, "Last Name," can afford to be somewhat small. Habitual users will learn that that two-word gray blob says "Last Name." Even new users, based on the context of the form on which it appears, will have a pretty good guess that it says "Last Name." The actual last name entered/displayed, however, must be clearly readable. This becomes even more important for numbers. Human languages are highly redundant, enabling people to "heal" garbled messages. Numbers, however, unless they follow a very strict protocol, have no redundancy, so people need the ability to examine and comprehend every single character.

- **Principle: Menu and button labels should have the key word(s) first, forming unique labels**

  Experienced users read only as much of an item name to differentiate among items. Highly experienced users actually trigger on the difference in the external shapes of the entire first word(s) without ever actually reading anything.

… TOG'S FIRST PRINCIPLE …



… (16. pravilo) Readability

- **Principle: Test all designs on your oldest expected user population**

  Presbyopia, the condition of hardened, less flexible lenses, coupled with reduced light transmission into the eye, affects most people over age 45. Do not trust your young eyes to make size and contrast decisions. You cannot.

- **Principle: There's often an inverse relationship between the "prettiness" of a font and its readability**

  Specifically, anti-aliasing softens the edges of a font, giving it a much smoother appearance on the digital page. The problem is that the human vision system responds to sharp edges, so, in smaller font sizes, an anti-aliased font, while often appearing more attractive, can be quite difficult to comprehend.There are anti-alias techniques that specifically increase the sharpness of the edges the eye is seeking, so this is not strictly a black and white issue (so to speak), but it is definitely something of which you should be aware and not something in which every graphic designer has been schooled. You will want to run some reading speed and comprehension tests on proposed font changes.

… TOG'S FIRST PRINCIPLE …



(17. pravilo) Simplicity …

- **Principle: Balance ease of installation vs. ease of use**

  As designers, we need to strive to simplify users' lives. That often requires a delicate balance between our effort to make installation of a product easier and making subsequent use of that product easier or better.

  Consider the autofill feature for browsers: The user is required to enter and maintain a database of information the browser can then drop into a form at the user's command. It takes time to set it up, and every time anything changes, it's just one more record that has to be altered. What's more, it often fails to work, either not responding or putting incorrect data all over the place.

  Apple has simplified the setup process by enabling the user to link Safari's autoFill to the user's contact card in his or her address book. However, the ability of Safari to actually fill out a form is just as dismal as it's ever been, largely because there is no standardization of labels, locations, or anything else in forms.

  I have solved the autoFill problem with a more technically complex solution: I use an app called Keyboard Maestro that sits in the background looking for certain key combinations. When it finds one I've programmed, it automatically replaces the text I've typed with a string of text I've previously stored. Setup was definitely more difficult, but now when I open a form and a field calls for my first name, I type, "bbbb" and it's replaced with "Bruce." I type "aaaa" and my address appears, "pppp" and my phone number pops into place, etc. It takes me 30 seconds to fill out a form, longer than autoFill would if it actually worked, but this method works on every single form every single time. It saves me time, effort, and frustration.

… TOG'S FIRST PRINCIPLE …



… (17. pravilo) Simplicity …

Often, you can assume that one user in the house will be technically inclined. When you have a trade-off between simplicity of installation/set-up and ease-of-use, get together with your marketing people. If they tell you that you can depend on at least one reasonably clever or sophisticated user, do make life a bit more difficult at first if it will make subsequent use a lot simpler for everyone else. However, expend effort making both installation and operation as simple as possible. That's the approach that Nest took, where one person in the house must go through a complex and confusing process to tie their products to the Internet, but thereafter the lives of everyone in the house become simper.

(In Nest's defense, they are doing their best to overcome a major flaw in the way Wi-Fi set up to work. It requires users to leave their normal Wi-Fi and log in to a new "network" with a gobbledygook name which is, in fact, their Nest device. This is a weird, backwards activity that throws most users the first time they encounter it. It also requires their going into the "basement area" of their phone or tablet, a place most users avoid whenever possible. It's all-around bad, and the committees that oversee the Wi-Fi protocol need to address the issue if connected devices are to take off on an expanded basis.)

… TOG'S FIRST PRINCIPLE …



… (17. pravilo) Simplicity …

- **Principle: Avoid the "Illusion of Simplicity"**

  In the early years of this century, Apple became so focused on generating the illusion of simplicity for the potential buyer that they began seriously eroding the productivity of their products. They thought they had a good reason: They wanted new products to look bright, shiny, and simple to potential users. That's an excellent goal, but actual simplicity is achieved by simplifying things, not by hiding complexity. (See Visibility.)

  It's just fine to make your showroom products look simple, but, to the extent you want to hide complexity to avoid scaring away buyers, do so in the showroom, not in the home or office of the purchaser now trying to accomplish real work. I started putting a special Dealer Mode into Apple software in 1978, so that the product would look and act differently in the showroom than in the buyer's home. Computers allow that. Somewhere along the line, people forgot.

… TOG'S FIRST PRINCIPLE …



… (17. pravilo) Simplicity …

- **Principle: Use Progressive Revelation to flatten the learning curve**

  It is OK to make the user's environment simpler when they are learning by hiding more advanced pathways and capabilities, revealing them when users come to need them and know how to handle them. This is distinct from the illusion of simplicity, where necessary controls are made invisible or hidden in obscure and unusual places so that users have to go on treasure hunts to find the tools they need to use right now.

  Progressive revelation can cut down on support costs by eliminating calls from users trying to understand advanced capabilities before they have learned enough about the task domain to need them. It can also raise costs if advanced features are not introduced when they are needed or are too well hidden.

… TOG'S FIRST PRINCIPLE …



… (17. pravilo) Simplicity …

- **Principle: Do not simplify by eliminating necessary capabilities**
  - This became another Apple problem after the release of their mobile devices. In 2014 on a Mac, you can set an alarm that will trigger 90 minutes before a calendar event. On the iPad, you can set a trigger for either one hour or two, but not 90 minutes. If a person needs a warning 90 minutes before the event, that's when they need the warning. Apple has "simplified" the interface by giving the user no way to set an arbitrary time. No weakness or defect in the underlying interface would prevent Apple from giving users this capability. It is a conscious decision to limit what people can do with the product.
  - The way you set a 90 minute warning on an iPad is to create a second event, 30 minutes before the real event, and set a 60 minute warning.
  - How is that simpler?

… TOG'S FIRST PRINCIPLE …



… (17. pravilo) Simplicity

Likewise, they have a very simple interface for finding photographs in your collection: You look through all your folders of photographs, one at a time, until you find the picture you're looking for. Apple will neither display nor allow you to sort/search on the title, caption, or keywords you've carefully associated with your images. One can argue that the interface is simple: If you want to find a specific photo among that 20,000 on your iDevice, just look through your 73 folders you've created in iPhoto or Aperture to hold them until you find it. You don't have to learn about searching, you don't have to remember the name, you just have to have 10 to 20 minutes on your hands to spend the time looking.

How is that simple?

Fortunately, after many years, help is at hand: Apps like Photo Shack HD (the HD is important) enable you to search on all the criteria that Apple is importing but refusing to show you. That's OK for really advanced features. However a remarkably high percentage (100% to be exact) know how to search.

… TOG'S FIRST PRINCIPLE …



(18. pravilo) State …

– **Principle: Because many of our browser-based products exist in a stateless environment, we have the responsibility to track state as needed**

Our systems should "know":

- **Whether this is the first time the user has been in the system**
- **Where the user was when they left off in the last session**
- **What the user has found of interest based on time spent with a pointing device moving, objects being touched, etc., in different areas**
- **Where the user has been during this session**
- **Where the user is right now and what they are doing**

and myriad other details. In addition to simply knowing where our users have been, we can also make good use of what they've done.

One site with which you are familiar is so involved in and good at tracking state that it could be described as a state-tracking system that happens to do other stuff. That site is amazon.com. Their uncanny ability to make suggestions on what we might want to explore and buy is the result of their understanding our full history on their site. They know what expensive items we've come back to repeatedly in the past, what we've lingered over recently, and what would go well with what we just or recently purchased based on like-minded individuals.

… TOG'S FIRST PRINCIPLE …



… (18. pravilo) State …

- **Principle: State information should be stored in encrypted form on the server when they log off**

  Users should be able to log off at work, go home, and take up exactly where they left off. Following the principle of Protect Users' Work, whatever they were last working on should be preserved in its current condition.

  A private service for doctors, *Physicians On Line,* does an excellent job with this. Doctors can be 95% of the way through a complex transaction, log off, log in again six weeks later from another part of the world, and the service will ask them if they want to be taken right back to where they were.

  "Track State" came late to this list, in 1996. Up until then, everyone had been tracking state on their own, without question. Because the web browsers failed to provide any tools beyond the purple color of a link indicating that link had been previously visited, engineers took this to mean they no longer needed to concern themselves with state at all. To the contrary, what it meant is that, from that day until this, applications engineers and designers have had to take over the full responsibility for tracking state that had historically been shared with the systems engineers, making the job that much harder.

… TOG'S FIRST PRINCIPLE …



… (18. pravilo) State

- **Principle: Make clear what you will store & protect the user's information**

  State data is neither good nor evil, but it can be put to both uses. You should make clear in your privacy policy that you will be saving data, making your case for why it is in the user's interest. Any data from the user, including state data, should be encrypted and safeguarded.

… TOG'S FIRST PRINCIPLE …



(19. pravilo) Visible Navigation …

- **Principle: Make navigation visible**
  - Most users cannot and will not build elaborate mental maps and will become lost or tired if expected to do so.
  - The World Wide Web, for all its pretty screens and fancy buttons, is, in effect, an invisible navigation space. True, you can always see the specific page you are on, but you cannot see anything of the vast space between pages. Once users reach our sites or web-based applications, we must take care to reduce navigation to a minimum and make sure the remaining navigation is clear and natural. Ideally, present the illusion that users are always in the same place, with the work brought to them, as is done with the desktop metaphor. This not only eliminates the need for maps and other navigational aids, it offers users a greater sense of mastery and autonomy.

… TOG'S FIRST PRINCIPLE



… (19. pravilo) Visible Navigation

- **Principle: Limit screen counts by using overlays**
  - In designing complex apps, strive for a minimal number of screens, each representing a separate and distinct task the user will be performing. When a user needs to perform a subtask, bring up an overlay that is smaller than full screen, so that users can see a darkened image of the main screen still present in the background. What is seen need not be memorized, so users need not remember how that overlay maps onto the screen behind it.

NORMAN'S PRINCIPLES…



- Don Norman's 'Design of Everyday Things'
  1. Visibility
  2. Feedback
  3. Affordance
  4. Mapping
  5. Constraints
  6. Consistency

… NORMAN'S PRINCIPLES …



(1. pravilo) Visibility

Users need to know what all the options are, and know straight away how to access them. In the case of websites, this is an easy win.

For example, use intuitive iconography that clearly indicates there are more options hiding deeper down (for example, the Hamburger). This is even more important for mobile design, when screen space is at a premium and there's a strong (STRONG) desire to get rid of everything except your CTA.

(2. pravilo) Feedback

Every action needs a reaction. There needs to be some indication, like a sound, a moving dial, a spinning rainbow wheel, that the user's action caused something.

Google Chrome does a great job of this when they're loading pages. The little spinning circle starts as soon as you hit enter, so you know something's happening, and goes faster when the page is about to load, so you know you're about to do something again. It's simple and effective feedback.

… NORMAN'S PRINCIPLES …



(3. pravilo) Affordance

Affordance is the relationship between what something looks like and how it's used.

For designers, it means that as soon as someone sees something, they have to know how to use it. For example, a mug has high affordance: it's easy to figure out intuitively how to use it. For web designers, affordance is even more important. Users need to be able to tell how to access information they want from a website, or else they'll just leave.

(4. pravilo) Mapping

Mapping is the relationship between control and effect. The idea is that with good design, the controls to something will closely resemble what they affect.

A great example of mapping is the vertical scroll bar. It tells you where you are in a page, and as you drag it down, the page moves down at the same rate; control and effect are closely mapped.

… NORMAN'S PRINCIPLES



(5. pravilo) Constraints

Constraints are the limits to an interaction or an interface. Some are really obvious and physical, for example the screen size on a phone. Others are more nuanced, like a single, continuous page website having an image peeking onto the main page. It is logical for the user to scroll down to see the next image, and thus the rest of the website.

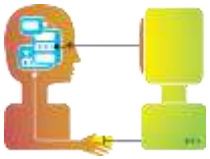
(6. pravilo) Consistency

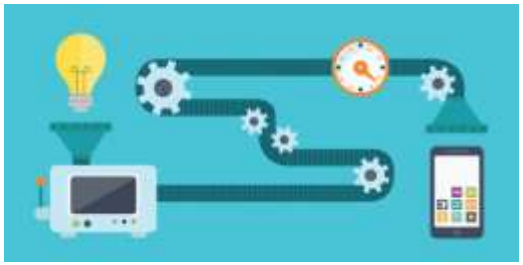
The same action has to cause the same reaction, every time.

If a website has a back button that sometimes turns a computer off, it becomes very hard to navigate around the web. The same applies for visual consistency. Logos and brands have power and recognition because they use the same logo. Twinings has used theirs for 227 years, and now everyone knows that a box with 'Twinings' written on it has tea inside it.

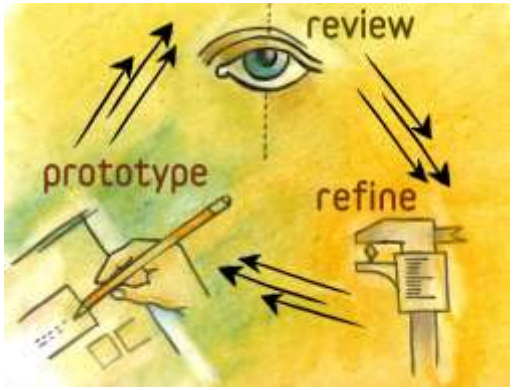
DIZAJN INTERFEJSA – PROTOTIPOVI

ŠTA JE TO PROTOTIP





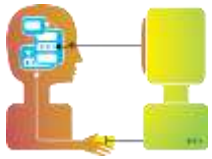
- Eksperimentalan, nedovršen dizajn
- Jednostavna verzija (simulacija) sistema koja se može brzo napraviti
- Osnovni alat za istraživanje, komunikaciju i evaluaciju dizajna
- Koristi se

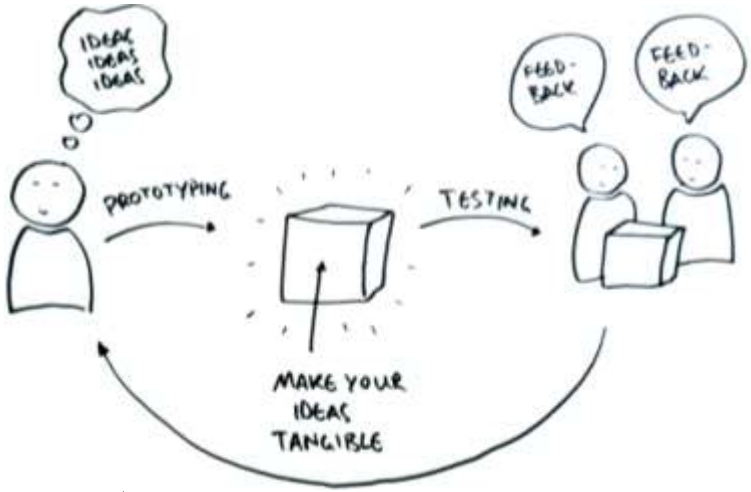


  - u ranoj fazi dizajniranja kako bi se komunicirale (prenele) i razmenile ideje između dizajnera kor. interfejsa i korisnika (svih zainteresovanih strana)
  - kako bi se razjasnile nedoumice iz prikupljenih zahteva
  - u kasnijim fazama razvoja kor. interfejsa da se razjasne nedoumice i demonstriraju koncepti interakcije i dizajna

- Najkorisniji su oni prototipovi koji se mogu brzo izmeniti

ZAŠTO KORISTITI PROTOTIPOVE ...



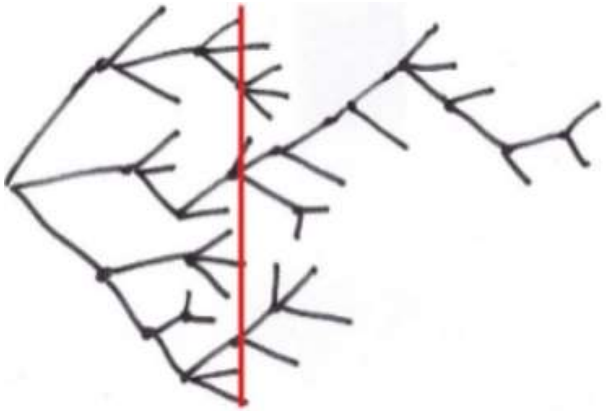


- Kako bi se proverile ideje sa korisnicima
- Kako bi se odredila upotrebljivost kor. interfejsa
- Kako bi se omogućilo korisnicima da doprinesu dizajnu kor. interfejsa
- Kako bi se omogućilo testiranje ideja i koncepata
- Kako bi se validirali prikupljeni zahtevi
- Kako bi se pregovaralo oko zahteve i funkcijalnosti sistema

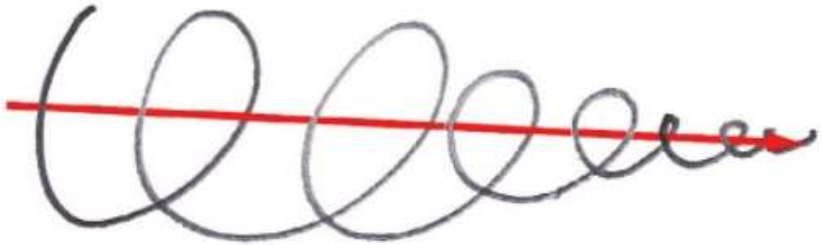


- Kako bi se na jeftin način dobile povratne informacije od korisnika u ranoj fazi dizajna
- Kako bi se eksperimentisalo sa alternativnim rešenjima
- Lako ih je izmeniti i odbaciti

... ZAŠTO KORISTITI PROTOTIPOVE



v



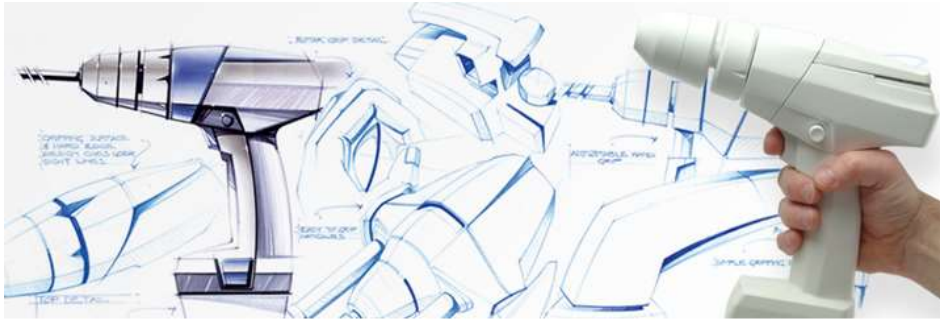
Design

"Branching Exploration"

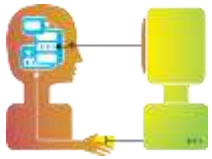
Prototyping

"Incremental iterative refinement"

Bill Buxton, Sketching User Experiences



FIDELITY (VERODOSTOJNOST PROTOTIPA) ...

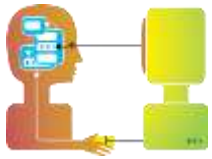


- Jedna od važnih osobina prototipa je njegova verodostojnost, tj. koliko je njegov izgled blizak izgledu konačnog dizajna

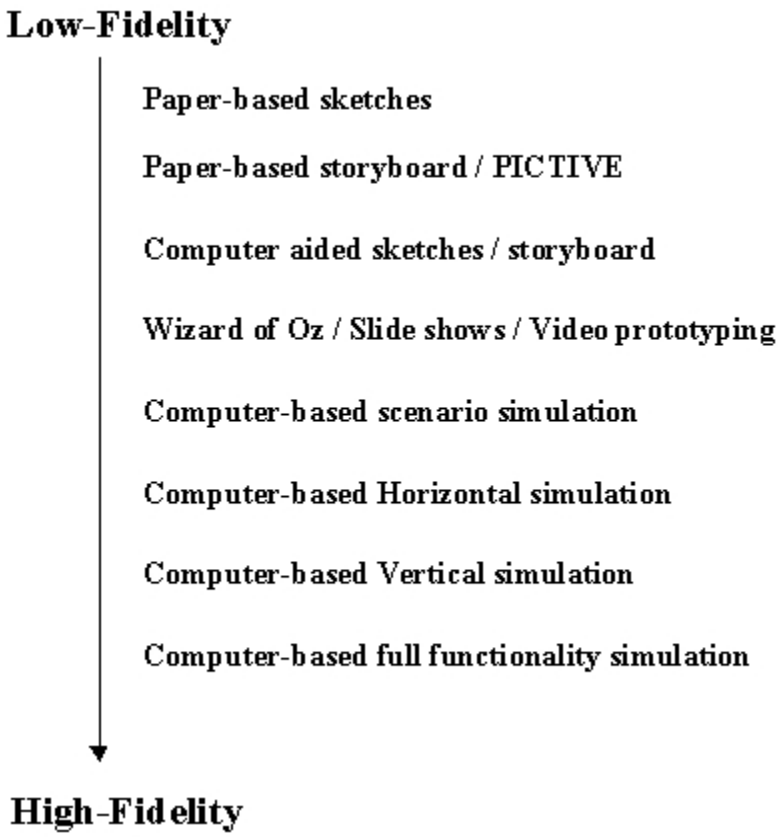
- Shodno tome, prototipovi se mogu podeliti na:
  - Low-fidelity prototipove – sa malo detalja, koriste jeftinije materijale i/ili koriste drugačije interakcione tehnike
  - High-fidelity prototipove – koji su vrlo slični krajnjem proizvodu



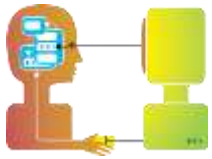
... FIDELITY (VERODOSTOJNOST PROTOTIPA) ...



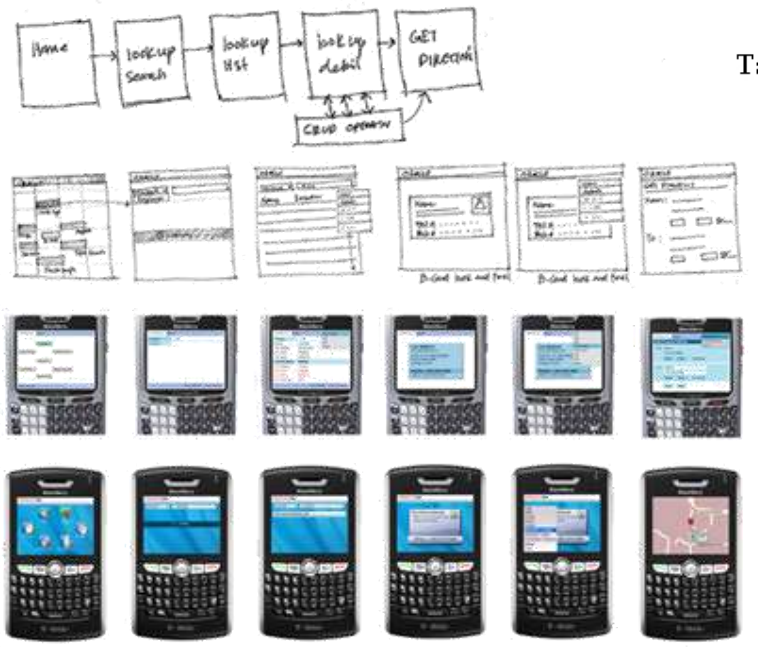
- Fidelity je multidimenzionalne prirode
  - Vertikalnost – koliko funkcija ima prototip (koliko je operativan prototip)
    - limitirani izbori, ograničeni odgovori, bez kontrole greške
    - samo vertikalnost implicira jedan deo feature-a implementiran skroz
  - Horizontalnost – koliko funkcionlanosti je podržano (koliko feature-a)
    - dovoljno feature-a za određen zadatak)
    - samo horizontalnost implicira samo front end bez backend-a
  - Presek se naziva scenario – pokazuje kako će front end izgledati (raditi) za jedan zadatak



... FIDELITY (VERODOSTOJNOST PROTOTIPA)



- Izgled prototipa – da li je crtano rukom, pomoću grafičkog alata (photshop npr.) ili generisano programskim razvojnim okruženjem (krajnji izgled)
- Fizičke metode za interakciju sa prototipom – pokazivanjem (papirni prototip) ili simulacijom kliktanja

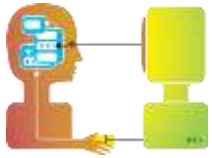


Early Design

Brainstorm different representations	Low fidelity paper prototypes
Choose a representation	
Rough out interface style	
Task centered walkthrough and redesign	
Fine tune interface, screen design	Medium fidelity prototypes
Heuristic evaluation and redesign	
Usability testing and redesign	High fidelity prototypes / restricted systems
Limited field testing	
Alpha/Beta tests	Working systems

Late Design

LOW-FIDELITY PROTOTIPOVI



- Genaralno papirnati, mada mogu biti nacrtani i u nekom alatu kao što je Paint / Power Point i naknadno odštampani za diskusiju sa korisnikom
- U ovo spadaju:
  - Skice
  - Mokapi ekrana
  - Storibordovi (scenarii korišćenja)
- Veoma su korisni u fazi prikupljanja zahteva i kao alat za komunikaciju između korisnika (stakeholders) i dizajnera kor. interfejsa
- Dobri za ilustrovanje ideja, rasporeda ekrana/kontroli i dizajnerskih alternativa
- Iako korisnici mogu da steknu osećaj o izgledu/funkcijama aplikacije (look and feel), daju ograničen pogled na izgled i funkciju kor. interfejsa, kao i ograničavajući pogled na to kako će se on koristiti

SKICE (PAPIRNATI PROTOTIP)

- Odličan izbor za rani dizajn kor. interfejsa kako bi se proverile ideje
- Može da pomogne dizajneru da shvati šta se očekuje od sistema i šta je potrebno da sistem uradi/podrži
- U primeru na slici dizajner je napravio skicu toga kako je on razumeo kor. zahteve (proizvodnja metalnih cevi) kako bi proverio sa korisnikom da li je dobro razumeo zahteve
- Dizajner je iskoristio skicu da na njega stavi i ono što on misli da su esencijalne informacije
- Dobro je i za proveru alternativa, nakon što su predstavljene alternative kroz skicu, prelazi se na razradu odabranog dizajna/interakcije

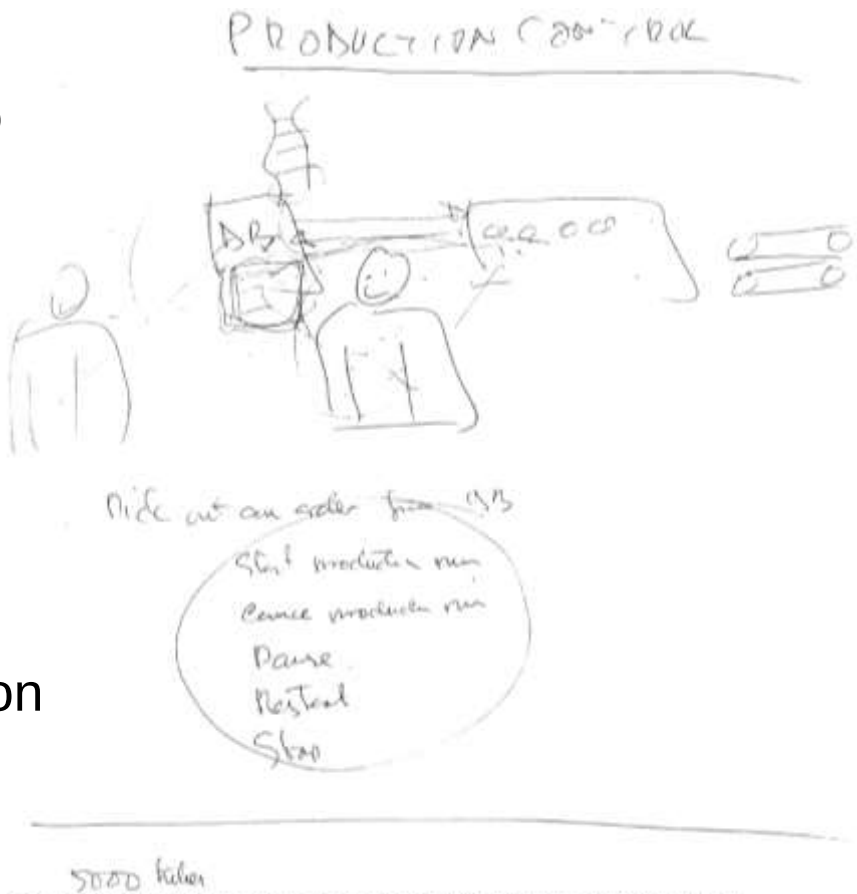


Figure 6.2 Initial high-level sketch of a UI for production control. (From Stone, 2001.)

MOKAPI EKRANA (PAPIRNATI PROTOTIP) ...

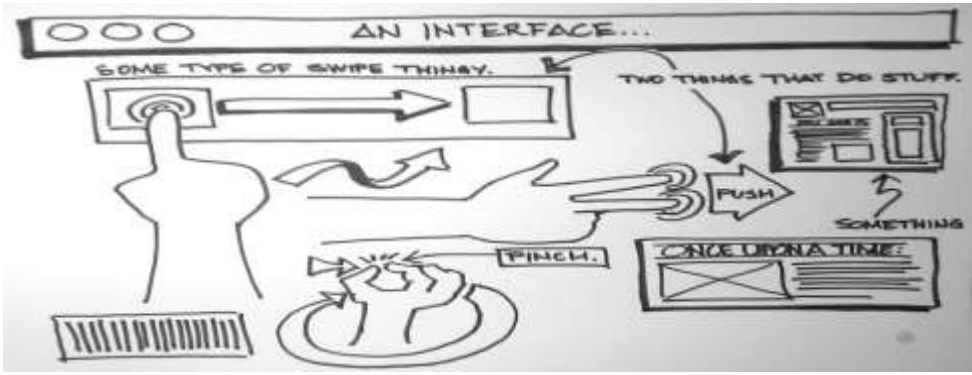
- Interaktivni papirnati prikazi ekrana
- Na više komada papira prikazuju se delovi ekrana, prozori, meniji, dijalozi sa elementima interakcije (zavisi od detalja mokapa)
- Interakcija je prirodna
  - prstom se prikazuje ono na šta će se kliknuti mišem
  - pisanje odgovara kucanju
- Čovek simulira operacije na računaru
  - Uzimaju se delovi papira (koji su u
  - fokusu), vraćaju i preuzimaju novi
  - Ispisuju se odgovori na "ekran"
  - Verbalno se opisuju efekti koje je teško prikazati (simulirati) na papiru



- Vrlo su niskog novoa detalja u pogledu izgleda, ali su vrlo detaljni u pogledu verodostojnosti interakcije (osoba simulira backend)

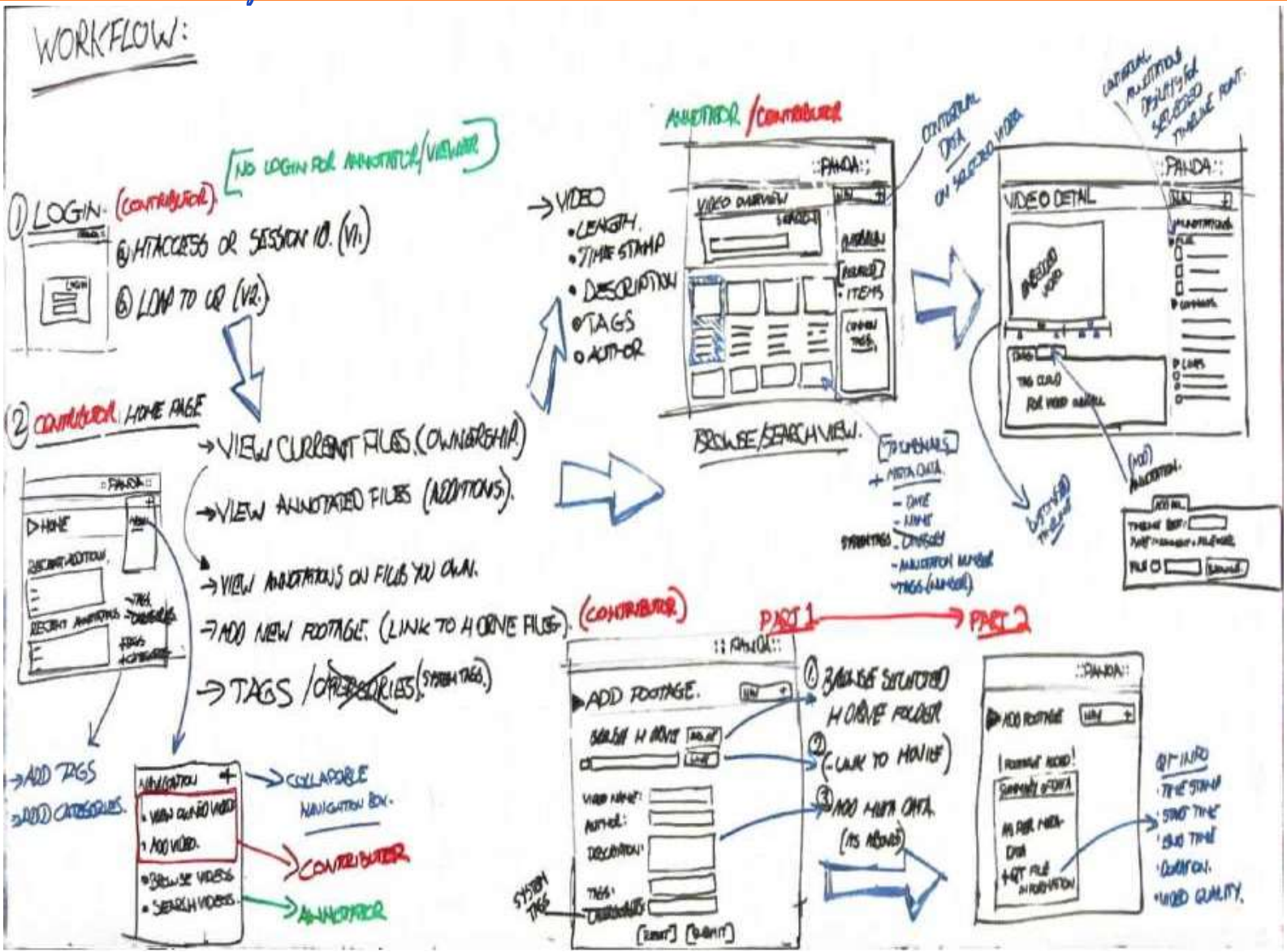
... MOKAPI EKRANA (PAPIRNATI PROTOTIP) ...

- Mogu se jako brzo izgraditi (mnogo brže od programiranja)
- Lako se menjaju
  - Izmene se mogu napraviti brzo nakon evaluacije ili, čak, tokom evaluacije
  - Nema investice u kod, pa se sve lako (bez žaljenja) može odbaciti (osim samog dizajna ili onog što je naučeno o dizajnu)



- Akcenat je na celokupnog priči (big picture) a ne na detaljima
  - Ne gubi se vreme na nepotrebnim detaljima
  - Korisnici daju komentare na bitne stvari, dajući kreativnije sugestije, ne gubi se vreme na sitničarenju oko detalja (da li plava ili crvena boja, levo ili desno, i sl.)
- Čak i ne-programeri mogu pomoći (potrebne samo veštine iz vrtića)

... MOKAPI EKRANA (PAPIRNATI PROTOTIP) ...

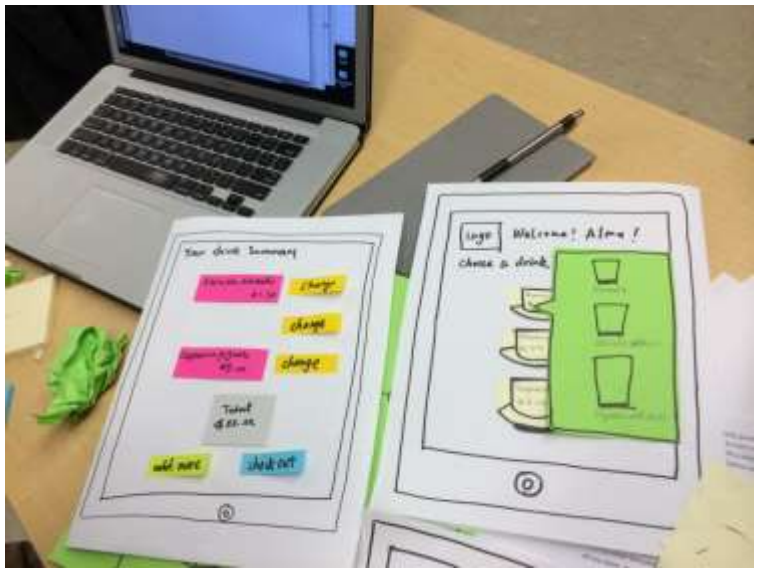


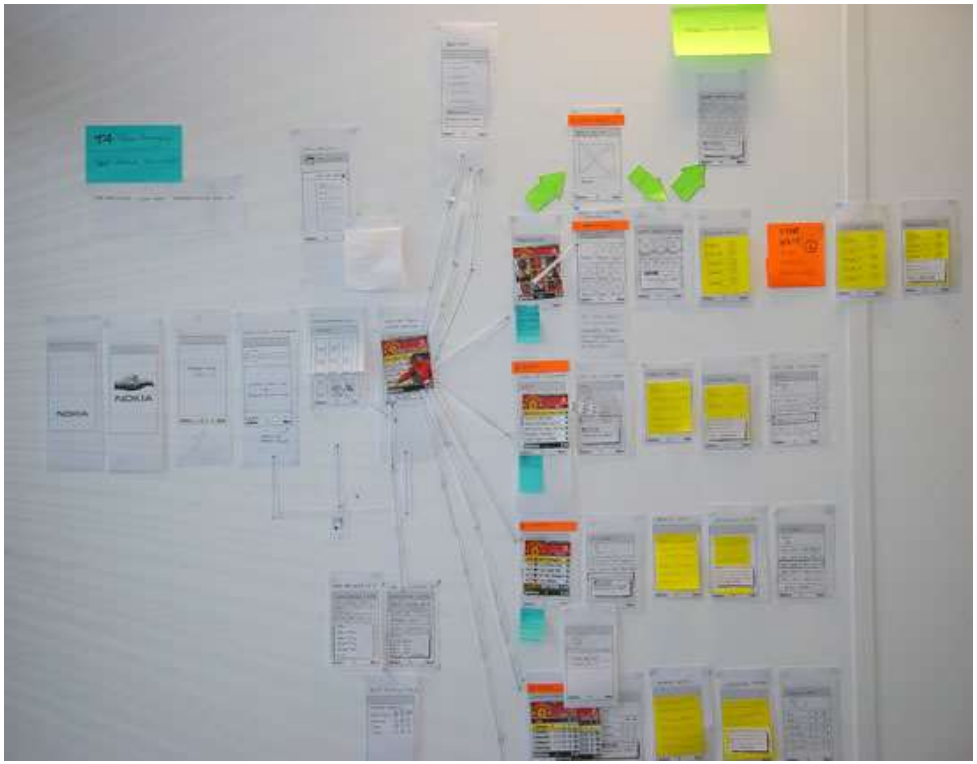
... MOKAPI EKRANA (PAPIRNATI PROTOTIP) ...

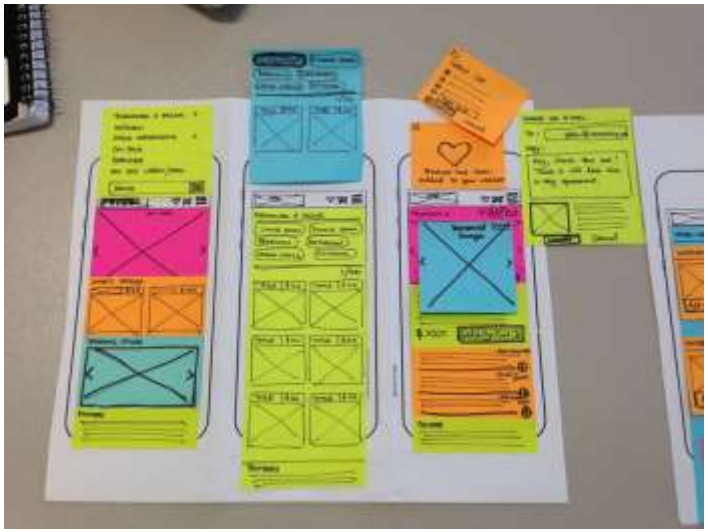
- Altke za pravljenje mokapa (MIT kurs):
  - Posteri sa belom pozadinom
  - Velike indeks kartice za menije, za sadržaj prozora i dijaloga
  - Leljive kartice koje se mogu odlepiti/zalepiti
  - Bele lepljive trake (papir) za ispravke (tekst koji se kuca) za tekst polja, čekboksove, kratke poruke
  - Transparentni papir ili neki sličan materijal
  - Fotokopir mašina, makaze, lepak, olovke&markeri
- Smernice za dobar papirnati prototip (MIT kurs):
  - Paziti da bude dovoljno veliko i jasno
  - Monohromatsko je dobro
  - Sve feedback metode koje je teško simulirati zamenuti verbalnim opisom
  - Pobrinuti se da sve bude dobro organizovano

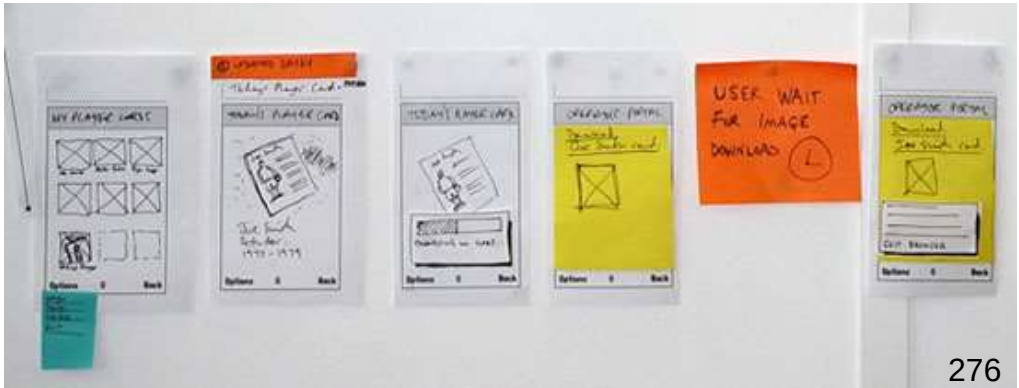


... MOKAPI EKRANA (PAPIRNATI PROTOTIP)

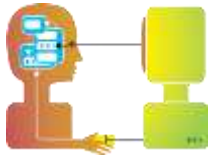








STORIBORD (PAPIRNATI PROTOTIP) ...



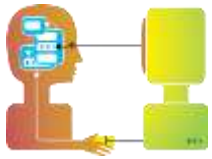
- Sekvenca skica ili mokapa čiji je fokus na svim akcijama, tj. celoj interakciji koju korisnik pređe kroz jedan svoj zadatak
- Storibord uzima tekstualni opis toka zadatka (kao što je scenario) i pretvara ga u vizuelne ilustracije interakcije
- Količina informacija koje će storibord sadržati zavisi od toga u kojoj se fazi razvoja softvera se pravi
  - Ako se koristi u tokom modelovanja zadataka i toka interakcije imaće daleko više detalja, nego ako se pravi tokom faze konceptualnog dizajna interfejsa
- Najveća prednost je što je njihovo stvaranje jeftino, brzo i lako se modifikuje



... STORIBORD (PAPIRNATI PROTOTIP) ...

Action	Illustration
1. Margaret is an account manager for a marketing company. One of the accounts she oversees is for Great Groceries! which is a large foodstore chain. The Great Groceries!, supermarket in Premiumville has recently been extended and now includes a home appliance section. The supermarket manager wants everyone in Premiumville to know that Great Groceries! is having a Grand Re-Opening Day, and that there will be lots of bargains. Margaret has created a colorful flyer to be sent out that contains all the details. She gives this to Susan, who is creating the address list.	
2. Susan is Margaret's secretary. She is helping Margaret to organize the Great Groceries! marketing campaign. Using Hey-Presto, a software database containing names and addresses, Susan does a postcode search and compiles a list of all the street addresses in Premiumville. She gives this list to Amy, who will do the mail merge.	

... STORIBORD (PAPIRNATI PROTOTIP)



3. Amy is responsible solely for mail merging. She is expert at her job, but takes great care as the software is often unreliable. Because of the expense, the company frowns on wasting marketing materials due to mail merging errors. No matter how long the list, Amy generally runs her merged documents in small batches. This is time consuming, and fiddly, but Amy can put up with this inconvenience if in the end she gets an accurate result with few spoiled documents. She then gives the documents to Linda.



4. Linda is responsible for quality control. She collects the mail merged documents from the printer and inspects a certain percentage of them for quality and accuracy. She then prepares the documents/flyers for feeding into a computer-controlled machine that folds them and inserts them into windowed envelopes. Finally, Linda collects them and has them sent to the mailroom at the appropriate point in the campaign.



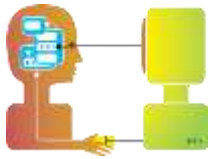
KAKO TESTIRATI PAPIRNATI PROTOTIP

- Nakon što se izgradi prototip, može se testirati tako što se stavi pred korisnike i posmatra se kako ga oni koriste
- Dizajn tim treba da ima tri uloge:



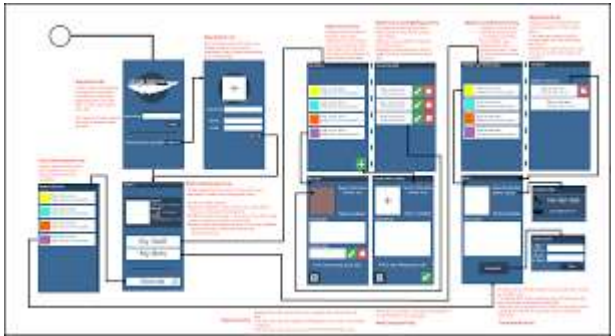
1. Kompjuter – osoba koja je zadužena da simulira rad računara, da oživi prototip; kreće se oko delova prototipa, podiže ih i spušta, ispisuje poruke, unosi poruke; esencijalno radi sve što i računar, ali nikako više od toga (mora se razmišljati mehanički)
2. Orator (predavač) – glas dizajnerskog tima; osoba koja prezentuje interfejs korisniku; opisuje korisniku cilj studije, zadatke i sve ostalo vezano za studiju; on je zadužen da od korisnika dobije odgovore i ohrabruje korisnika da glasno priča dok prati prototip i rešava zadatak
3. Posmatrač – svi ostali pored kompjutera, oratora i korisnika a koji prisustvuju testiranju (evaluaciji); njihov zadatak je da ćute i prate (beleže) šta radi korisnik; ne pričaju i ne daju savete korisniku, niti ulaze u diskusiju sa njim

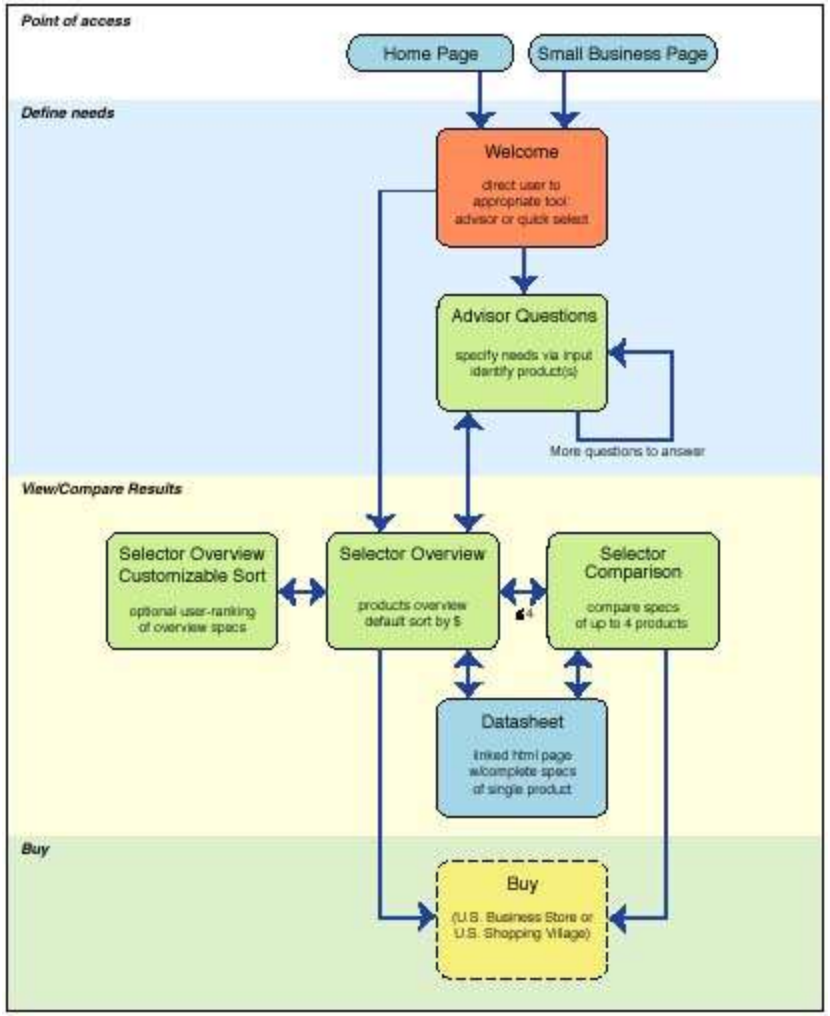
DIJAGRAM TOKA INTERFEJSA ...



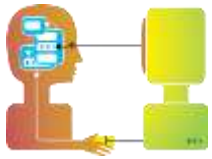
- Omogućuje modelovanje veza visokog nivoa između elemenata korisničkog interfejsa i time nam daju uvid u moguće pobojšanje po pitanju korisnosti i interaktivnosti sistema

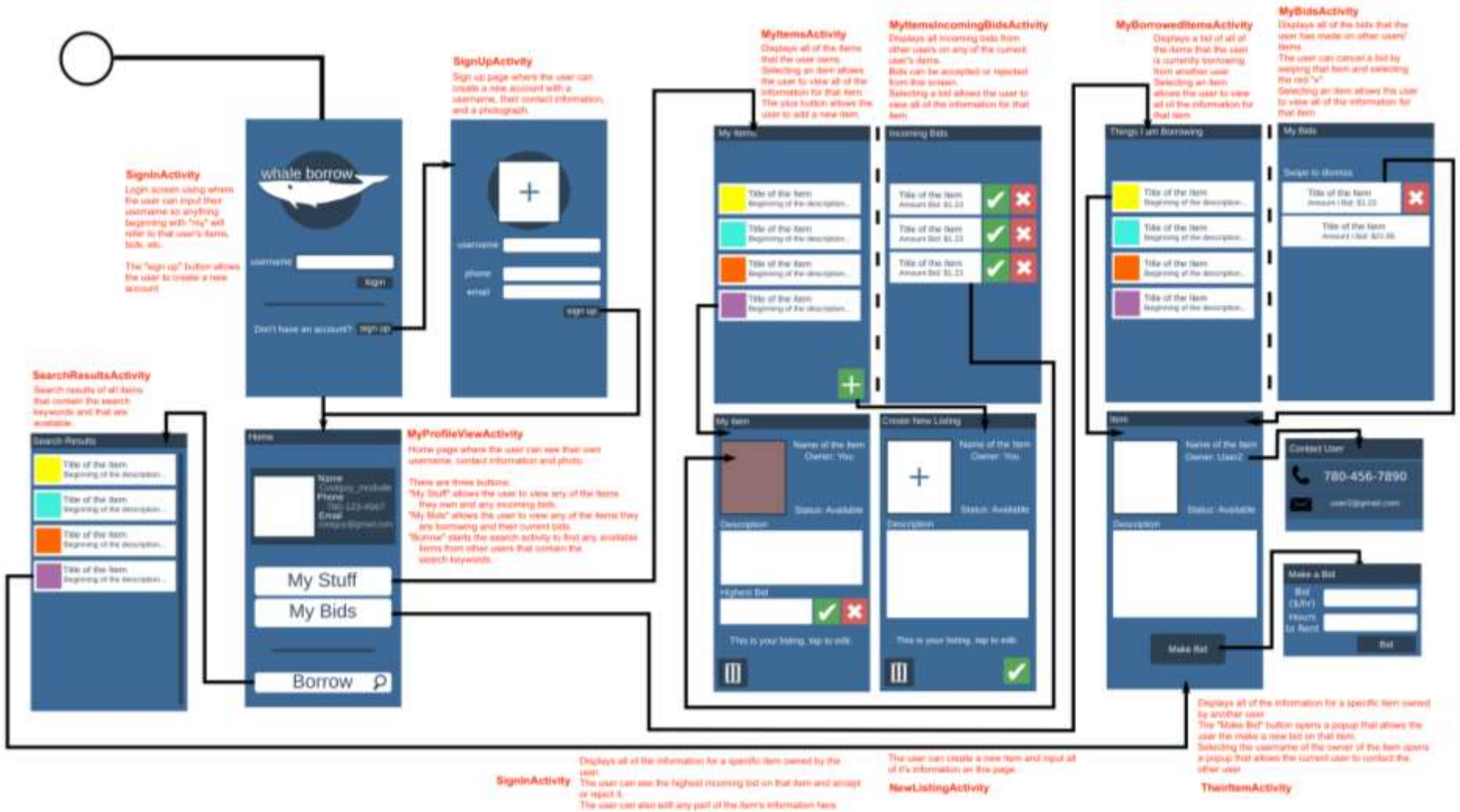
- Dijagram toka interfejsa koristan je kako projektantima, tako i menadžerima ali i samim korisnicima (na zidu, kao podsetnik)



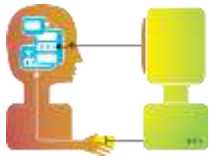


... DIJAGRAM TOKA INTERFEJSA ...

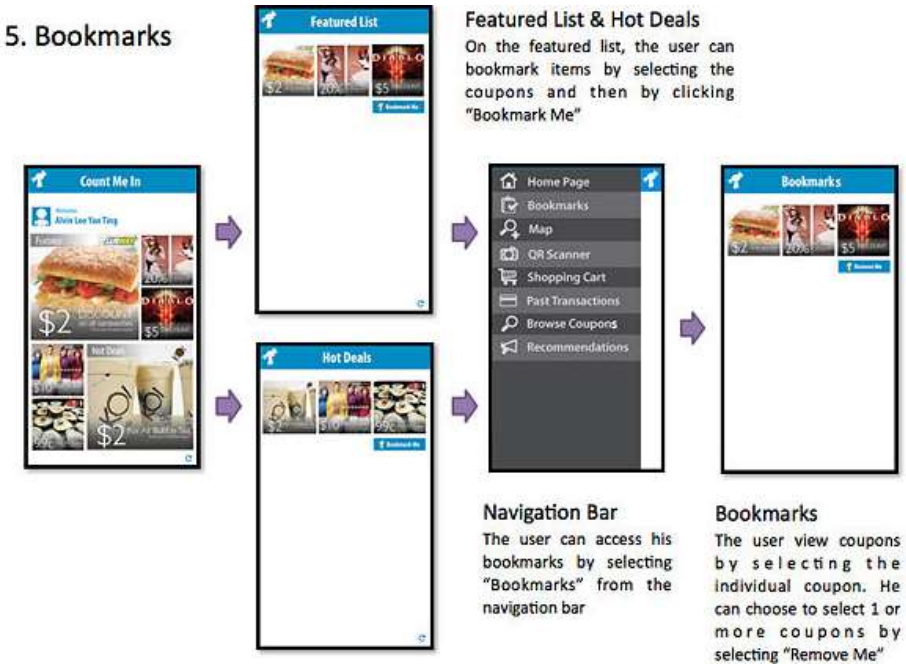




... DIJAGRAM TOKA INTERFEJSA ...

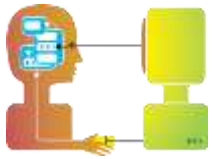


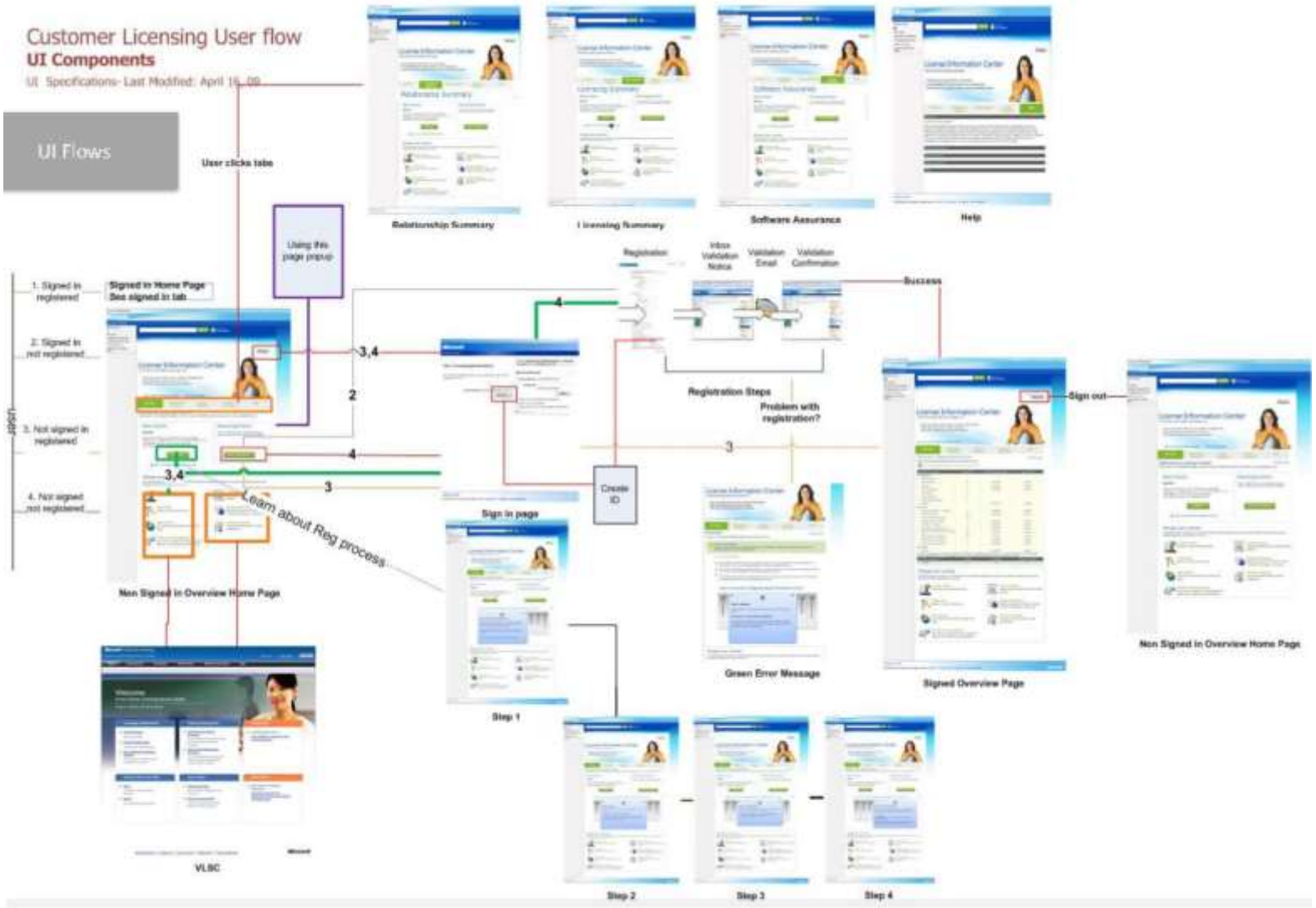
- DTI se primenjuje radi sticanja dobrog pregleda, opšte slike i uočavanja svih mogućih tokova sa jednog mesta
- Na ovaj način moguće je modelovati interakciju koju će korisnik imati sa sotverom, ali i stvoriti dobar uvid u to kako će korisnički interfejs aplikacije izgledati i kakav će biti po pitanju korisnosti



- Zahvaljujući tome brzo se dolazi do razumevanja kako bi sistem trebao da radi i moguća je ocena toka podataka u korisničkom interfejsu aplikacije
- Recimo, ukoliko ima mnogo elemenata interfejsa i mnogo veza između njih to može da ukaže na prevelik sistem težak za razumevanje i upotrebu

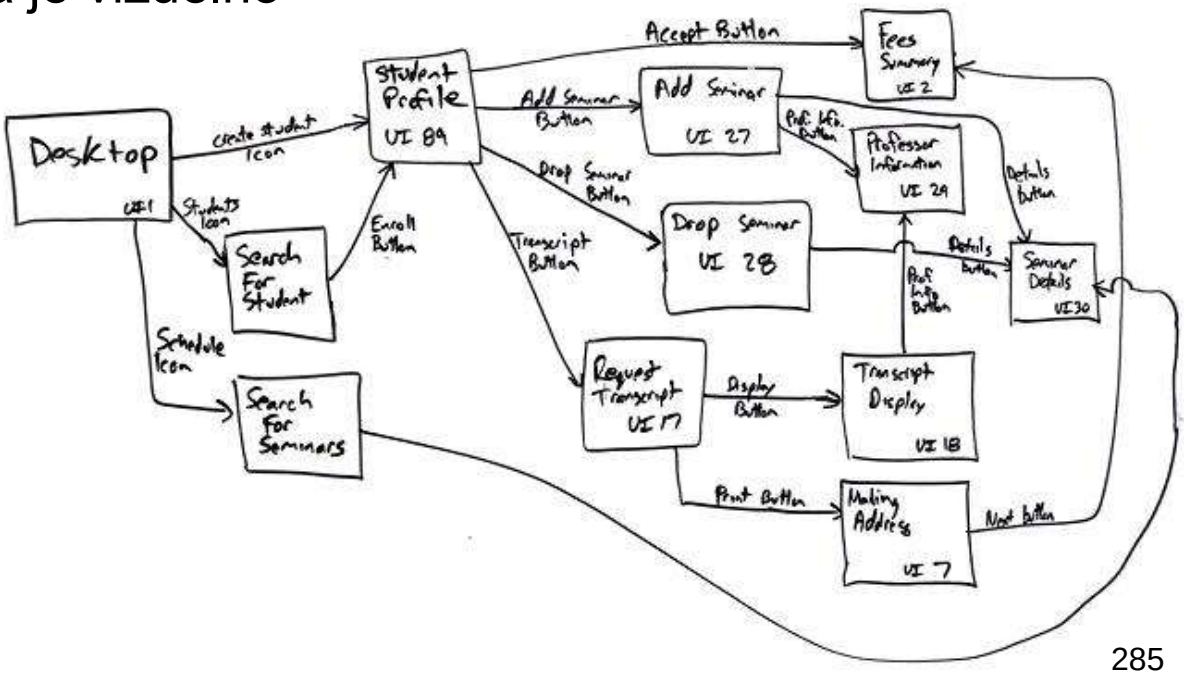
... DIJAGRAM TOKA INTERFEJSA ...



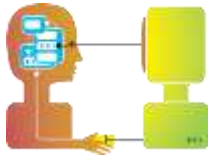


... DIJAGRAM TOKA INTERFEJSA

- DTI spada u grupu grafičkih metoda
- Poseduje hijerarhiju stabla i opisuje tok interfejsa (modelovan strelicama) referišući se na objekte interfejsa (modelovane pravougaonicima, kao što bi smo modelovali instance, odnosno objekte u use-case dijagramima)
- Na ovaj način interakcija je vizuelno predstavljena što u mnogome olakšava njen pregled
- Slabost DTI je manjak samih detalja korisničkog interfejsa, pre svega onih dizajnerskih

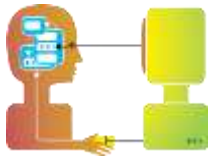


XUAN ...



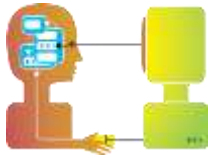
- Omogućuje opis feedback-a korisničkog interfejsa i modelovanje ponašanja korisnika
- Ove specifikacije prate stanje interfejsa i opisuju zadatke kao evolucije stanja
- UAN notacija je orijentisana ka korisnicima i zadacima
- Zadatke prezentuje u formi tabele sa ciljem da razjasni njihovu funkciju i njihove međuzavisnosti (primer: dodavanja teksa na spoljnu ivicu objekta u CorelDraw aplikaciji)
- Notacija je korisna za izražavanje vremenskih problema i asinhronih zadataka u interakciji
- Koristi se kod direktne manipulacije jer prati korisnika, feedback sistema i reakciju sistema i prezentuje ih u tabelarnoj formi

... XUAN ...



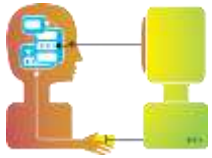
Korisnik			Računar	
Unutrašnja akcija	**Vidljiva akcija**	**Stanje interfejsa**	**Vidljiva akcija**	**Unutrašnja akcija**
Lociraj objekat				
	Pritisni levo dugme miša iznad objekta			Selekcija objekta
		Objekat selektovan	Objekat je selektovan (oko njega su se pojavili markeri)	
Lociraj Text opciju na glavnom meniju (sl. 3.3)				Otvaranje popup menija
	Pritisni levo dugme miša iznad Text opcije glavnog menija	Otvoren padajući meni Text na glavnom meniju I selektovan objekat	Otvara se padajući meni Text	
Lociraj Fit Text To Path opciju u okviru Text padajućeg menija (sl. 3.4)				Prelazak na režim za unos teksta
	Pritisni levo dugme miša iznad Fit Text To Path opcije glavnog menija	Unos teksta na ivici objekta	Pojavljuje se kursor za unos teksta na ivici objekta	
Priprema za unos teksta				
	Unos teksta na tastaturi	Tekst poravnat po ivici objekta	Prikazuje se konačni izgled unetog teksta, poravnatog po ivici objekta. Tekst je selektovan (uokviren markerima)	

... XUAN



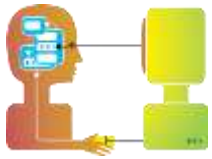
- Mana ove notacije je što za neke aplikacije ima tendenciju da više puta ponavlja opise za česte operacije
- Glavni problem UAN specifikacije je labava povezanost zadataka, feeedback-a sistema i sistemskih operacija
- Zbog toga neke važne vremenske povezanosti mogu biti predstavljane implicitno
- Vremenske povezanosti koncentrisale su se primarno na akcije korisnika, zanemarujući vremenske aspekte onih kategorija zadataka koji se ne svrstavaju u akcije korisnika

HIGH-FIDELITY PROTOTIPOVI ...



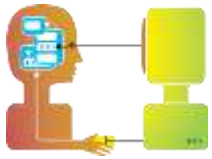
- Low-fidelity prototipovi ne demonstriraju funkcionalnost, niti svu lepotu i detalje budućeg kor. interfejsa
- Bazirani na softveru, ovi prototipovi daju dobar uvid u sistem, jer obezbeđuju funkcionalnu verziju sa kojom korisnici mogu da intereaguju
- Prikazuju raspored (layout) elemenata interakcije i njihovu navigaciju
- Korisnici mogu da dožive look&feel krajnjeg (budućeg) sistema
- Dobro sa usability testove i kao marketinški alat budućeg proizvoda
- Nisu više tako skupi kao nekada jer postoje alati koji omogućuju horizontalnost, veliku verodostojnost, ali bez backend-a
  - PowerPoint, HTML
  - Photoshop, Balsamiq Mockups, Mockingbird
  - VisualStudio, QtDesigner, Eclipse (Flash Builder), Silverlight, Mac Interface Builder

... HIGH-FIDELITY PROTOTIPOVI



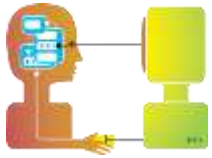
- Mogu isto što i low-fidelity prototipovi, iste se stvari mogu saznati, ali još i:
  - Informacije o screen layout
    - Da li je pregledan, prenatrpan, komplikovan?
    - Da li odvlači pažnju?
    - Da li korisnik može da pronađe važne elemente?
  - Da li su dobro odabrane boje, fontovi, ikonice i preostali vizuelni elementi
  - Informacije o feedback-u
    - Da li korisnici primećuju i reaguju na statuse, promene kursora i druge povratne informacije
  - Koliko je interfejs efikasan
    - Da li su kontrole dovoljno velike?
    - Da li su kontrole suviše blizu jedna drugoj?
    - Da li skrolovanje suviše dugo traje?

ALATI ZA PRAVLJENJE PROTOTIPA



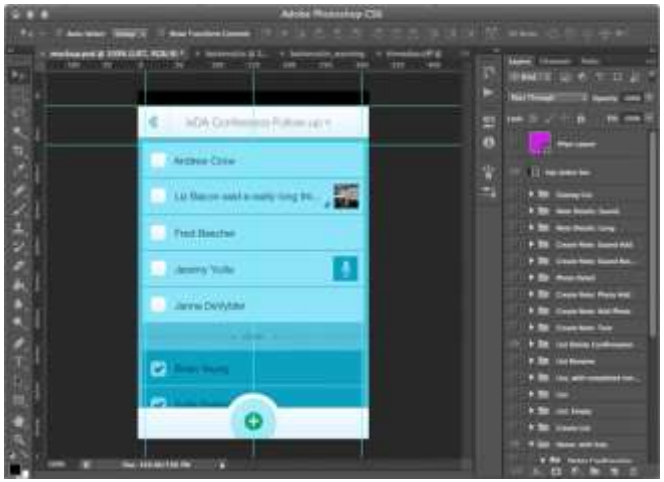
- Moguće je napraviti prototip tako što se on iskodira upotrebom Jave ili C++ programskog jezika i user interface toolkita, kao što su Swing ili MFC (ako ste skloni mazohizmu)
- Naravno, daleko je bolje upotrebiti neki alat za pravljenje high-fidelity prototipa zato što je
  - Brže i lakše napraviti prototip nego kroz kodiranje
  - Nema potrebe za debagiranjem koda
  - Lakše se menja i lakše ga je odbaciti
- Kod kodiranja kompjuterski prototip može postati tako složen da on postane finalni proizvod iako ima loš dizajn i propuste u interakcije (jer nam je žao da ga odbacimo posle toliko truda)
- Mora se paziti da softver (njegov UI deo) ne određuje kor. interfejs, tj. da ne diktira on njegov razvoj
- Čak i pored podrške softv. alata, razvoj kompj. prototipa može zahtevati jako puno posla (primer: drag&drop u Excel-u)

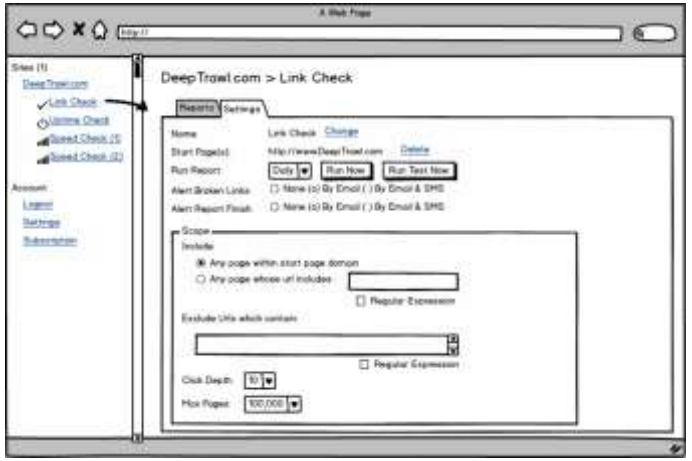
STORIBORD (HIGH-FIDELITY)



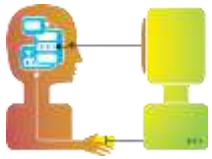
- Slično papirnatim, ali se koriste screenshot-ovi u boji
- Sekvenca (graf) screenshot-ova
- Mogu biti povezani hiperlinkovima (hotspots)
- Neka prelaz između prozora može biti animiran da bi se prikazali dinamički efekti (mouse feedback ili drag-drop feedback)
- Alati koji se mogu koristiti:
  - Photoshop, Balsamiq Mockups, Mockingbird, ...
- Prednosti:
  - Može da se nacrta sve
- Mane:

  Nema unosa teksta;
  Widget-i nisu aktivni;
  Treba pronaći hotspot

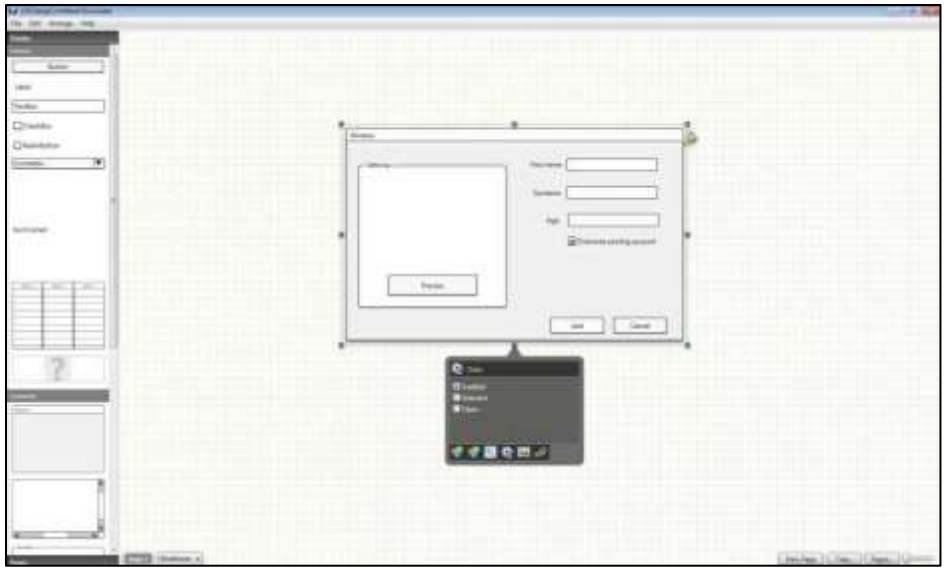




GENERATORI FORMI (HIGH-FIDELITY)



- Različiti alati za pravljenje formi
  - VisualStudio, Qt Designer, Eclipse (Flash Builder), Silverlight, Mac Interface Builder
- Prednosti:
  - Stvarne kontrole, ne samo njihove slike
  - Moguće je po potrebi dodati backend ako je to potrebno (ali postoji opasnost da se neće odbaciti)



- Mane:
  - Ograničava razmišljanja i mogućnosti na standardne widget-e
  - Manje korisno za bogate grafičke interfejse

WIZARD OF OZ (HIGH-FIDELITY)

- Softverska simulacija sa čovekom u pozadini, slično papirnom mokapu, gde čovek simulira odzive sistema
- Čovekov uticaj može i ne mora biti skriven od korisnika
- Često se koristi da simulira buduće (future) tehnologije
  - Prepoznavanje govora
  - Učenje (AI)

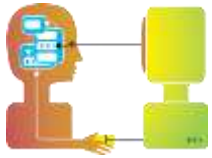




- Mane:
  - Dva kor. interfejsa o kojima treba voditi računa
  - Wizard mora da se ponaša mehanički

ELEMENTI INTERAKCIJE – HARDVERSKE I SOFTVERSKE KOMPONENTE

ELEMENTI INTERAKCIJE



- Svaki kor. interfejs predstavlja miks elemenata interakcije
  - Hardverske komponente – koje se odnose najčešće na interakcione uređaje za ulaz i izlaz, poput džojstika, miša, kontrolera, tastature, ekrana, zvučnika



  - *Softverske komponente – aspekti kor. interfejsa koje generiše softver kao što* su slike na ekranu, organizacije komponenti na ekranu, boja, tekst, animacija, video i zvuk



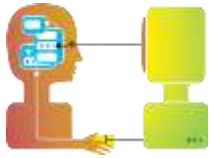
HARDVERSKE KOMPONENTE ...

- Izbor adekvatnih hardverskih komponenti je važan
  - Svodi se na izbor odgovarajućeg alata za dati problem (nećete pomoću miša napisati dokument)
  - Ako na raspolaganju postiji više uređaja, vodi se računa o ograničenjima raspoložive tehnologije, ali i zadacima koji se trebaju rešiti tehnologijom na raspolaganju
  - Nije uvek moguće birati
  - Zahtevi za ulazne i izlazne uređaje se razlikuju (veličina i oblik tastera, robusnost tastature, preciznost i lakoća morišćenja pointerskog uređaja, koliko je lako koristiti ponterski uređaj, koliko je mesta potrebno za rad point. uređaja, koliko detaljna treba da bude slika, koliko nijansi boja treba da ima, kolika je potrebna veličina i prenosivost ekrana



"Darn these hooves! I hit the wrong switch again! Who designs these instrument panels, raccoons?"

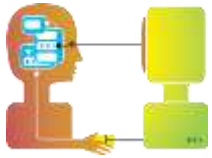
... HARDVERSKE KOMPONENTE



There are circumstances in which choosing the correct device is essential. Users may have particular requirements, so it is important to understand the users of the system and the way in which they use the technology. For example, the messaging company Pitney Bowes carried out a survey of office workers in North America, Britain, and Germany. One of the findings was that Europeans use communications technology very differently from the way North Americans do. In particular, Europeans prefer direct contact using a mobile telephone rather than leaving voicemail messages. The opposite is true for North Americans. The same difference is reinforced by the use of pagers: 20 percent of North Americans use pagers compared with about five percent of Europeans. The implications of these cultural differences are that the technology that a company purchases needs to take into account the way in which the employees will use it. Thus, a European company may prefer to employ a secretary, so that its clients can speak directly to a person; but for North Americans, a voicemail system might suffice.

Or the task may have particular requirements. For example, in a large warehouse, if the boxes were not bar coded, then the employees would need to type in all the details every time a box arrived or was dispatched. With a large warehouse having thousands of these transactions every day, doing this would be time consuming and prone to error.

SOFTVERSKE KOMPONENTE



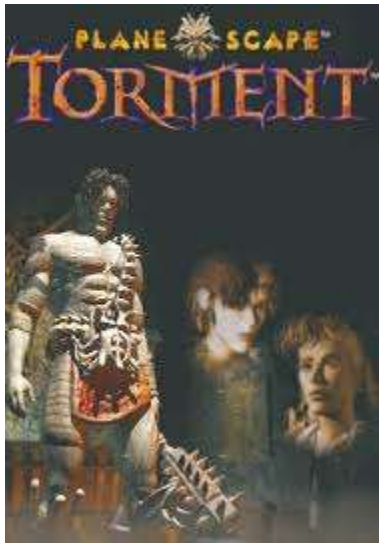
- Adekvatan izbor softverskih komponenti omogućuje maksimalno iskorišćenje hardverskih komponenti
- Ne može se puno uticati na sam izgled ili rad ulaznih uređaja, ali se može uticati na reakcije sistema na akcije tih uređaja
  - kako će se sistem odazvati na akcije ulaznih uređaja i kako će se to predstaviti na izlaznim uređajima
  - kako će se kodirati sadržaj tokom interakcije



- Softverske komponente su:
  - Tekst – kako da obezbedimo da je tekst čitljiv? Koji font koristiti? Koliko dugačke da budu linije?
  - Boja – koje boje idu zajedno? Kako preneti informacije pomoću boje?
  - Slike – koje vrste slika postoje? Kako odabrati edekvatnu sliku?
  - Video – kada je korisna animacija? Kada se sa video dobija dobar efekat?
  - Zvuk – kada je zvuk koristan? Koje kategorije zvuka postoje i kada primeniti koju kategoriju?

TEKST

- Tekst dominantan element gotovo svakog kor. interfejsa
- Predstavlja fleksibilan i moćan oblik komunikacije
- Njegove tehničke prednosti su:
  - Tekst fajlovi su mali za razliku od svih ostalih softverskih komponenti
  - Tekstom se može lako manipulisati, npr. može se lako pretraživati što ne važi za preostale softv. komponente
  - Tekst je najlakše interpretirati, iako je moguće protumačiti jedan te isti tekst na različite načine, nivo nejjasnoće je daleko manji nego kod drugih softv. komponenti



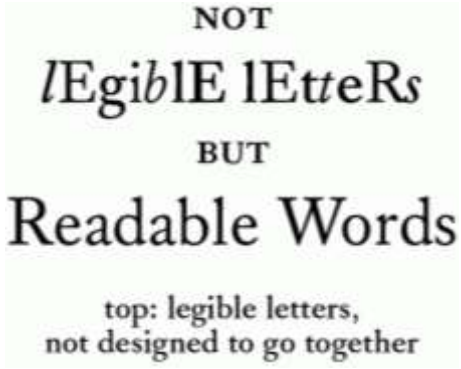
- Prilikom dizajniranja kor. interfejsa potrebno je istražiti kakav je stav korisnika prema tekstu
  - Moguće je odrediti starosnu čitalačku grupu
  - Da li čitaju knjige/novine
  - Da li samo prolaze kroz veće količine teksta ili ga zaista čitaju



ČITLJIVOST TEKSTA ...

- Zavisi od nekoliko faktora:
  - typeface,
  - veličina fonta (size),
  - razmaku između linija (line spacing),
  - dužini tekst linije (line length),
  - poravnanju (justification), završetku linije (line endings)

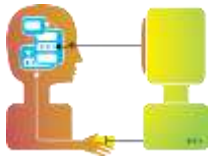




- Ne postoji upustvo za upotrebu teksta koje je univerzalno primenjivo, zato uvek treba uzeti u obzir ko je korisnik (da li je korisnik slabovid?)
- Termin font nije opšte definisan ali se gotovo uvek odnosi na:
  - typeface (vrstu fonta) – Arial, Microsoft Sans Sarif, i sl.
  - veličinu fonta – od 8 do 72 (ovo je vrlo relativno i zavisi od rezolucije ekrana)



... ČITLJIVOST TEKSTA ...



Typeface	Serif or sans serif Familiar or unfamiliar	A serif is the finishing stroke at the end of a letter. See Figure 13.1. Sans serif typefaces are more suitable than serif typefaces for use on a screen: the resolution on screen is likely to be poorer than on paper, and the details of serif typefaces may be lost. Familiar typefaces such as Times or Arial are easier to read; unfamiliar ones, such as weird are more difficult.

Pack my box with five dozen liquor jugs!

Pack my box with five dozen liquor jugs!

Sans Serifs Have Low-Contrast Obliques

Pack my box with five dozen liquor jugs!

Pack my box with five dozen liquor jugs!

Serifs Have High-Contrast Italics

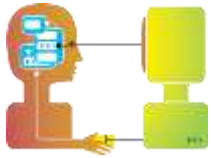
Serif

Purus et etiam, pulvinar odio? Facilisis, rhoncus purus? Duis et lorem tortor dolor ridiculus sociis risus, enim. Non odio cras elit ac. Enim quis, hac, enim enim nisi! Est sed, adipiscing augue! Nec cursus? Nec egestas cras sagittis turpis platea egestas ultrices elit a, ac, integer tristique integer scelerisque magna.

Sans-Serif

Purus et etiam, pulvinar odio? Facilisis, rhoncus purus? Duis et lorem tortor dolor ridiculus sociis risus, enim. Non odio cras elit ac. Enim quis, hac, enim enim nisi! Est sed, adipiscing augue! Nec cursus? Nec egestas cras sagittis turpis platea egestas ultrices elit a, ac, integer tristique integer scelerisque magna.

... ČITLJIVOST TEKSTA ...



Type size

Too small is harder to read

Too big is also hard

Most screens have around 72 dots per inch; ordinary office printers have around 300 dots per inch; high-quality printing (such as this book) typically has 1200 dots per inch. The lower the resolution, the larger the type size required for comfortable reading.
For text for continuous reading, 11- to 14-point type is a good range to work with. Headings will stand out better if they are 3 to 5 point sizes larger. It is possible to use smaller type sizes for areas of the screen that are read episodically, rather than continuously, such as menu bars or captions for icons.

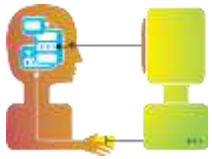
Letter spacing

Letters too close together are hard to read

Well-designed type faces have a pleasant amount of space between the letters. Too little or too much and legibility is affected.

L e t t e r s t o o f a r a p a r t a r e h a r d t o r e a d

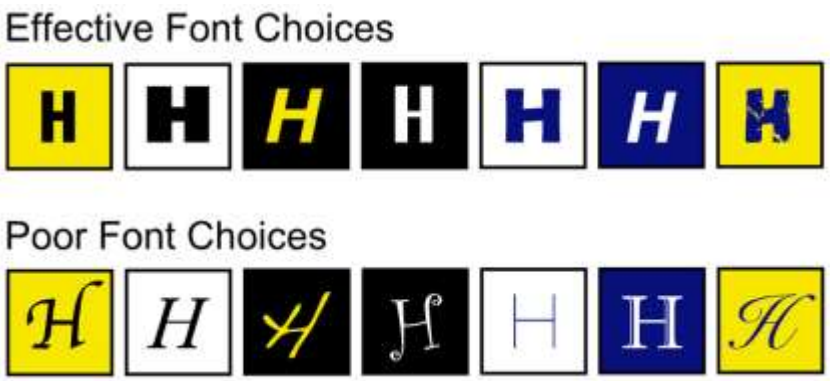
... ČITLJIVOST TEKSTA ...



Line spacing (leading)	If you have small type sizes, then you can increase legibility by increasing the leading. This text is at the default leading If you have small type sizes, then you can increase legibility by increasing the leading. This text has extra leading.	The legibility of smaller type size can be improved by increasing the line spacing. The longer the line, the wider the line spacing should be (and the wider the line spacing, the longer the line can be). However, if the line spacing is too wide, then the lines may not be perceived as being related to each other (see the Gestalt laws of perceptual organization in Chapter 5).
Line length		The maximum line length on a screen should be around 60 characters (or eight to twelve words). This allows a meaningful unit of text to appear in most lines. You should avoid very short lines, as they fragment the text and it is more difficult to construct the meaning.
Justification		Fully justified text (left- and right-justified) can create uneven gaps between the words on a page. It is usually best to left-justify blocks of text, as this gives a predictable place for the eyes to start from when they jump back to the beginning of the next line. On screen the right margin is best kept unjustified (ragged). If you are placing very short pieces of text next to other items on the screen (for example, putting captions on buttons or under icons), then you need to consider the relationship of the text to the item it belongs with. For instance, text on buttons usually looks neater if it is centered.

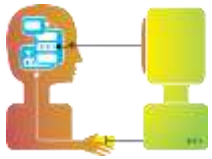
... ČITLJIVOST TEKSTA

Line endings	Very short lines are easier to read if the line endings coincide with grammatical boundaries. Very short lines are harder to read if the line endings come at any arbitrary place.	If you are using short line lengths — for example, in tightly-packed screen designs — it is easier for the reader to understand a line if it has a distinct thought in it, so, where possible, line endings should coincide with grammatical boundaries.

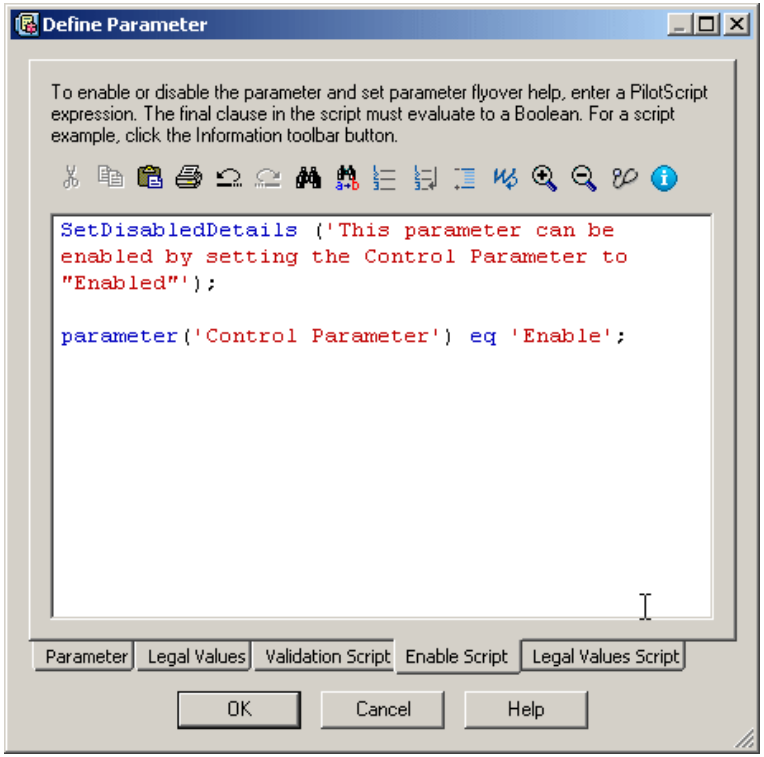




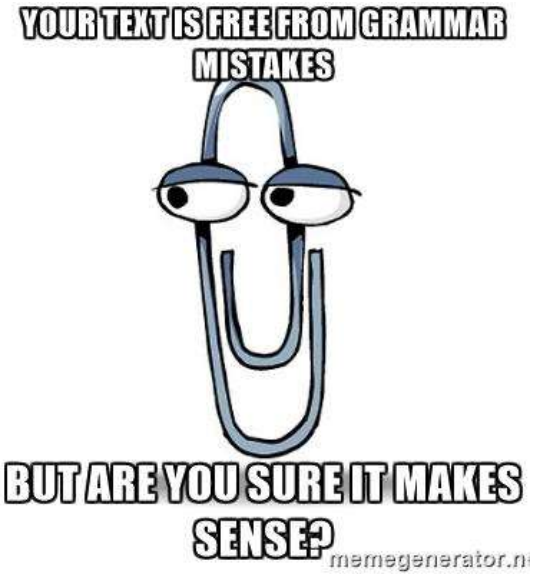
SMISAO TEKSTA



- Iako je tekst čitljiv i lep za oko, to ne znači i da ima smisla za korisnika
- Prilagoditi tekst i izraze krajnjem korisniku
- Koristiti skraćenice, ali one koje imaju smisla za korisnika







BOJA



- Boja se koristi
  - da privuče korisniku pažnju – važni delovi ekrana (softv. dugmad) su često obojena drugačiji od ostatka ekrana da privuku pažnju
  - da prikažu status – kako status postaje važniji (kritičniji) tako se može menjati njegova boja (npr. svetlo na semaforu)
  - da se informacija na ekranu prikaže jasnije – boja se koristi kako bi se organizovali elementi na ekranu ili da se prikaže perspektiva
  - da se prikaz napravi atraktivnijim – pored usability-a, važna je i estetika prikaza; džabe sva upotrebljivost kor. interfejsa ako je on korisniku ružan



KARAKTERISTIKE BOJE

- Ekran može da prikaže samo deo boja koje su vidljive ljudskom oku
- Preciznost ekrana u reprodukciji boja je ograničena
- Između različitih računara postoji varijacija u paleti boja koje mogu prikazati





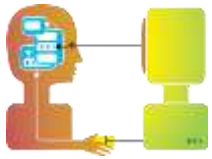
- Postoji razlika u tome kako percipiramo boje na ekranu i boje na papiru
  - boje na ekranu mogu delovati bledo i nerealno spram boja odštampanih na papiru
- Određene kombinacije boja izazivaju osećaj treperenja na ekranu

ZNAČENJE BOJE ...



- Boje imaju svoju konotaciju (značenje)
- Ali različite boje asociraju različite ljude na različite stvari
- Na šta boje asociraju delimično zavisi od kulture kojoj korisnik pripada
  - **Crvena boja** u zapadnoj kulturi znači opasnost (upozorenje), dok kod Kineza ono znači radost/sreću
  - Srećom kulturološko značenja boje se može naučiti
  - Zbog izrazite globalizacije ove razlike u značanju boja sve više nestaju
- Čak i kada je značenje neke boje opšte prihvaćeno i rasprostranjeno, mora se koristiti oprezno
  - Npr. često se u kolima signalizira da je ručna kočnica povučena crvenom bojom, ali to u isti mah znači da možete pustiti i nožnu kočnicu na uzbrdici
  - Ovako pomešani signali mogu biti zbunjujući, zato je možda bolje koristiti neku drugu boju

... ZNAČENJE BOJE ...



BLUE
- (+) tranquility, love, loyalty, security, trust, intelligence
- (−) coldness, fear, masculinity

RED
- (+) love, energy, power, strength, passion, heat
- (−) anger, danger, warning

BROWN
- (+) friendly, earth, outdoors, longevity, conservative
- (−) dogmatic, conservative

GREEN
- (+) money, growth, fertility, freshness, healing
- (−) envy, jealousy, guilt

PINK
- (+) healthy, happy, feminine, compassion, sweet, playful
- (−) weak, femininity, immaturity

TAN/BEIGE
- (+) dependable, flexible, crisp, conservative
- (−) dull, boring, conservative

PURPLE
- (+) royalty, nobility, spirituality, luxury, ambition
- (−) mystery, moodiness

YELLOW
- (+) bright, energy, sun, creativity, intellect, happy
- (−) irresponsible, unstable

GRAY
- (+) security, reliability, intelligence, solid
- (−) gloomy, sad, conservative

TURQUOISE
- (+) spiritual, healing, protection, sophisticated
- (−) envy, femininity

ORANGE
- (+) courage, confidence, friendliness, success
- (−) ignorance, sluggishness

BLACK
- (+) protection, dramatic, classy, formality
- (−) death, evil, mystery

SILVER
- (+) glamorous, high tech, graceful, sleek
- (−) dreamer, insincere

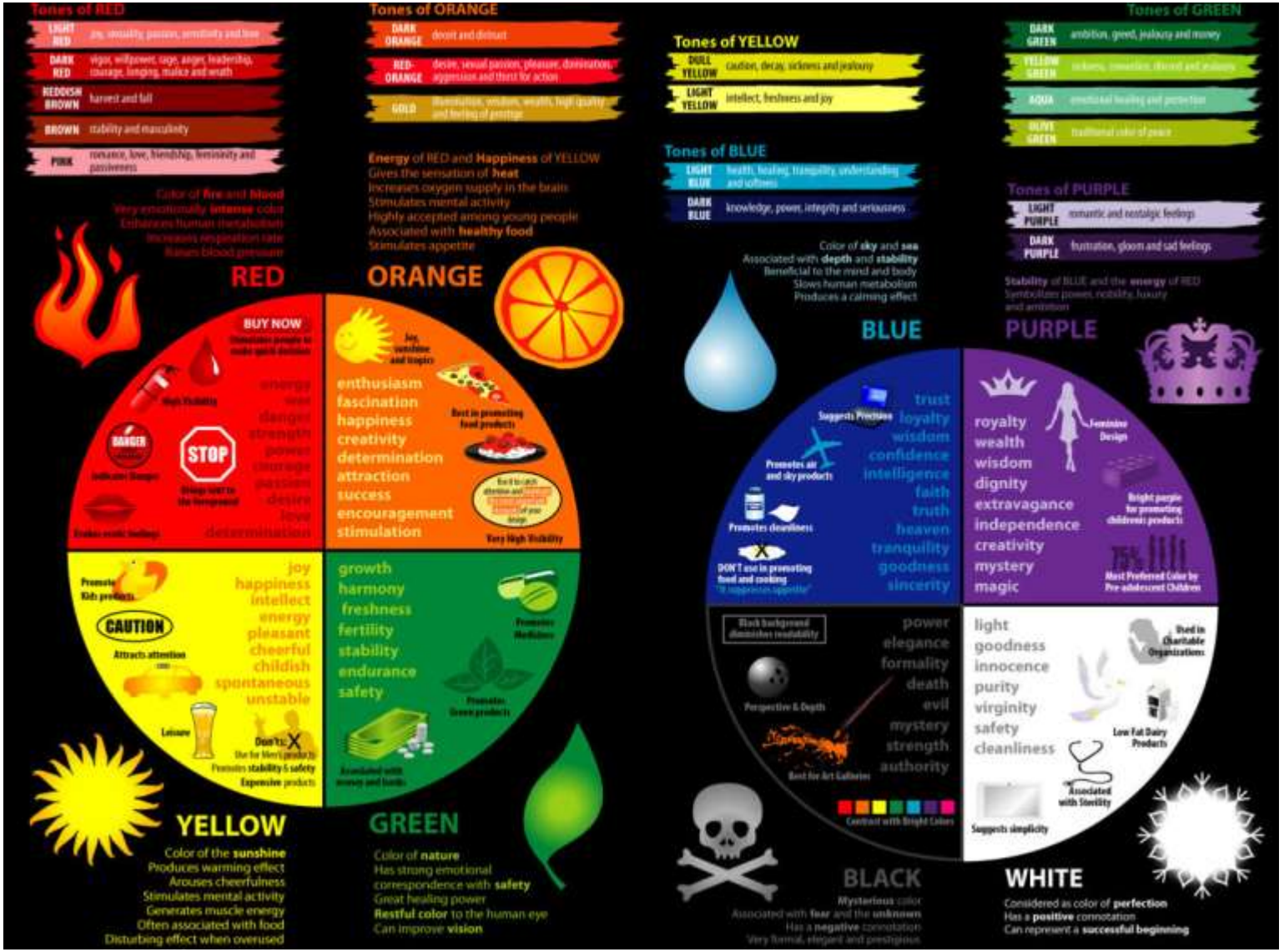
GOLD
- (+) wealth, prosperity, valuable, traditional
- (−) greed, dreamer

WHITE
- (+) goodness, innocence, purity, fresh, easy, clean
- (−) winter, cold, distant

blue	red	black	green
TRUST SMART CALM FAITH NATURAL STABLE POWER	LOVE IMMEDIACY ENERGY SALE PASSION ANGER HUNGER	BOLD RICH POWER MYSTERY ELEGANCE EVIL STRENGTH	SOOTHING ECO-FREINDLY NATURAL ENVY JEALOUSY BALANCE RESTFUL

yellow	orange	pink	purple
CHEER ATTENTION CHILDISH FRESH WARMTH ENERGY OPTIMISM	HEALTH ATTRACTION STAND OUT THIRST WEALTH YOUTHFUL HAPPINESS	TENDERNESS SENSITIVE CARING EMOTIONAL SYMPATHETIC LOVE SEXUALITY	ROYAL MYSTERIOUS ARROGANT LUXURY CHILDISH CREATIVE SADNESS

... ZNAČENJE BOJE ...

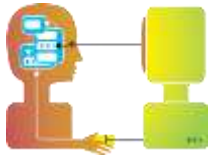


... ZNAČENJE BOJE

- Saturacija boje može imati takođe značajan efekat
  - Boje namenjene mladim ljudima mogu biti čiste i svetle (bright)
  - Boje namenje starijim znaju biti dubokih tonova (tamnije)
- Moda ima veliki uticaj na izbor boje, naročito kod mladih ljudi
  - Kako se boje u modi menjaju trebalo bi izbegavati izbor boja koju su demode (outdated)
  - Kad god je moguće, uključiti grafičkog dizajnera u projektovanje kor. interfejsa
- Boje primenjivati uvek konzistentno uzimajući u obzir karakteristike korisnika, njihovog zadatka i okruženje u kojem će se raditi



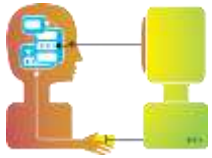
EFIKASNA UPOTREBA BOJA ...



- Vrlo kompleksna i tehnički zavisna oblast
  Kako kombinovati boje a da se dobije željeni efekat?

Number of colors	If you are using colors to organize the screen, then it is better to limit their number: too many colors can be confusing and unpleasant to look at. Some guidelines recommend no more than six colors, in addition to black and white, for any one screen, and fewer is often better. Some experimental evidence suggests that using many colors arbitrarily can result in users suffering eyestrain, fatigue, and disorientation.
Design for monochrome	Designing in black and white first can help to focus attention on the layout of the UI. Also, there may still be some users who don't have color screens. Do not forget that monochrome displays can be extremely effective, like the display in a railway station that indicates when the next train is about to arrive.
Color perception	Color perception varies greatly. About eight percent of the male population is color blind. Red, orange, and green are often confused, as are purple, blue, and magenta, and white, gray, and cyan.
Color for reinforcement	Color should not be used in isolation. For example, speed limit signs on UK roads have a red border, but they also have black text on a white background stating the maximum speed allowed.

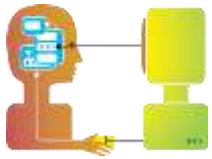
... EFIKASNA UPOTREBA BOJA ...



- Kada se kombinuju boje mora se voditi računa o njihovoj usađenoj osvetljenosti (intrinsic brightness)
- Ukupna osvetljenost kombinacije boja zavisi od osvetljenosti svake boje koja se kombinuje kao i od relativne količine svake pojedinačne boje u kombinaciji
  - npr. ako se tamno plava pomeša sa belom, dobija se svetlo plava koja je po prirodi svetlija (osvetljenija) od tamno plave boje

Achromatic colors	Colors	Intrinsic brightness
White		Very high
	Yellow, yellow–green	High
	Orange	High to medium
Medium gray	Red / Green	Medium
	Violet	Medium to low
	Blue	Low
Black		None

... EFIKASNA UPOTREBA BOJA ...



- Usađena osvetljenost može imati značajan efekat (implikaciju)
  - Da bi tekst bio čitljiv, kontrast između osvetljenosti pozadine i naličja mora biti dovoljno izražen (iste kombinacije boja je teško čitati)
  - Svetle boje sa značajnom osvetljenošću, moraju se koristiti obazrivo, jer velike površine svetlih boja mogu postati teške za (duže) gledanje; iako se crni tekst najbolje čita sa bele površine, ako korisnik čita sa bele površine tokom celog dana to može postati naporno, zato je bolje zameniti belu površinu sivom, krem ili magnolijom
  - Kontrast u osvetljenosti ne bi trebala da bude ekstremna, jer može dovesti do vizualnog efekta vibriranja
- Kako je percepcija personalna karakteristika, dobra je praksa dozvoliti korisniku da može da modifikuje kolor šemu kor. interfejsa

White on yellow is hard to read

White on black is easy to read

Red on blue

Blue on red

Green on pink

Pink on green

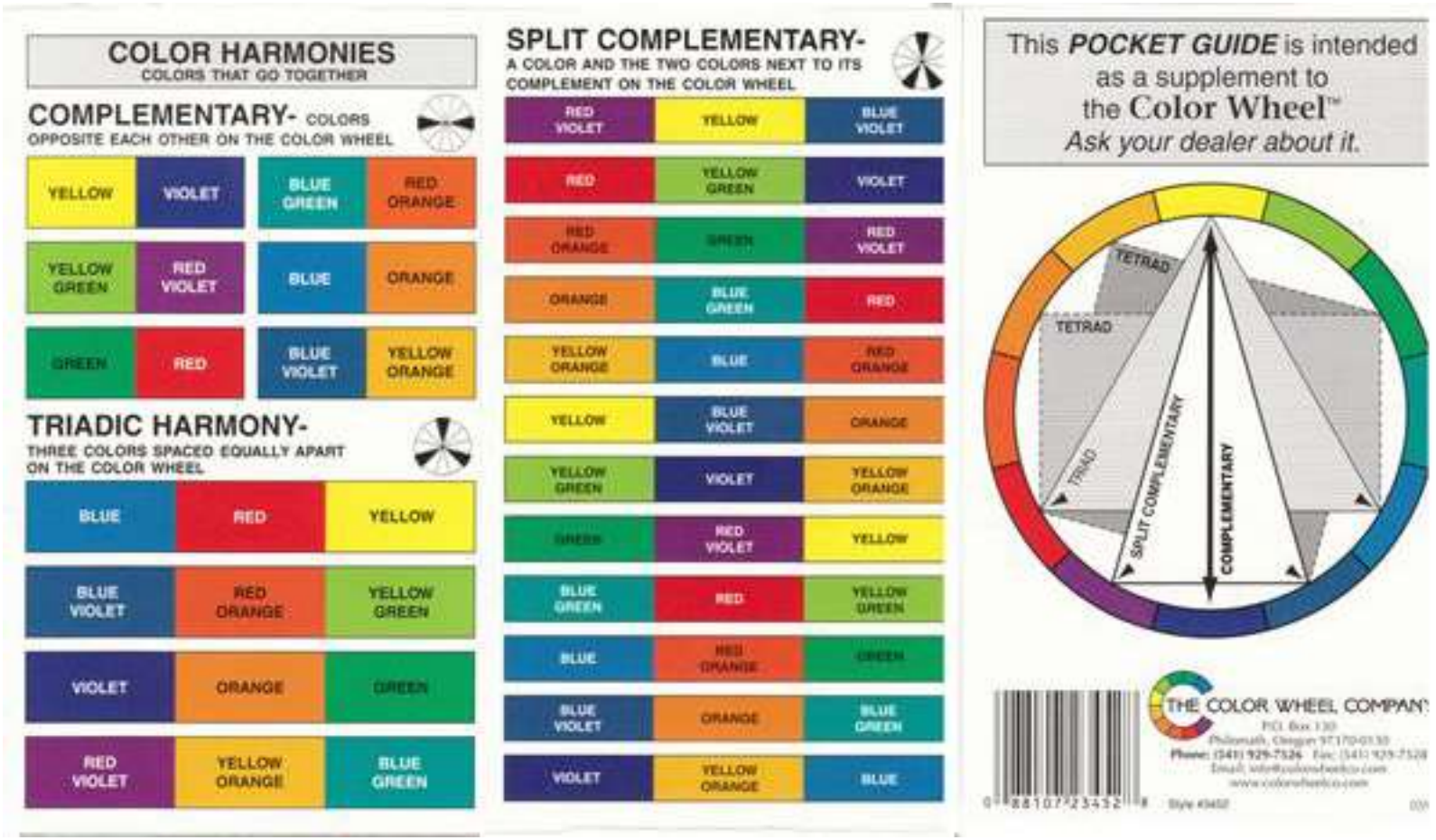
Dark blue on yellow

Yellow on dark blue

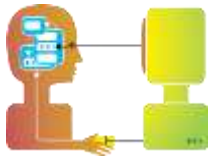
... EFIKASNA UPOTREBA BOJA ...



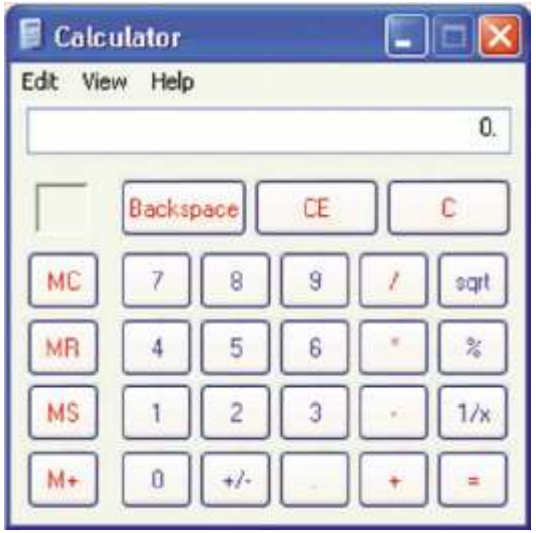
... EFEKTIVNA UPOTREBA BOJA



REPREZENTACIJA INFORMACIJA BOJOM

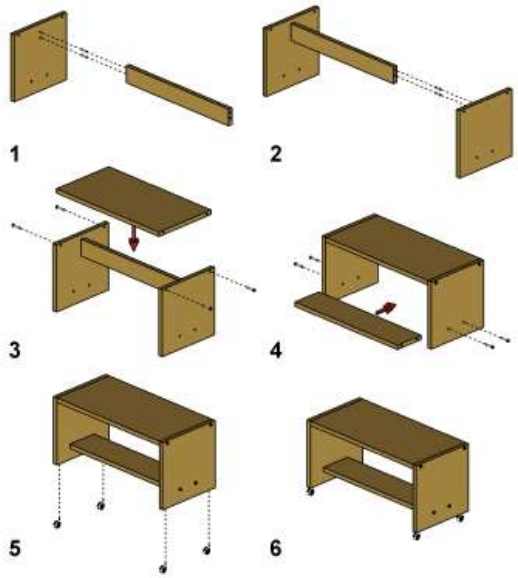


Technique	Description	Example
Color for emphasis	Color can be used to emphasize the important areas of the screen or the key parts of a diagram.	Figure 13.5 shows two contrasting representations of marshaling signals. The one that uses color to emphasize the lights is much easier to understand than the one in black and white. Box 13.1 shows the use of color to emphasize oceanic features.
Color for grouping	Color can be used to organize the screen.	Areas of the screen containing different types of information may have a different background color.
Color coding	Color can be used to represent a particular object or status; this is known as color coding.	In an accounts package, the overdue accounts could be indicated by a red symbol.
Perspective	Color can be used to reinforce perspective on the screen.	You should use dark or dim colors for the background and brighter colors for the foreground. Thus, the title bar of an active window is usually a brighter color than that of an inactive window.
Layering	Related to the use of perspective is the use of color to represent different layers within a diagram.	In an air traffic control system, high-flying planes could be a different color from low-flying planes. The colors should be naturally related to each other, such as different shades of blue.

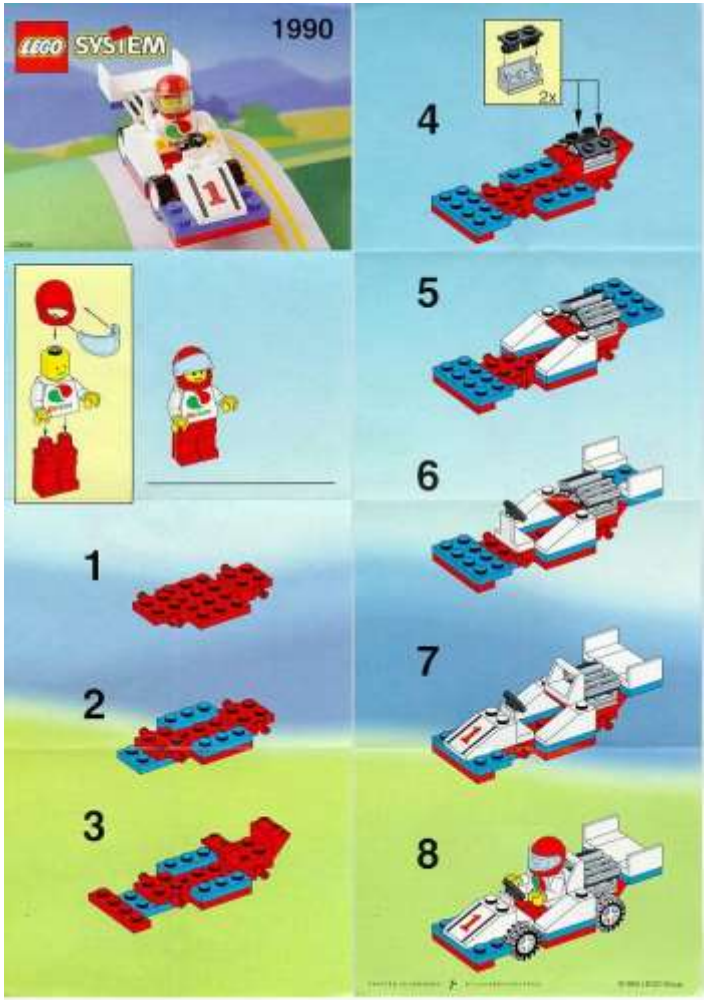


SLIKA

- Slike se mogu koristiti na više načina
  - Da motivišu, da privuku pažnju korisnika, da zabave, da ubede. Ova primena je vrlo važna u marketingu.
  - Da prenesu informaciju, naročito prostornu informaciju. Koristit se u računarom podržanom učenju.



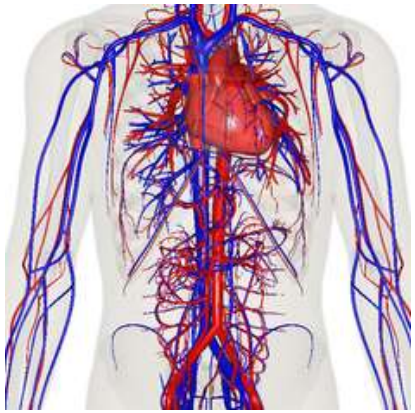
  - Da pomognu da se prenebregnu jezičke barijere. Vrlo rasprostranjeno u korisničkim upustvima.
  - Da podrže interakciju. Metafore ekrana i ikonice.



EFIKASNA UPOTREBA SLIKA ...



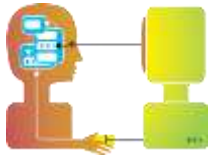
- Osnovni tipovi slika
  - Slike (pictures) – fotografije, crteži, crtaći. Prenose informacije koje je teško opisati rečima.
  - Dijagrami (diagrams) – mape, i druge reprezentacije relacija između objekata kao npr. porodično stablo, Venovi dijagrami. Koriste karakteristike dvodimenzionalnog prikaza da ilustruju relacije i procese.
  - Grafici i grafikoni (graphics and charts) – vizuelne reprezentacije brojeva i sl., kao što su pite, histogrami, bar grafovi i sl.





- Svaki tip slike reprezentuje drugačiji tip informacije

... EFIKASNA UPOTREBA SLIKA



- Uobičajeno je da se u sliku uključi određena funkcionalnost – npr. mapa regije ima ugrađenu funkcionalnost da kada korisnik klikne na neki njen deo, otvara se sadržaj sa informacijama o kliknutom području
- Kada se slike koriste u kor. interfejsu, treba voditi računa o sledećem:
  - Izabrati tip slike koji najviše odgovara vrsti informacije koja se želi preneti i efektu koji se želi postići
  - Dizajnirati slike tako da budu u skladu sa zahtevima najviše što mogu (previše neesencijalnih dijagrama mogu da odvuku pažnju korisniku)

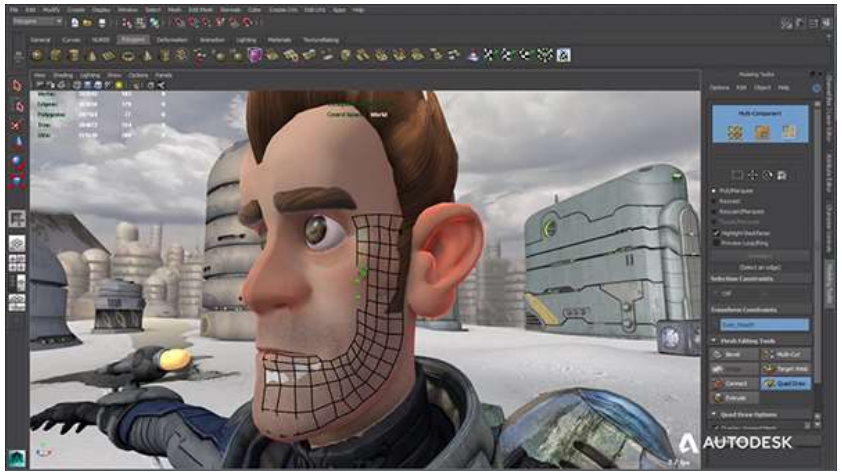


  - Pratiti sve relevantne konvencije kako bi se ostvarila konzistencija sa drugim slikama istog tipa (identični simboli, boje sa istim značenjem)
  - Kombinovanje slika i teksta može biti izrazito efikasno
  - Voditi računa o rezoluciji ekrana kako bi se ne bi izgubili detalji na slikama
  - Voditi računa koliko slika zauzima prostora na hard disku i ako se radi o kor. interfejsu u Web prezentaciji koliki je vreme njenog download-a

SEKVENCA SLIKA ...



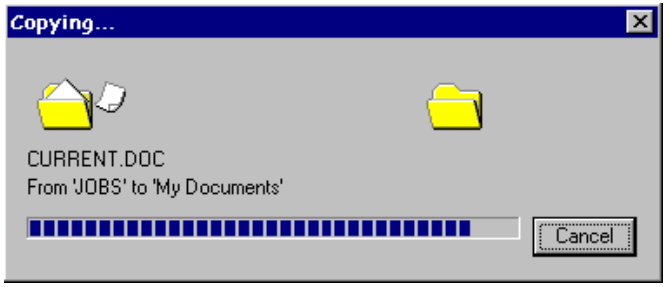
- Prikaz sekvence slika čak i na papiru može biti vrlo korisno jer omogućuje korisniku da sagleda čitav proces (postupak)
- Sekvanca slika se na računaru realizuje pomoću animacije ili videa
- Da bi se kreirala animacija potreban je odgovarajući softver i razumevanje principa na kojima počiva animacija



- Za video su potrebna kamera, softver za editovanje videa, ali i poznavanje principa na kojima se kreira dobar video (scenario, režija..)
- Zbog svega toga, najčešće generisanje animacije/videa treba prepustiti profesionalcima iz odgovarajuće oblasti

... SEKVENCA SLIKA

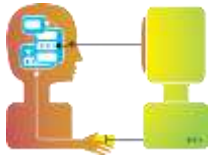
- Animacija može biti deo samog kor. interfejsa ili sadržaja koji se prikazuje u okviru softvera
  - Animacija prenosa fajlova koja prati operaciju kopiranju iz jednog foldera u drugi je očigledno deo kor. interfejsa
  - Reklama na veb sajtu očigledno nije deo kor. interfejsa ali je deo sadržaja koji se prikazuje korisniku





  - U velikom broju slučajeva nije lako napraviti jasnu razliku. Na ovo utiče i činjenica da se većina animacija/videa pokreće bez bekog naročitog učešća ili uticaja korisnika
  - Teško je naći jasne primere u kojem su animacije/video upotrebljene samo kao deo kor. interfejse, a ne kao sadržaj ekrena
  - Kako tehnologija napreduje, tako treba očekivati sve više pokretnih slika kao deo kor. interfejsa

... SEKVENCA SLIKA



- Kada se koriste pokretne slike, treba voditi računa o tome da je većina korisnika naviknuta da gleda animacije/video visokog kvaliteta na Tvu, te očekuju sličan kvalitet i u samo kor. interfejsu
- Ako standard korisnika nije zadovoljen, teško da će ih animacija ili video motivisati



- Kako animacije/video zauzimaju mnogo prostora (high bandwith) i kako njihova produkcija puno košta, njihovo uključivanja mora da bude dobro promišljeno i opravdano, a ne da budu samo skupi wallpaper-i
- Takođe se mora voditi računa da animacija/video može:
  - Izazvati određene vrste epilepsije
  - Odvući pažnju korisnika od osnovnog zadatka
  - Otežati fokusiranje korisnika na zadatak



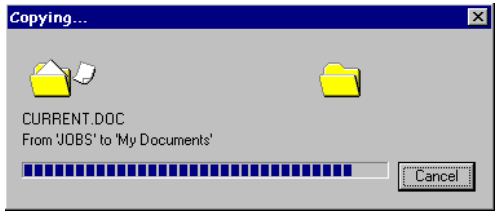
- U slučaju da se upotrebi animacija/video, korisniku se mora omogućiti zaustavljanje, pokretanje, pauziranje sekvence, kao i kontrola brzine reprodukcije

ANIMACIJA





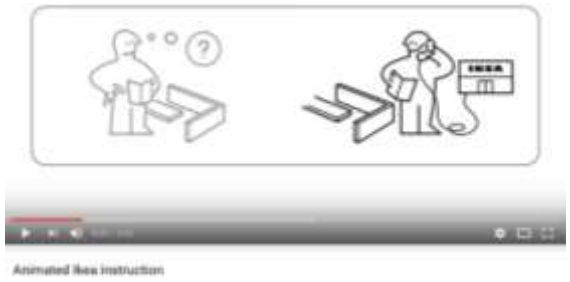
- Animacija se može iskoristit da:
  - ilustruje pokret. Npr. animacija u edukativnom programu koja služi da pokaže korisniku kako da uradi nešto pravilno, npr. dvokorak u košarci



  - prikaže dinamički feedback kao potvrda da se nešto dešava (da postoji reakcija sistema). Npr. animacija kopiranja fajlova



  - privuče pažnju korisnika. Npr. promena sata ili rezultata da privuče pažnju na semafor. Može postati iritantno



  - objasni na koji način se neki sistem koristi



  - prikaže da je računarski sistem aktivan (funkcionalan) tokom operacija koje se dešavaju u pozadini a traju određeni vremenski period

VIDEO

- Video klipovi se mogu koristiti u mnogim situacijama u kojima se koristi i animacija, ali i u dodatnim slučajevima
  - da prenesu ljudsko ponašanje i emocije. Npr. video tutorijal za zaposlene u banci kako treba da se ponašaju sa klijetnima u raznim situacijama može pokazivati situaciju kada korisnik dođe ljut u banku, tako da zaposleni mogu videti izraze lica ljutog klijetna







  - da prikažu događaje koje korisnik ne može direktno pogledati. Npr. unutrašnjost reaktora ili neki istorijski događaj
  - da motivišu. Npr. prikaz nekog ko obavlja posao bolje od korisnika, može ga motivisati da i on sam radi bolje
  - da pruže dodatnu kontekstnu informaciju. Npr. dodatne informacije o nekom događaju. Npr. avatar rođaka koji pomaže savetima po kući. Mora da se redovno ažurira kako se ne bi videla starost videa



ZVUK

- Vrlo je koristan kao izlaz, naročito kada korisnici ne mogu lako da vide vizuelnu informaciju
  - Aplikacije u kojima pogled i pažnja nisu sve vreme usmereni ka ekranu. Npr. medicina, fabrika, avion i sl. Monitorisanje trkača i njegovog srca, pa signaliziranje ako je broj otkucaja prešao preko dozvoljenog praga
  - Aplikacije koje monitorišu (upravljaju) procesom gde postoji potreba za neprikinut nadziranjem stanja sistema. Alarm u ovom slučaju signalizira promenu stanja. Npr. praćenje stanja pacijenta, upravljanje nuklearnom elektranom, štampači kada dođe do zagušenja papira i sl.

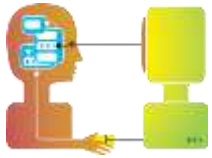






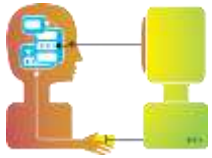
  - Aplikacije za slabovide ili slepe korisnike. U ove aplikacije spadaju čitači ekrana (screen readers)

VRSTE ZVUKA ...



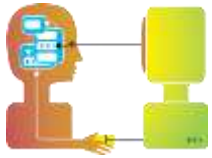
- Tri kategorije
  - Ambijentalni zvuk i zvučni efekti. Ambijentalni zvuk je prirodni produkt nekog okruženja ili uređaja (zvuk rada računara, kiše, zvuk kliktanja tastera). Potreba za ambijentalnim zvukom je takva da se u nekim situacijama mora rekreirati (npr. zvuk kliktanja po tasterima na virtuelnoj tastaturi mobilnog telefona). Zvučni efekti se koriste i kao potvrda da je operacija završena (zvuk gužvanja papira prilikom slanja u kantu za otpatke)
  - Muzika. Mnogi programi koriste muziku kao pratnju ili dodatni oblik alarma
  - Govor. Najkomplikovanija od svih kategorija, zbog potrebe za prepoznavanjem govora i generisanjem govora.

... VRSTE ZVUKA ...



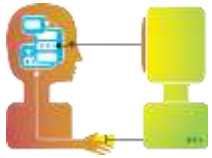
- Zvučni efekti se koriste da
  - Pojača vizuelne komponente kor. interfejsa (zvuk okretanje stranice knjige u virtuelnoj knjizi)
  - Potvrdi izvršenje neke operacije (kada se ostvari konekcija, čuje bip, tj. zvuk kao potvrda)
  - Privuče pažnju (zvuk poruke o grešci kako bi se privukla pažnja korisnika na postojanje greške)
- Muzika se koristi retko osim u kompj. igrama i specijalnim aplikacijama koje koriste kompozitori i muzičari
  - Posebne muzički tonovi za različite operacije / akcije u nekim operativnim sistemima (zvuk podizanja Windows-a)

... VRSTE ZVUKA



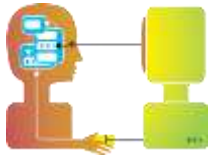
- Upotreba govora je vrlo moćan alat za prenos / prezentaciju informacija
- Najkorisnije za aplikacije namenjene slabovidima / slepima
- Koristi se i za iskazivanje poruka, najefikasnije kad su sledeći uslovi ostvareni
  - Jednostavna poruka
  - Kratka poruka
  - Poruka neće biti do slovca ispričana
  - Poruka prati događaje u vremenu
  - Poruka zahteva trenutnu akciju
  - Vizualni kanali je preopterećen
  - Okruženje je suviše svetlo, suviše tamno, podložno vibracijama ili iz nekog drugog razloga nepovoljno za prenos vizuelne informacije
  - Korisnik nije obavezan da sedi ispred ekrana

UPOTREBA ELEMENATA INTERAKCIJE



- Elemente interakcije treba upotrebiti tako da se:
  - Smanji vizuelni napor (reduce visual work)
  - Smanji intelektualni napor (reduce intellectual work)
  - Smanji potreba za memorisanjem (reduce memory work)
  - Smanji motorna aktivnost (reduce motor work)
  - Smanji ili eliminiše bilo koje opterećenje ili instrukcije koje namaće tehnologija (Minimize or eliminate any burdens or instructions imposed by technology)
- Ljudi će provoditi puno vremena gledajući u vaš interfejs
- Očekivati da će se korisnici navići na sve je pogrešno!
- Najbolji interfejsi čine sve na ekranu očiglednim

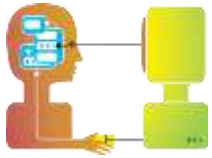
ELEMENTI INTERAKCIJE – NEKE SMERNICE



- Obezbediti jasno uočljivu početnu tačku (u gornjem levom ćošku ekrana)
- Fokusirajte pažnju korisnika na najvažnije delove ekrana
  - Kako će korisnici skenirati informacije zavisi od kompoziciji prikaza
- Ljudi teže ka tome da prvo pogledaju tekst, posle slike
- Velike stvari dominiraju nad manjim
- Prvo se pažnja posveti informacijama koje se menjaju
- Ljudi posmatraju vrlo često u takozvanom F paternu

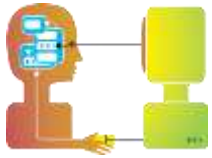
Graphical User Interface (GUI)

KOMBINOVANJE ELEMENATA INTERAKCIJE



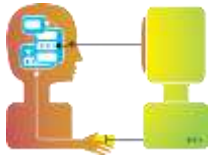
- Kompletan grafički korisnički interfejs nastaje kombinovanjem interakcionih uređaja (hardverskih elemenata interakcije) i softverskih komponenti u jednu jedinstvenu celinu
- Ako se odabere previše različitih komponenti, koje se međusobno nadmeđu za prostor, rezutat može biti konfuzan (prikaz preplavljen nizovima ikonica, menija i prozora ...)
- Ključ je u jednostavnosti (simplicity) – razmisliti o poziciji svakog elementa i zapitati se da li on baš mora da bude tu
- Nekad se kompleksnost ne može izbeći zbog prirode zadatka ali i zbog iskustva korisnika koji zahtevaju tu kompleksnost
- Kada se dizajnira GUI uvek postoje konfliktni zahtevi i tu je potrebna veština i iskustvo kako bi se odredilo šta treba uključiti a šta ne u kor. interfejs

PRINCIPI DOBROG LAYOUT-A ...



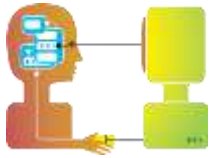
- Da bi se napravio upotrebljiv kor. interfejs potrebno je kombinovati različite elemente kako bi korisnici obavili svoje zadatke lako, brzo i efikasno
- To zahteva dobro raspoređivanje (good layout) softverskih komponenti i interakcionih uređaja
- Za softverske komponente to uobičajeno znači njihovo dobro raspoređivanje na ekranu (prozoru)
- Za interakcione uređaje to znači dobro mapiranje ključnih tastera (keymaping and keypad desing) i raspoređivanje softv. dugmadi

... PRINCIPI DOBROG LAYOUT-A ...



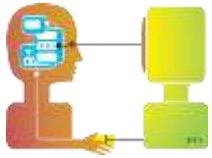
- **Obezbediti prirodno grupisanje elemenata**
  - Razmisliti o prirodnoj strukturi informacija
  - Npr. podaci o zaposlenom se mogu grupisati na:
    - deo sa kontakt podacima,
    - deo sa istorijom zapošljavanja,
    - deo sa detaljima o trenutnom zaposlenju, i
    - deo sa podacim o plati.
  - Isti princip važi i za grupisanje kontrola
  - Npr. grupisati sve menije ili ikonice koje se odnose na isti kontekst ili koncept

... PRINCIPI DOBROG LAYOUT-A ...



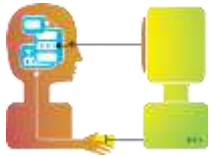
- **Razdvojiti trenutno aktivne komponente**
  - Jako je važno naglasiti šta korisnik trenutno radi
  - Ako je to ispoštovano, ako im nešto odvuče pažnju, lako će moći da nastave
  - Takođe, to omogućuje fokusiranje korisnikove pažnje na ono što je trenutno važno i aktuelno u zadatku koji obavlja
  - Uzima se u obzir psihološko ograničenje da se ljudi teško fokusiraju na više od jednog zadatka u istom trenutku
  - Npr. aktivni prozor u Windowsu je uvek iznad svih ostalih i jasno označen (naglašen)

... PRINCIPI DOBROG LAYOUT-A ...



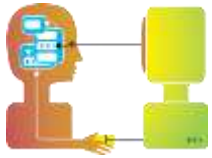
- **Naglasiti važne komponente**
  - Isticanjem se omogućuju lakša identifikacija i pronalaženje stvarno važnih komponenti
  - Npr. ***Panic*** dugmad su izražene crvene boje (zašto?)
  - Dobro i pažljivo identifikovati koje su komponente važne
  - Koristiti vrlo rezervisano, samo kad ima smisla
  - Ako je sve naglašeno, gubi se smisao naglašavanja i postaje zbunjujuće
  - Boja, veličina fonta i kurzitiv, pozicioniranje na ekranu, zvuk i animacija mogu se upotrebiti za naglašavanje komponononti
  - Dobra praksa je kobinovati ove komponente kako bi se istakao efekat (npr. alarm u kolima i trepće i pišti)

... PRINCIPI DOBROG LAYOUT-A ...



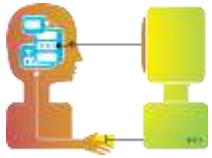
- **Koristiti prazan prostor na efikasan način**
  - Prazan prostor (white space) je ekstremno važan za dizajniranje dobrog kor. interfejsa
  - Prazan prostor na ekranu na kojem ne postoje elementi interakcije
  - Često je daleko efikasnije od linije za grupisanje elemenata, jasnije ističe layout od linije
  - Uvek uključiti prazan prostor i razmak, po cenu da se informacije moraju rasporediti na više prozor, bolje više neiskorišćenog prostora, nego previše kompleksan prenatrpan sadržaj
  - Isto važi i za softverske komponente – bolje smanjiti njihov broj na prozoru (upotrebom menija razbijenih na više nivoa), nego ih sve smestiti na jedan prozor i napraviti gužvu
  - Informacije i komponente koje će se prikazati direktno na prozoru odabrati pažljivo

... PRINCIPI DOBROG LAYOUT-A ...



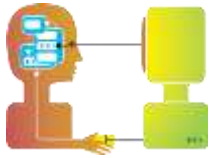
- **Pojačati vidljivost komponenti (jasno istaći njihovu funkciju)**
  - Da bi komponenta bila vidljiva, njena funkcija mora biti jasno naglašena
  - Ovo se oslanja na princip da je lakše se nečeg podsetiti, nego zapamtiti
  - Komponente postavljene na prozoru moraju da imaju jasno nazančenu funkciju uzimajući u obzir korisnikovo poznavanje računarskih sistema i sveta uopšte (domensko znanje, kulturološku i socijalnu pozadinu, obrazovanje, i sl.)

... PRINCIPI DOBROG LAYOUT-A



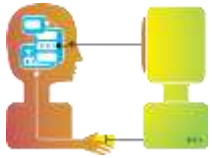
- **Balansirati estetiku i upotrebljivost**
  - Estetski izgled kor. interfejsa dobija na sve većem značaju
  - Pojava Apple Macintosh računara sa širokom paletom raspoloživih boja promenila je kako korisnici doživljavaju kor. interfejs i njihova očekivanja – **korisnici više nisu zadovoljni sivim kutijama!!!**
  - Od tada mnogo je pažnje posvećeno redizajniranju GUIa kako bi se ispunila ta očekivanja i kako bi kor. interfejs postao atraktivniji
  - Međutim, mora se voditi računa da se sa estetikom ne preterana
  - Gotovo nikada ne treba žrtvovati usability zarad estetike (na duge staze usability pobeđuje)
  - Npr. *A vivid background may look stunning, but it could be tiring for the eyes. You need to ask yourself which is more important. You may have some arguments with your marketing manager over this point.*

DIZAJNERSKI PROSTOR INTERAKCIJE ...



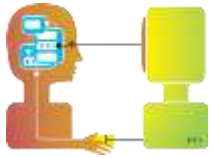
- Pre nego što se pristupi dizajniranju kor. interfejsa, potrebno je odabrati prostor u kojem će korisnik biti u interakciji sa sistemom
- Dizajnerski prostor predstavlja hardverskih i softverskih komponenti
- Svaki ovaj prostor je specifičan sa svojim problemima i principima (ne može se tek tako seliti iz jednog prostora u drugi što je Microsoft nekoliko puta demonstrirao)
- Neki od dizajnerski prostora su:
  - GUI
  - Web stranice
  - Embedded sistemi (poput uređaja nemenjenih držanju u ruci – handheald devices)
  - Sveprisutno računarstvo (ubiquitous computing)
- Postoje i drugi dizajnerski prostori
- Granica između dizajnerskih sistema nije jasna

... DIZAJNERSKI PROSTOR INTERAKCIJE



- Kada se dizajnira kor. interfejs treba voditi računa o sledećem:
  1. Identifikovati relevantni dizajnerski prostor i uzeti u obzir karakteristike, principe i smernice tog prostora
  2. Razmisliti dobro o konceptualnom dizajnu napravljenom kao rezultat faze prikupljanja zahteva
  3. Kombinovati dizajnerske komponente ali uzimajući u obzir dizajnerski prostor i šta se želi da se desi u okviru kor. interfejsa kako bi se ostvarili zahtevi
- Oblast razvoja kor. interfejsa se konstantno menja, te se samim tim stalno menjaju i dizajnerski prostori (njihove karakteristike)
  - Svaka nova verzija Windows-a donosi nove GUI komponente (widget-e)
  - Funkcionalnost Web stranica se menja svakog dana kako se unapređuje sama tehnologija
  - Nemogući opisati kako se svaka pojedinačna komponenta ponaša i očekivati da će to biti trajno upustvo

DIZAJNIRANJE GUI-A



- Uključuje razne tehnike interakcije, menije, forme, direktnu manipulaciju, ali i primenu različitih metafora
- Vrlo je rasprostranjen i kompleksan dizajnerski prostor
- Ko god je koristio računar, koristio je i GUI
- GUI je prisutan već neko vreme, ali problemi uopšte nisu rešeni i nije trivijalno koristiti ga uz svo postojeće iskustvo
- Nećemo opisivati kako se GUI koristi (to bi trebalo da je poznato)
- Widget-i su elementi GUI-a: prozori, dijalozi, tabovi, meniji, toolbar-ovi, softv. dugmad (komandna i za izbor – radio dugmad), ček boksovi, list boksovi, tekstualna polja, …
  - Kako izabrati odgovarajući widget?
  - Kako koristi widget-e efikasno?
  - Kako kombinovati widget-e?

WIDGET

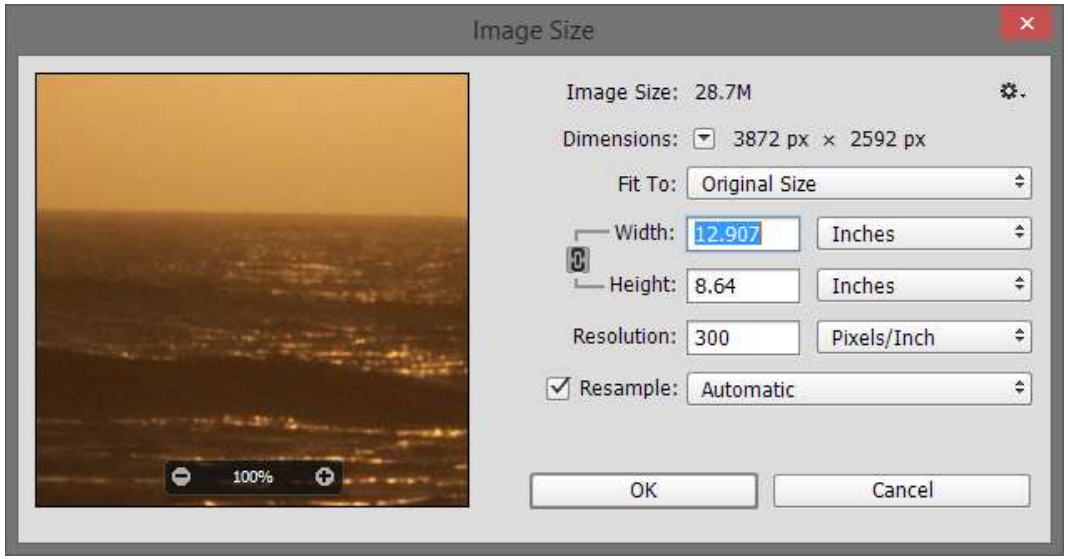
- Po Wikipedia-i samo za softver:
  - Software widget, a generic type of software application comprising portable code intended for one or more different software platforms
  - Window widget, a control element (sometimes called a control or widget) in a graphical user interface is an element of interaction, such as a button or a scroll bar

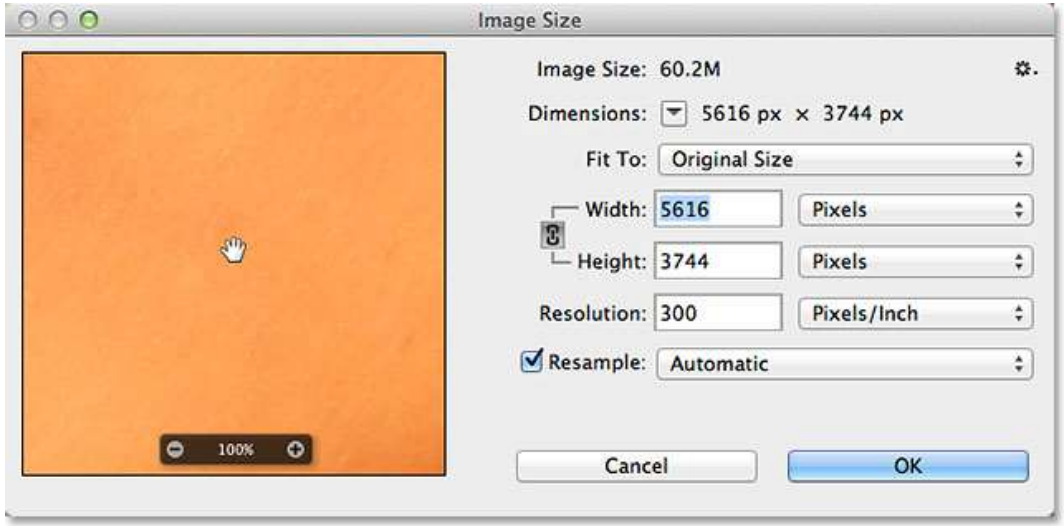


  - Widget toolkit, a software library containing a collection of GUI widgets that collaborate when used in the construction of applications
  - Web widget, an applet intended to be used within web pages
  - Widget drive, a hard drive used only in the Apple Lisa computer system

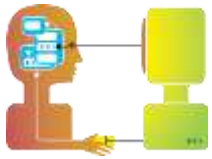
IZGLED WIDGET-A NA RAZLIČITIM PLATFORMAMA

- Različiti softveri i platforme prikazuju widget-e na različite načine

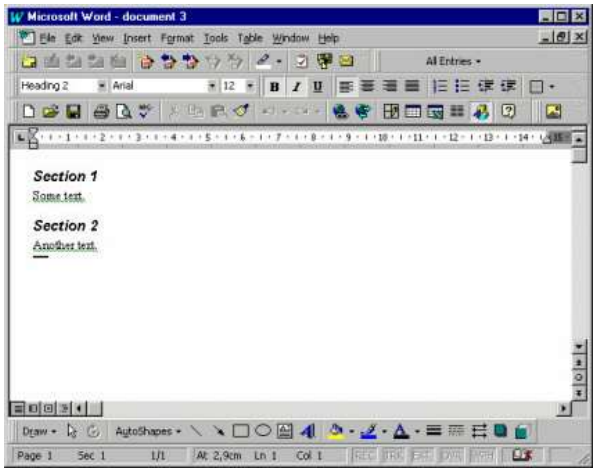




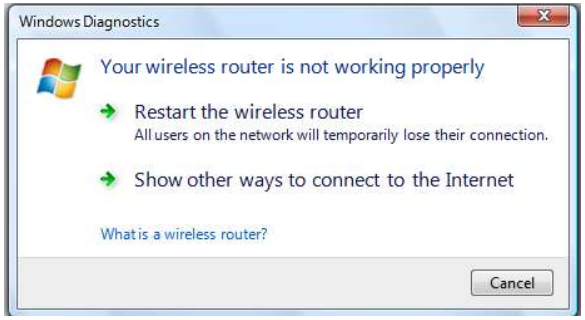
WIDGET-I ZA STRUKTURIRANJE INTERAKCIJE ...

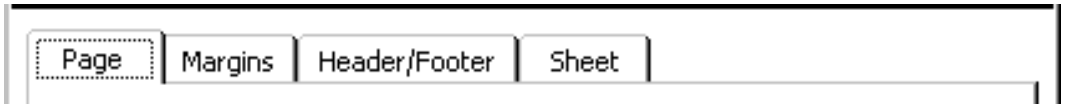


- GUI je uglavnom raspoređen preko „high-level“ widget-a kao što su
  - prozori, primarni ili sekundarni,
  - dijalozi, ili
  - tabovi
- Omogućuju strukturiranje kor. interfejsa tako da reflektuje konceptualni dizajn

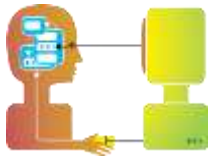








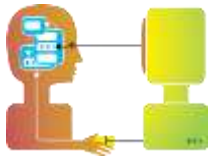
PRIMARNI PROZORI (PRIMARY WINDOWS)



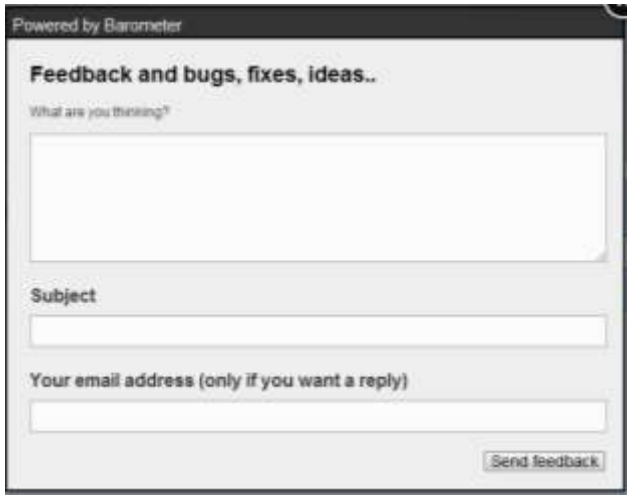
- Prozori uglavnom sadrže okvir, naslovnu liniju, menije, skrol barove, ...
- Najčešće odgovaraju objetktima oko kojih se realizuje osnovni (primarni) zadatak konceptualnog dizajna
- Npr. Word ima jedan primarni prozor koji odgovara word procesoru (objekat osnovnog zadatka je dokument koji se piše)
- U jednoj aplikaciji broj primarnih prozora je uobičajeno mali
- Uobičajeno, svaki primarni prozor predstavlja osnovnu tačku kojoj se korisnik vraća
- Često postoji jedan primarni prozor koji je relej ka ostalim primarnim prozorima

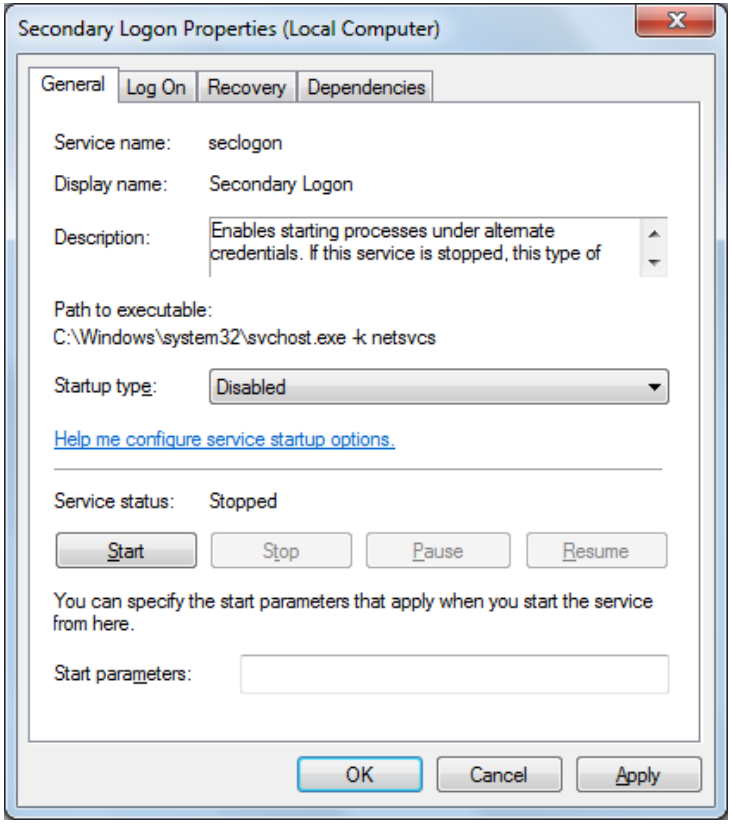


SEKUNDARNI PROZORI (SECONDARY WINDOWS)

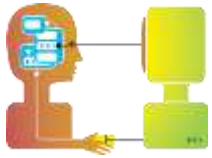


- Dopunjuju primarne prozore
- Pružaju dododatnu funkcionalnost i podršku korisnicima
- Postoji nekoliko vrsta sekundarnih prozora
- Vrlo korisni i često upotrebljavani su:
  - Poruke (message boxes)
  - Dijalozi (dialog boxes)



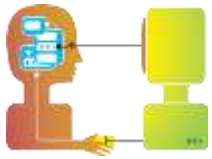


PORUKE (MESSAGE BOXES) ...



- Prenose poruke korisniku (kao što kaže njihov naziv)
- Često se odnose na neki nastali problem ili situaciju koji korisnik mora hitno da razreši
- Često su modalnog karaktera (modal dialog) – da bi privukli korisnikovu pažnju time što ga teraju da klikne kako bi nastavio sa poslom
- Treba odrediti da li je poruka uopšte potrebna, a zatim jako dobro formulisati sadržaj poruke
- Modalni dijalog može da izazove frustraciju ako korisnik misli da nije adekvatan za datu situaciju (ili nije jasno zašto se on pojavio tu) i/ili ako sadržaj poruke nije adekvatan ili razumljiv
- Izbor između modalnog ili nemodalnog dijaloga (modeless dialog) zavisi od situacije i potreba korisnika (da li se njegov rad želi zaustaviti ili ne) – modalni je bolji za početnike, za iskusne korisnike nemodalni

... PORUKE (MESSAGE BOXES) ...



- Beskorisne poruke
  Zašto?

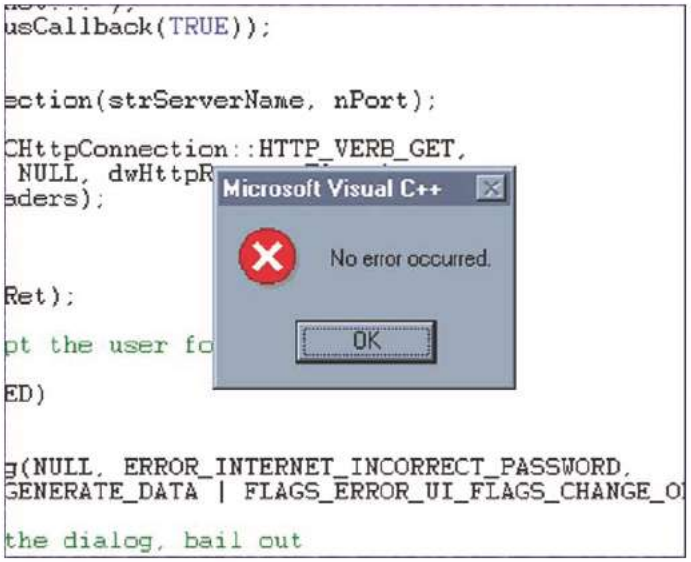
(a)



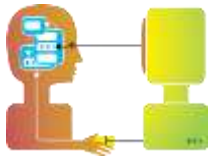
(b)



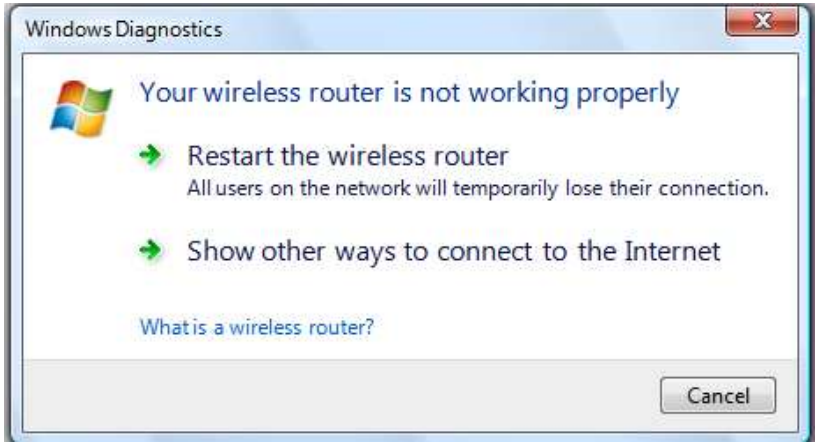
(c)

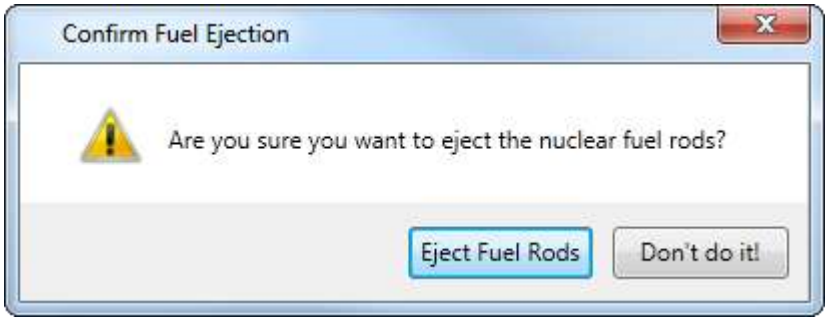


... PORUKE (MESSAGE BOXES)

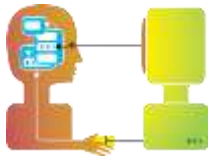


- Poruke u primeru su beskorisne zato što:
  a) Greška generisana prilikom instalacije Visual Basic-a – korisnik ne razume poruku, najverovatnije zaostalo kao rezultat debagiranja
  b) Nastala prilikom kopiranja fajlova – pokušaj pomoći korisniku procenom vremena trajanja kopiranja (u primeru je zanačajno promašena ocena trajanja kopiranja)
  c) Pozitivna ocena da nema greške je prikazana korisniku u formatu greške

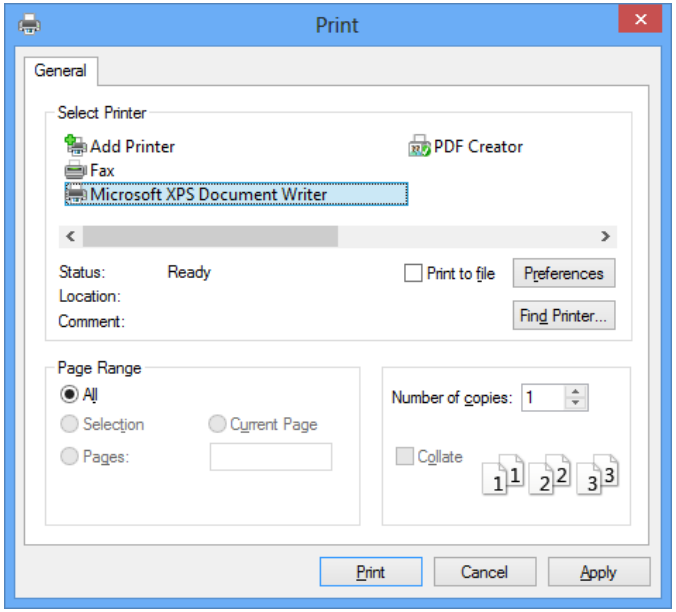




DIJALOZI (DIALOG BOXES)

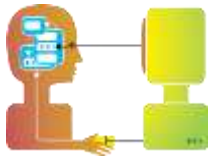


- Pokreće ih korisnik
- Koriste se često kada je potrebno doći do datnih informacija kako bi se ostvario zadatak/funkcija
- Mogu biti vrlo kompleksni eniteti na ekranu – sadržati tekstualna polja, softv. dugmad ...
- Većinom su nemodalna kako bi se korisnici mogli kretati između prozora ako je potrebno

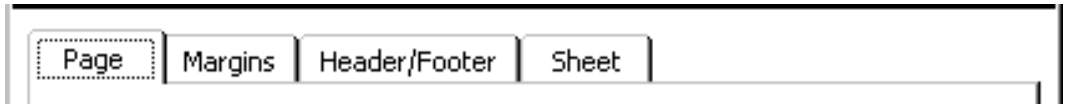


- Nekada korisnik mora da prođe kroz seriju dijaloga u striktno definisanom redosledu, te se tada koriste modalni dijalozi
- Serija modalnih dijaloga kroz koje se prolazi u striktnom redosledu naziva se Wizard

TAB DIJALOZI (TABS)

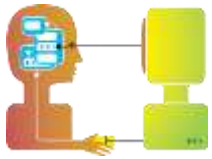


- Korisni za klasifikaciju/grupisanje karakteristika objekta zadatka koji je reprezentovan prozorom
- Baziraju se na metafori kartica (index card metaphor)

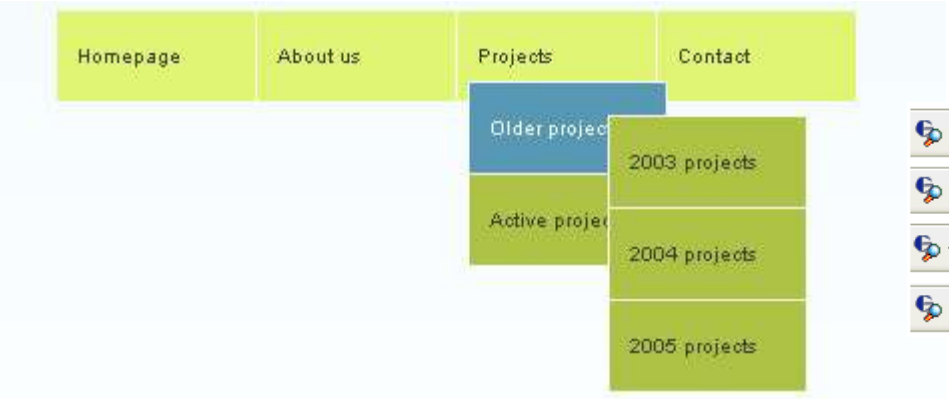


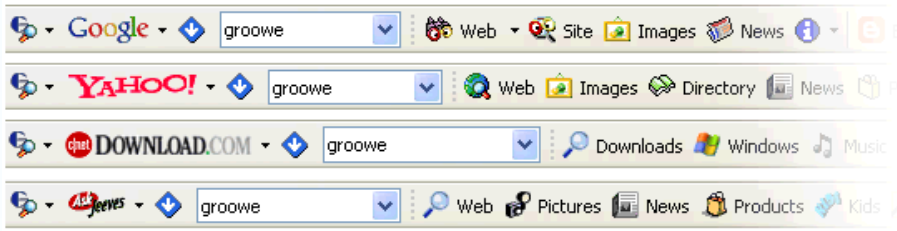
- Mora da se pazi da se ne pretera sa broj tabova (kada ih je previše, mora da se deli u više redova, šta je sa pomeranjem usled selekcije?)
- Kada se koriste tab dijalozi treba voditi računa o:
  - ***Da li su informacije na različitim tabovima nezavisne***? – trebale bi da budu, u suprotnom korisnik će se zbuniti
  - ***Koliko tabova je zaista potrebno***? – previše će pogoršati kor. interfejs umesto da ga unapredi; bolje koristiti više dijaloga
  - ***Da li tabove treba popuniti u specifičnom redosledu***? – ako je odogovor da, onda to ne odgovara metafori (nije namenjeno tab dijalogu) i treba odabrati alternativni pristup
- Moguće je da korisnik zaboravi da popuni neki tab (jer nije sve na jednom mestu

WIDGET-I ZA KONTROLU INTERAKCIJE

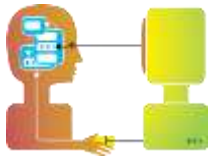


- Postoje različiti widget-i koji omogućuju kontrolu interakcije (u smislu da se preko njih realizuje interakcija i izdaju naredbe), poput:
  - Menija,
  - Tulbarova,
  - Softv. dugmadi
- Omogućuju korisniku da izvrše odgovarajući zadatak/akciju predviđenu konceptualnim dizajnom

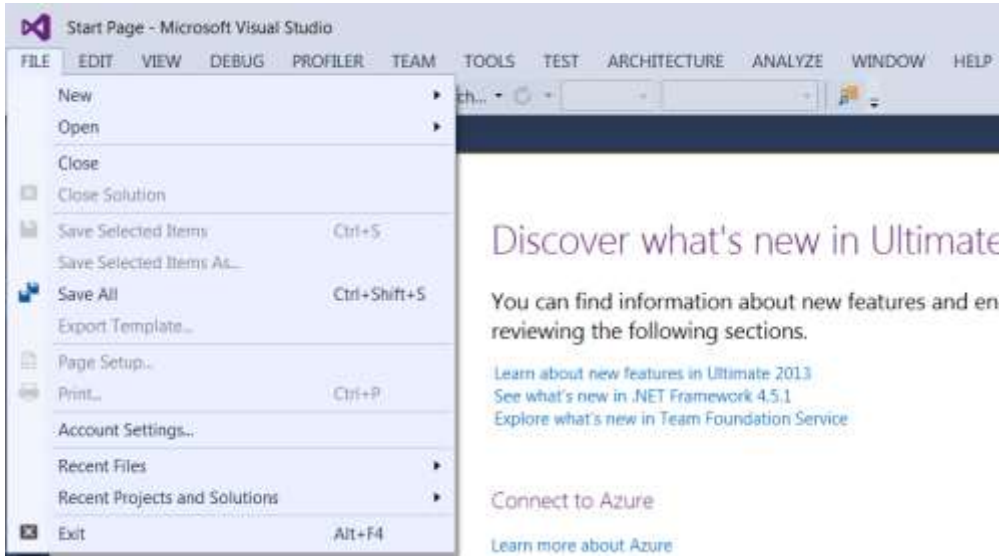




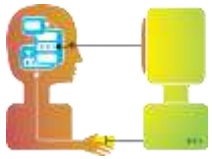
MENIJI (MENUS) ...



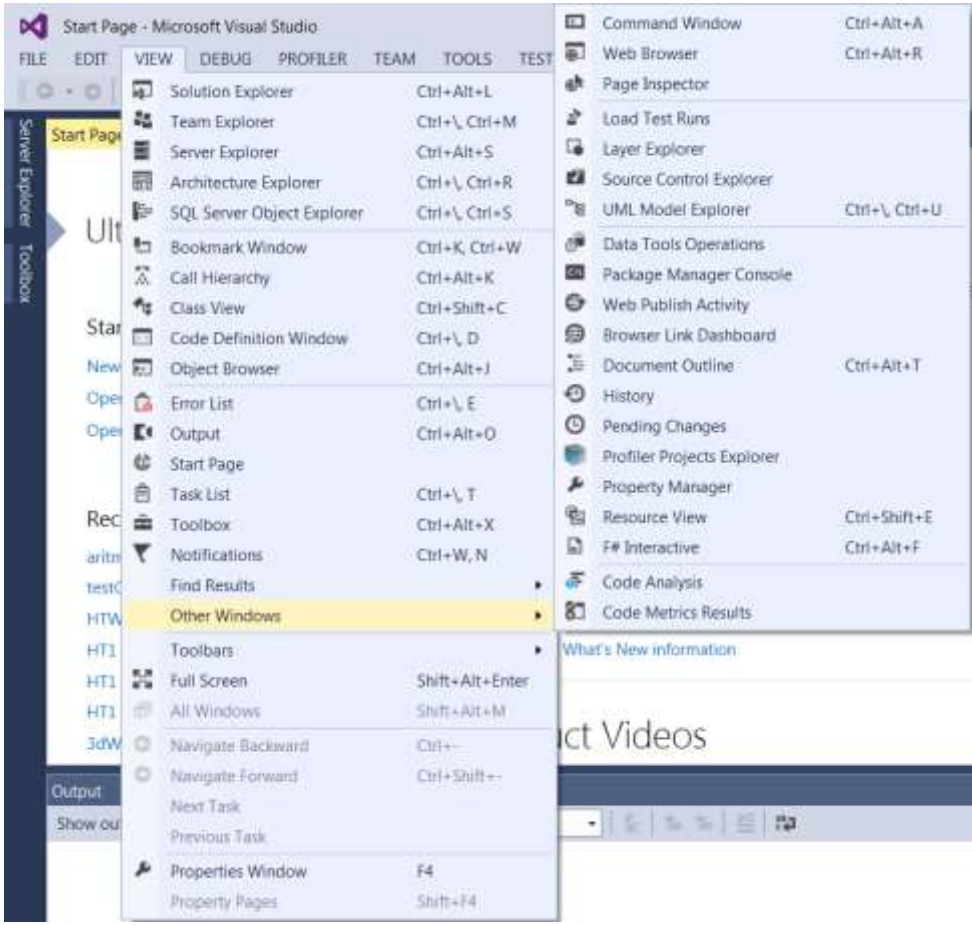
- Većina kor. interfejsa sadrži neku vrstu menija
- Skup stavki unutar kojih korisnik pravi izbor
- Klikom na stavku menija bira se jedna akcija
- Stavke/kategorije moraju biti razumljive i različite tako da korisnik ima jasnu ideju šta će se desiti kada nešto odabere
- Mogu voditi i do dijaloga koji predstavlja seriju koraka kako bi se realizovao zadatak
- Padajući meni (drop-down menus) – prikazuju se kada se pritisne labela na menu bar-u ili ikonica na tool bar-u (nekada su označeni strelicom)
  - najčešći oblik menija
  - konstanto dostupan na vrhu (može i drugde, ali nije uobičajeno)



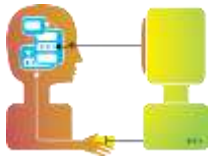
... MENIJI (MENUS) ...



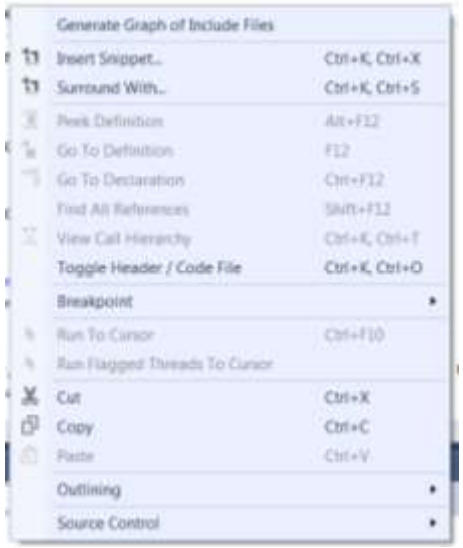
- Kaskadni meniji (cascading menus) – podmeniji koji se prikazuju kada se pozicionira mišom iznad stavke menija (koja ima strelicu kao oznaku pored teksta)
  - rezultuje istovremenim prikazom određenog broja podmenija
  - zahtevaju veštinu dobrog rukovanja pointerskim uređajem
  - ekskluzivne grupe disjuknih naziva
  - ako je grupisanje prirodno i razumljivo, lako se koristi i stiže se do željene akcija u zadovoljavajuće kratko vreme
  - ako grupisanje nije prirodno i razumljivo, korisnik ima samo maglovitu predstavu o tome šta mu je cilj, tada će se izgubiti

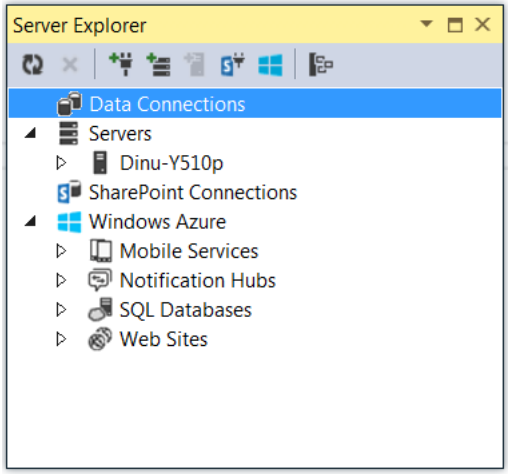


... MENIJI (MENUS) ...



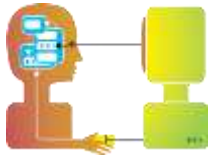
- Konteksni meniji (contex or pop-up menus) – plutajući meniji koje poziva korisnik (uobičajeno kliktanjem na određeni objekat) čija pozicija i sadržaj zavise od toga na šta se kliknulo
  - uobičajeno je da sadrže akcije i zadatke povezane sa objektom na koji se kliknulo





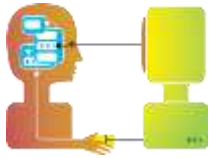
- Sažimajući meniji (roll-up menus) – samostalni (plutajući) meniji koji se mogu pomerati i stalno su vidljivi
  - mogu se pokrenuti slično konteksnim menijima ali ostaju vidljivi dok se eksplicitno ne zatvore
  - mogu se sažete kako bi uštedeli prostora

... MENIJI (MENUS)

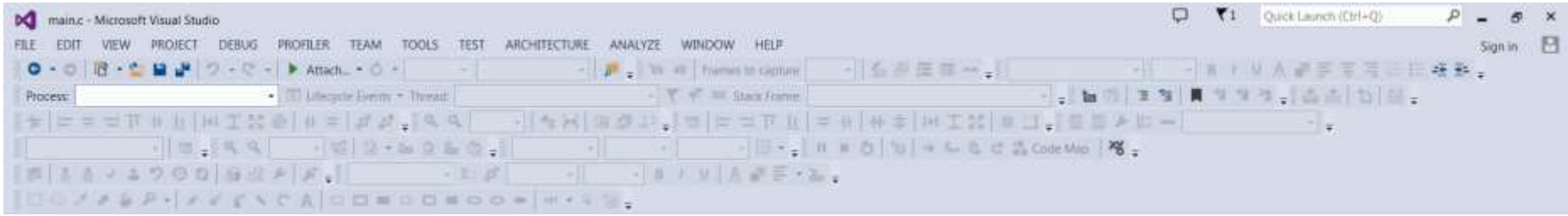


- Prilikom rada sa menijima mora se voditi računa
  - o nazivima stavki menija – ako ne razumete svrhu stavke menija, ne znate kada da koristite datu stavku menija
  - o tome da li meniiji omogućuju korisniku da radi to što želi, ali ne i više od toga – meni se ne pravi oko funkcija sistema, već oko zadataka korisnika; treba da reflektuju zadatke e ne funkcije programa
  - o uređenju stavki menija i njihovom organizovanju – većina korisnika čita sa leva u desno i od gore na dole; iako bi se dalo očekivati, najvažnije stavke menija nisu najčešće pozicionirane na početku u gornjem levom ćošku, čak ni najčešće korišćene
- Važno je da korisnici razumeju zašto su meniji organizovati tako kako su organizovani (postoje određene konvencije, ali se treba odlučiti da li ih pratiti ili ne)
- Do rasporeda stavki menija može se doći testiranjem korisnika pomoću metode koja se naziva sortiranje karata (card sort) – korisniku se ponude funkcije na kartama i traži se od njega da ih grupiše

TOOL BAR-OVI ...

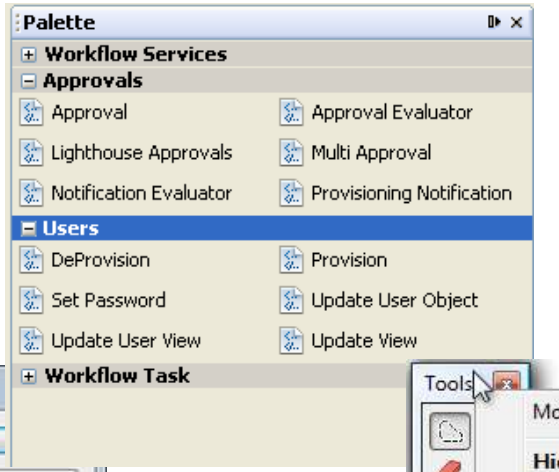


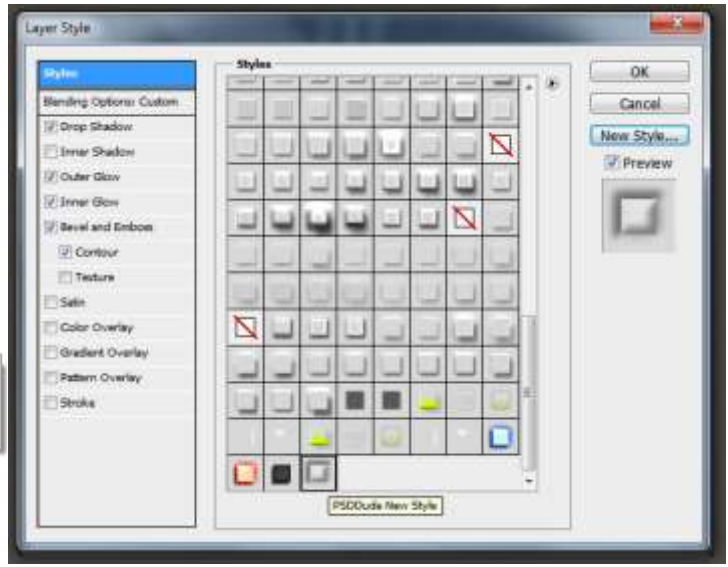
- Dopunjuju menije (u neku ruku još jedna vrsta menija), uvek vidljivi
- Sadrže frekvento korišćene komande reprezentovane ikonicama
- Ikonice su često objašnjenje tool tip-ovima (pop-up prozori sa tekstom koji dodatno opisuju svrhu akcije iza ikonice)
- Omogućuju korisnicima brzu upotrebu softvera
- Uobičajeno je da postoji izbor između više tool bar-ova, ali i mogućnost njihove selekcije i kastomizacije
- Grupisanje komandi u tool bar-ove prema domenu (funkcionalnostima programa)
- Često su tool bar-ovi nevidljivi ako nisu potrebni u datom trenutku

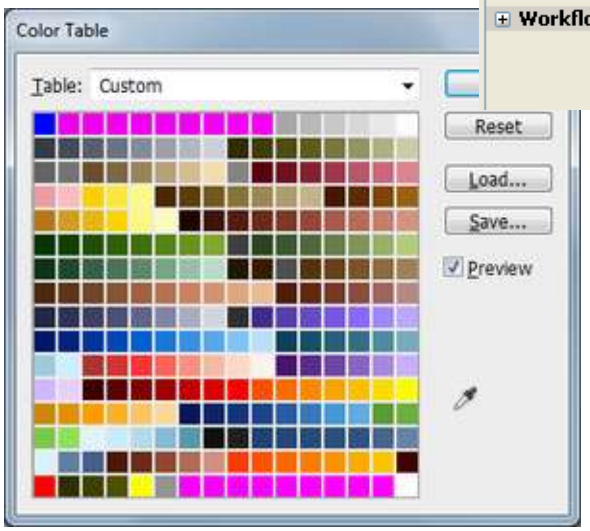


… TOOL BAR-OVI

- Pozicioniraju se ispod glavnog menija aplikacije na primarnom prozoru, mogu se nalaziti i na sekundarnim prozorima (palette windows)
- Ključan je izbor adekvatne ikonice (šta god odabrali, testirani sa korisnicima)









IKONICE ...

- Slika ili lik koji reprezentuje neki koncept
- Mala reprezentacija objekta ili akcije
- Ikonice bi trebalo da imaju nekoliko poželjnih
  - ***Lako ih je međusobno razlikovati*** – korisnik treba da odabera pravu ikonicu čak i ako mu se žuri; ako su ikonice suviše slične, lako može da pogreši



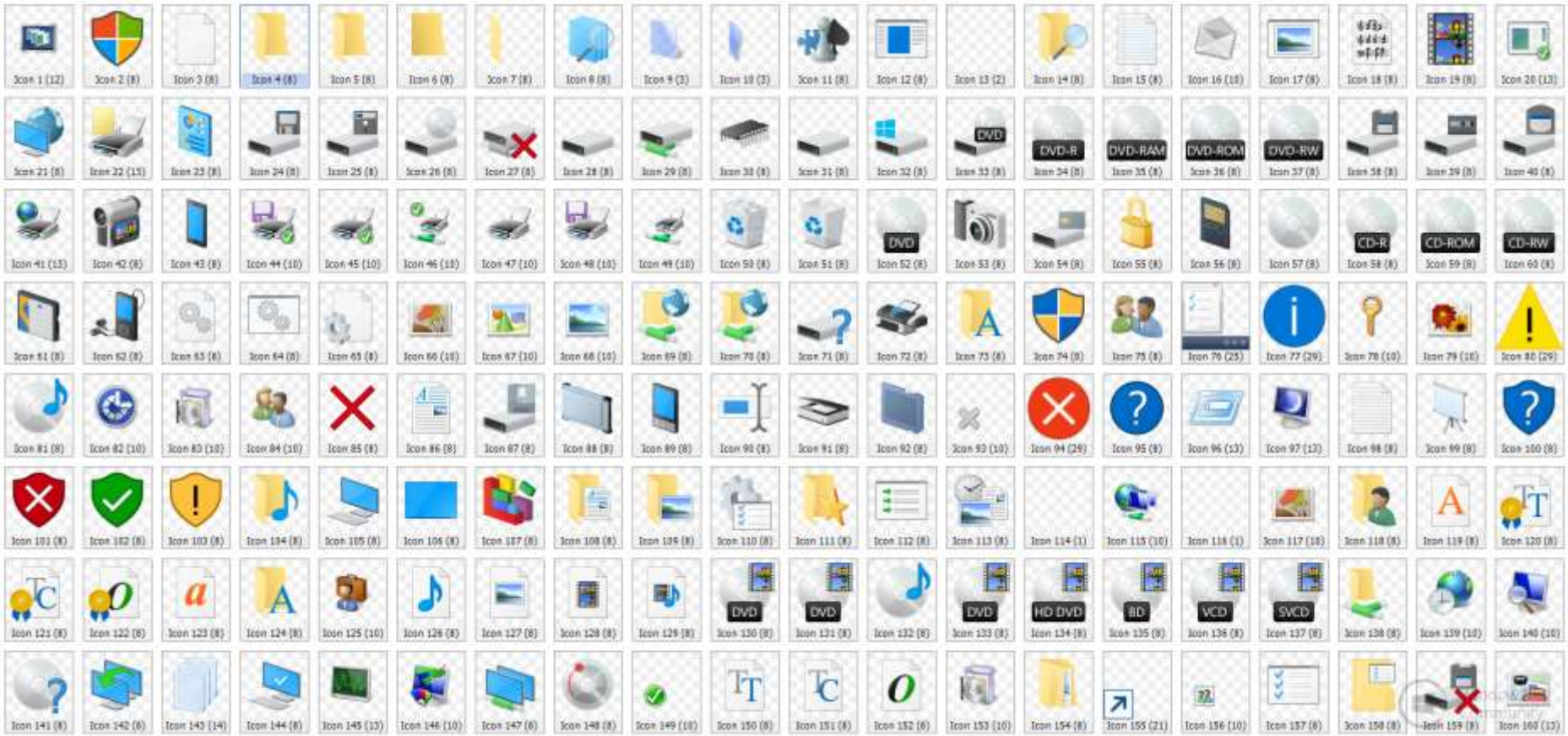
  - ***Lako ih je prepoznati i razumeti*** – korisnik mora biti familijaran sa ilustrovanim objektom i u stanju da poveže ikonicu sa konceptom koji se nalazi iza nje
  - ***Vizuelno su jednostavne*** – ikonice ne bi trebale da sadrže nepotrebne detalje (da li je potrebno prikazati baš svaki detalj na ikonici štampača)
  - ***Informativne su*** – iz njih je moguće saznati potrebnu informaciju (npr. ikonice za poravnavanje teksta)

... IKONICE ...

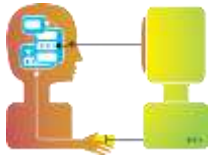
- ... Ikonice bi trebalo da imaju nekoliko poželjnih karakteristika
  - ***Reprezentuje konkretne objekte*** – čak i ako je koncept apstraktan, često je bolje reprezentovati ga konkretnim objektom koji je čvrsto povezan sa apstraktnim konceptom
    - zbog toga ikonica kuće može da reprezentuje home stranicu
    - slično je i sa procesima (zato sličica diskete može da reprezentuje save opciju)
  - ***Lako ih je uočiti*** – izbor boja je jako važan i mora se birati pažljivo izbegavajući suviše kompleksnosti (detalja) kako ikonica ne bi bila nejjasna a njen sadržaj neprepoznatljiv



... IKONICE


DVD
DVD-R
DVD-RAM
DVD-ROM
DVD-RW
CD-R
CD-ROM
CD-RW
DVD
DVD
DVD
HD DVD
BD
VCD
SVCD

SOFTV. DUGMAD ZA KOMANDE ...



- Tipično se koriste na dijalozima kako bi se korisniku omogućila kontrola dijaloga
- Kao i tool bar-ovi, uvek su vidljivi

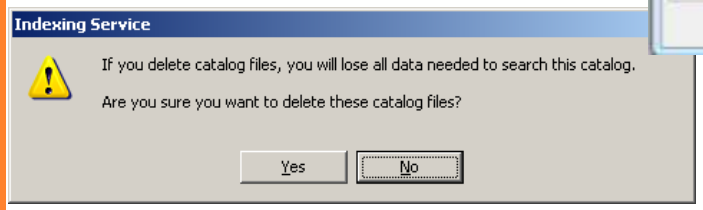


- Prilikom korišćenja softv. dugmadi za komande treba voditi računa
  - ***o tekstu koji će se na njima ispisivati*** – jako je važno da korisnik razume koja akcija će se desiti kada klikne na neko dugme
  - ***o poziciji na kojoj će se smestiti dugme*** – često se nalaze na dnu dijaloga, ali to nije pravilo; uobičajeno je da se stavljaju u kontekst čitavog dijaloga, te bi trebali da slede iza sadržaja na koji se odnose (ispod njega ili sa desne strane posle njega)

… SOFTV. DUGMAD ZA KOMANDE

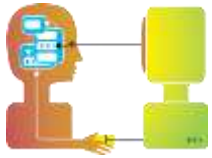
- Prilikom korišćenja softv. dugmadi za komande treba voditi računa
  - ***u kom redosledu se prikazuju dugmad*** – uobičajeno je da prvo ide najvažnije dugme (najvažnija akcija koju dugme reprezentuje), tj. redosled ide po važnosti u skladu sa redosledom u kojem korisnici čitaju
  - ***o veličini dugmadi*** – da bi se vizuelno grupisali, dugmad bi trebala da su iste veličine i oblika; različita dužina dugmadi je prihvatljiva ako su dužine teksta različite i ako dugmad idu u nizu





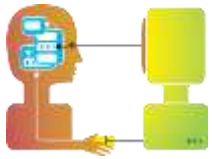


WIDGET-I ZA PREUZIMANJE PODATAKA ...

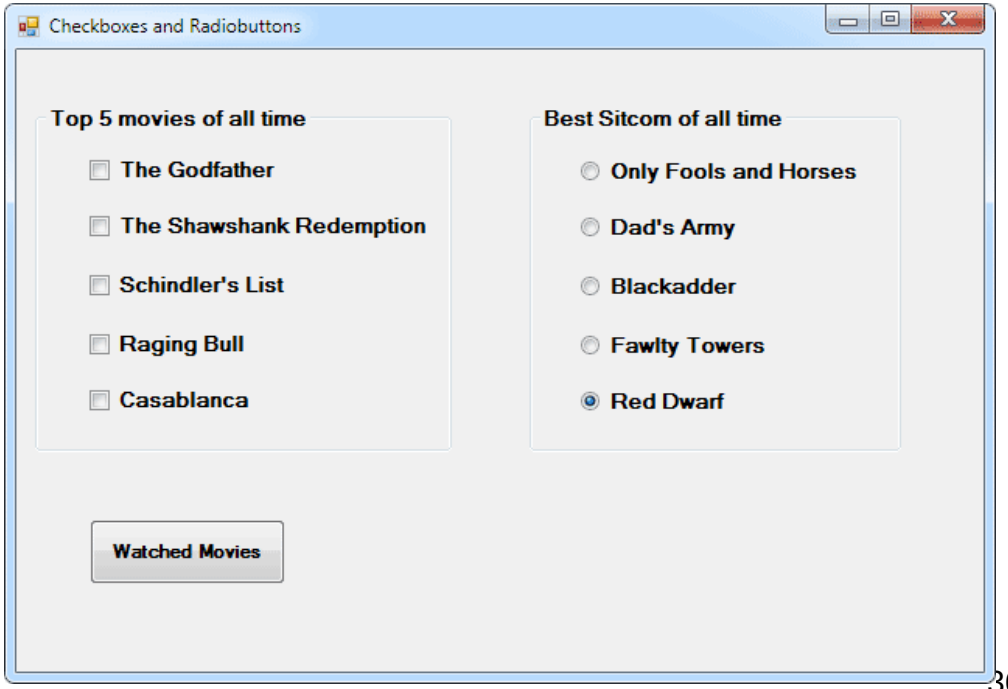


- Postoje različiti widget-i koji omogućuju korisniku unos informacija u program poput:
  - Soft. dugmadi za izbor opcije,
  - Check boksova,
  - List boksova,
  - Tekst polja,
  - Tabele
- Ako ovi widget-i nisu dovoljni, ili nisu zadovoljavajući moguće razviti widget-e posebne namene

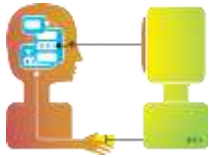
OPCIJE I CHECK BOX KOMPONENTE ...



- Softverska dugmad za izbor opcije i check boksovi obavljaju sličnu funkciju i često se brkaju
- Predstavljaju brz precizan način unosa informacija u rač. sistem
- Softverska dugmad za izbor opcije (radio buttons, slider buttons, on/off buttons, i sl.) se koriste onda kada je potrebno odabrati ***samo jednu opciju*** iz skupa ponuđenih opcija
- Check boksovi se koriste kada je potrebno da korisnik ***odabere jednu ili više opcija*** iz skupa ponuđenih opcija



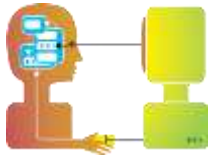
... OPCIJE I CHECK BOX KOMPONENTE ...



- Postoji jako puno primera u praksi kada su upotrebljena softv. dugmad za izbor opcije umesto check boksova i obrnuto
- Kada se koriste ove vrste widget-a mora se voditi računa o nekoliko stvari:
  - ***Koliko se opcija može predstaviti na ovaj način*** – izbor jedne od tri opcija na ovaj način je prihvatljivo, čak i na malim dijalozima, ali da li bilo prihvatljivo upotrebiti ih za izbor jedne od tridest opcija?
  - ***Kako grupisati i rasporediti kontrole na ekranu*** – ako se organizuju unutar jedne kolone, onda neće zbuniti korisnika (biće jasno koja je grupa i koje su opcije unutar grupe), ali će zauzimati više prostora na ekranu
  - ***Kako ih urediti na ekranu*** – veoma je korisno da različite opcije iz ponuđenog skupa budu na neki način uređene: alfabetski, spram verovatnoće njihovog izbor (oni koji se češće biraju na početak), ili na neki drugi predvidiv način



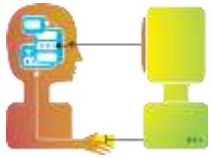
... OPCIJE I CHECK BOX KOMPONENTE



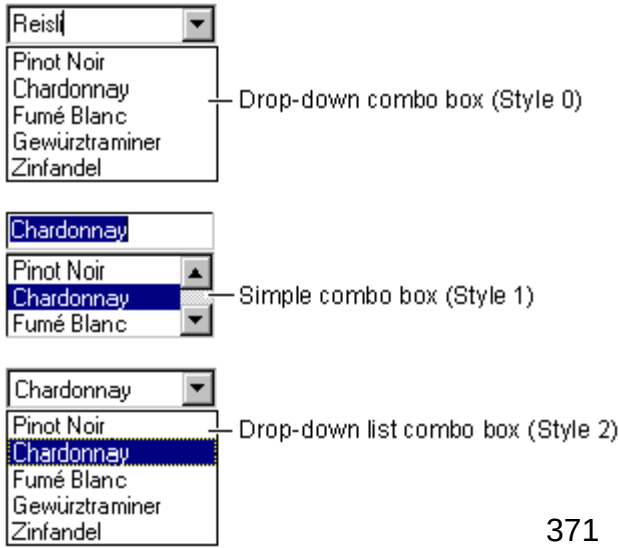
- Kada se koriste ove vrste widget-a mora se voditi računa o nekoliko stvari:
  - ***Kako izabrati nazive opcija*** – kao i sa menijima i dugmadima, tekst mora da bude takav da ga korisnik razume i da jasno ukazuje na to šta se bira i šta će se desiti ako se to izabere (tj. koja informacija će se uneti u sistem)
- Kako će se koja od navedenih stavki realizovati, zavisi od konteksta



LIST BOKSOVI ...

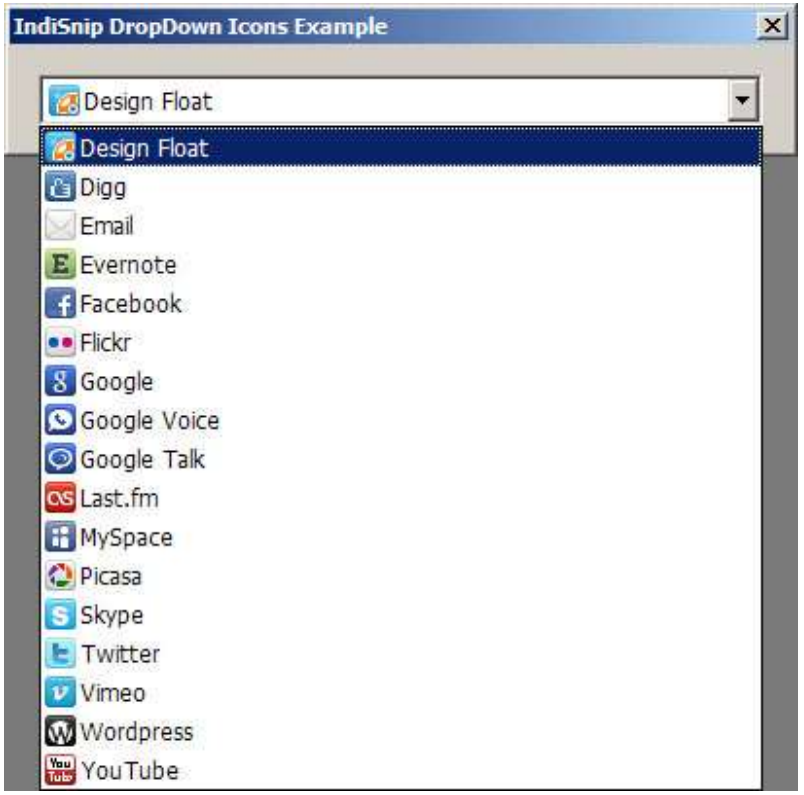


- Kada je potrebno da korisnik bira opciju iz skupa sa velikim brojem opcija ili skupa čiji se sadržaj dinamički menja (čitanjem iz baze podataka ili fajla), preporučuje se upotreba list boksova
- List boksovi mogu da omoguće izbor jedne ili više opcija (kombinovanjem pointerskog uređeja i tastera CTRL, ali ovo zahteva dodatno uputstvo), zavisno od svoje prirode
- Neki list boksovi predstavljaju padajuće liste kod kojih je samo selektovana stavka vidljiva, dok ostale postaju vidljive tek tokom akcije selekcije (čak ni tada ne moraju sve opcije biti vidljive, već se koristi skrol bar za kretanje kroz listu), dok je kod drugih list boksova prikazana cela lista ili jedan njen deo (do ostatka se dolazi skrolovanjem)

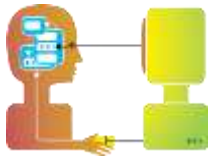


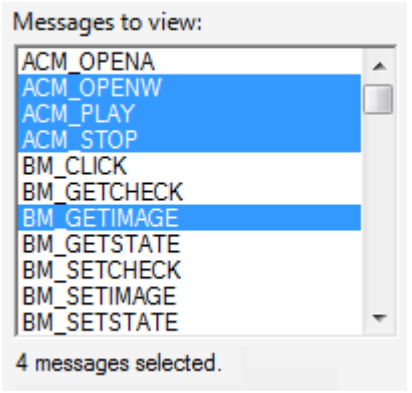
... LIST BOKSOVI ...

- Kada se koristi ova vrsta widget-a mora se voditi računa o sledećim stvarima:
  - ***Koliko opcija prikazati*** – neki autori preporučuju od tri do osam, ali ponekad neke opcije nisu vidljive i mora se skrolovati da bi one postale vidljivo (mora biti jasno korisniku da postoji još opcija za izbor)
  - ***Koliko je prostora na raspolaganju*** – da li postoji dovoljno prostora za stalno raširenu listu ili je potrebno koristiti padajuću listu
  - ***Da li će se opcije menjati*** – ako su opcije promenljive, ova vrsta widget-a je značajno bolja i fleksibilnija od check boksova i radio dugmadi

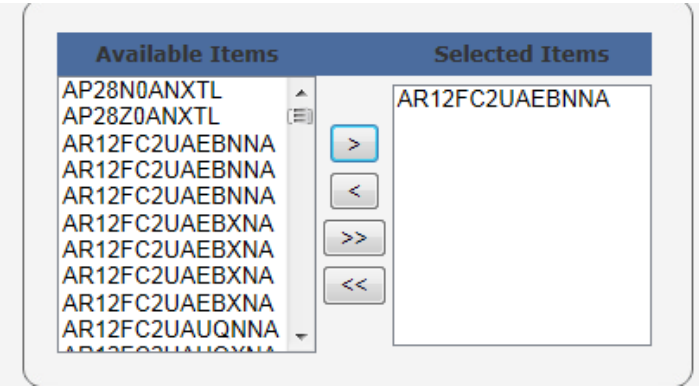


... LIST BOKSOVI



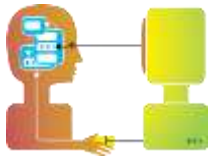


- Kada se koristi ova vrsta widget-a mora se voditi računa o sledećim stvarima:
  - ***Da li su potrebne višestruke selekcije*** – *ako su potrebne, list boks mora podržavati tu mogućnost*
  - ***Koje su početne vrednosti*** – početne vrednosti mogu značajno ubrzati programa; ako se neka opcija daleko češće bira od ostalih, onda bi ona morala biti početna (**mada i tu postoji opasnos. koja?**)
- Ovaj widget se često kombinuje sa tekstualnim poljem

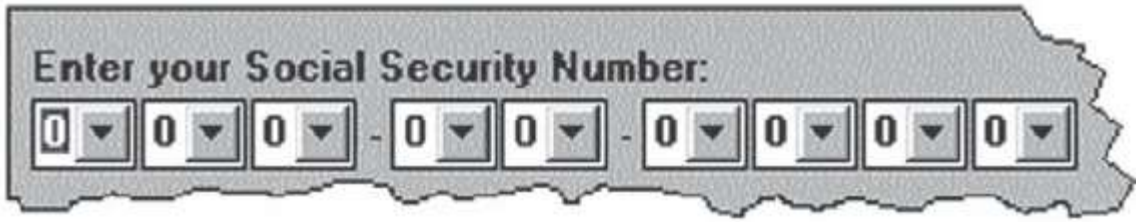




TEKSTUALNA POLJA ...

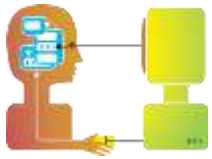


- Tekstualno polje je najfleksibilniji widget za unos informacija
- Ako nije moguće predvideti vrednost koju će korisnik uneti (ne može mu se ponuditi kao opcija) ili se korisnik ne želi ograničiti ni na koji način, onda je tekst polje najbolje rešenje (neki bi rekli i jedino)
- Ova sloboda nije uvek najbolja, jer utiče na standardizaciju informacija koje se unose GUI-jem – ne može se garantovati da će korisnik zaista uneti podatak u odgovarajućoj formi (muški, Muški, MUŠKI, M.)
- Tekstualno polje se često kombinuje sa list poljem (Combo box) – omogućuje koliko toliko zaobilaženje problema standardizacije

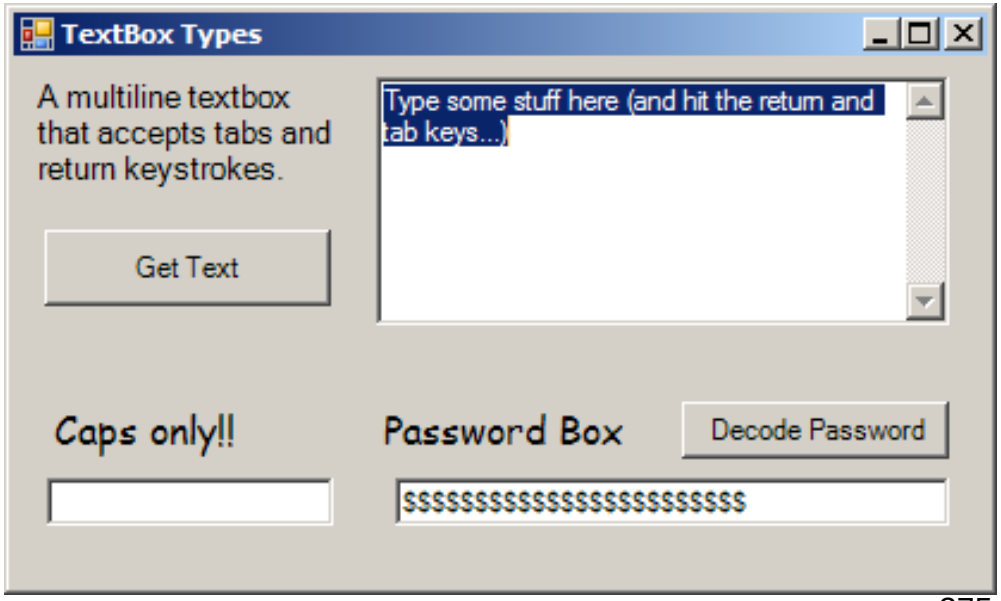




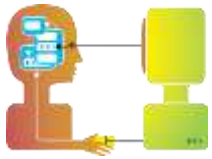
... TEKSTUALNA POLJA ...



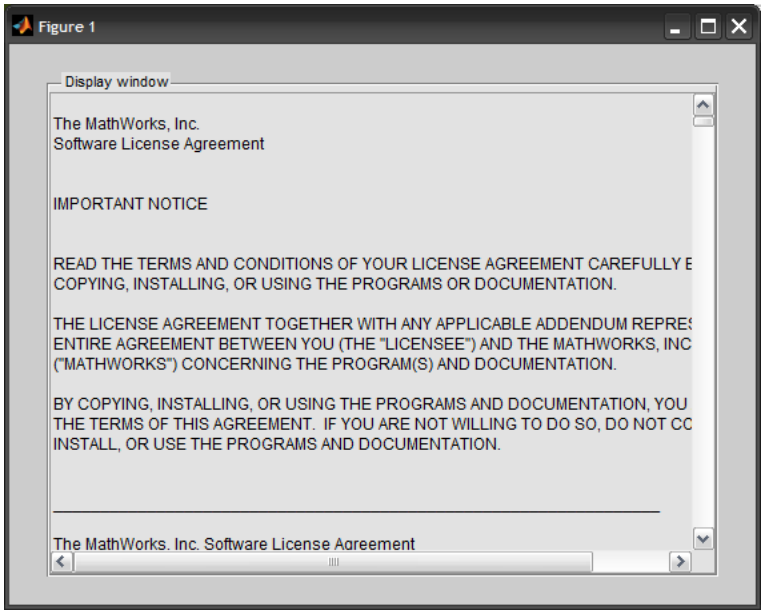
- Kada se koristi ova vrsta widget-a mora se voditi računa o sledećim stvarima:
  - ***Koje veličine i oblika treba da bude tekstualno polje*** – veličina tekstualnog polja treba da sugeriše dužinu teksta koji se može uneti, i to važi i za broj karaktera u redu i za broj redova; ako korisnik treba da unese ime, onda treba da koristi tekstualno polje koje omogućuje unos jednog reda; tekstualna polja za unos više redova teksta se koriste kada se npr. traže komentari
  - ***Koju količinu teksta će korisnik želeti da unese*** – ako se količina teksta ne može predvideti, onda tekst polje mora imati skrol barove a samim tim i podržavati više linija

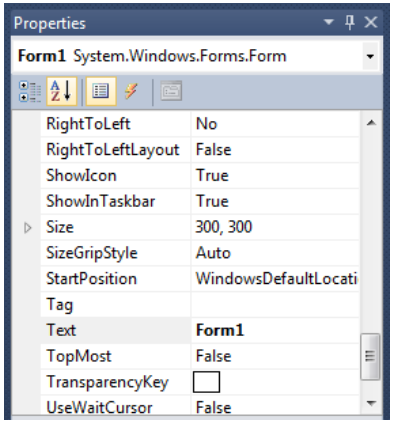


... TEKSTUALNA POLJA



- Kada se koristi ova vrsta widget-a mora se voditi računa o sledećim stvarima:
  - ***Da li se želi sprečiti izmena sadržaja tekstualnog polja*** – nekada se želi da se tekstualno polje posivi; mora se voditi računa o sementici tekstualnog polja, za unos se koristi tekstualno polje, za ispis se koriste labele ili tekstualno polja tako formatirana da je jasno da samo vrše prikaz informacije, tj. da se ne očekuje unos

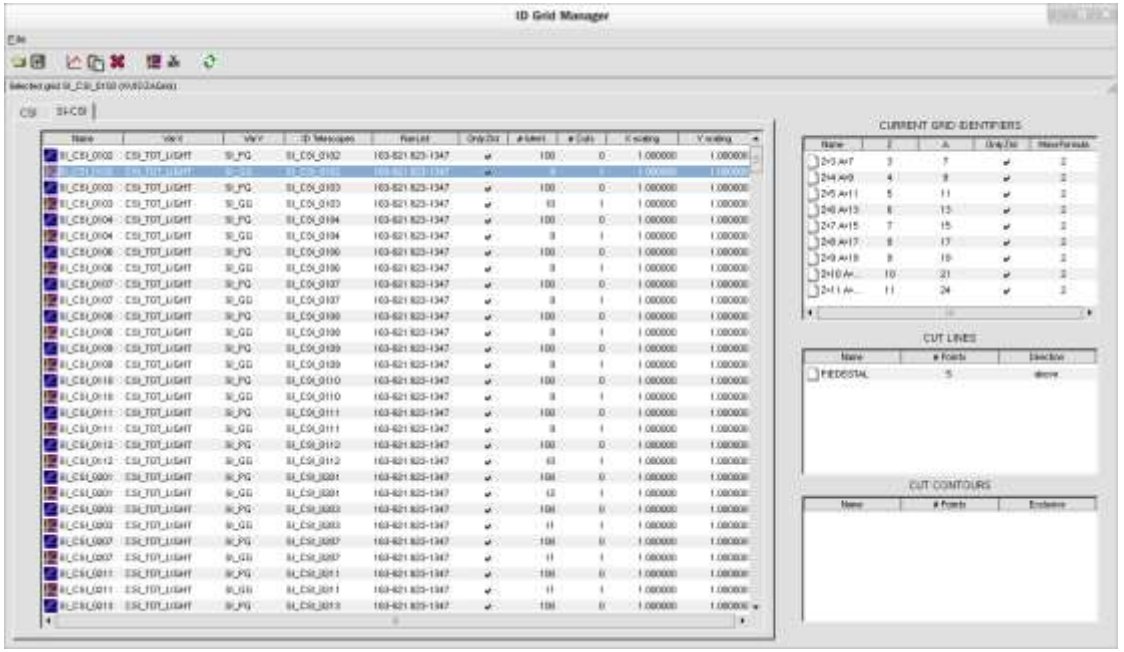




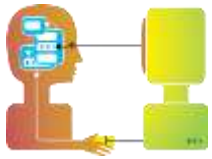
- Često se koriste labele kao dodatak da bolje objasne značenje tekstualnog polja (ili nekog drugog widget-a) ali i da predstave informaciju korisniku (koriste se za read only informacije)

TABELE ...

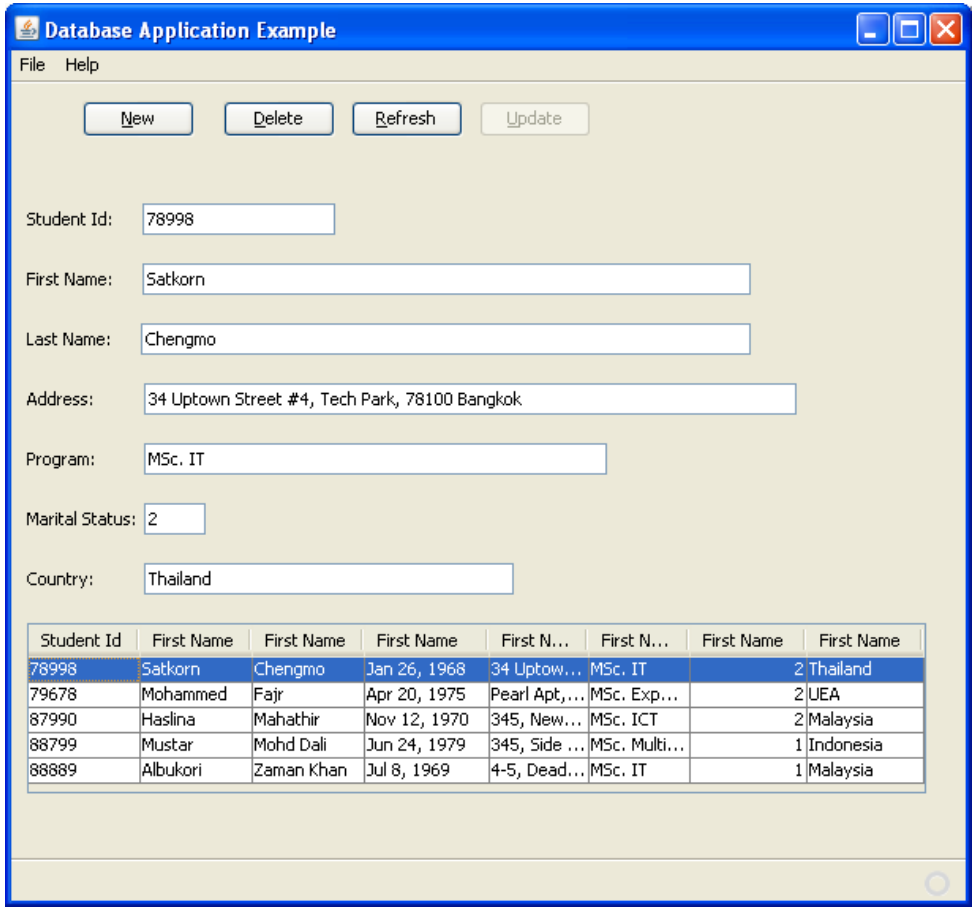
- Služe za prikaz velike količine podataka i njihovo poređenje
- Matrica informacija raspoređenih u redove i kolone
- Zauzima manje informacija od individualnih komponenti
- Može nastati kao rezultat upita iz baze
- Može se koristiti samo za prikaz podataka ali i za unos/izmenu podataka
- Zbog veličine može biti nezgodna za upotrebu, naročito ako je broj kolona jako velik (tumačanje polja se nalazi u naslovnoj liniji, pa je zbog skrolovanja nezgodno)



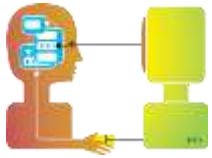
... TABELE ...



- Može se koristiti kao navigacija kroz velike skupove podataka do traženog podatka koji će se prikazati u strukturiranom obliku nakon selekcije polja u tabeli
- Uvek se selektuje ceo red bez obzira na selektovano polje (kada se koristi u režimu pregleda)

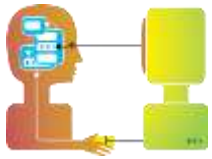


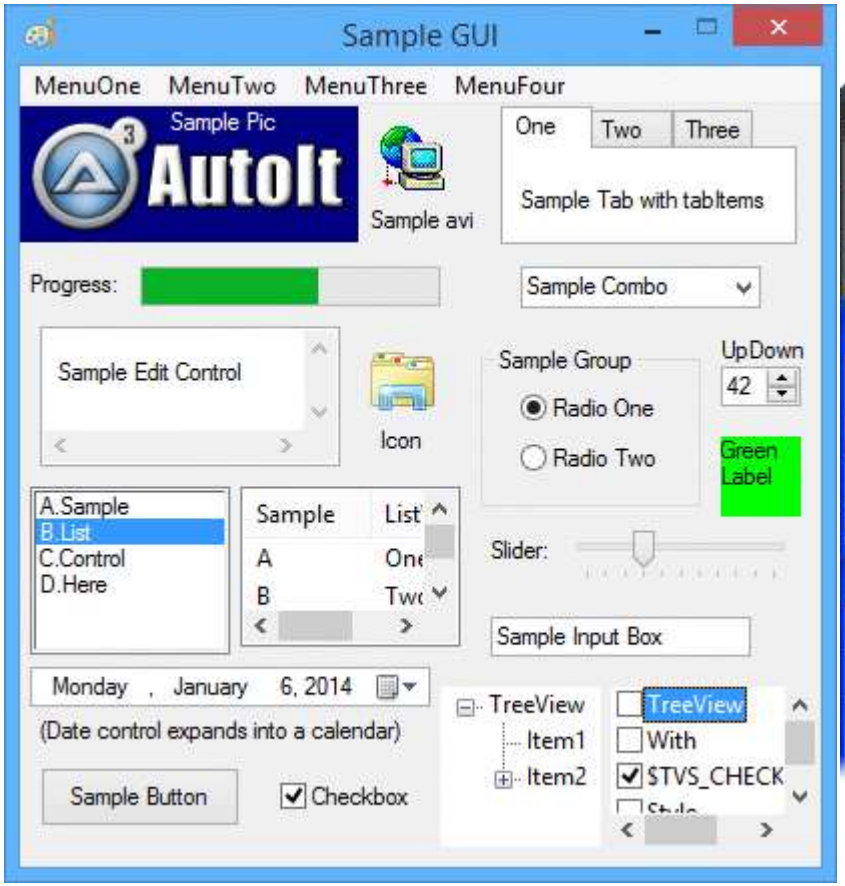
... TABELE

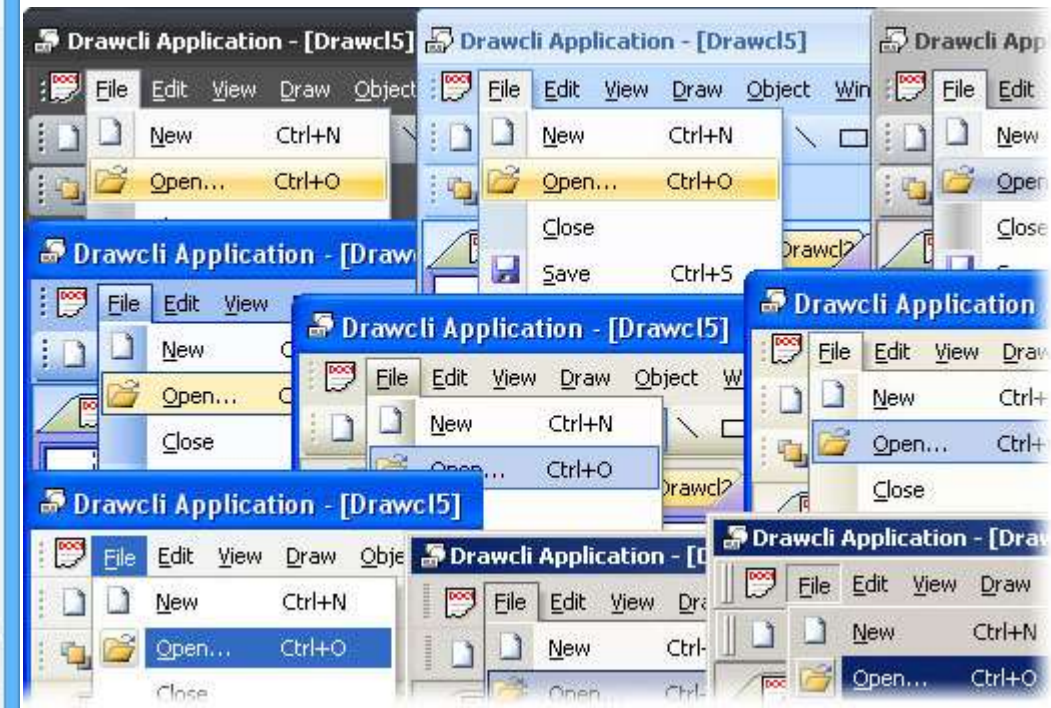


- Kada se koristi ova vrsta widget-a mora se voditi računa o sledećim stvarima:
  - ***Nazivi zaglavlja kolona*** – korisniku mora biti jasno šta koja kolona znači i koja se informacija nalazi u njoj; ne koristiti dvotačku; navoditi merne jedinice
  - ***Prezentacija podataka u tabeli*** – kako će se podaci poravnavati (leva ivica za tekst, desna za brojeve?); poravnavati na isti način zaglavlja kao i podatke u odgovarajućim kolonoma; obezbediti jasnu separaciju između susednih vrsta (najčešće senčenjem), odnosno kolona (linijom ili razmakom)
  - ***Sadržaj tabele i veličina ekrana*** – ako se pretera sa brojem kolona, sadržaj neće biti čitak/vidljiv (zbog toga se treba odlučiti koji podaci će se prikazati u tabeli)
  - Navigacija sadržaja – za tabele sa velikom količinom sadržaja (nije sve vidljivo odjednom) trebalo bi obezbediti opciju pretrage i filtriranja

KOMBINOVANJE WIDGET-A



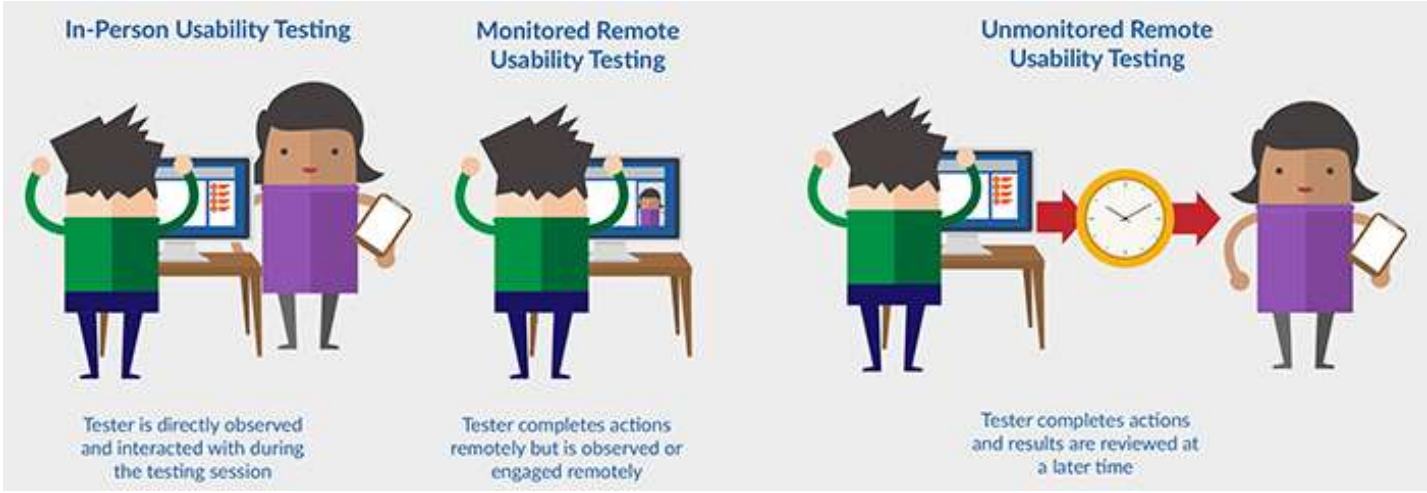




TESTIRANJE KORISNIČKOG INTERFEJSA

TESTIRANJE KORISNIČKOG INTERFEJSA

- U ovoj lekciji pod testiranjem se podrazumeva testiranje se korisnicima, tj. stavljanje korisničkog interfejsa ispred korisnika
- Podrazumeva se da su prošla sva interna testiranja vezana za funkcionalnost i izgled samog interfejsa (on radi i nema bagova)
- Ovaj test podrazumeva proveru da li interfejs radi onako kako korisnik (i sam dizajner) očekuju da će raditi
- Postoji nekoliko vrsta testova koji uključuju korisnike, ali u svima njima treba voditi računa da su u pitanju ljudi i da postoje određena etička pitanja koja moraju biti ispunjena



ŠTA OČEKUJEMO OD TESTIRANJA Uia …



- Da validiramo valjanost našeg prototipa.
- Da potvrdim da li naš proizvod odgovara korisnikovim očekivanjima i očekivanjima generalno.
- Da identifikujemo probleme sa kompleksnim tokovima interakcije.
- Da dopunimo nedostatke u podacima kao i da identifikujemo značajne tačke u interfejsu
- Da razvijemo empatiju (identifikujemo delove interfejsa koji su problematični za korisnike sa posebnim potrebama).
- Da se korisnici aktivno uključe u razvoj proizvoda (*Get buy-in for change*).
- Da ultimativno omogući dolazak do boljeg UX-a.

… ŠTA OČEKUJEMO OD TESTIRANJA Uia



- Dovođenje novih korisnika da testira naš UI ili posmatranje kako postojeći korisnici koriste naš UI je efikasan način da se utvrdi da li korisnici:
  - razumeju kako aplikacija (interfejs) funkcioniše,
  - se gube u interfejsu ili ih interfejs zbunjuje
  - mogu (uspevaju) da kompletiraju glavne akcije koje moraju sprovesti
  - nailaze na probleme sa upotrebljivošću aplikacije ili nailaze na bagove
  - imaju funkcionalno i efikasno iskustvo
  - primećuju bilo kakve usabiliti probleme ili bilo kakve druge probleme koje mi nismo uočili ili nismo uopšte o njima razmišljali

VRSTE TESTIRANJA

- Formativna evaluacija
  - Sprovodi se nad prototipovima, u kontrolisanom okruženju
  - Služi da se otkriju problem u usability-u kako bi se otklonili u sledećoj iteraciji dizajniranja
  - Ne zahteva potpunu, funkcionalnu implementaciju, ali se može primeniti na prototipovima različitog nivoa detalja
  - Rezultat testiranja su zapažanja o kvalitetu (***qualitative observations***)
- Evaluacija u realnom okruženju
  - Pronalazak problema radom u relanom okruženju
  - Mora biti funkcionalna verzija kor. interfejsa i softvera
  - Koriste se realni problemi
  - Rezultat uglavnom zapažanja o kvalitetu
- Kontrolisani eksperimenti
  - Za proveru određene hipoteze
  - Nad funkcionalnom verzijom softvera
  - Rezultat uglavnom zapažanja o kvalitetu
- Usability testing



Developer watching videotape of usability test.

KVALITATIVNO VS KVANTITATIVNO



- Prema vrsti podataka koji se prikupljaju, mogu se podeliti na:
  - **Kvalitativne** – fokus je na prikupljanju uvida, osećaja, otkrića, anegdota o tome kako korisnici koriste neku uslugu ili proizvod. Ove vrste testiranja su najbolja u otkrivanju grešaka u UX-u. Dominantniji oblik testiranja upotrebljivosti interfejsa.
  - **Kvantitativne** – fokus na prikupljanju podataka, različitih metrika koje opisuju UX. Najčešće metrike koje se koriste su procenat uspešnosti u realizaciji zadataka i vreme potrebno da se realizuje zadatak. Najbolje je koristiti prilikom benčmark testova.
- Broj učesnika zavisi pre svega od vrste studija. Kvalitativne metrike zahtevaju veći broj učesnika, dok se kod kvalitativnih metrika preporučuje najmanje 5 učesnika (videti Nielsenovo istraživanje: https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/).

UDALJENO VS UŽIVO



- Prema tome gde nalaze korisnici u odnosu na osobu koja sprovodi istraživanje, mogu se podeliti na:
  - **Testiranje uživo** – i osoba koja sprovodi test i osoba koja učestvuje u testiranju se nalaze na istoj lokaciji.
  - **Udaljeno testiranja** – podrazumeva da korisnik testira interfejs na lokaciji koja njemu odgovora. Može biti:
    - **nadzirano** – slično testiranju uživo, kad su i osoba koja sprovodi testiranje i učesnik testiranja u isto vreme prisutni onlajn, samo ne na istim lokacijama uz korišćenje čet alata i alata za deljenje ekrana, ili
    - **nenadzirano** – ne postoji direktna sinhrona interakcija između osobe koja sprovodi test i osobe koja se testira. Osoba koja testira, sprovodi unapred zadate zadatke ritmom koja njoj odgovara. Koristi se specijalizovan softver za praćenje i beleženje rezultata. Sva uputstva o zadacima i testu nalaze se stalno dostupna u okviru alata za testiranje.

CENA USABILITY TESTIRANJA

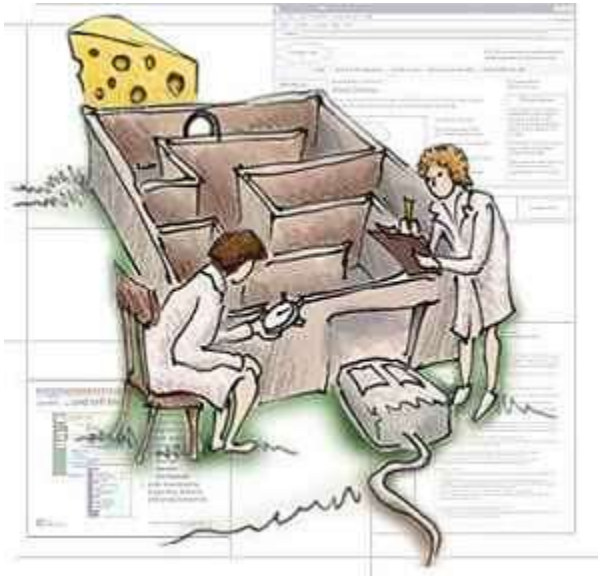


- Jednostavni testovi mogu biti zaista jeftini, iako se savetuje da se učesnicima testiranja plati njihov anganžman kako bi se obezbedila njihova puna saradnja
  - Testiranja se mogu sprovesti u jednostavnoj prostoriji za konferencije
  - Mogu trajati do 3 dana
    1. dan za planiranje studije
    2. dan za testiranje 5 učesnika
    3. dan za obradu i analizu dobijenih rezultata, pisanje izveštaja i preporuka za narednu iteraciju softvera
- Cena može biti izuzetno visoka za velike studije kada se testira na nekoj posebnoj lokaciji sa velikim brojem učesnika
- Šta povećava cenu: kompetetivne studije više dizajna, interno testiranje u više zemalja, testiranje više grupa korisnika (ili persona), kvantitativne studije, upotreba fensi opreme (eyetrackers), upotreba laboratorija za testiranje i monitoring, pisanje detaljnih izveštaja i preporuka

USABILITY TEST ...

- Testiranje usabilty-a sprovodi se implementacijom scenarija ili prototipa, nakon čega sledi snimanje ponašanja korisnika

- Kod snimanja ponašanja korisnika posebna pažnja se posvećuje posmatranju tipičnih upotreba ili kritičnih slučajeva, zatim na broj tastera koji korisnik mora da pritisne da bi obavio određeni posao



- Tokom snimanja ponašanja korisnik se ohrabruje da razmišlja naglas

- Nakon toga sledi intervju korisnika o njegovim subjektivnim impresijama, a zatim i analiza njegovog ponašanja

... USABILITY TEST ...

- Svrha ovoga je da se shvate korisnikove metode i problemi sa kojima se suočava prilikom rešavanja zadataka i interakcije sa softverom
- Ovim se otkriva proces razmišljanja korisnika
- Osnovna prednost ove metode jeste opservacija realne upotrebe realnog sistema



- Manu predstavlja činjenica da se često analiziraju samo limitirani aspekti sistema, zatim da se ispitivanje obično odvija u laboratoriji na “toy problems”, sa izmišljenim podacima
- Ova analiza ističe samo prvu upotrebu sistema zanemarujući šta će se desiti nakon dužeg perioda rada sistema

... USABILITY TEST ...

- Usability test se sadrži 3 elementa:
  1. Osoba koja sprovodi testiranje i prati rad korisnika („Facilitator“)
  2. Zadatak ili zadaci koja učesnik realizuje
  3. Učesnik testiranja interfejsa koji realizuje zadatke



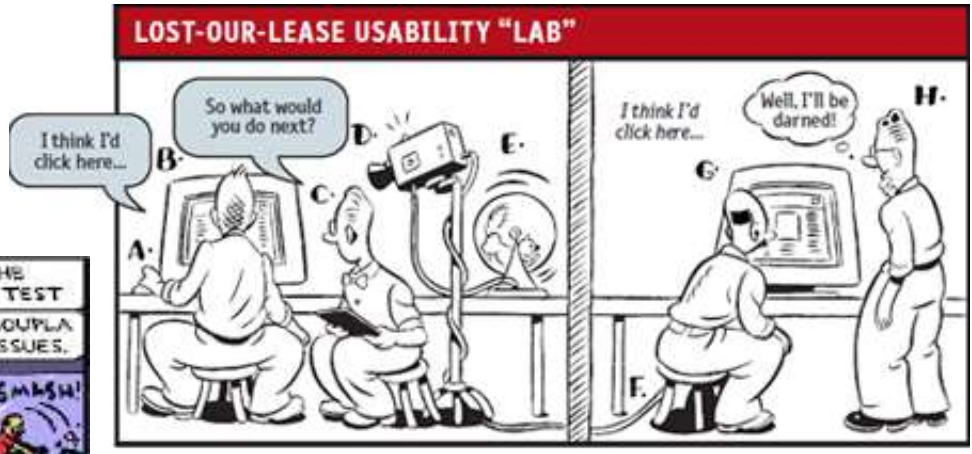
Nielse Norman Group – primer Usability testa

- Korisnici rešavaju realne zadatke određene od strane testera
- Mora se voditi računa o previše stvari i ne dobija se slika o tome kako će se korisnički interfejs ponašati u realnom okruženju
- Tri uloge: korisnici, izvršioci, posmatrači

... USABILITY TEST ...

- Standardni koraci u testiranju utilitarnosti/korisnika su:
  1. Podesiti ambijent za opservaciju,
  2. Opisati svrhu opservacije i saopštiti korisniku da ima pravo da odustane u svakom trenutku,
  3. Opisati i demonstrirati opremu u prostoriji,
  4. Objasniti korisniku kako da “misli naglas” i da mu/joj nećete pružati nikakvu pomoć tokom testa,





... USABILITY TEST ...

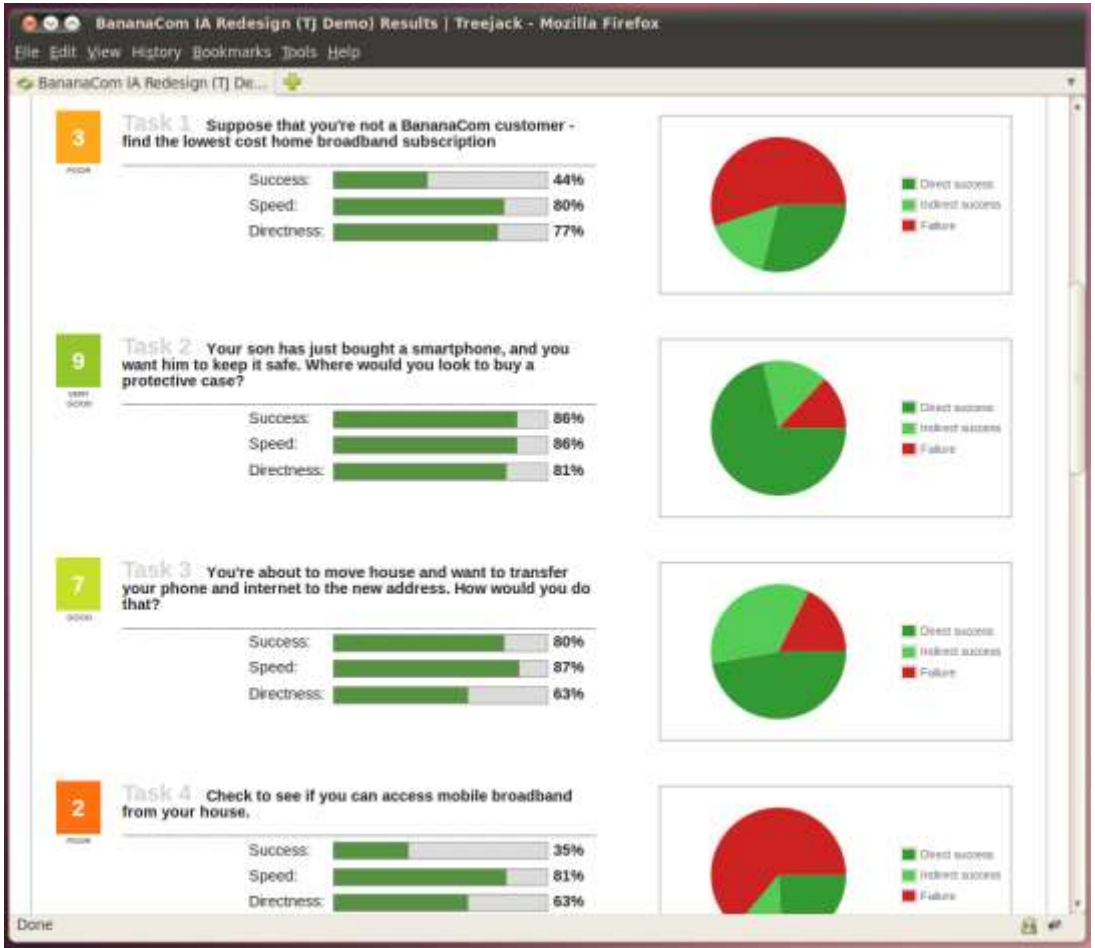


- Standardni koraci u testiranju utilitarnosti/korisnika su:
  5. Opisati zadatak i uvesti korisnika u sistem
  6. Insistirajte da vas korisnik pita pre opservacije
  7. Opservacija pa debrifing subjekta
  8. Analiza dobijenih podataka
  9. Tabelarna organizacija rezultata

		Task Completed	Time on Task	Comments / Observations / Improvement Ideas	Observer's Name
Feature Name / Item to Test	Usability Tester / Customer Persona	Yes OR No	Minutes	Where did the tester navigate / where did they click / interpret the UI elements? Any other observations / thoughts / ideas.	
1. Save the File	Joe / Technical Support	Yes	1	Did not see the save icon. Tried to right-click!!! We could put in right click menu. There is no help icon on the <blabla> screen	Jane
2. Create a new report	Joe / Technical Support	No	10	Did not see the save icon. Got very frustrated. Expected 'save' to be on the 'file' menu	Jane
3. <feature name 3	Joe / Technical Support	<yes or no>	<enter number>		Jane
1. Save the File	<tester 2 / customer persona>	<yes or no>	<enter number>		Jane
2. Create a new report	<tester 2 / customer persona>	<yes or no>	<enter number>		Jane
3. <feature name 3	<tester 2 / customer persona>	<yes or no>	<enter number>		Jane

... USABILITY TEST ...

- Standardni koraci u testiranju utilitarnosti/korisnika su:
  10. Interpretacija rezultata u kontekstu drugih rezultata

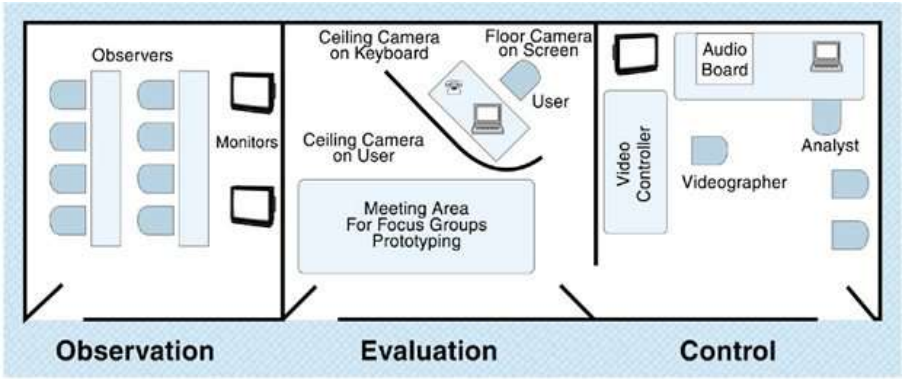


LABORATORIJE ZA TESTIRANJE USABILITY-A

- Potreba za testom utilitarnosti kulmirnirala je ranih 1980tih godina



- Sa razvitkom utilitarnosti razvijaju se posebne laboratorije utilitarnosti, prvo u okviru fakulteta da bi kasnije prešle u okvire firmi (IBM na Floridi, MS, SUN,...), a danas čak postoje i kao nezavisne kuće
- Ovakve laboratorije učestvuju u ranim fazama analize zadatka ili revizije projekta, zatim u potrazi za informacija o softverskim alatima i literaturom, kao i u razvoju skupa zadataka za testiranje utilitarnosti "u kući"
- Aktuelni lider u testiranju utilitarnosti trenutno je IBM



ETIČKA PITANJA

- Usability test uključuje i određena etička pitanja
- Osnovni principi su da nema fizičkog ili psihičkog povređivanja, da je participacija dobrovoljna i da je korisniku obezbeđeno pravo na privatnost, kao i mogućnost odustajanja u svakom momentu
- Studija, kao i njena svrha, mora biti adekvatno objašnjena kako bi korisnik dao svoj potpis na saglasnost
- Mora se voditi računa jer test, koliko god da je bezazlen, vrši određeni pritisak na učesnike
  - Postoji pritisak da se uradi dobro
  - Izgleda kao test inteligencije
  - Strah da se ne obruka pred ispitivačem
  - Poređenje/takmičenje sa drugim ispitanicima



VIZUELIZACIJA PODATAKA

VIZUELIZACIJA



- Čemu služi vizuelizacija?
  - Za razmišljanje, uočavanje i pronalaženje informacija
  - Za komunikaciju
- Vizuelizacija predstavlja formu usvajanja znanja na osnovu spoljnih nadražaja (extern cognition)
  - Skica (slika) nacrtana na pravi način pomaže da se uoče obrasci i veze koje bi inače bile nevidljive (ili ih je teško uočiti samo na osnovu teksta/tabela)
  - Analiza podataka upotrebom percepcije
  - Manipulacija kako bi se podaci što bolje razumeli
- Vizuelizacija je izuzetan način da informacije prenose drugima
  - Omogućuju zajedničko razmišljanje
  - Omogućuju zajedničko donošenje odluka



VIZUELIZACIJA – OSNOVNE SMERNICE



- Your data is only as good as your ability to understand and communicate it, which is why choosing the right visualization is essential
- Information can be visualized in a number of ways, each of which can provide a specific insight
- When you start to work with your data, it’s important to identify and understand the story you are trying to tell and the relationship you are looking to show
- Knowing this information will help you select the proper visualization to best deliver your message

ČOVEKOV VIZIONI SISTEM



- Čulo sa najvećim rasponom i mogućnostima, najviše se oslanjamo na njega
- Brzo je, i paralelno
- Jako dobro prepoznavanje obrazaca (pattern recognition)
- Vrlo osetljivo, uočavanje i najfinijih detalja
- Proširuje memoriju (vizuelno pamćenje) i kognitivni kapacitet
- Ljudi razmišljaju vizuelno!



- **Vizuelizacija je idealna, u čemu je onda problem?**

VIZUELIZACIJA – PRIMERI



- Pronađite crvenu tačku

... VIZUELIZACIJA – PRIMERI ...



- Tabela, da li je dobra ili loša?
- Koja USA država ima najveći prihod? Odnos između prihoda i stručne spreme? Autlajeri?

Table - StateData ()

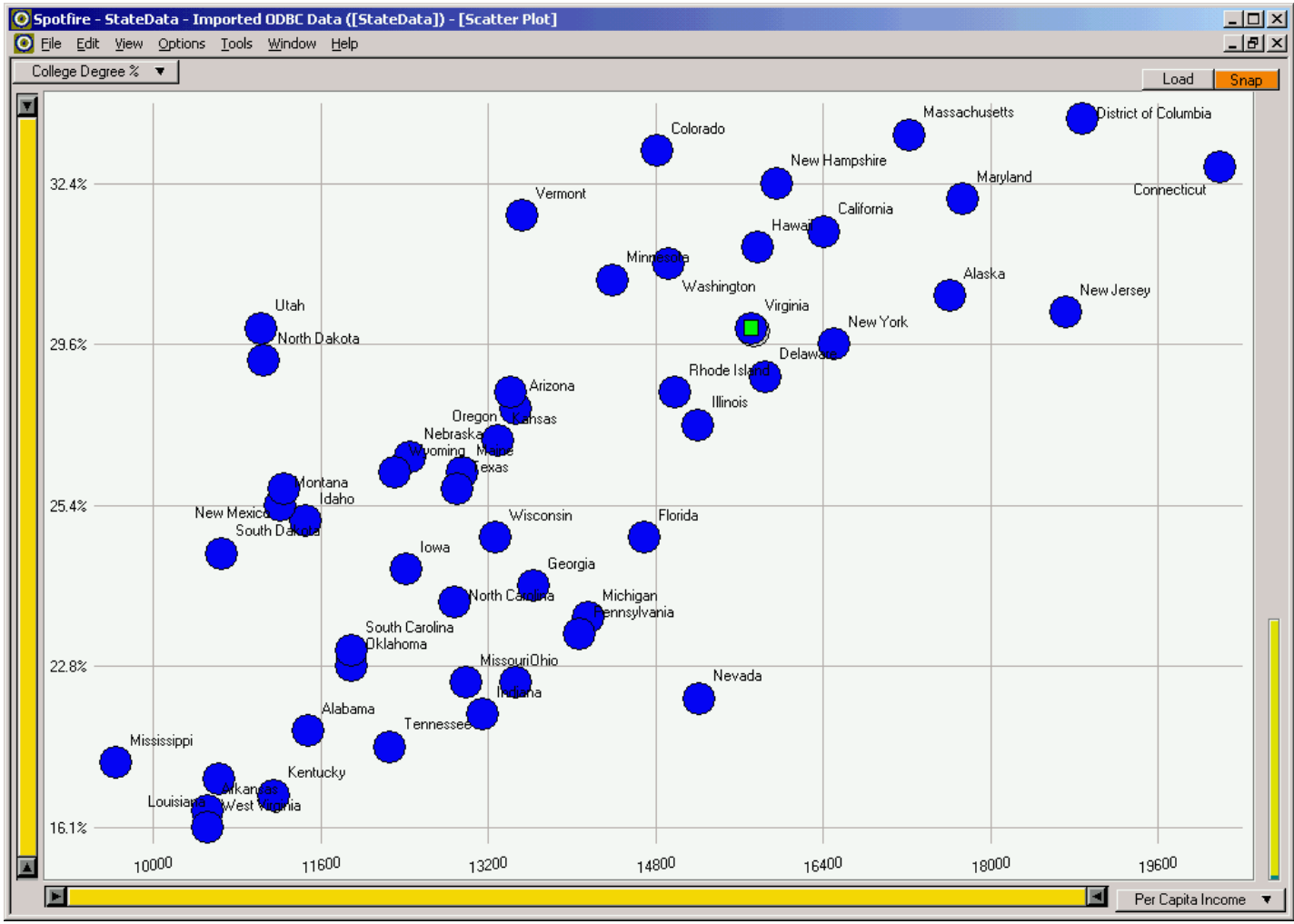
Load | Snap

State	College Degree %	Per Capita Income
Alabama	20.6%	11486
Alaska	30.3%	17610
Arizona	27.1%	13461
Arkansas	17.0%	10520
California	31.3%	16409
Colorado	33.9%	14821
Connecticut	33.8%	20189
Delaware	27.9%	15854
District of Columbia	36.4%	18881
Florida	24.9%	14698
Georgia	24.3%	13631
Hawaii	31.2%	15770
Idaho	25.2%	11457
Illinois	26.8%	15201
Indiana	20.9%	13149
Iowa	24.5%	12422
Kansas	26.5%	13300
Kentucky	17.7%	11153
Louisiana	19.4%	10635
Maine	25.7%	12957
Maryland	31.7%	17730
Massachusetts	34.5%	17224
Michigan	24.1%	14154
Minnesota	30.4%	14389
Mississippi	19.9%	9648
Missouri	22.3%	12989
Montana	25.4%	11213
Nebraska	26.0%	12452
Nevada	21.5%	15214
New Hampshire	32.4%	15959
New Jersey	30.1%	18714
New Mexico	25.5%	11246
New York	29.6%	16501
North Carolina	24.2%	12885
North Dakota	28.1%	11051
Ohio	22.3%	13461
Oklahoma	22.8%	11893
Oregon	27.5%	13418
Pennsylvania	23.2%	14068
Rhode Island	27.5%	14981
South Carolina	23.0%	11897
South Dakota	24.6%	10661
Tennessee	20.1%	12255
Texas	25.5%	12904
Utah	30.0%	11029
Vermont	31.5%	13527
Virginia	30.0%	15713
Washington	30.9%	14923
West Virginia	16.1%	10520
Wisconsin	24.9%	13276
Wyoming	25.7%	12311

... VIZUELIZACIJA – PRIMERI ...



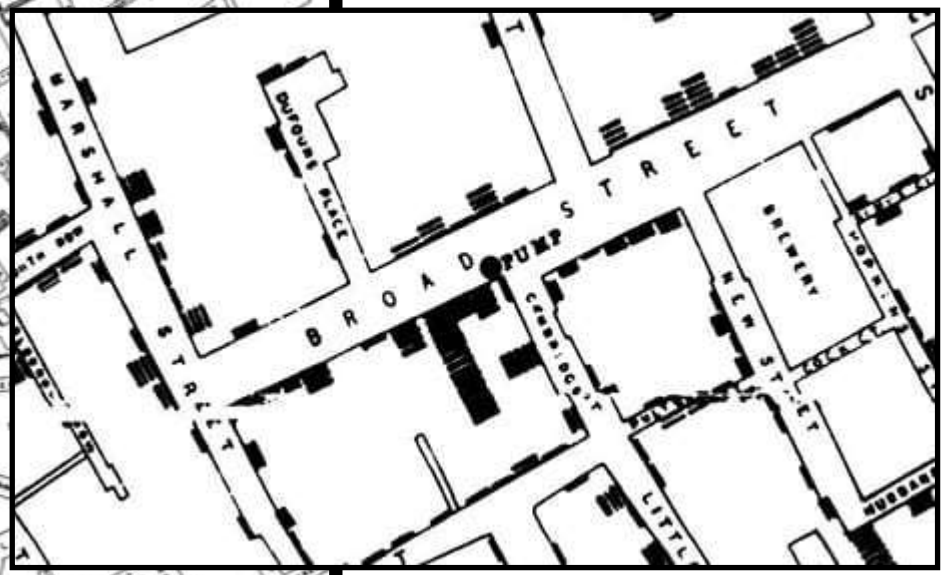
- Vizualizacija, bolja?



... VIZUELIZACIJA – PRIMERI ...







John’s Snow Cholera outbreak map

... VIZUELIZACIJA – PRIMERI ...

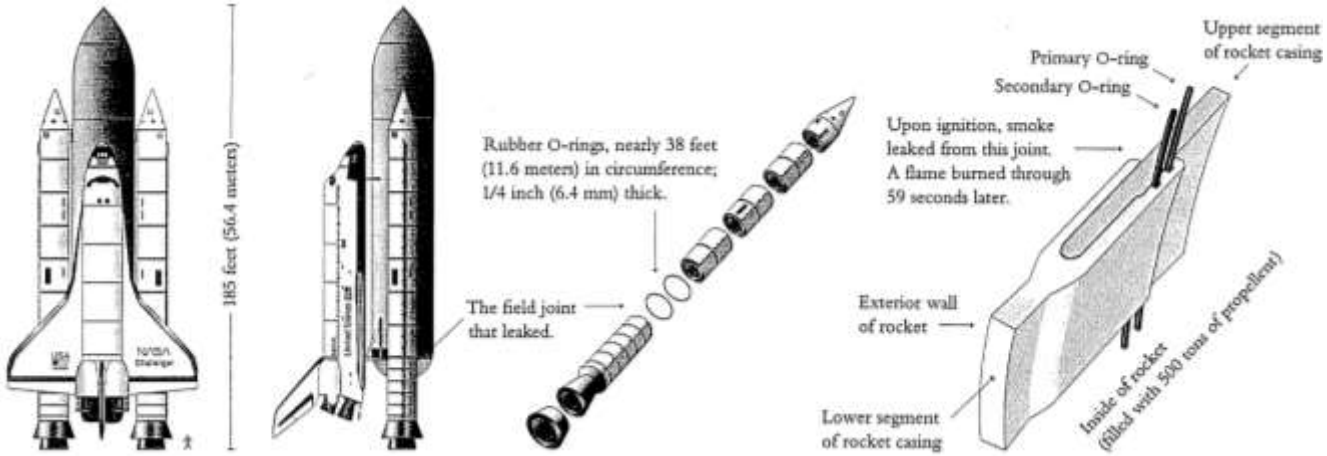


- John's Snow Cholera outbreak map (moderna verzija)



... VIZUELIZACIJA – PRIMERI ...

- Challenger disaster
  - Poleteli su iako vreme nije bilo povoljno suviše hladno za jednu vrstu materijala
  - *Many people assume it was because of poorly-functioning O rings on the booster rocket. However,*



those O rings didn't send that ship up on a cold winter's morn. People did, and those people drew their most critical information from two simple charts, screened by an overhead projector. The graphs displayed tiny pictures of each shuttle booster, lined up in chronological order, showing launch temperatures and any O ring damage.

  - Da su podaci bili u drugoj formi, da li bi se nezgoda izbegla?

... VIZUELIZACIJA – PRIMERI ...



- Challenger disaster (nastavak)
  - Podaci u tabeli

BLOW BY HISTORY
SRM-15 WORST BLOW-BY
- 2 CASE JOINTS (80°), (110°) ARC
- MUCH WORSE VISUALLY THAN SRM-22

SRM 22 BLOW-BY
- 2 CASE JOINTS (30-40°)

SRM-13A, 15, 16A, 18, 23A 24A
- NOZZLE BLOW-BY

HISTORY OF O-RING TEMPERATURES (DEGREES-F)

MOTOR	M&T	AMB	O-RING	WIND
DM-1	68	36	47	10 MPH
DM-2	76	45	52	10 MPH
QM-3	72.5	40	48	10 MPH
QM-4	76	48	51	10 MPH
SRM-15	52	64	53	10 MPH
SRM-22	77	78	75	10 MPH
SRM-25	55	26	29 27	10 MPH 25 MPH

HISTORY OF O-RING DAMAGE ON SRM FIELD JOINTS

SRM No.	Cross Sectional View: Erosion Depth (in.)	Cross Sectional View: Perimeter Affected (deg)	Cross Sectional View: Nominal Dia. (in.)	Top View: Length Of Max Erosion (in.)	Top View: Total Heat Affected Length (in.)	Clocking Location (deg)
61A LH Center Field**	22A	None	None	0.280	None	None	36° -66°
61A LH ~~CENTER~~ FIELD**	22A	NONE	NONE	0.280	NONE	NONE	338°-18°
51C LH Forward Field**	15A	0.010	154.0	0.280	4.25	5.25	163
51C RH Center Field (prim)***	15B	0.038	130.0	0.280	12.50	58.75	354
51C RH Center Field (sec)***	15B	None	45.0	0.280	None	29.50	354
41D RH Forward Field	13B	0.028	110.0	0.280	3.00	None	275
41C LH Aft Field*	11A	None	None	0.280	None	None	--
41B LH Forward Field	10A	0.040	217.0	0.280	3.00	14.50	351
STS-2 RH Aft Field	2B	0.053	116.0	0.280	--	--	90

*Hot gas path detected in putty. Indication of heat on O-ring, but no damage.
**Soot behind primary O-ring.
***Soot behind primary O-ring, heat affected secondary O-ring.

Clocking location of leak check port - 0 deg.

OTHER SRM-15 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY AND NO SOOT NEAR OR BEYOND THE PRIMARY O-RING.

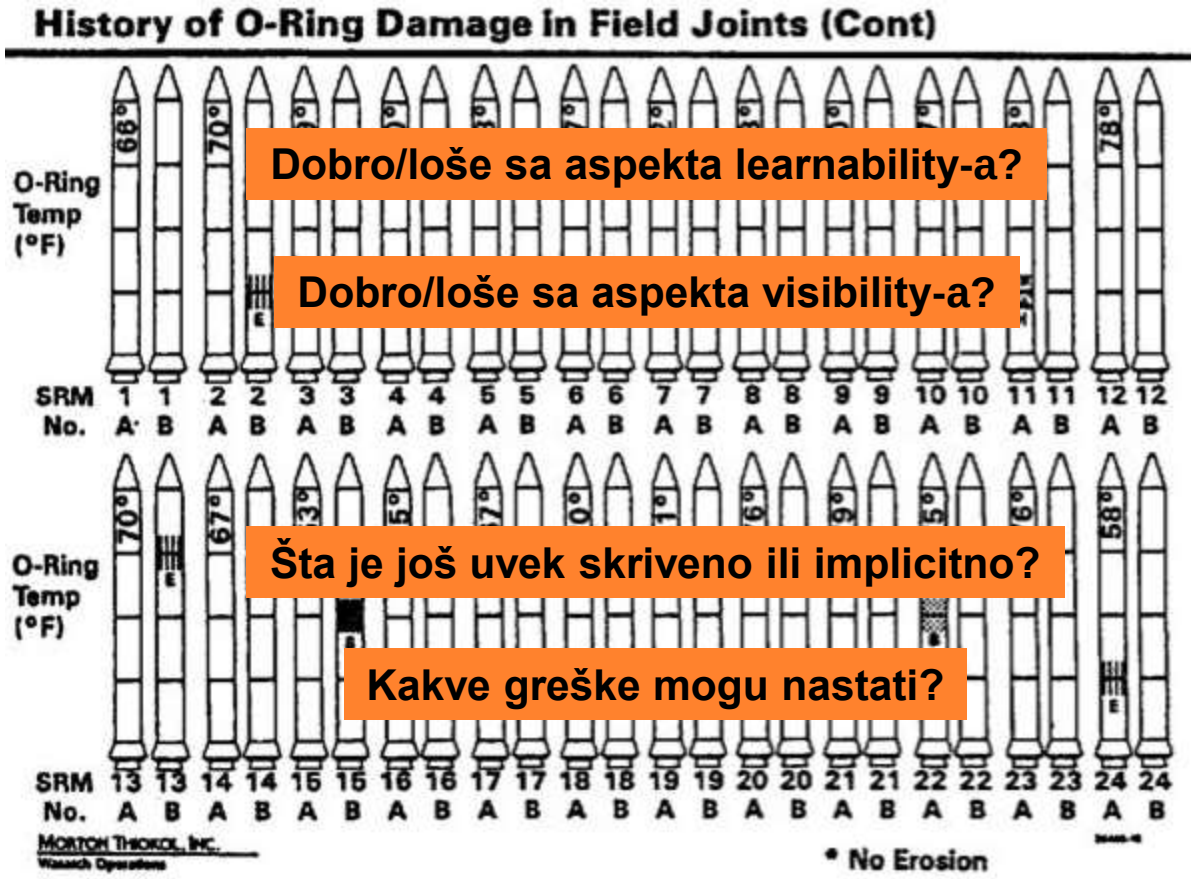
SRM-22 FORWARD FIELD JOINT HAD PUTTY PATH TO PRIMARY O-RING, BUT NO O-RING EROSION AND NO SOOT BLOWBY. OTHER SRM-22 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY.

  - Tabele su vrlo slab oblik iskazivanja vizuelnih informacija jer ne iskorišćavaju sve sposobnosti našeg vizelnog sistema (ne oslanjaju se na sve dimenzije dostupne vizuelnom sistemu

... VIZUELIZACIJA – PRIMERI ...



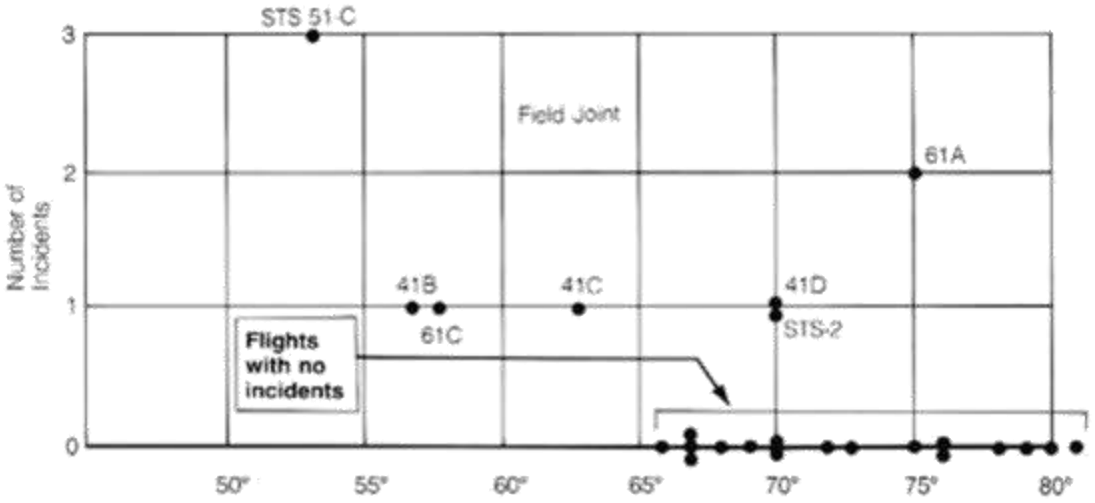
- Challenger disaster (nastavak)
  - Vizuelizacija koja je zaista upotrebljena prilikom odlučivanja u Challenger misiji
  - Svaki par raketa reprazentuje stvarni par raketa lansiranih u ranijim misijama
  - Linija predstavljaju O rings
  - Žvrljotine predstalvljaju primećena oštećenja
  - Temperature su temperature sa prethodnih lansiranja

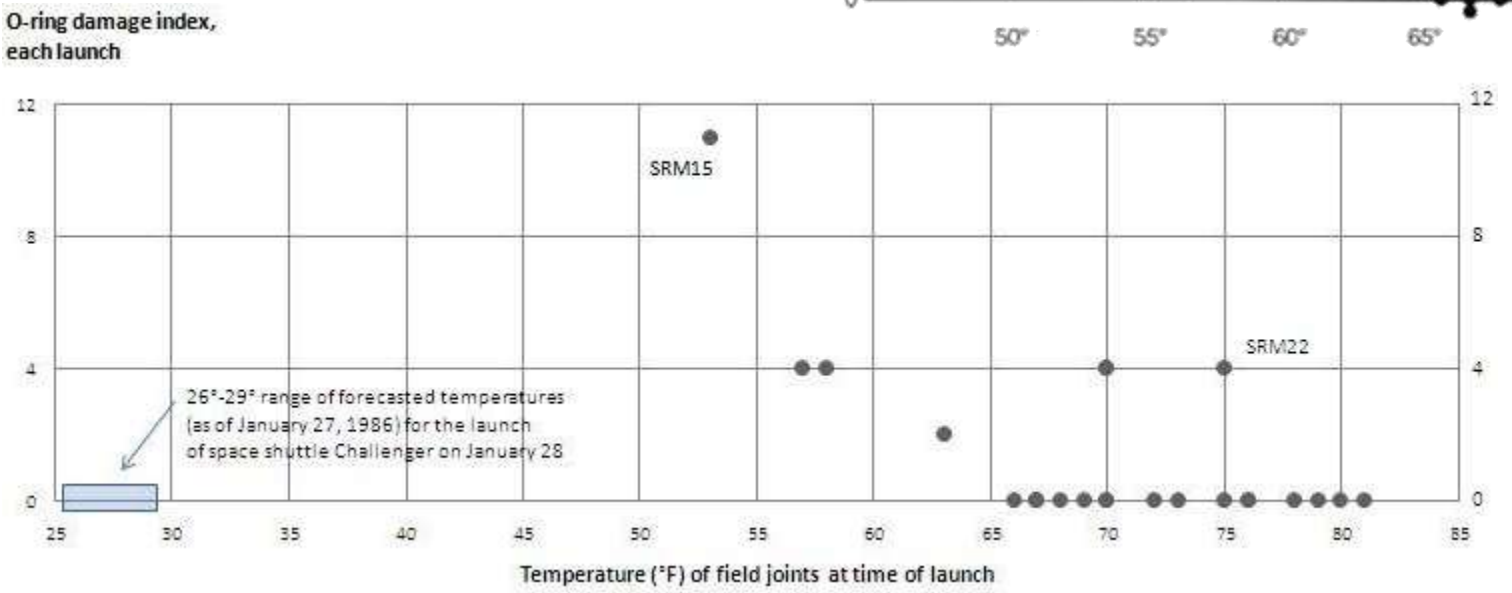


... VIZUELIZACIJA – PRIMERI ...



- Challenger disaster (nastavak)
  - Vizuelizacija koja je upotrebljena prilikom analize šta se dogodilo u Challenger misiji

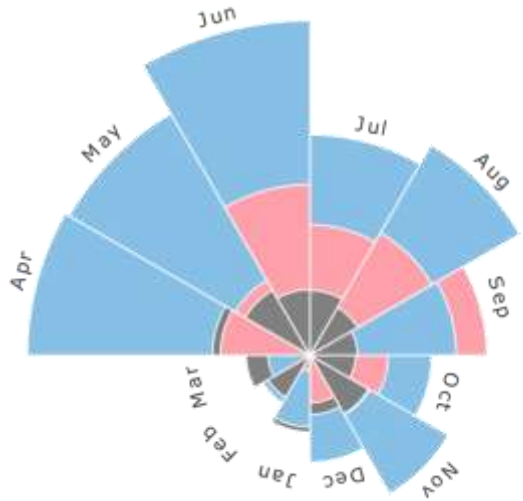


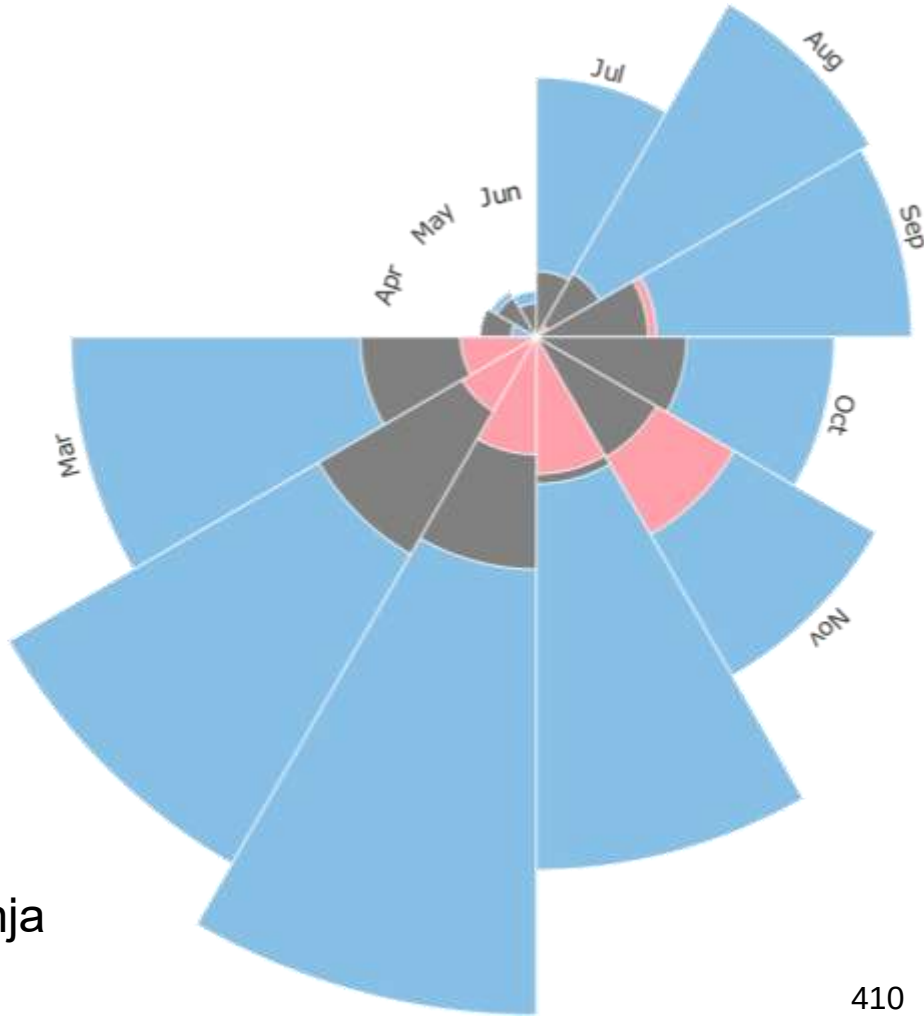


... VIZUELIZACIJA – PRIMERI ...



- Florence Nightingale, crveni krst



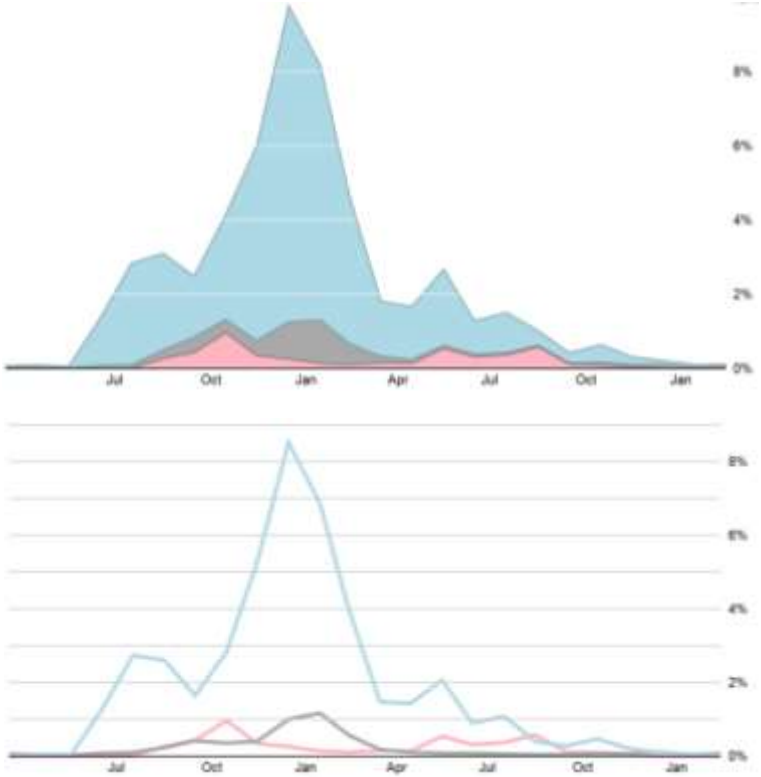


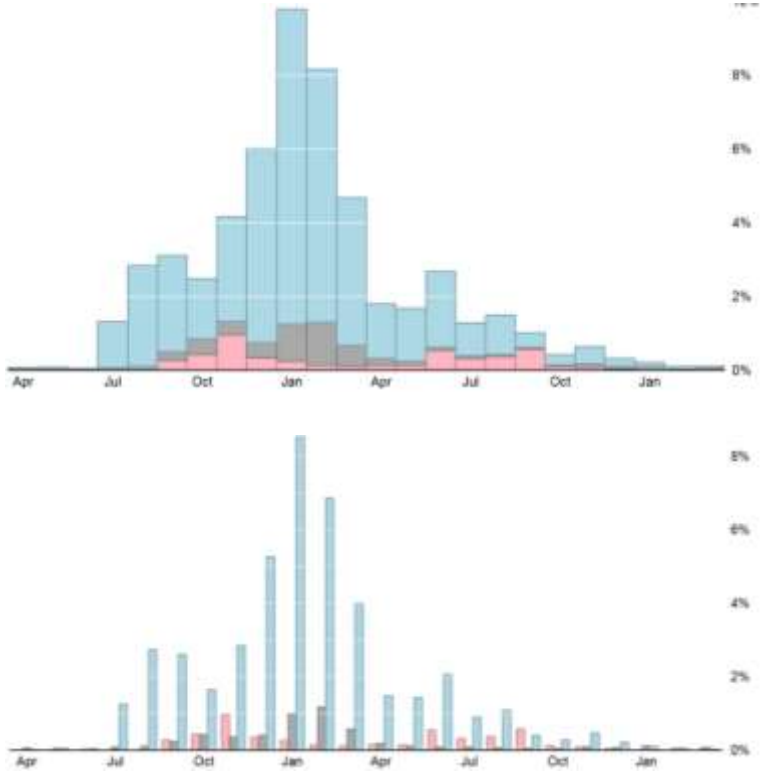
  - "Nightingale rose" dijagram
  - Procenat smrtnosti u engleskoj vojsci (Army of the East) od posledica ranjavanja u borbi (roze) i od nehigijenskih uslova (plavo) pre (desno) i posle (levo) poboljšanja higijenskih uslova

... VIZUELIZACIJA – PRIMERI ...



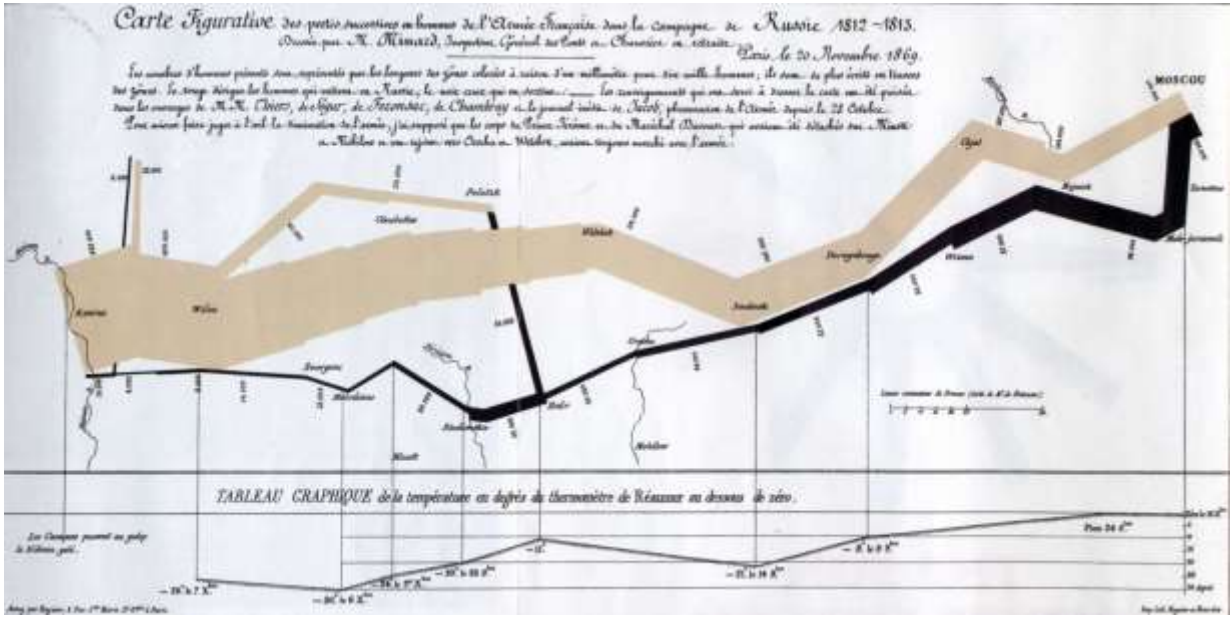
- Florence Nightingale, crveni krst (nastavak)
  - Dijagram obmanjuje, jer je broj smrti proporcionalan poluprečniku svakog segmenta; čovek interpretira celu oblast kao nosioca informacije, tako da će ono što je na obodu uvek delovati većim nego što je zaista





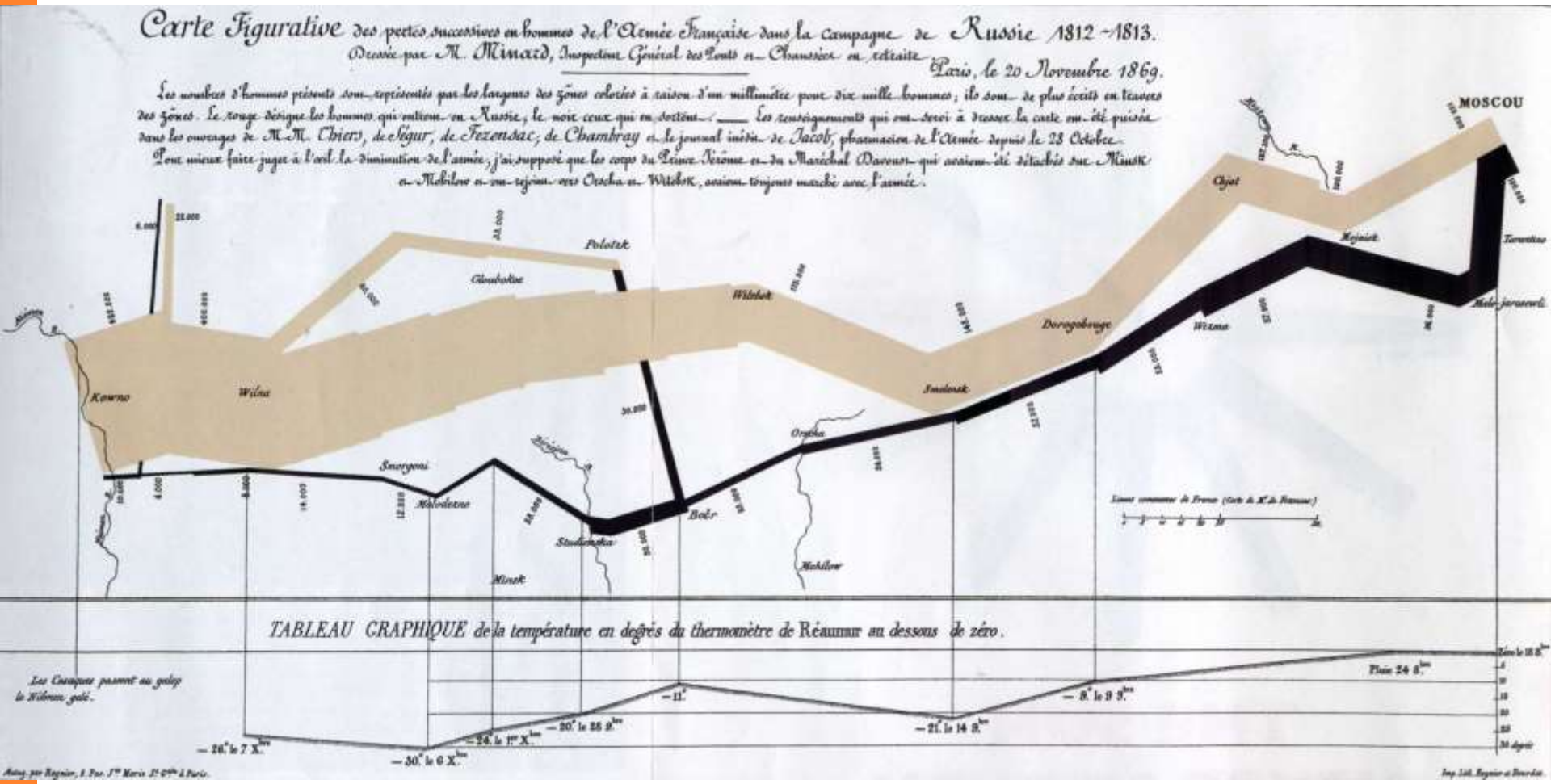
... VIZUELIZACIJA – PRIMERI ...

- Napoleonova kampanja u Rusiji
  - Charles Minard, 1869
  - Smeđa je marš ka Moskvi, crna označava povlačenje
  - Veličina armije kroz debljinu linije i broj
  - Putanja kroz veće gradove i kako se menjao broj vojnika, zajedno sa delovima vojske koji su odvajani na drugu stranu
  - Temperatura koja je bila u datom trenutku (jer je ona uticala dešavanja i osipanje armija)



... VIZUELIZACIJA – PRIMERI

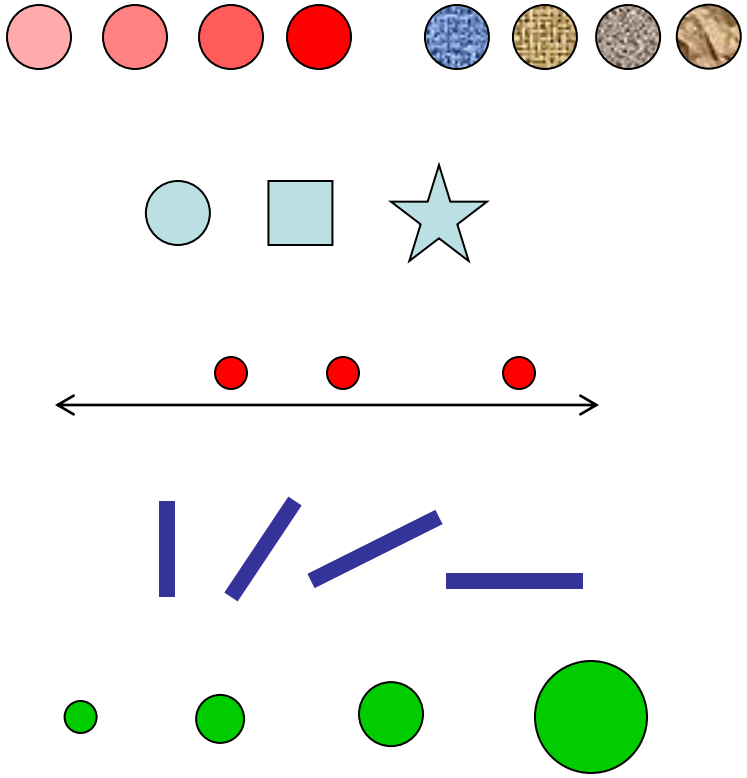
- Napoleonova kampanja u Rusiji (nastavak)



VIZUALNO MAPIRANJE ...



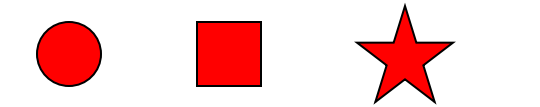
- Vizualne promenljive:
  - Boja (vrednost, nijansa), tekstura
  - Oblik
  - Pozicija: x, y, z
  - Orijentacija, ugao, nagib
  - Veličina, dužina, površina, volumen
  - Animacija, vreme, blinkanje, pokret



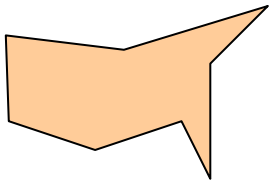
... VIZUALNO MAPIRANJE ...

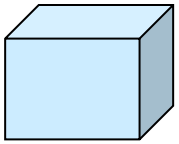


- Vizualne oznake :
  - Tačke
  - Linije
  - Oblast (površina)
  - Volumen
  - Glifovi (glyphs)











... VIZUALNO MAPIRANJE



- Mapiranje:
  1. Kategorije podataka → vizuelne oznake
  2. Atributi kategorija podataka → vizuelne promenljive oznaka

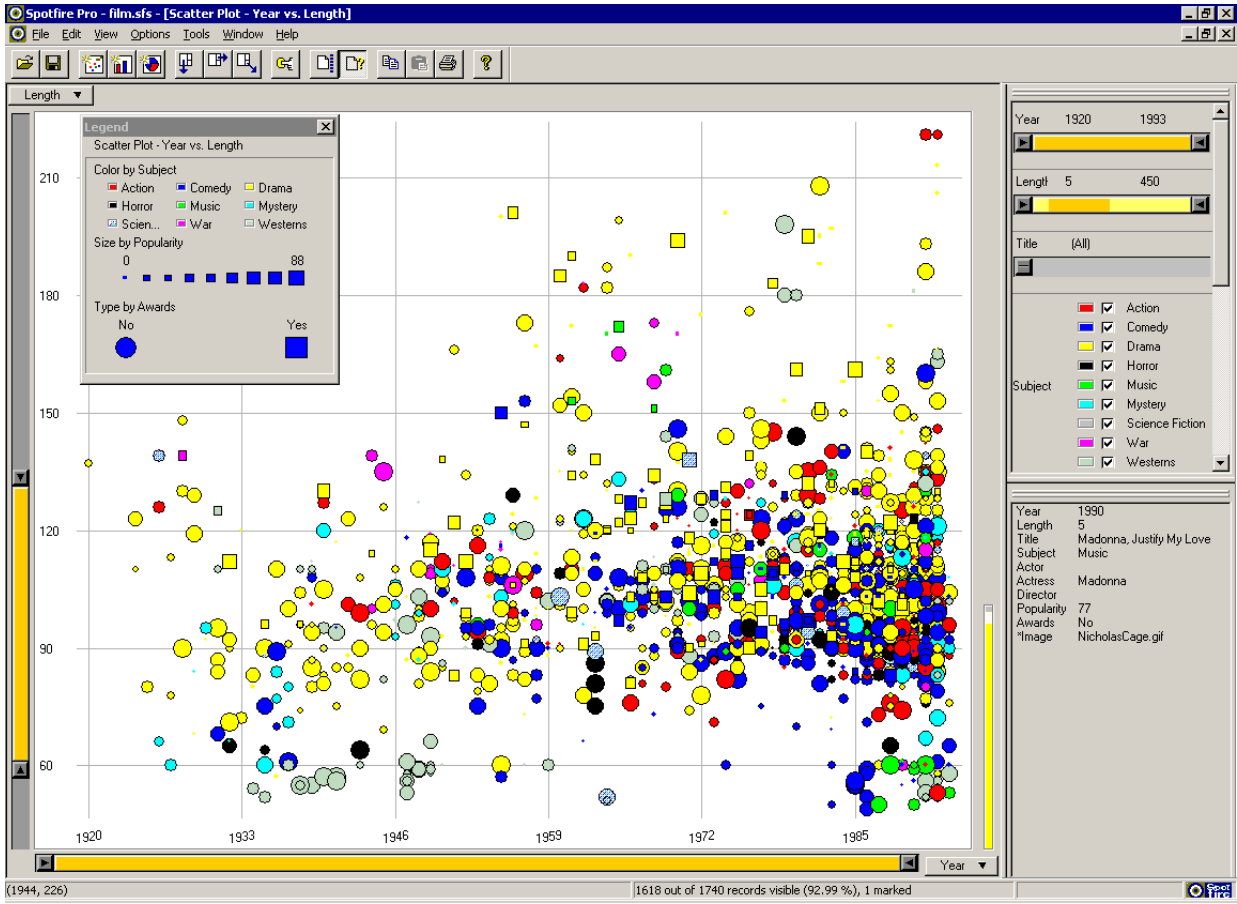


- U vizuelizaciji podataka se koriste neke ili sve vizuelne promenljive
- Koje su vizuelne promenljive korišćene u prethodnim primerima?

VIZUALNO MAPIRANJE – PRIMER



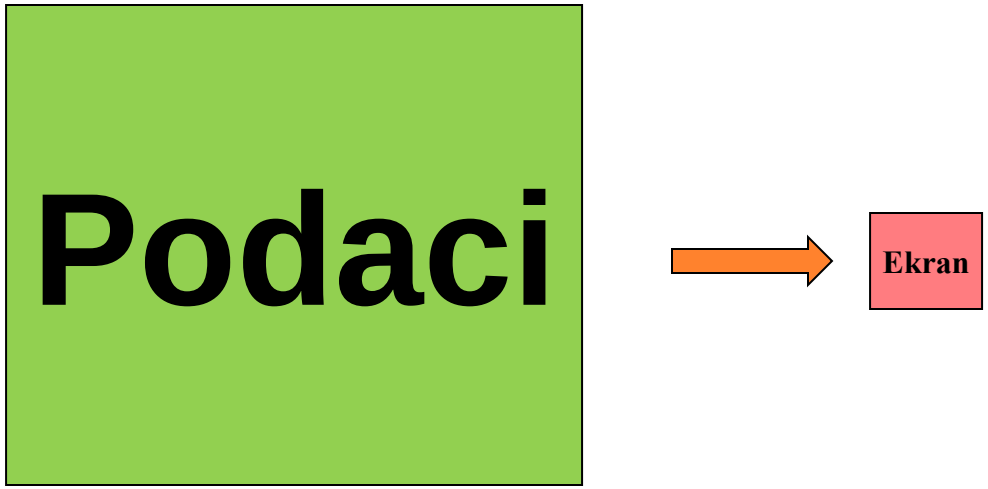
- Baza filmova u SpotFire programu
  - Film ⟶ oznaka (geometrijksa slika)
  - Godina ⟶ X osa
  - Trajanje ⟶ Y osa
  - Popularnost ⟶ veličina
  - Žarn ⟶ boja
  - Nagrada ⟶ oblik



VIZUALNO MAPIRANJE – POZICIJA ...

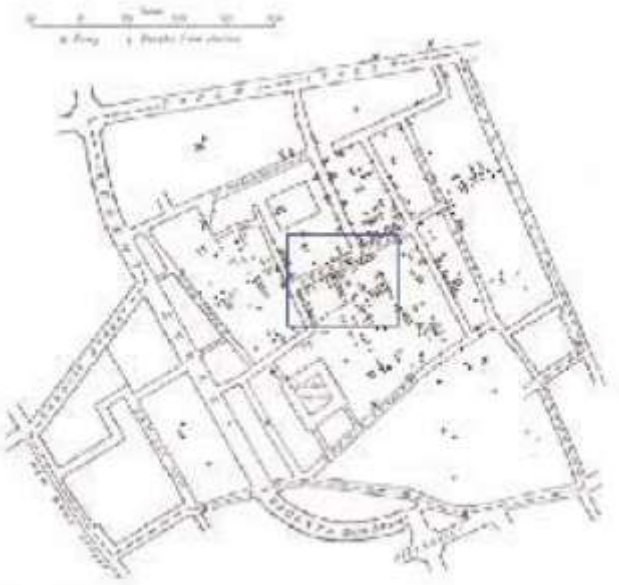


- Pozicija je osnovna vizuelna promenljiva, jer je
  - vrlo raznovrsna (i po dužiini i po tipu),
  - izrazito selektivna (jer je lako kretati fokus kroz vizuelno polje), i
  - lako iskače (vrednosti koje odskaču se lako uočavaju)
- Ujedno je i vrlo zahtevan resurs (za koga često nema dovoljno mesta na prikazu)



... VIZUALNO MAPIRANJE – POZICIJA ...

- Nekada je najjednostavniji način korišćenja pozicije, onaj direktan – mapiranje pozicije iz stvarnog sveta na vizuelizaciju (mapa)
  - Ako podaci zaista imaju prostorne koordinate i ako fizička lokacija utiče na razmišljanje i prenošenje informacija o zadatku koji korisnik treba da sprovede, onda ovo radi posao

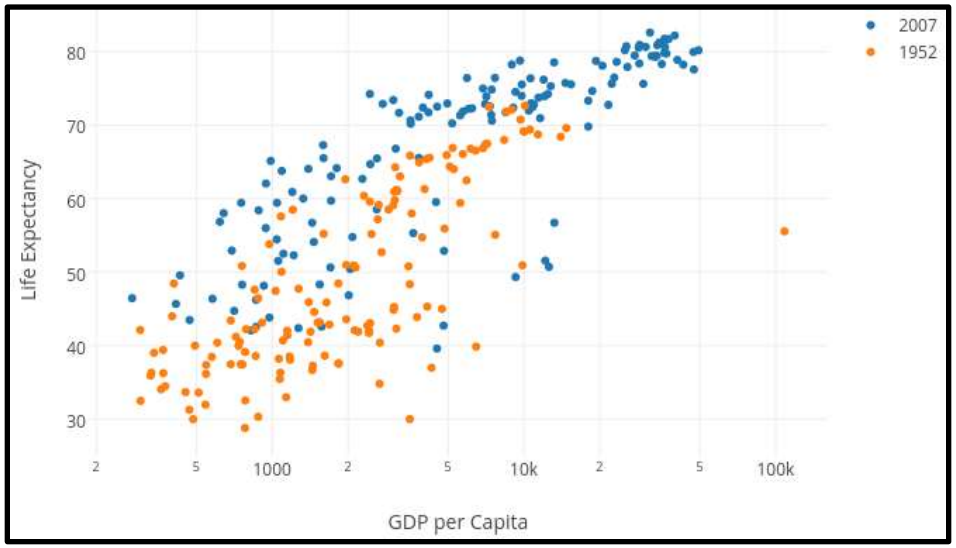


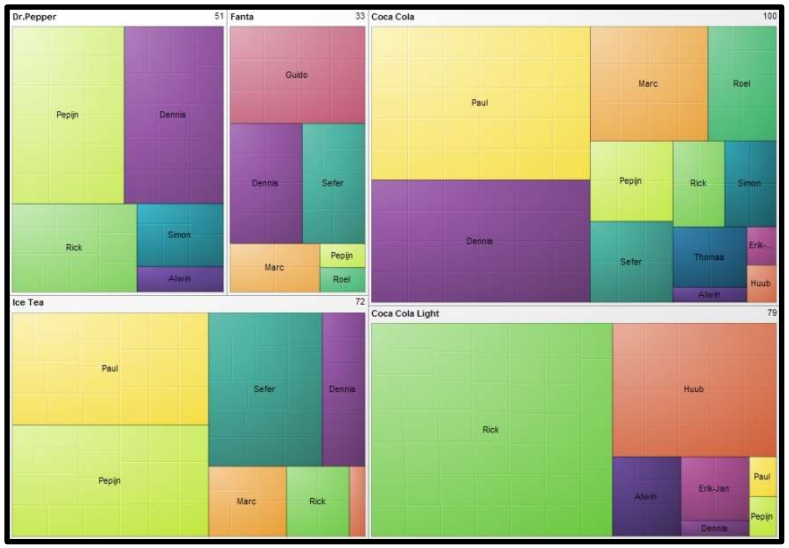


... VIZUALNO MAPIRANJE – POZICIJA



- Pozicija nije relevantna, onda se mora pronaći apstraktno mapiranje
  - Koordinate x i y se koriste za prikazivanje drugih promenljivih, tj. drugih atributa kategorije podataka (kao u scaterplot grafiku), ili
  - Koordinate x i y se koriste za pakovanje (prikazivanje) srodnih informacija jedna pored druge (kao na primer u stablo-mapi)





- I kod ostalih vizuelnih promenljive se koristi stvarno ili apstraktno mapiranje

KORISNIČKI ZADATAK



- Proste stvari (najčešće 1, 2 dimenzije):
  - Minimum, maksimum, procenat, srednja vrednost
  - Precizni upiti, poznati elementi pretrage

Excel

- Složene stvari (mnogo dimenzija, ne zna se šta se očekuje):
  - Obrasci, trendovi, distribucije, promene tokom vremena
  - Otlajeri, izuzeci
  - Odnosi, relacije, višestruka povezanost
  - Kombinacija minimuma i maksimuma, tradeoffs
  - Klasteri, grupe, poređenja, konteksti
  - Anomalije, greške u podacima, odstupanja
  - Putanje
  - ...

Vizuelizacija

TRADICIONALNE TEHNIKE VIZUALIZACIJE



- Tradicionalne tehnike:
  - Bar grafovi
  - Linijski grafovi
  - Dijagrami oblasti (area)
  - Pite
  - Scatter-plot dijagrami
  - Bubble dijagrami
  - Heat mape



- Imaju prednost da se lako uče i da su poznate, jer se koriste još iz škole i sastavni su deo prezentacija u medijima

BAR GRAFOVI ...

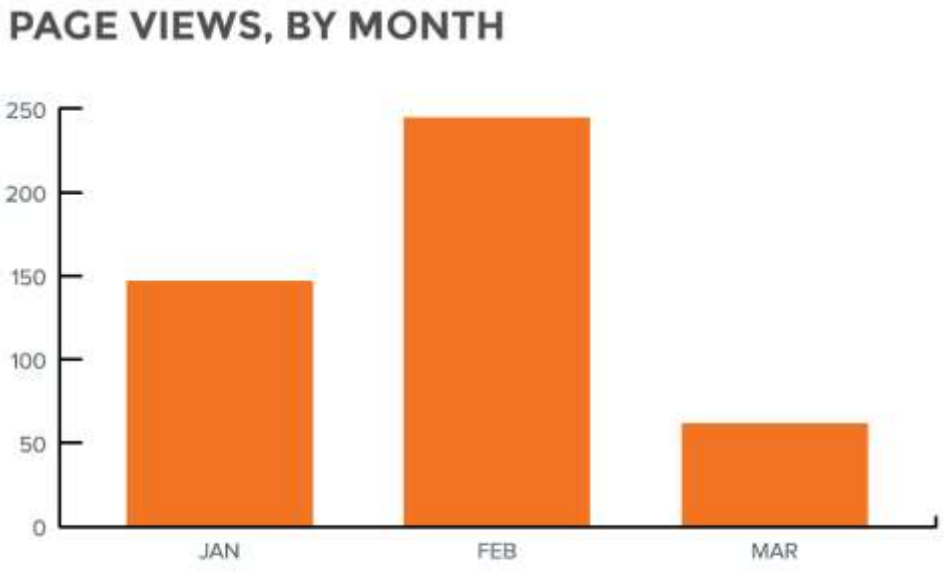


- Vrlo raznovrsni
- Koriste se
  - da prikažu promenu tokom vremena
  - za poređenje različitih kategorija
  - za poređenje delova celine
- Mogu biti:
  - Horizontalni
  - Vertikalni
  - Nadovezani (stacked)

... BAR GRAFOVI ...



- Horizontalni i vertikalni



VERTICAL (COLUMN CHART)

Best used for chronological data (time-series should always run left to right), or when visualizing negative values below the x-axis.



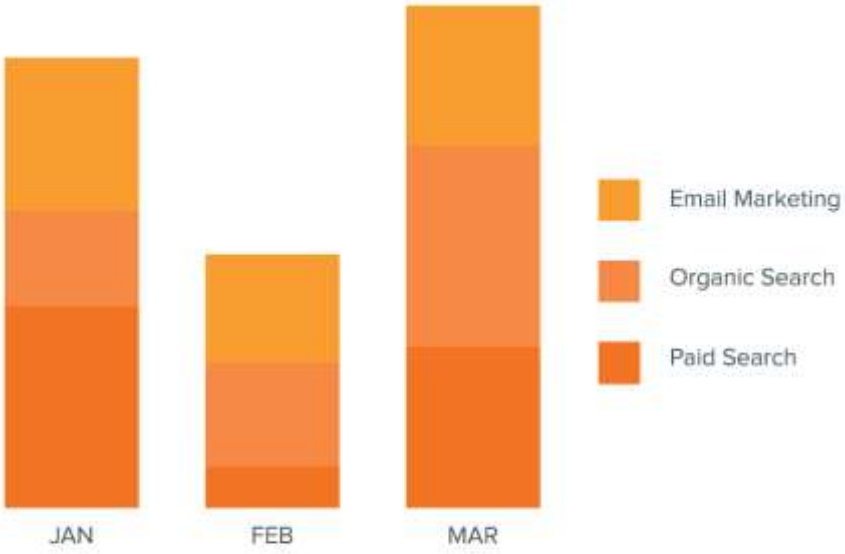
HORIZONTAL

Best used for data with long category labels.

... BAR GRAFOVI ...

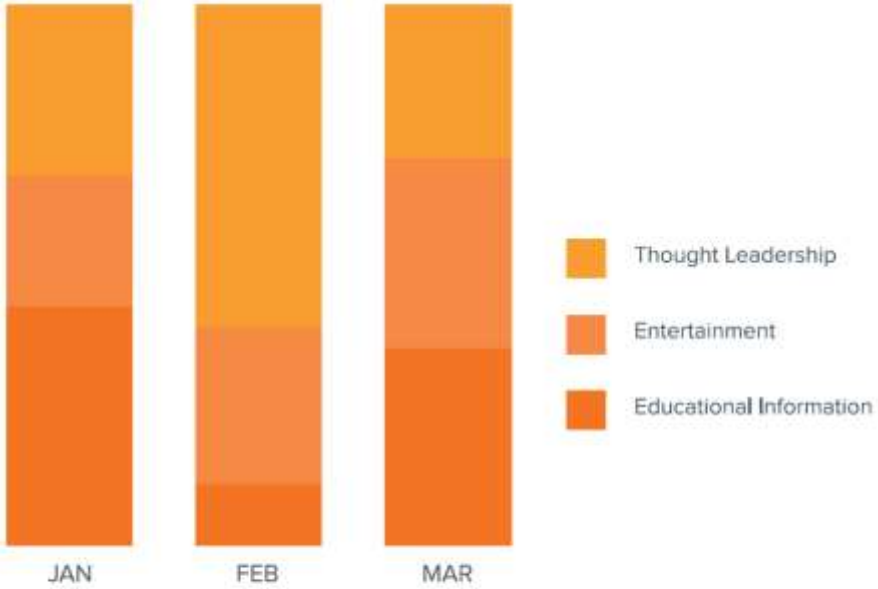


- Nadovezani (stacked)



STACKED

Best used when there is a need to compare multiple part-to-whole relationships. These can use discrete or continuous data, oriented either vertically or horizontally.



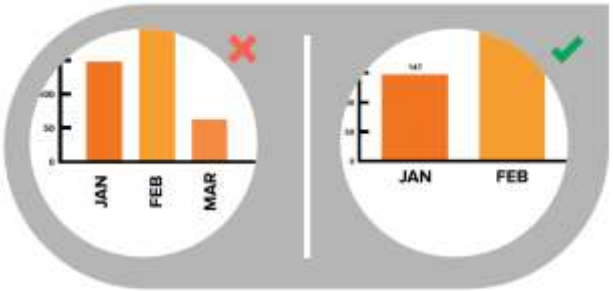
100% STACKED

Best used when the total value of each category is unimportant and percentage distribution of subcategories is the primary message.

... BAR GRAFOVI



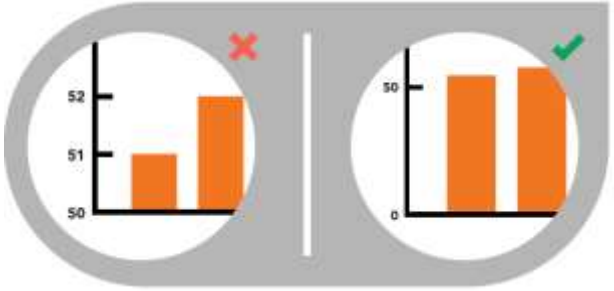
- Osnovne smernice



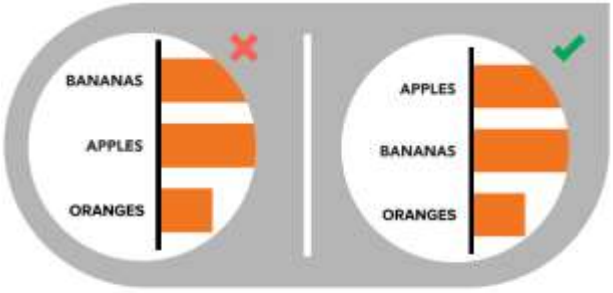
USE HORIZONTAL LABELS
Avoid steep diagonal or vertical type, as it can be difficult to read.



SPACE BARS APPROPRIATELY
Space between bars should be ½ bar width.



START THE Y-AXIS VALUE AT 0
Starting at a value above zero truncates the bars and doesn't accurately reflect the full value.



ORDER DATA APPROPRIATELY
Order categories alphabetically, sequentially, or by value.



USE CONSISTENT COLORS
Use one color for bar charts. You may use an accent color to highlight a significant data point.

LINE GRAFOVI ...



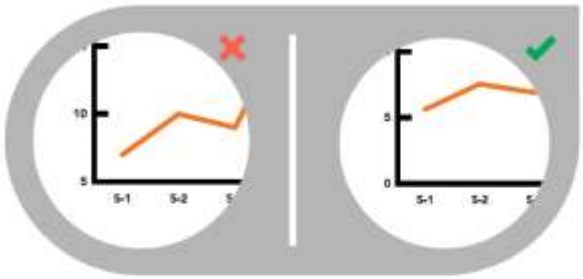
- Koriste se da prikažu kontinualne podatke u vremenu
- Pomažu u uočavanju
  - trendova,
  - rasta,
  - opadanja,
  - pikova,
  - volatillity



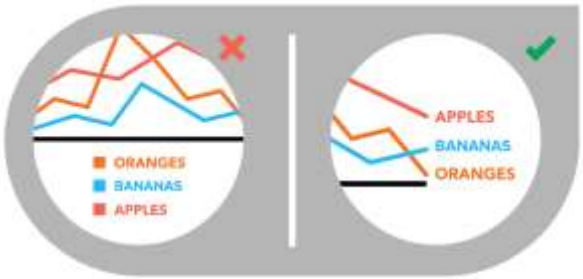
... LINE GRAFOVI



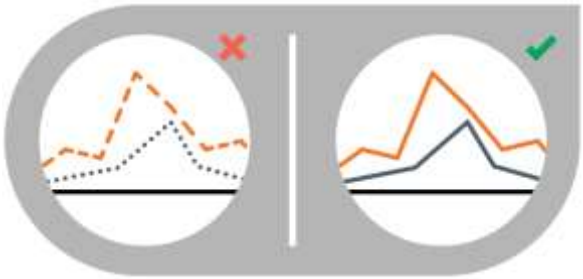
- Osnovne smernice



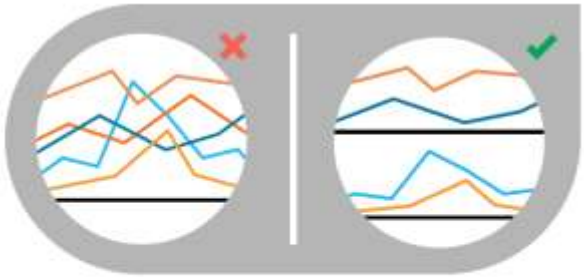
INCLUDE A ZERO BASELINE IF POSSIBLE
Although a line chart does not have to start at a zero baseline, it should be included if possible. If relatively small fluctuations in data are meaningful (e.g., in stock market data), you may truncate the scale to showcase these variances.



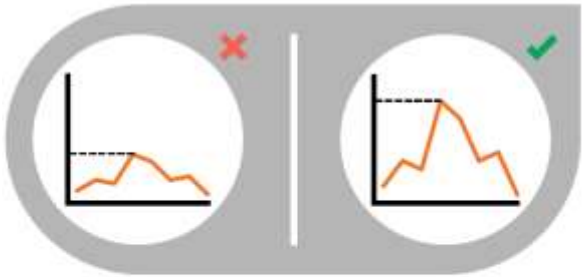
LABEL THE LINES DIRECTLY
This lets readers quickly identify lines and corresponding labels instead of referencing a legend.



USE SOLID LINES ONLY
Dashed and dotted lines can be distracting.



DON'T PLOT MORE THAN 4 LINES
If you need to display more, break them out into separate charts for better comparison.

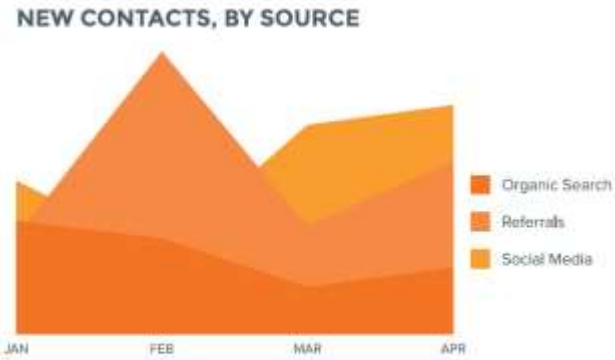


USE THE RIGHT HEIGHT
Plot all data points so that the line chart takes up approximately two-thirds of the y-axis' total scale.

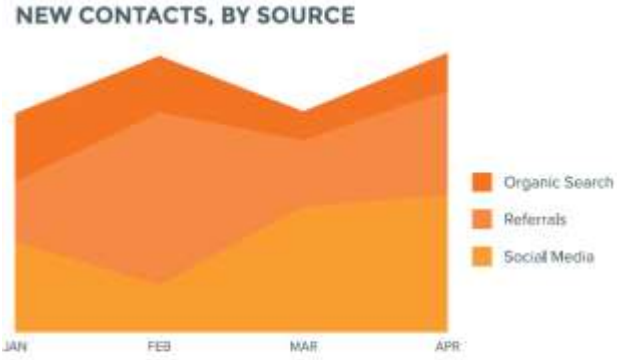
DIJAGRAMI OBLASTI (AREA) ...



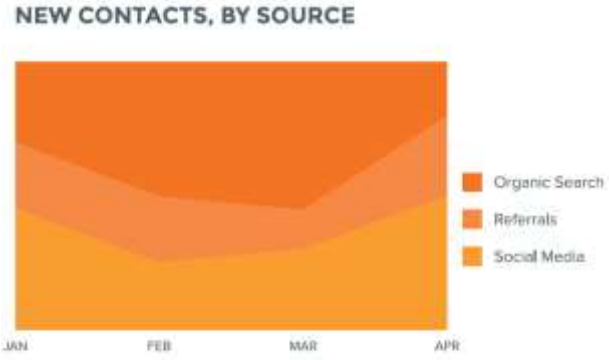
- Slični line dijagramima
- Mogu biti:
  - Standardni
  - Nadovezani (stacked)



AREA CHART
Best used to show or compare a quantitative progression over time.



STACKED AREA
Best used to visualize part-to-whole relationships, helping show how each category contributes to the cumulative total.

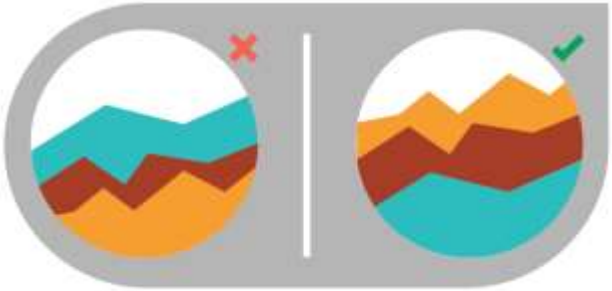


100% STACKED AREA
Best used to show distribution of categories as part of a whole, where the cumulative total is unimportant.

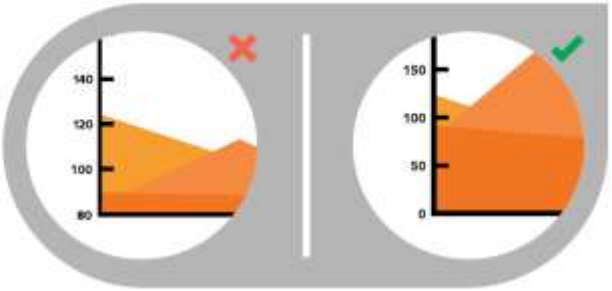
... DIJAGRAMI OBLASTI (AREA)



- Osnovne smernice



MAKE IT EASY TO READ
In stacked area charts, arrange data to position categories with highly variable data on the top of the chart and low variability on the bottom.



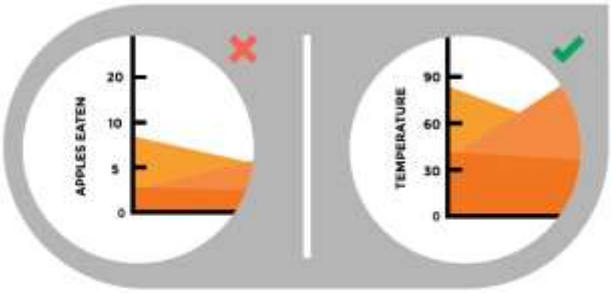
START Y-AXIS VALUE AT 0
Starting the axis above zero truncates the visualization of values.



DON'T DISPLAY MORE THAN 4 DATA CATEGORIES
Too many will result in a cluttered visual that is difficult to decipher.



USE TRANSPARENT COLORS
In standard area charts, ensure data isn't obscured in the background by ordering thoughtfully and using transparency.



DON'T USE AREA CHARTS TO DISPLAY DISCRETE DATA
The connected lines imply intermediate values, which only exist with continuous data.

PITE ...

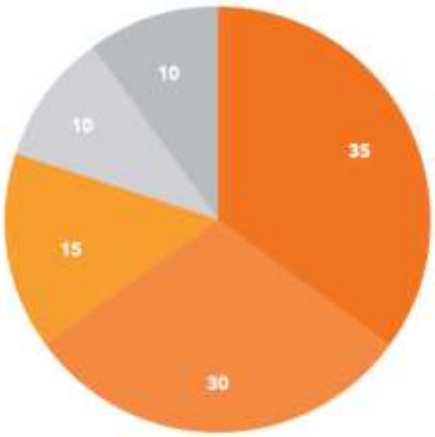


- Jedan od najpopularnijih tehnika vizuelizacije
- Za poređenje delova spram celine kod diskretnih ili kontinualnih podataka
- Najveći efekat imaju na malim skupovima podataka (mali broj dimenzija)
- Međutim, pokazano je da imaju efekat samo ako su delovi pite u poznatim odnosima (25%, 50%, 75%, 100%) i na poznatim lokacijama (prepoznatljivi uglovi od 30°, 45°, 90°, i sl.), sve ostalo ljudi prepoznaju na nekonzistantan način (svaki čovek drugačije), što otežava poređenje relativnih veličina i čini ovaj tip dijagram manje efikasnim
- Mogu biti:
  - Standardni
  - U obliku krofne

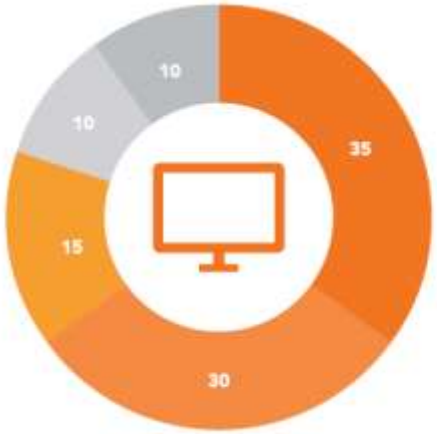
... PITE ...



- Standardne i u obliku krofne



STANDARD
Used to show part-to-whole relationships.



DONUT
Stylistic variation that enables the inclusion of a total value or design element in the center.

... PITE



- Osnovne smernice



ORDER SLICES CORRECTLY
There are two ways to order sections, both of which are meant to aid comprehension:

OPTION 1
Place the largest section at 12 o'clock, going clockwise. Place the second largest section at 12 o'clock, going counterclockwise. The remaining sections can be placed below, continuing counterclockwise.

OPTION 2
Start the largest section at 12 o'clock, going clockwise. Place remaining sections in descending order, going clockwise.



VISUALIZE NO MORE THAN 5 CATEGORIES PER CHART
It is difficult to differentiate between small values; depicting too many slices decreases the impact of the visualization. If needed, you can group smaller values into an "other" or "miscellaneous" category, but make sure it does not hide interesting or significant information.



DON'T USE MULTIPLE PIE CHARTS FOR COMPARISON
Slice sizes are very difficult to compare side-by-side. Use a stacked bar chart instead.

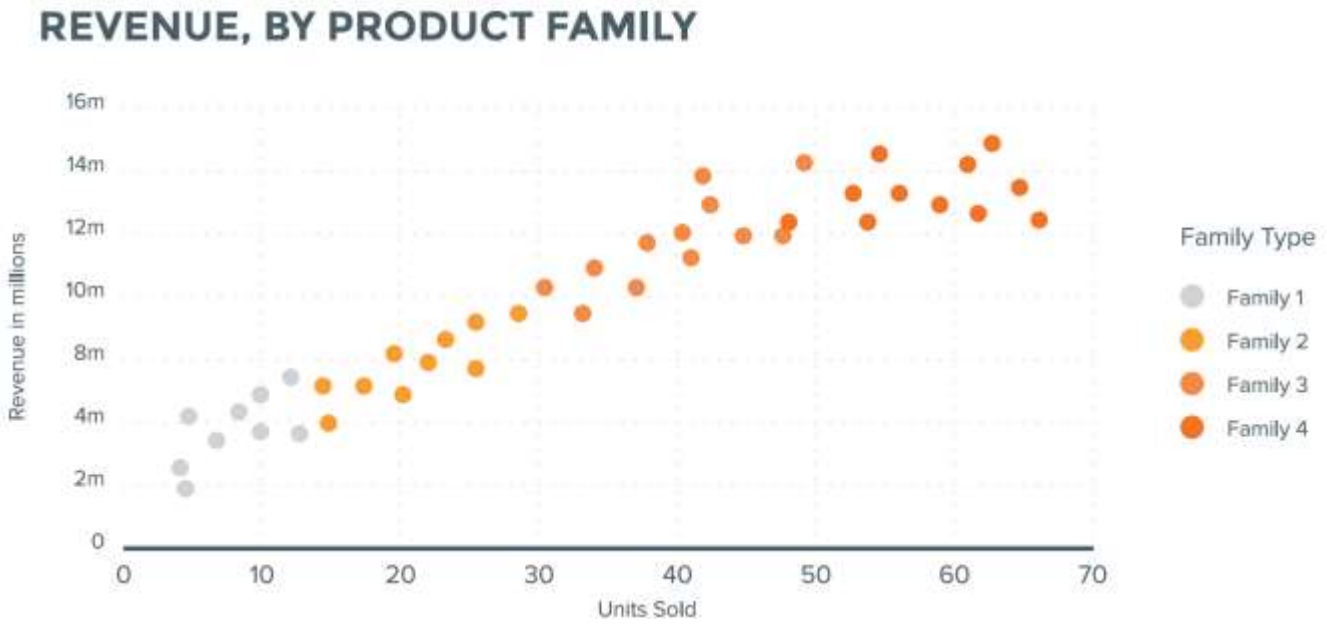


MAKE SURE ALL DATA ADDS UP TO 100%
Verify that values total 100% and that pie slices are sized proportionate to their corresponding value.

SCATTER PLOT ...



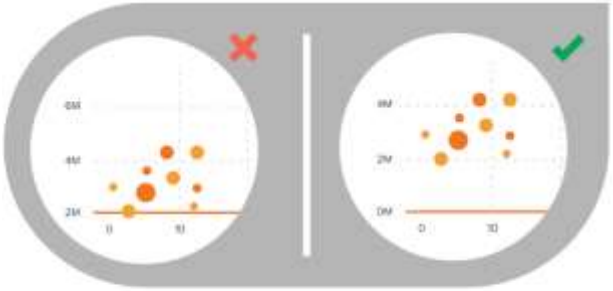
- Prikazuju relacije između elemenata na osnovu dve dimenzije koje su najčešće nekako povezane
- Najbolji su za prikaz relacija u velikim skupovima podataka
- Mogu se dodatno mapirati još neki atributi pomoću boje, veličine, oblika



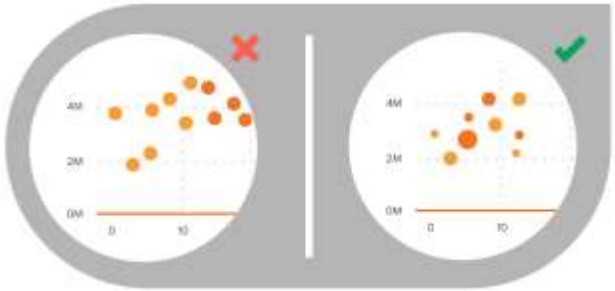
... SCATTER PLOT



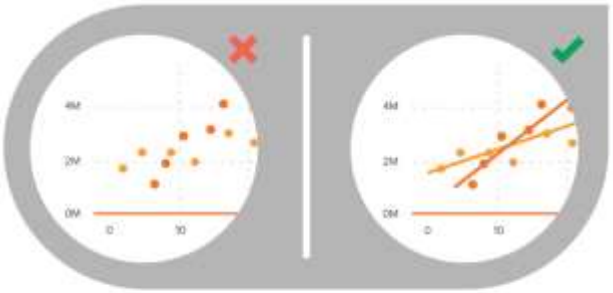
- Osnovne smernice



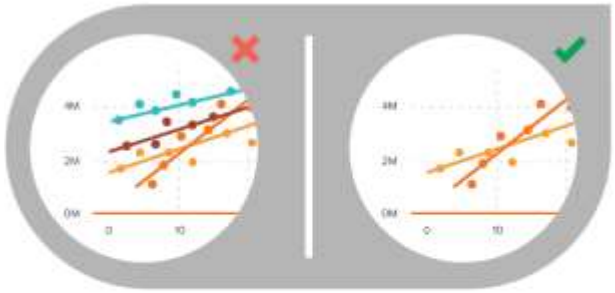
START Y-AXIS VALUE AT 0
Starting the axis above zero truncates the visualization of values.



INCLUDE MORE VARIABLES
Use size and dot color to encode additional data variables.



USE TREND LINES
These help draw correlation between the variables to show trends.

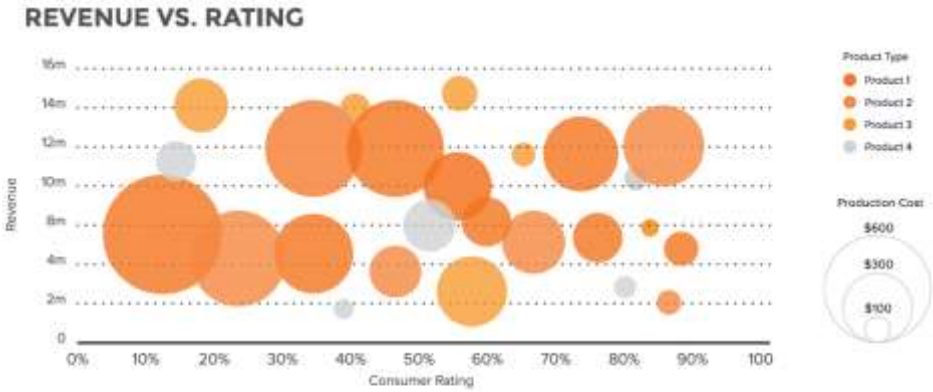


DON'T COMPARE MORE THAN 2 TREND LINES
Too many lines make data difficult to interpret.

BUBBLE CHART ...



- Dobre su za predstavljanje i poređenje nominalnih vrednosti ili za rangiranje kategorija
- Mogu biti:
  - Dijagrami
  - Mape



BUBBLE PLOT

This is a scatter plot with bubbles, best used to display an additional variable.



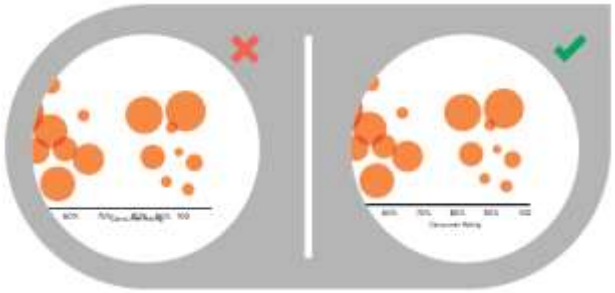
BUBBLE MAP

Best used for visualizing values for specific geographic regions.

... BUBBLE CHART



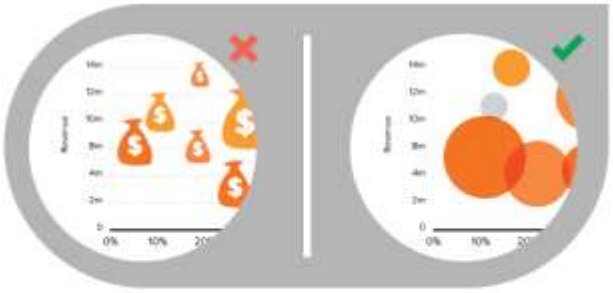
- Osnovne smernice



MAKE SURE LABELS ARE VISIBLE
All labels should be unobstructed and easily identified with the corresponding bubble.



SIZE BUBBLES APPROPRIATELY
Bubbles should be scaled according to area, not diameter.

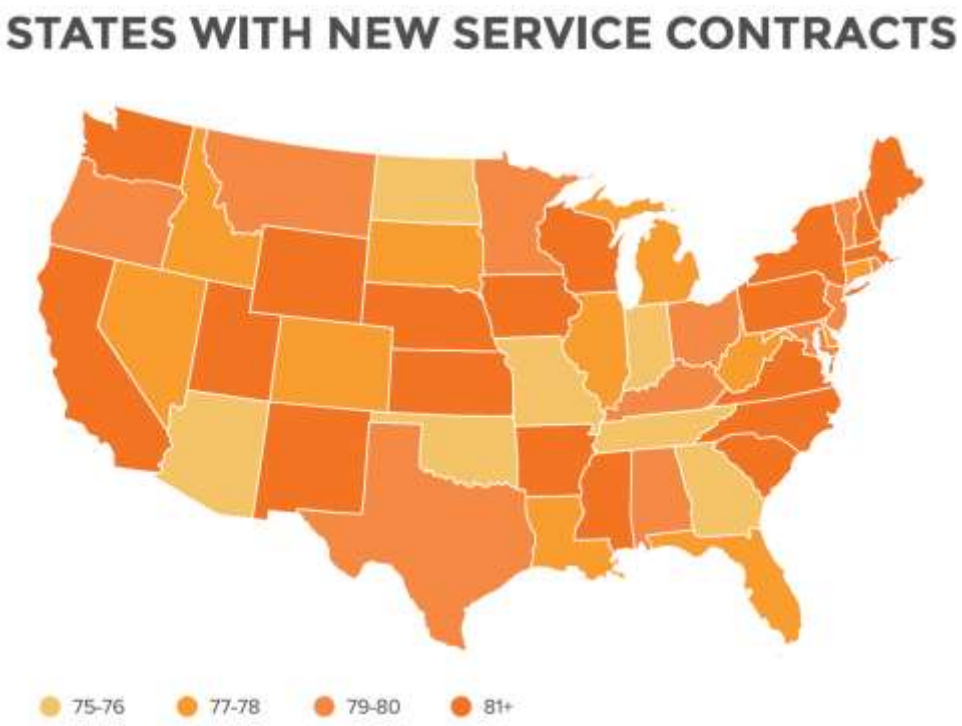


DON'T USE ODD SHAPES
Avoid adding too much detail or using shapes that are not entirely circular; this can lead to inaccuracies.

HEAT MAP ...



- Heat mape se koriste da pomoću intenziteta boje prikažu podatke podeljenje u kategorije, geografske regije ili tabele sa podacima



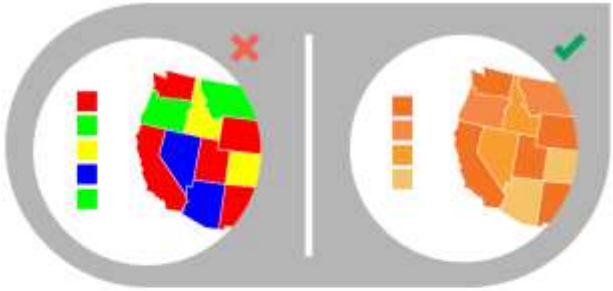
... HEAT MAP



- Osnovne smernice



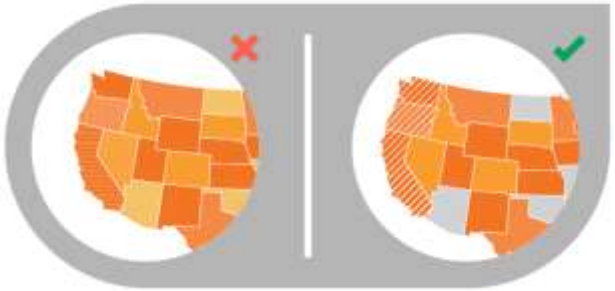
USE A SIMPLE MAP OUTLINE
These lines are meant to frame the data, not distract.



SELECT COLORS APPROPRIATELY
Some colors stand out more than others, giving unnecessary weight to that data. Instead, use a single color with varying shade or a spectrum between two analogous colors to show intensity. Also remember to intuitively code color intensity according to values.



CHOOSE APPROPRIATE DATA RANGES
Select 3-5 numerical ranges that enable fairly even distribution of data between them. Use +/- signs to extend high and low ranges.



USE PATTERNS SPARINGLY
A pattern overlay that indicates a second variable is acceptable, but using multiple is overwhelming and distracting.

NAPREDNIJE TEHNIKE VIZUALIZACIJE



- Ne-tradicionalne tehnike:
  - Stablo-mape (TreeMaps)
  - Vizuelizovane tabele (TableLens)
  - Hiperboličko stablo (star tree)
  - Cloud tag
  - Star plot (radial diagram)
  - Parallel coordinates
  - Parallel coordinates 3D

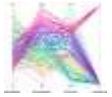














TREEMAP ...



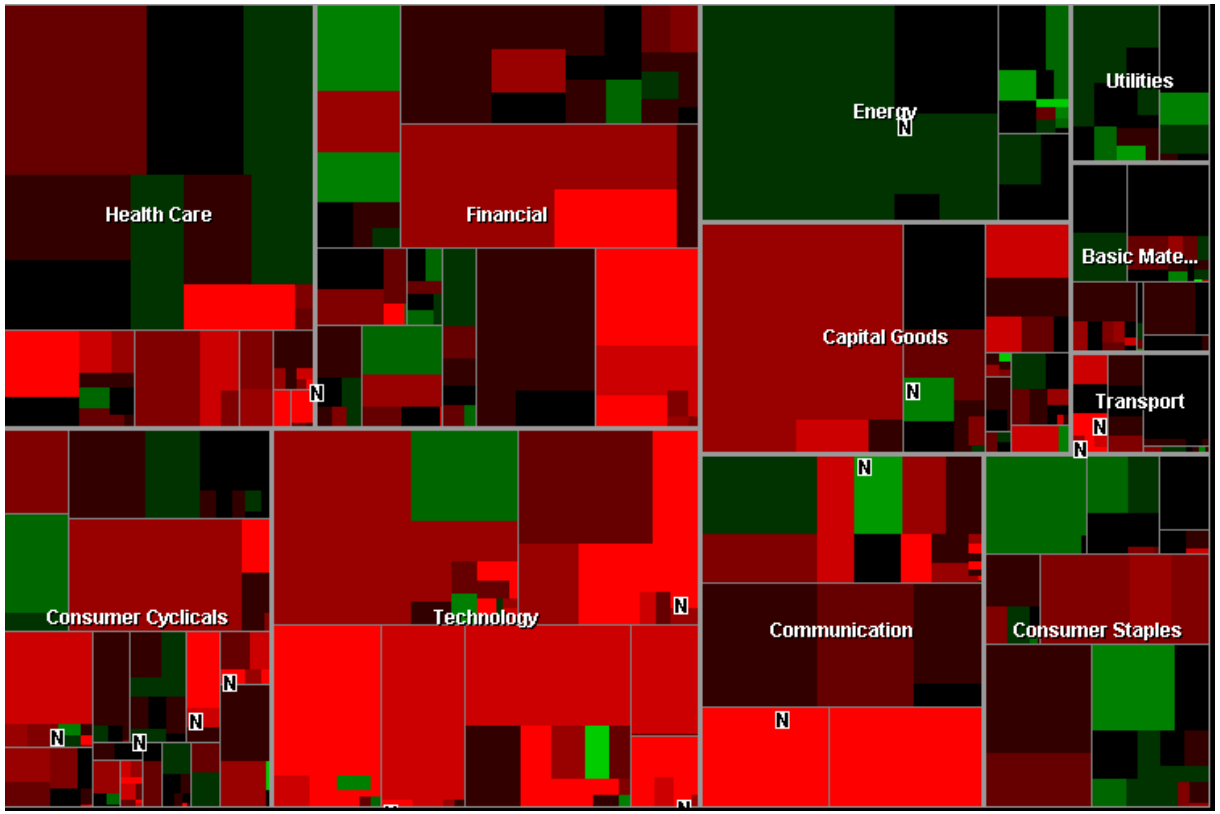
- Vizualizacija stabla koje kompaktno popunjava čitav pravougaoni prostor
- Svaki list predstavlja jedan kvantitativni atribut



... TREEMAP



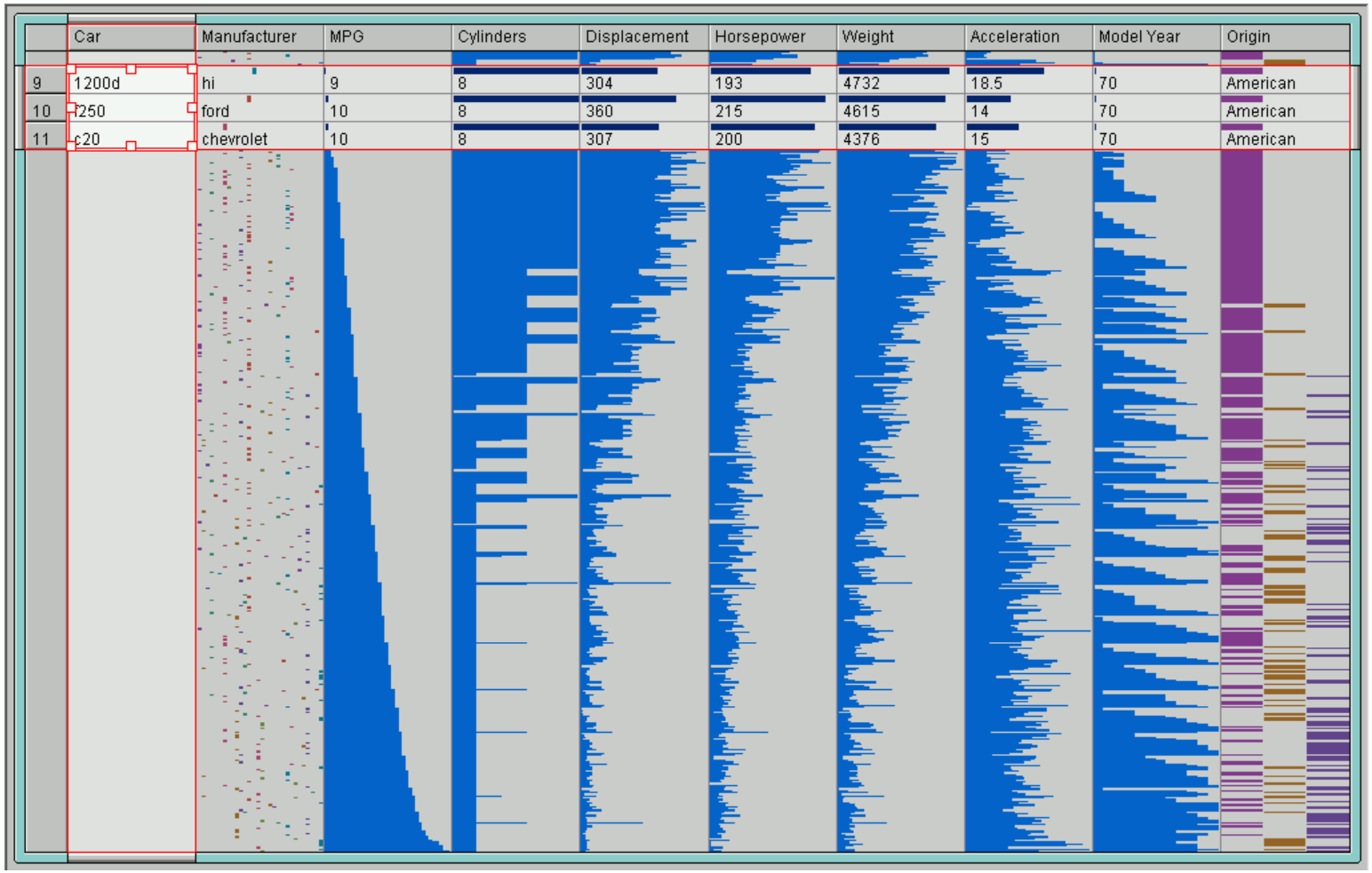
- Atributi se mapiraju na veličinu, boju i nijanse
- Teži se da elementi imaju što je više moguće kvadratni oblik, jer se time olakšava poređenje



TABLELANS …



- Spoj tabelarnog prikaza podataka i vizuelizacije



… TABLELANS

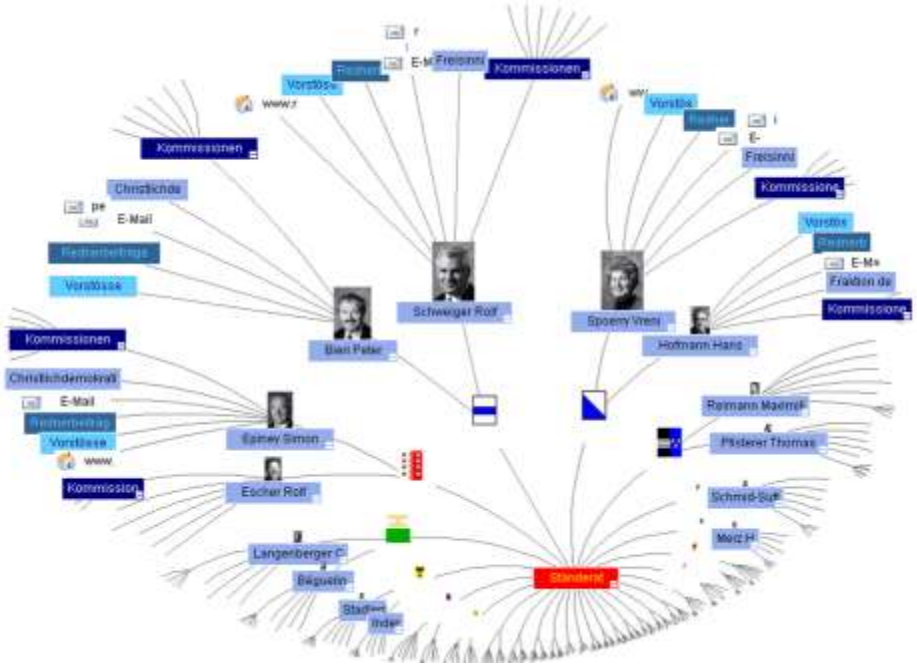


- Istraživački projekat
- Prikazuje hiljade redova pomoću vizuelnog prikaza širine samo jednog piksela (ili manje) po redu
  - Kvantitativni podaci se mapiraju na liniju određene dužine
  - Nominalni podaci se mapiraju na poziciju i boju
- Podržava sortiranje po kolonama
- Uspostavljanje višestrukih relacija između različitih kolona u isto vreme
- Zumiranje i detaljniji pregled pojedinačnih elemenata/kolona

STAR TREE



- Hijerarhija stabla koje sažima/širi i smanjuje/povećava u zavisnosti od toga šta je u fokusu
- Cikličan oblik
- Smanjuje natrpanost tako što se delovi stabla sažimaju i smanjuju ako nisu u fokusu
  - što je element dalje od centra to je on manji (osim ako nije u fokusu, kada se povećava)
  - elementi na grani u fokusu se povećavaju, dok se ostali smanjuju

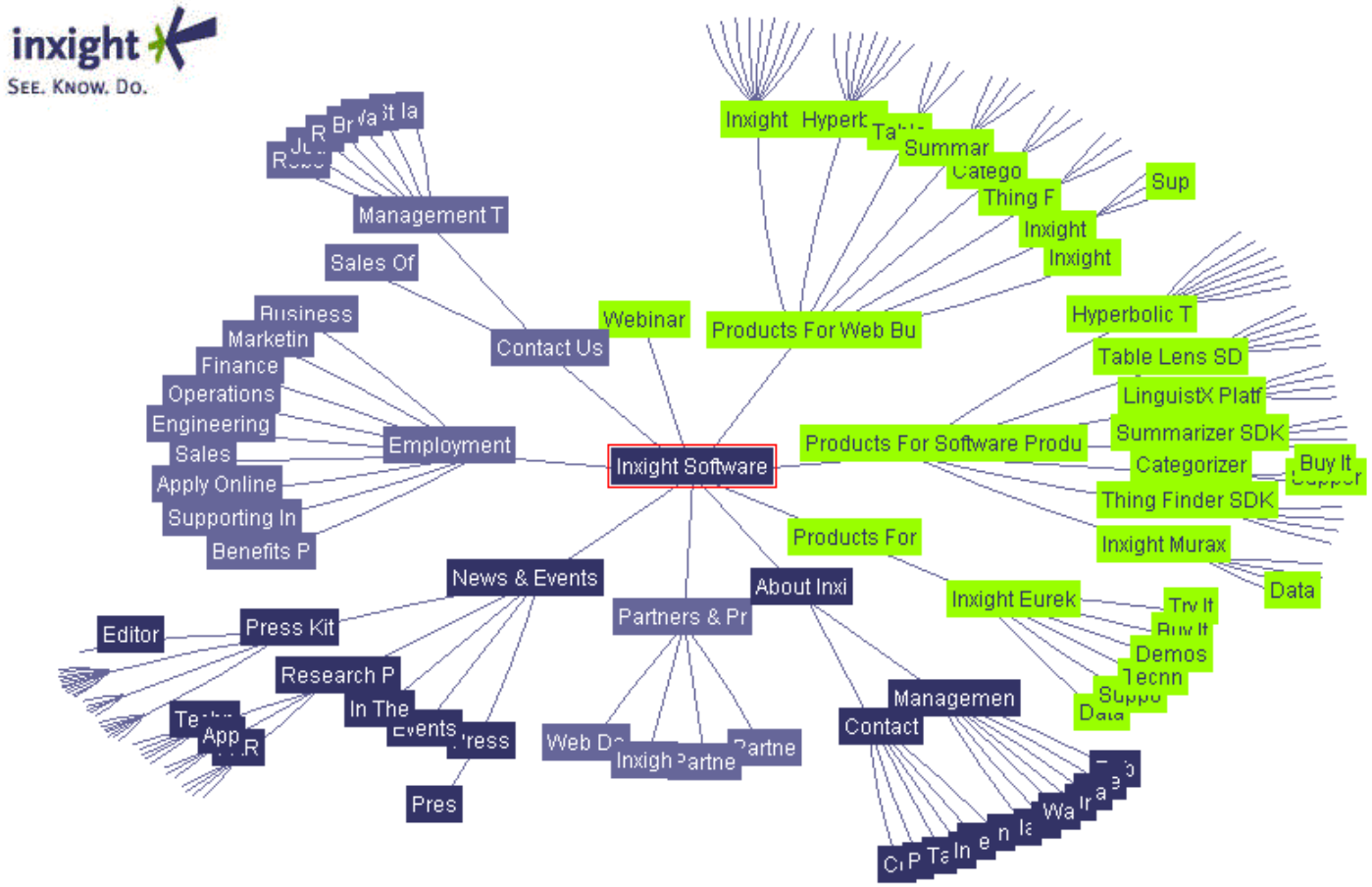


- Označavanje putanje u zavisnosti od konteksta
- Elementi se mogu povlačiti

... STAR TREE



- Xerox PARC, Inxight



TAG CLOUD



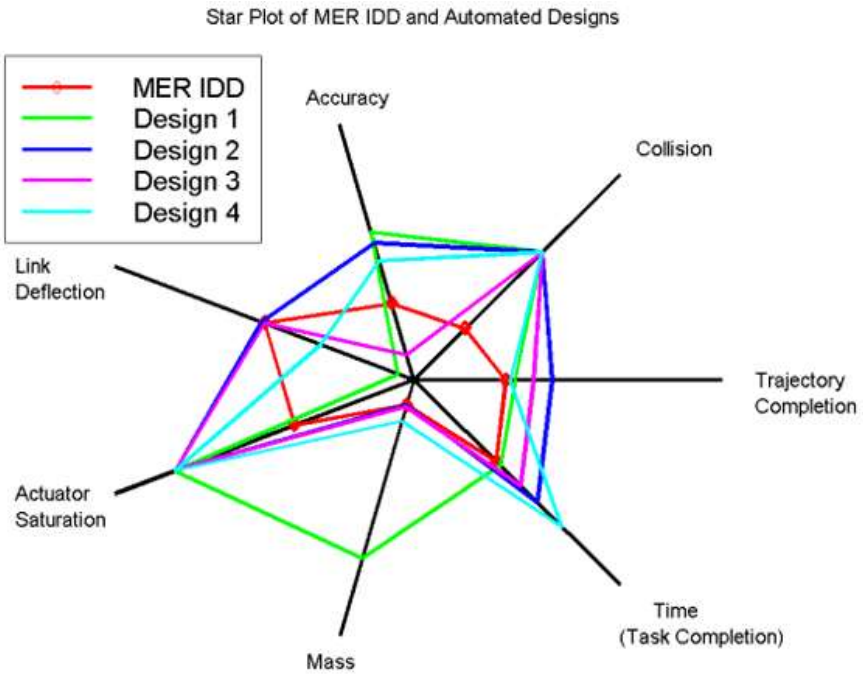
- Koristi se tekst za prikaz pojma, dok se veličina, boja koriste za kodiranje vrednosti još dva atributa
- Boja se nekad koristi čisto za razlikovanje pojmova, ne mora uvek imati ukodiranu neku informaciju
- Koristi se često za isticanje ključnih reči
- Problem: keyword spamming!



STAR PLOT ...



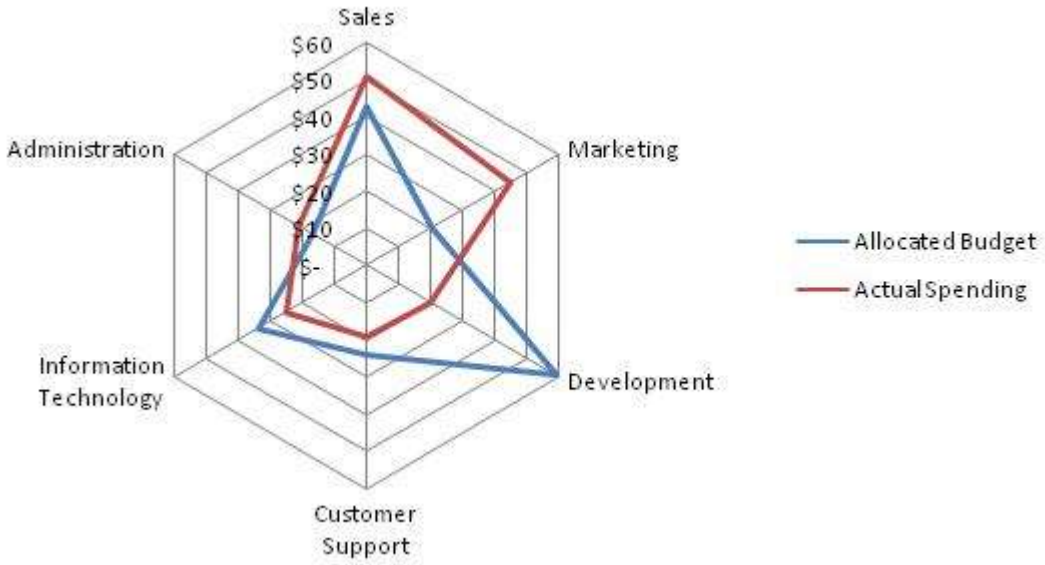
- Koristi se za prikazivanje podataka sa više atributa (dimenzija)
- Relativna pozicija koordinata i uglova obično ne nose nikakvu informaciju
- Koordinate su linije raspoređene u obliku zvezde u jednoj ravni
- Minimalna i maksimalna vrednost dimenzije preslikavaju se na početnu (unutrašnju) i krajnju (spoljašnju) vrednost na koordinati
- Veza se uvek uspostavljaju između susednih koordinata
- N vrednosti povezuje se u poliliniju od N delova



... STAR PLOT



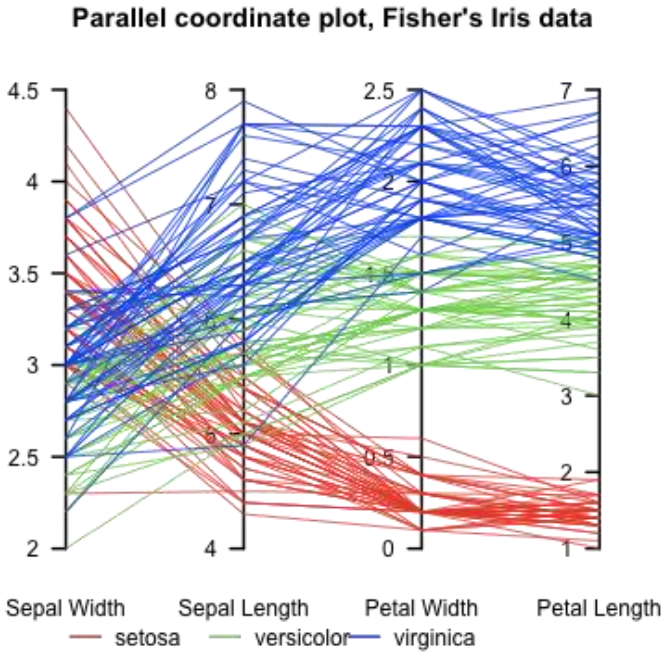
- Vrlo su dobre u predstavljanju
  - outlajera i trendova
  - kada jedan element značajno bolji od svih ostalih
- Loši su za donošenje odluka kada postoji trade off, tj. kada su različiti elementi koji se porede dobri po različitim atributima
- Zbog cirkularne prirode teško je porediti dužine vrhova sa različitih koordinata



PARALLEL COORDINATES ...



- Koristi se za prikazivanje velikog broja podataka gde je svaki element opisan sa više (N) atributa (dimenzija)
- Dimenzije su reprezentovane paralelnim, vertikalnim ili horizontalnim, linijama
- Broj koordinata koji se može prikazati ograničen je veličinom prikaznog medija
- Minimalna i maksimalna vrednost dimenzije preslikavaju se na početnu (donju) i krajnju (gornju) vrednost na koordinati
- Svaka koordinata, osim prve i poslednje, ima dva suseda

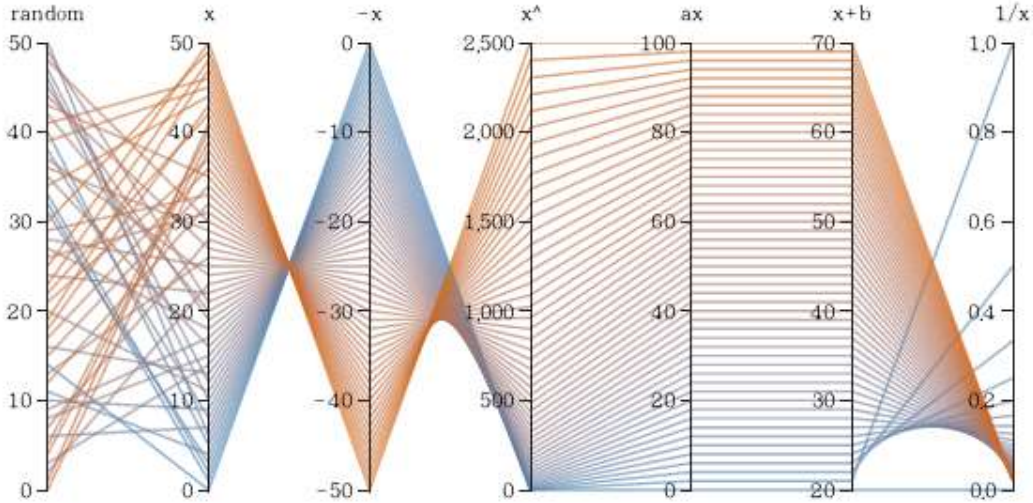


- Veze se uvek uspostavljaju između susednih koordinata
- N ustanovljenih vrednosti povezuje se N – 1 linijom koje formiraju jednu poliliniju

... PARALLEL COORDINATES



- Može da se koristi za prikaz vremenskih serija kada koordinate imaju tačno određenu poziciju (postoji uređenost koordinata)
- Uobičajeno je da koordinate nemaju uređenost i da se redosled koordinata može preuređivati
- Cilj je da se na 2D koordinate mapiraju N-dimenzionalne relacije
  - Služe za uočavanje outlajera i trendova

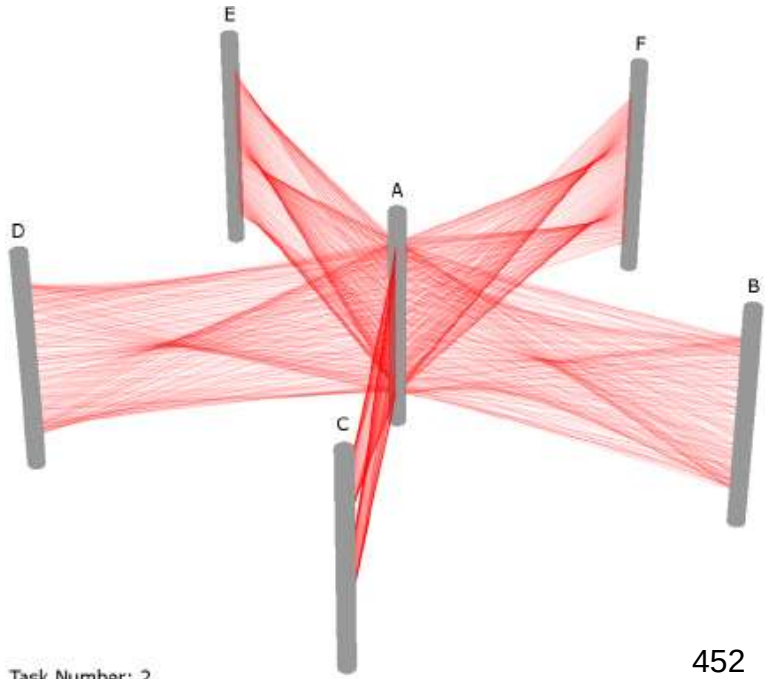


- Ako je relativno mali broj atributa (10tak), onda može i mapiranje atributa na koordinate, za uočavanje konkretnih vrednosti i pronalaženje najboljih vrednosti (nezgodan za donošenje odluka kada postoji trade off)

PARALLEL COORDINATES U 3D



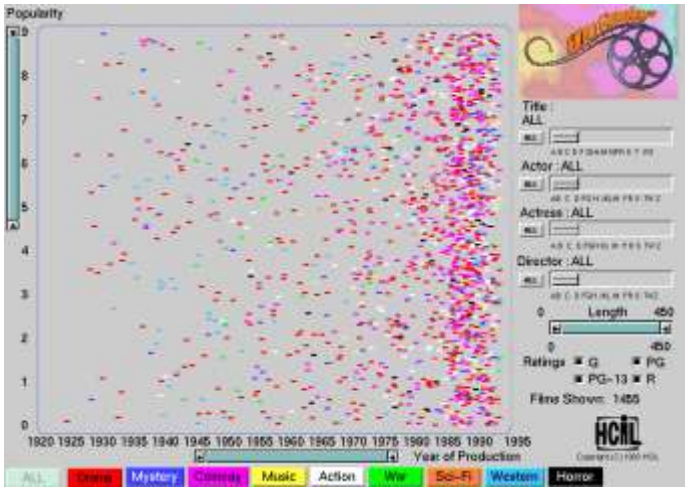
- Clustered Multi-Relational Parallel coordinates
- Koristi se za predstavljanje višestrukih relacija između koordinata
- Vizuelizuju odnos između jedne koordinate i svih preostalih koordinata
- Koordinata od interesa (centralna koordinata) predstavljena je linijom koja prolazi kroz osu prizme. Prikazuju se sve veze centralne koordinate sa spoljašnjim koordinatama
- Rastojanje između centralne koordinate i spoljašnjih koordinata definiše korisnik
- Spoljašnje koordinate su simetrično raspoređene u odnosu na centralnu koordinatu kako bi se korisniku olakšalo sagledavanje veza ka centralnoj koordinati

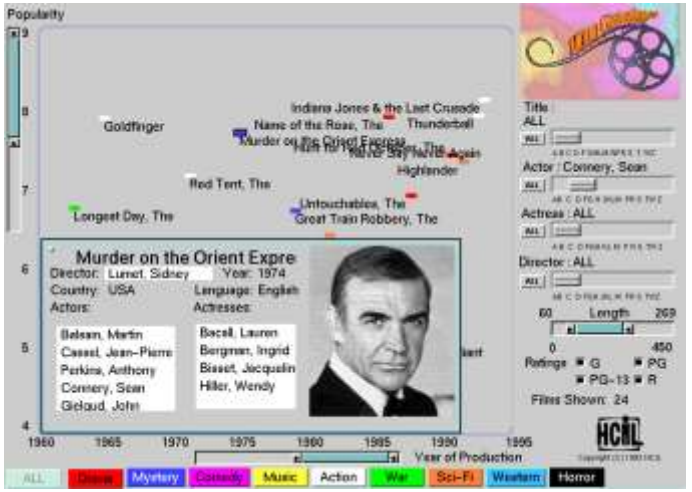


SHNEIDERMAN MANTRA



- **Overview first, zoom and filter, then details on demand**
- Vizualizacija pored samog prikaza informacija, treba da obezbedi i interaktivnost
- Overview – pregled čitavog skupa podataka („Big Picture“), da se vidi gde je korisnik i da uoči najvažnije relacije odmah („at a glance“)
- Zoom and filter – mogućnost da korisnik smanji pregled i količinu podataka na razumnu meru, fokusiranje na podskup podataka
- Details on demand – prikaz detalja samo na korisnikov eksplicitan zahtev

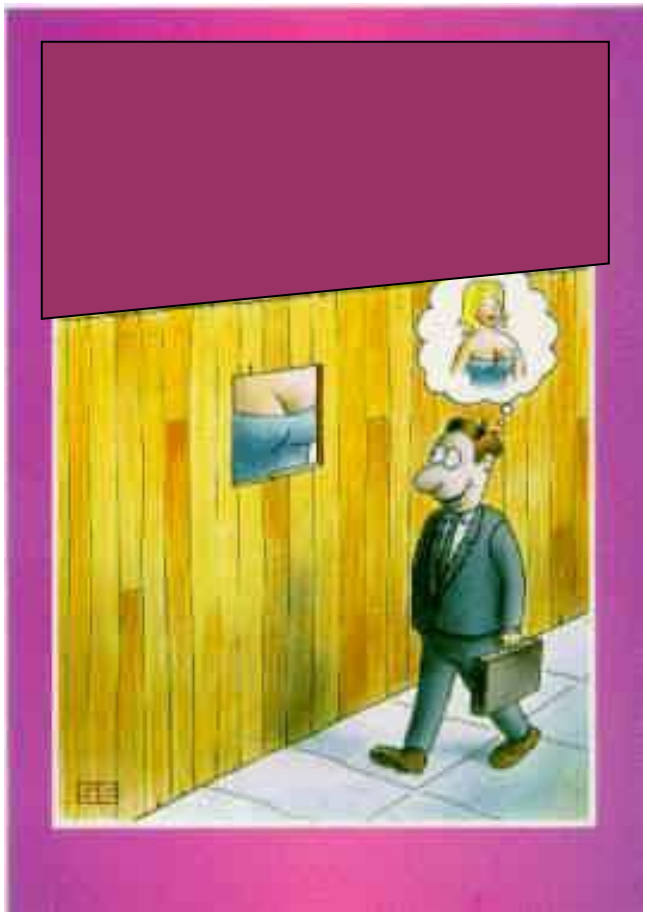




KONTEKST PODATAKA



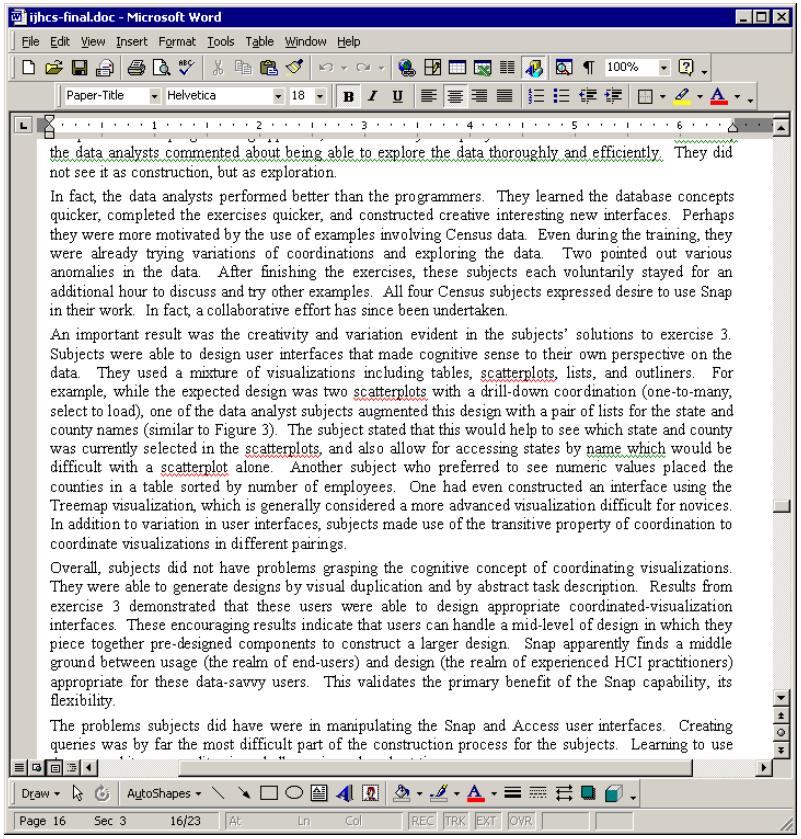
- Kontekst je jako važan za interpretaciju podataka

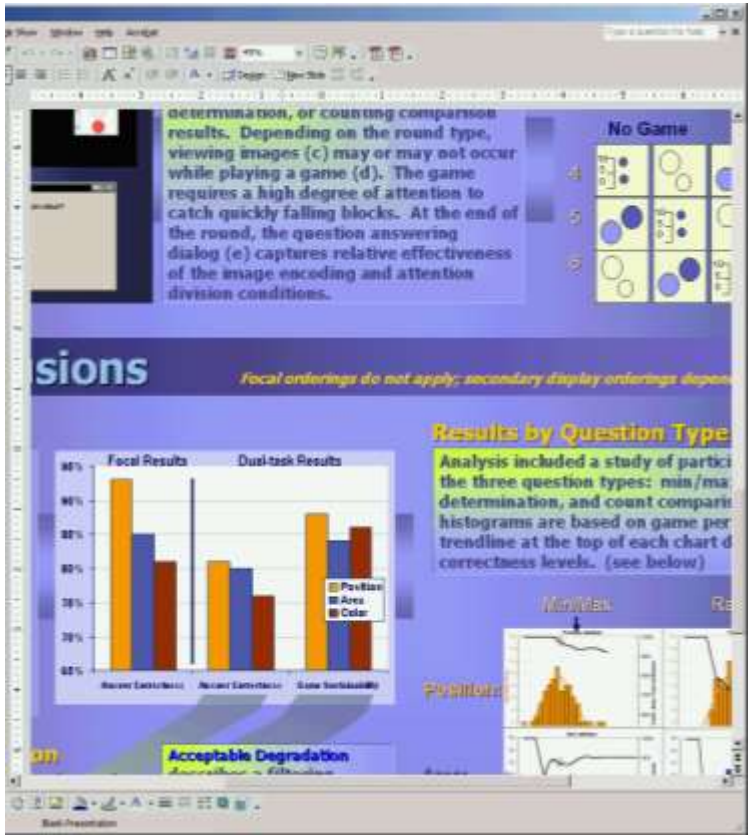


NAVIGACIJA ...



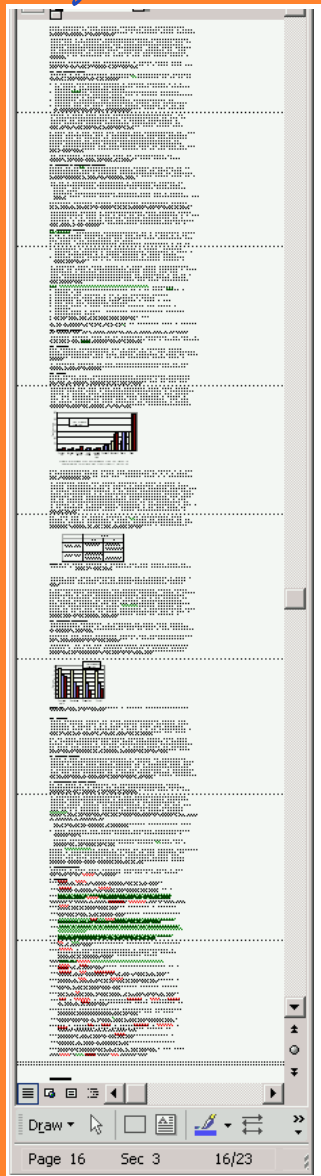
- Navigacija je važna kako bi se razumeo kontekst i kako bi se snalazilo u podacima

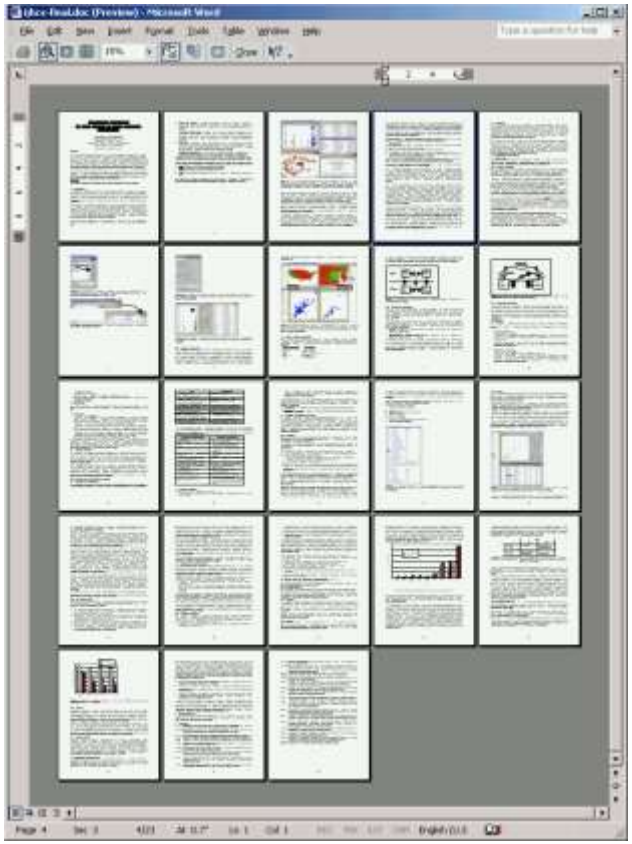


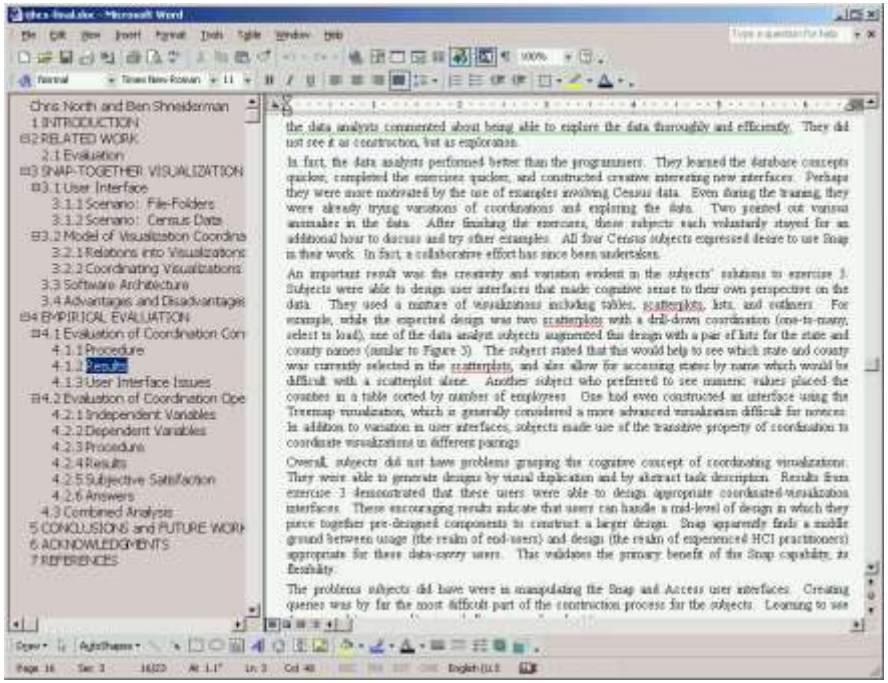


... NAVIGACIJA




Draw
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Chris North and Ben Shneiderman
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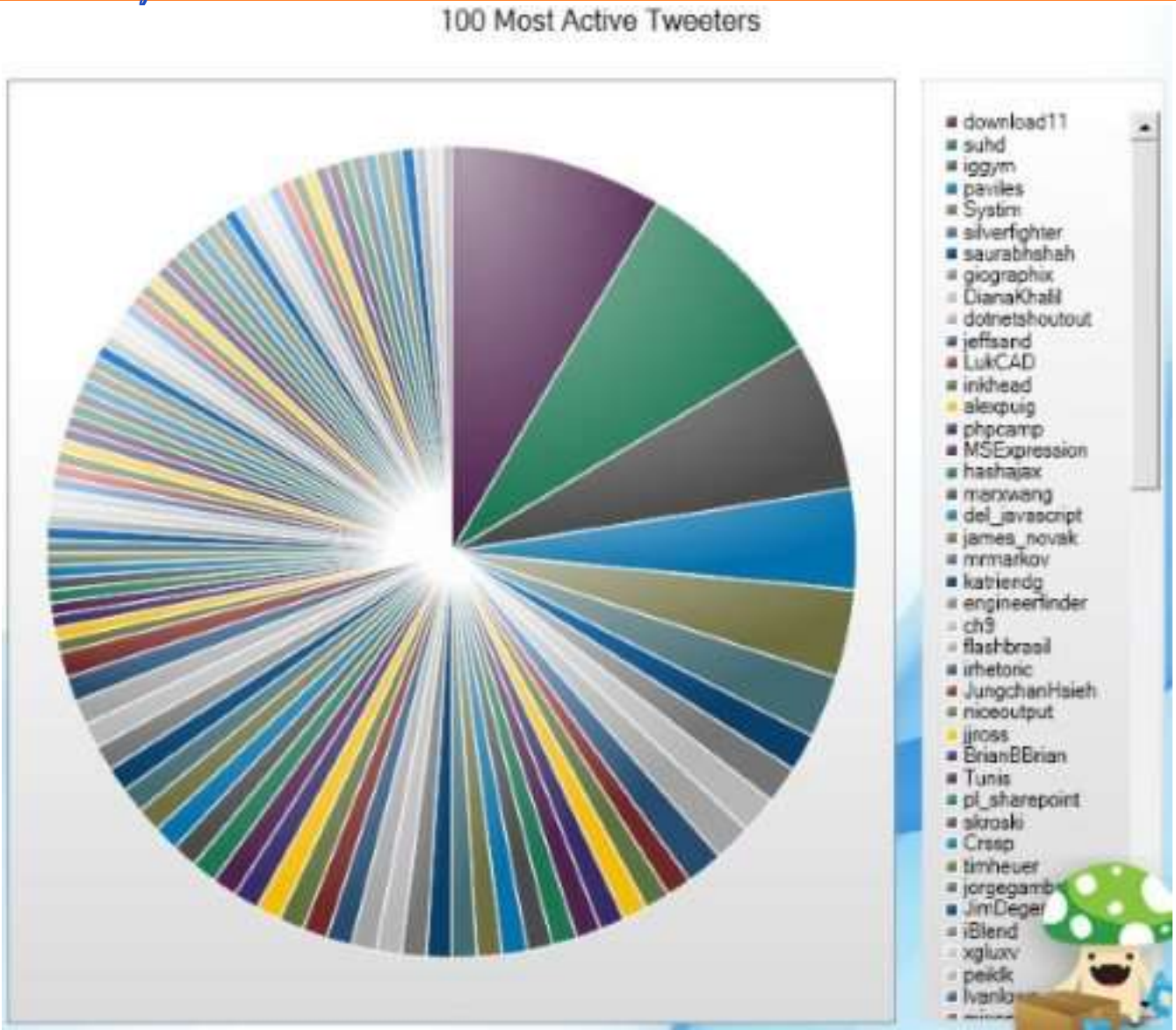
KAKO NAPRAVITI LOŠU VIZUELIZACIJU …



- Postoji nesrećna tendencija da se izabere tehnika i da se onda ta tehnika nahrani ogromnim skupom podataka i rezultujuća slika *šta god da je* pokaže korisniku
- Još gore, postoje *manipulativne* vizuelizacije koje su način da se slikom efektno laže

… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

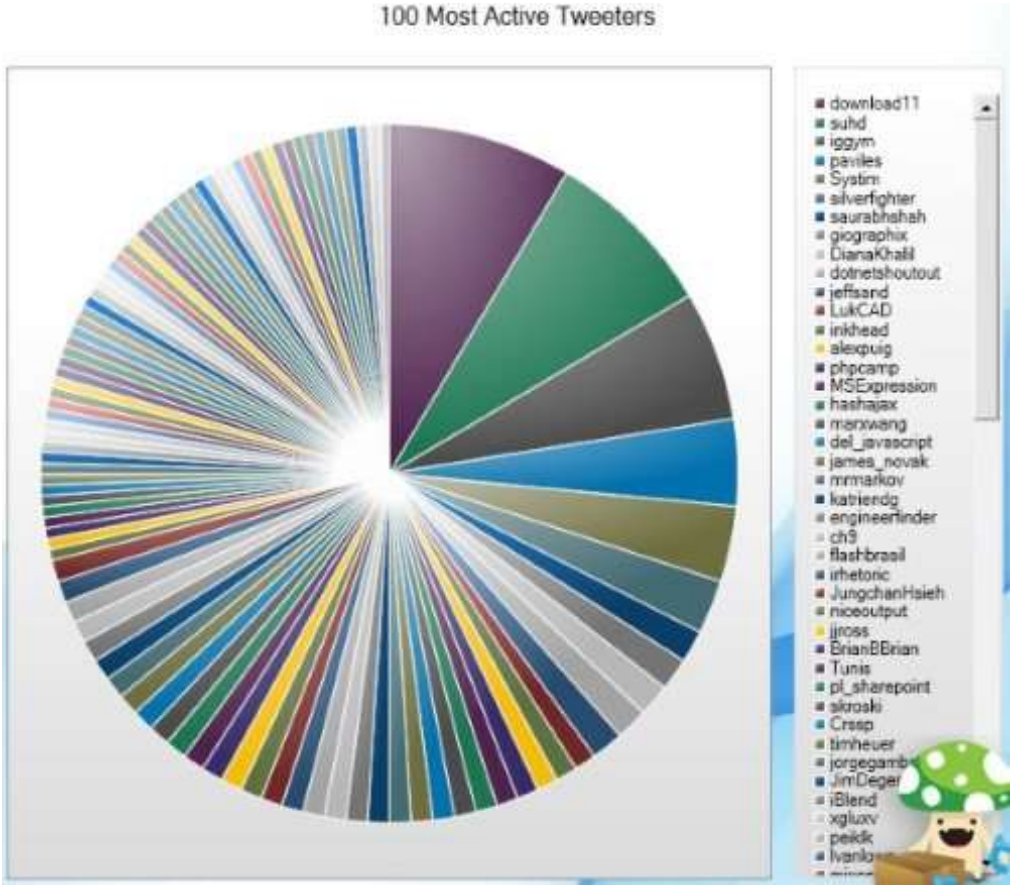




… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

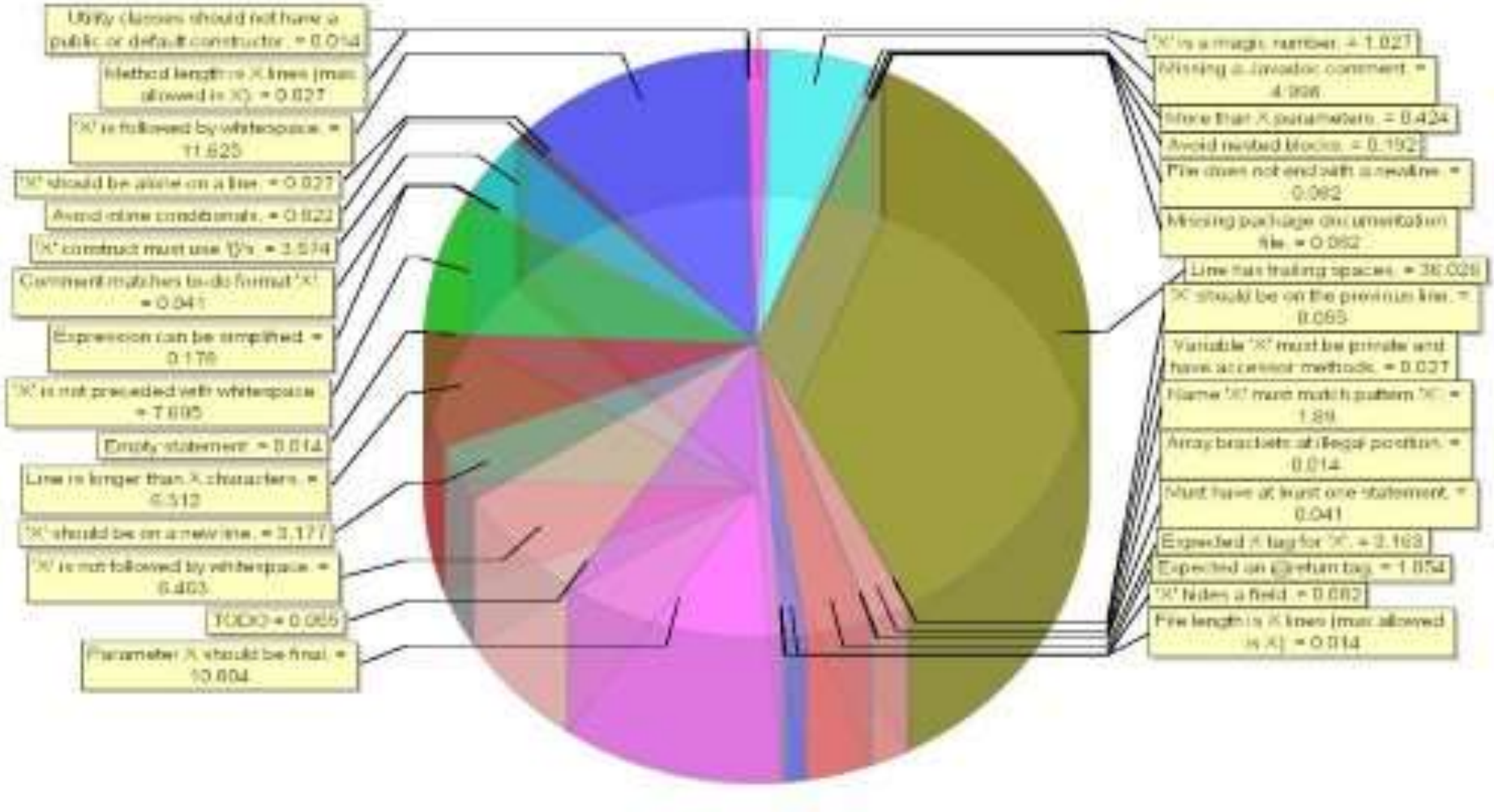


- Pravila HCI ne ostavljaju na miru:
  - Ljudski um jednostavno ne može da istovremeno poredi više od negde između 5 i 11 stvari
  - To što je prikazano korisniku 100 stvari ne znači da korisnik vidi 100 stvari
  - Još crnje, jako je loša ideja koristiti vizuelizacije bazirane na podeli celine (čuvena "pita") ako nije odmah jasno šta je celina
  - Ovde se, recimo, prikazuje uvid u… šta?



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

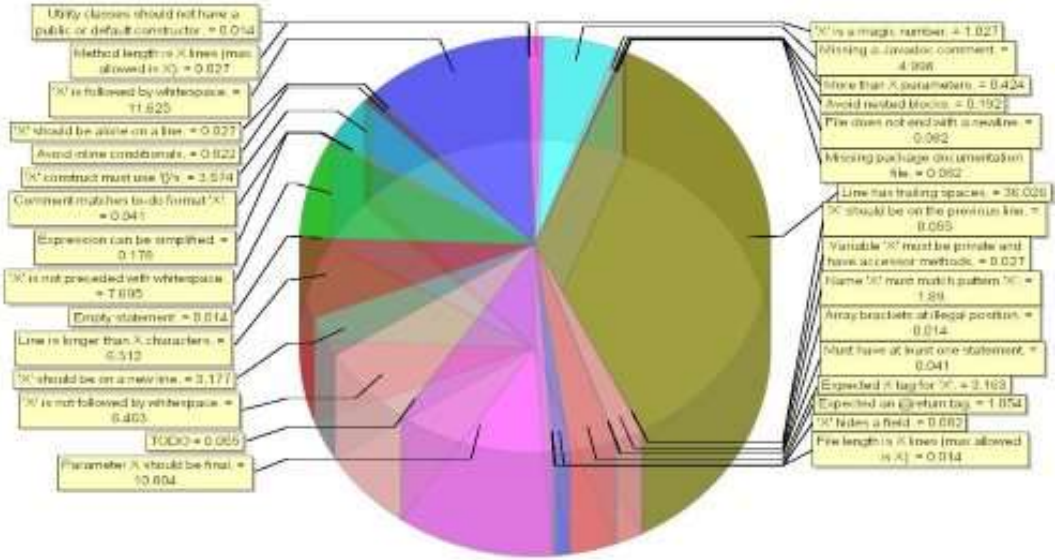




… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

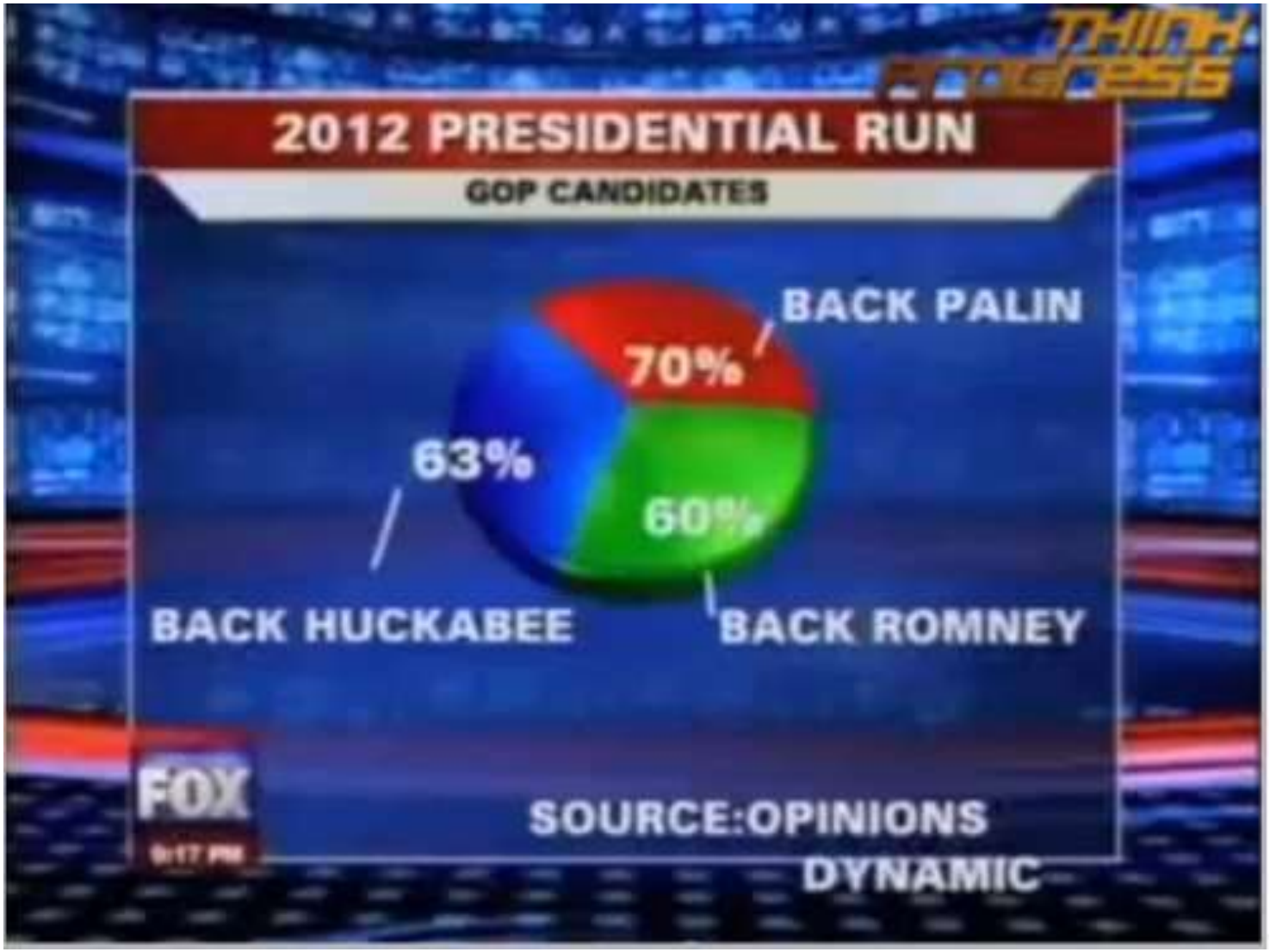


- To što ga Microsoft podržava ne znači i da treba se *koristi*
- 3D pie chart je *spektakularno* loša ideja
- Prvo, zašto koristiti sekcije kruga (što ljudi teže procenjuju), a ne sekcije štangle (što ljudi lakše procenjuju?)
- Mnogo bitnije, zašto onda da zahtevamo od korisnika da ceni veličine objekata u tri dimenzije
  u perspektivi?



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

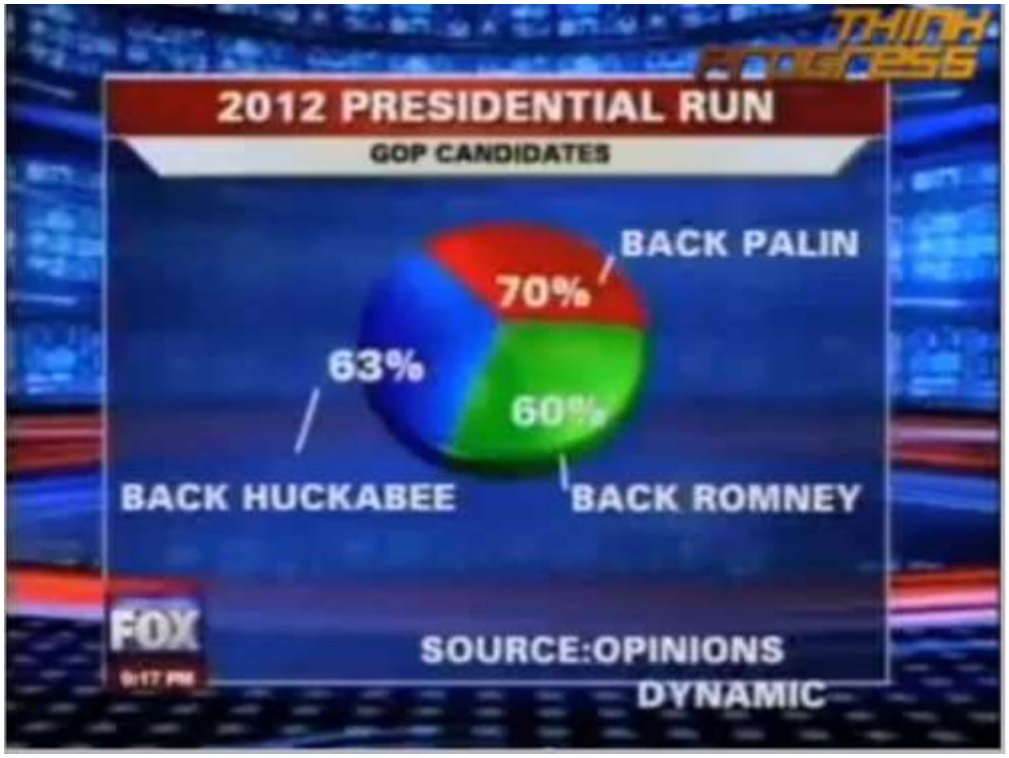
- Stil američke politike



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …



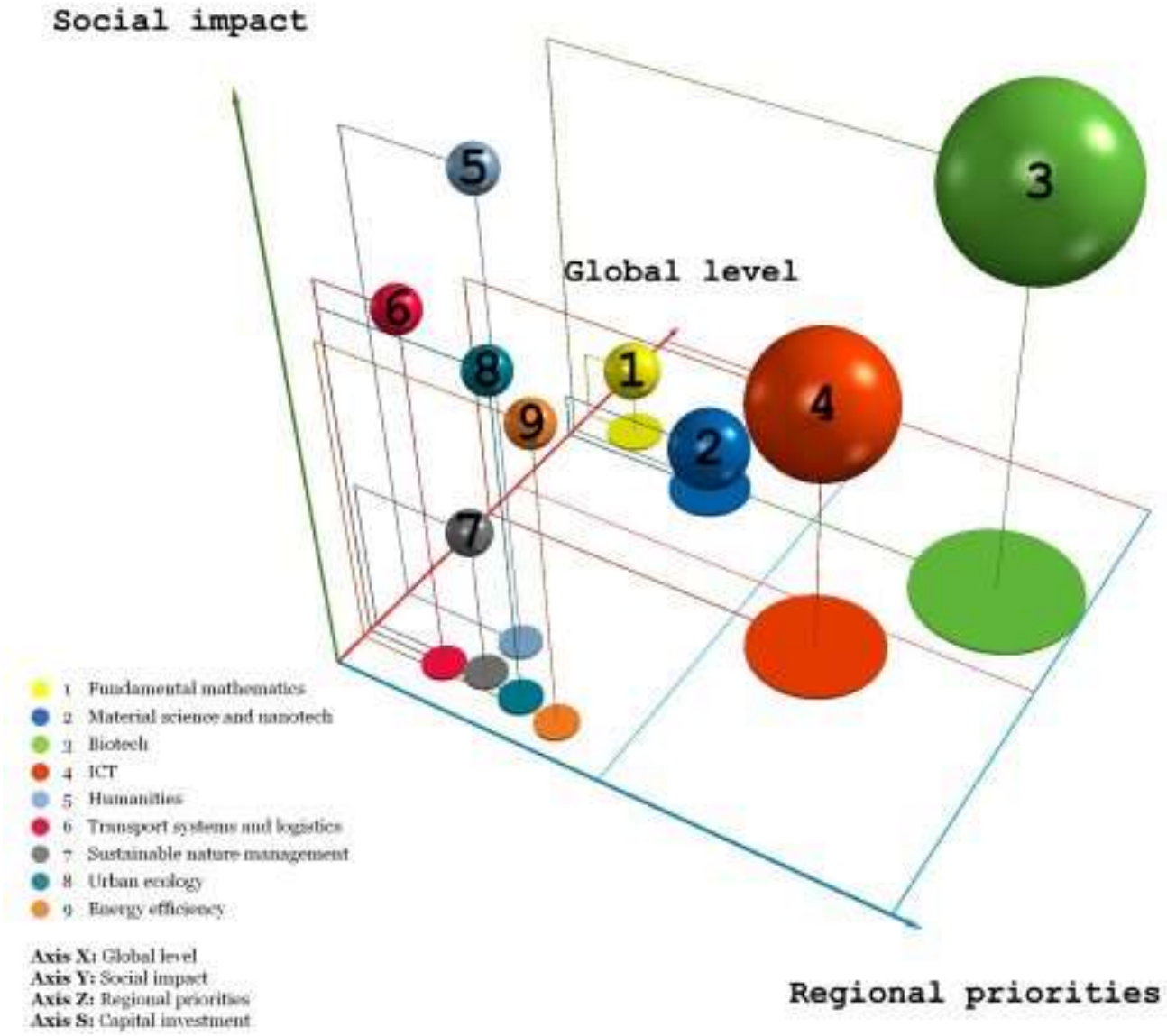
- Stil američke politike
- Kako rade procenti???
- Postoje dve mogućnosti da objasne prethodni… *kreativni* crtež:



– U pitanju je bila anketa sa više mogućih odgovora koja nije adekvatno obrađena pred vizuelizaciju
– Bavljenje američkom političkom scenom je korozivno za ljudski um

… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

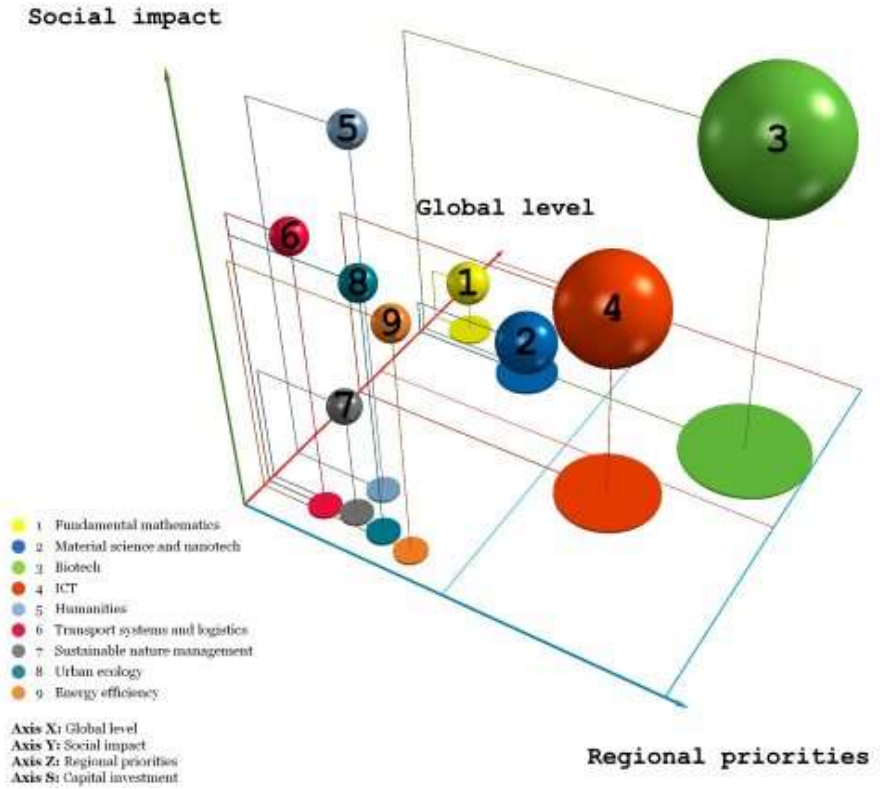




… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

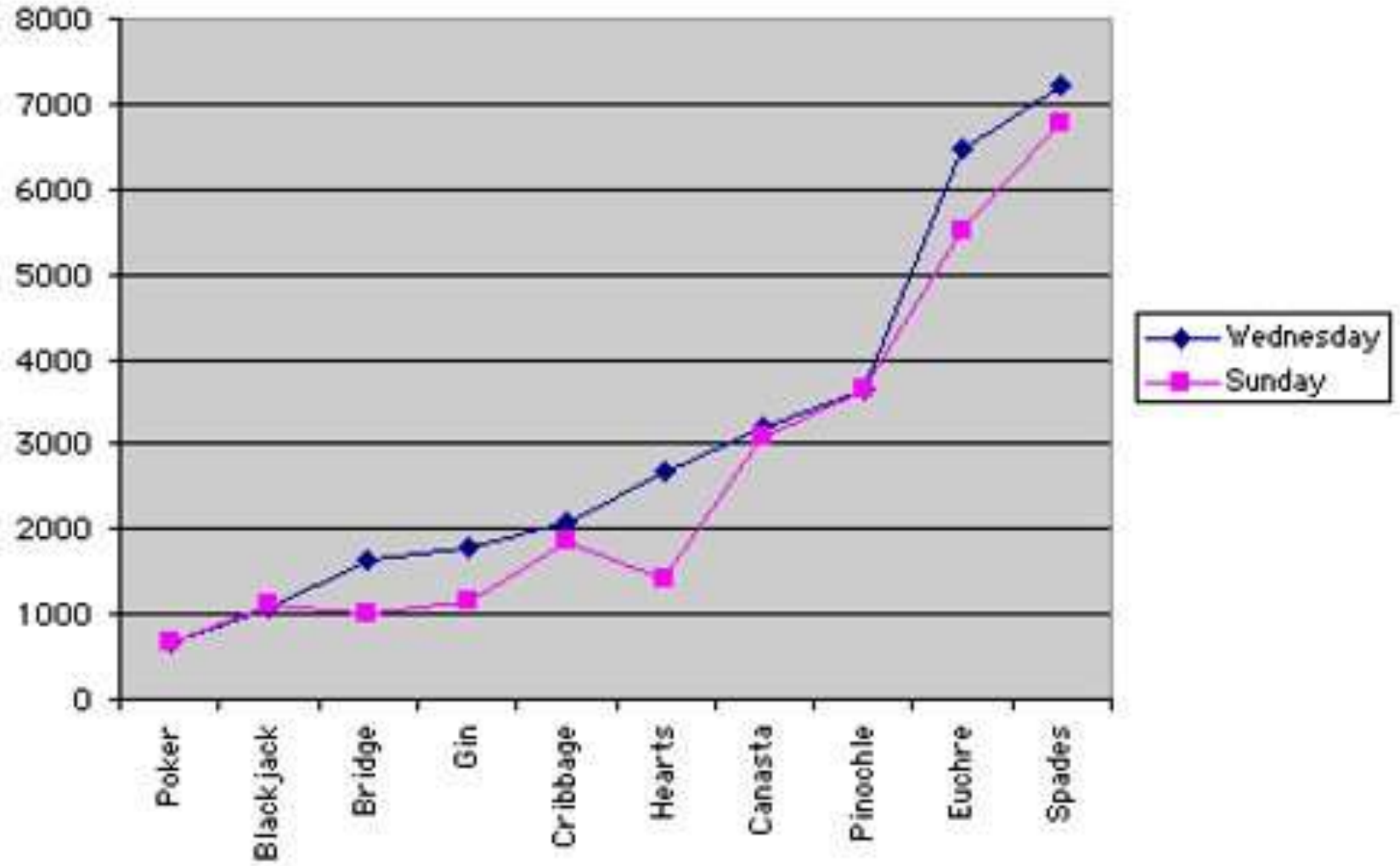


- Ima 7 vizuelnih promenljivih, ali to ne znači da treba i sve da se koriste.
- Prvo, ovo stavlja previše stvari na isti dijagram—iako je moguće uraditi vrlo zanimljive stvari sa multidimenzionalnom vizuelizacijom, ona mora biti *pažljiva.*
- Ovo nije pažljivo.
- Naročito nezgodno jeste što se koriste veličine *sfera* da predstave neku dimenziju. Što je to problem?
- Da li poredimo *prečnike* sfera, *površine* sfera, površine 2D projekcija sfera, ili zapremine sfera?



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

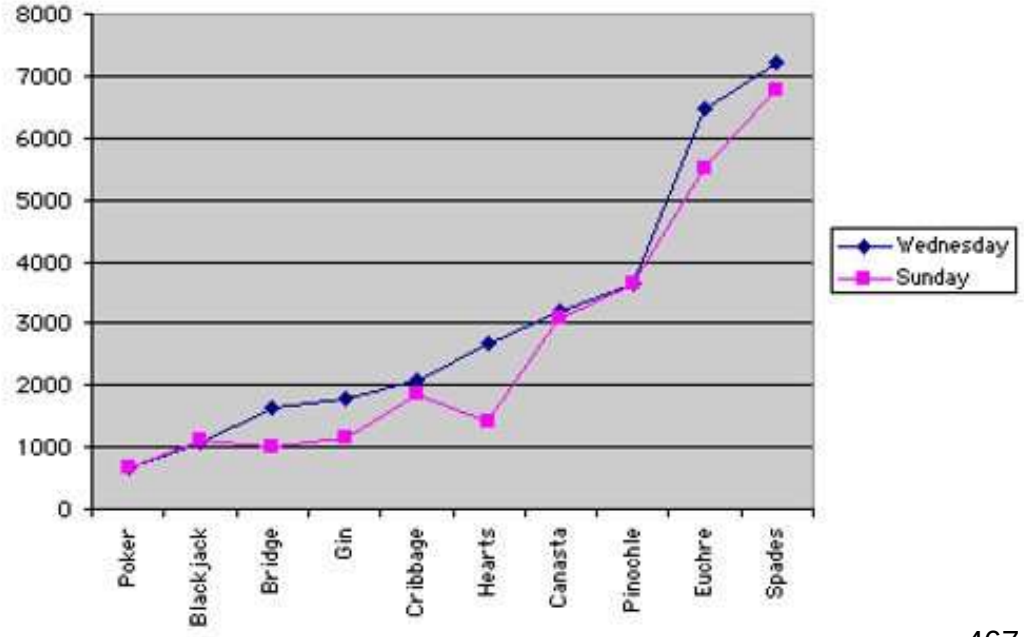




… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

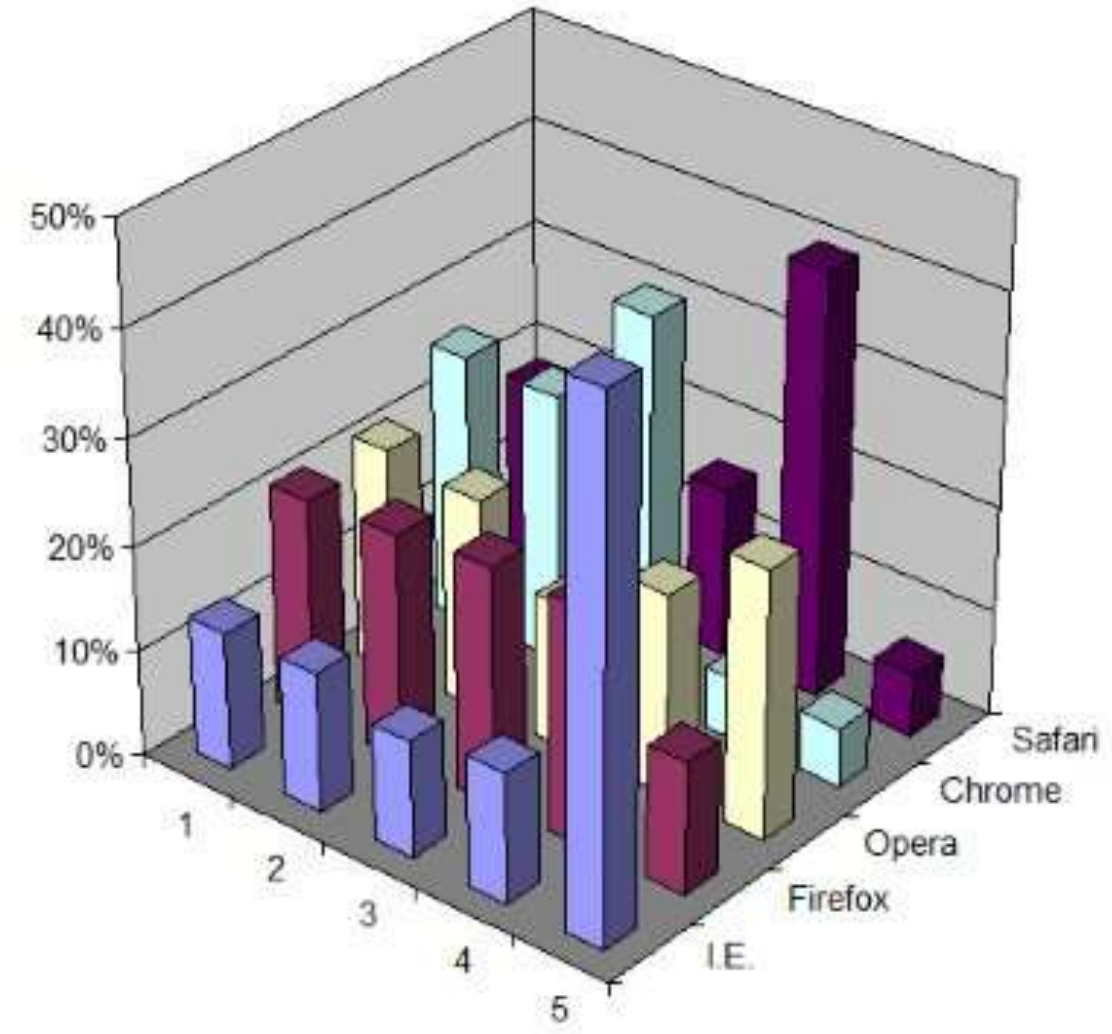


- Osovna nekompetentnost
- Ovde je slučaj vrlo česte greške, tzv. "Excel sindrom."
- Da, ja mogu da selektujem manje-više šta god hoću u Excel-u i pritisnem dugme koje generiše bilo koji grafikon
- To ne znači da *treba* da to radim
- U ovom slučaju izabran je način prikaza koji zahteva da obe ose imaju smisleno uređenje ne bi li se prikazala linearna-u-delovima veza između dve nezavisno varirajuće promenljive



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

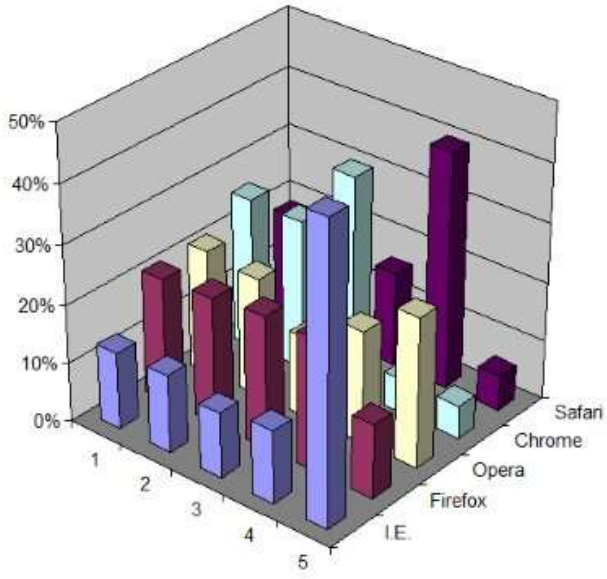




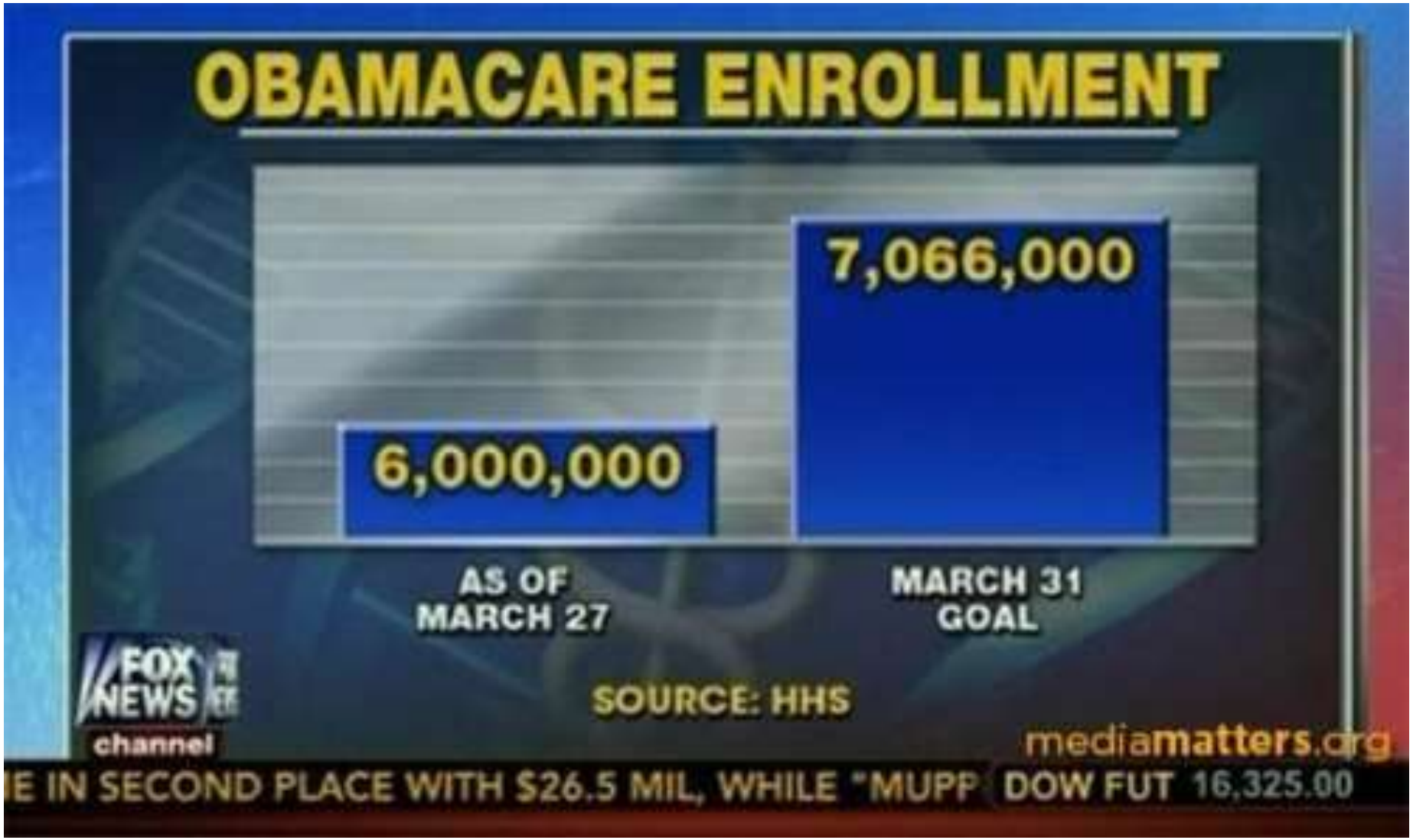
… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …



- Jako je privlačno raditi vizuelizaciju u 3D-u
- Izgleda bolje, impresivnije
- Daje nam celu jednu ekstra dimenziju da se sa njom igramo
- Problem? Mi ne koristimo volumetrijske prikaze
- Korisnik neće videti tri dimenzije, videće 2 koje su rezultat projekcije
- Drugim rečima, uveli ste kompresiju na mala vrata, i to na način koji ne možete da koristite.
- Kada koristiti 3D?
  - Napredan prikaz (VR?)
  - Kada prikazujemo nešto što je stvarno u 3D
  - Kada, minimalno, omogućavamo kvalitetnu interaktivnost.



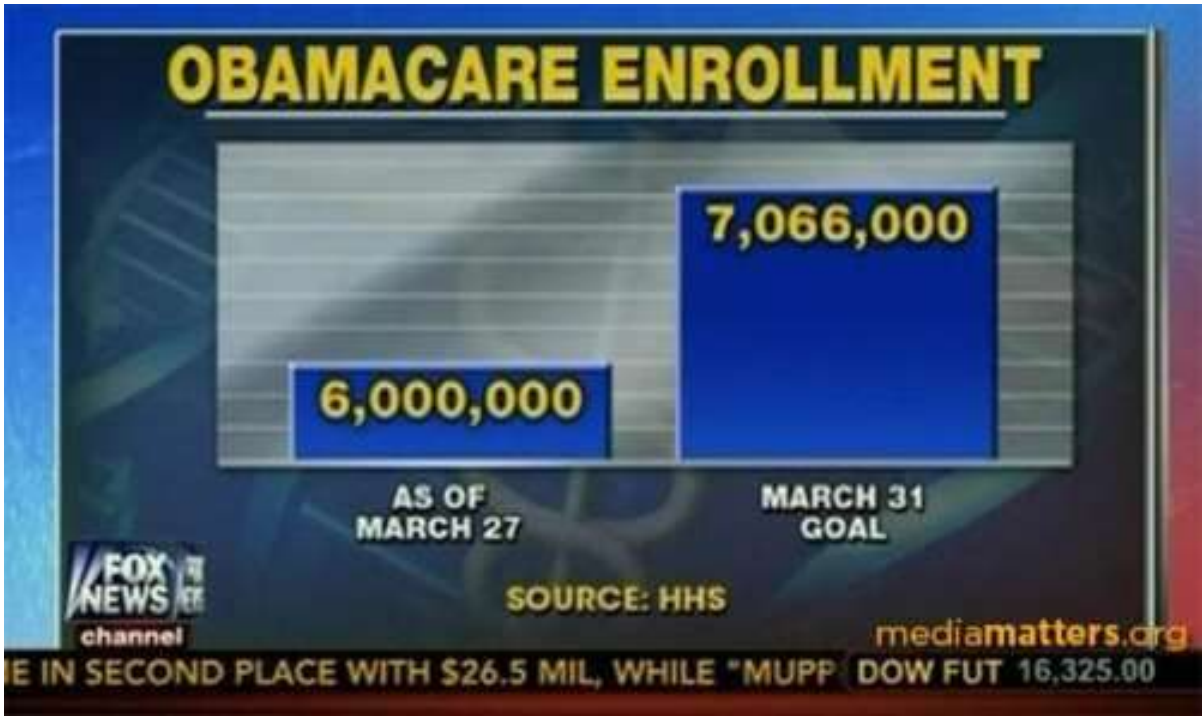
… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

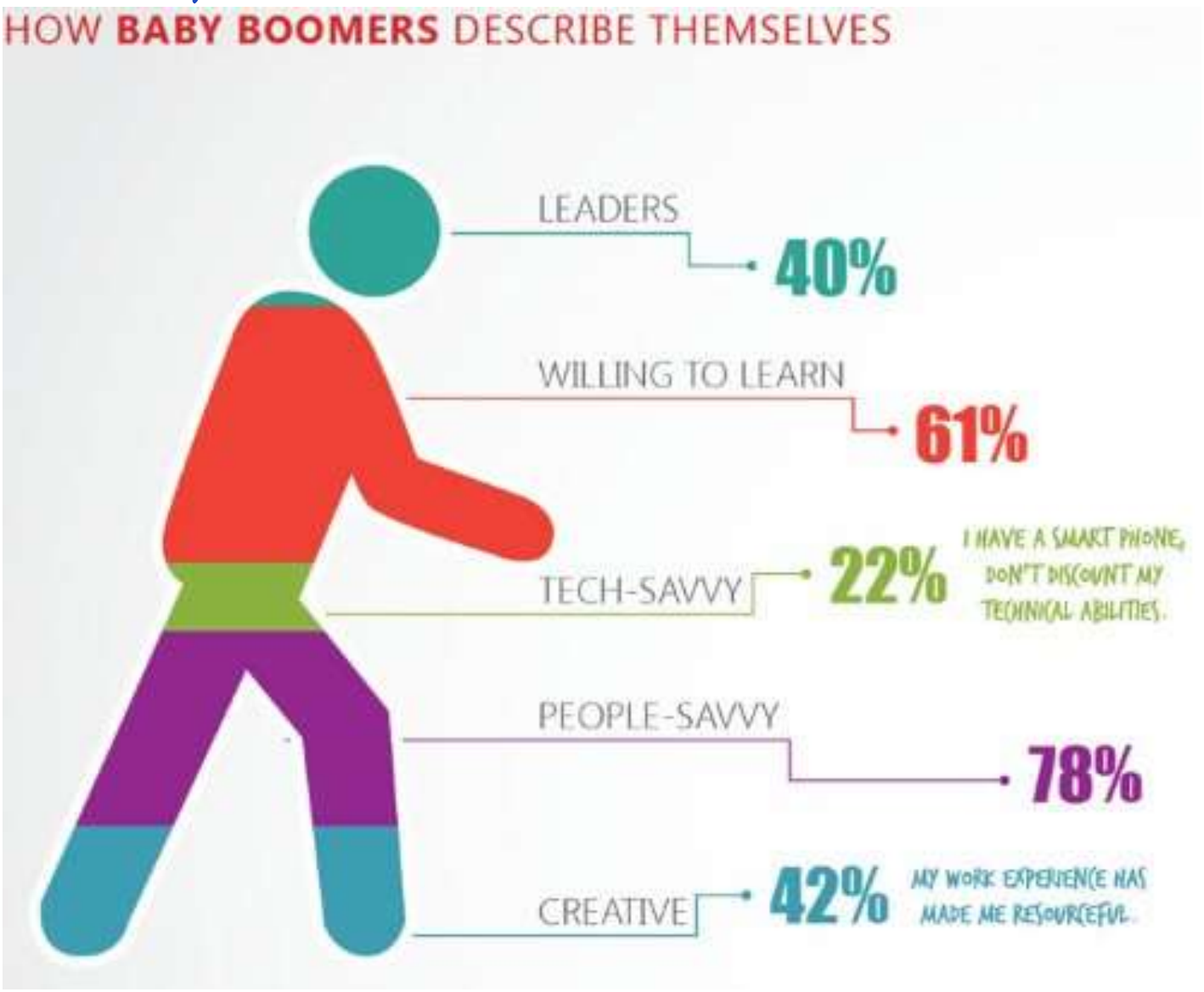


- Laži, proklete laži, i vizuelizacije.
- Pravilo koje može da vam pomogne u životu: Ako vam neko prikaže vizuelizaciju gde ose nisu jasno obeležene *laže vas*
- Čak i da su obeležili da je 0 y-ose u stvari… rekao bih negde oko 5.5 miliona ili tako nešto, to je i dalje pokušaj prevare da učini razliku proporcijalno većom



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

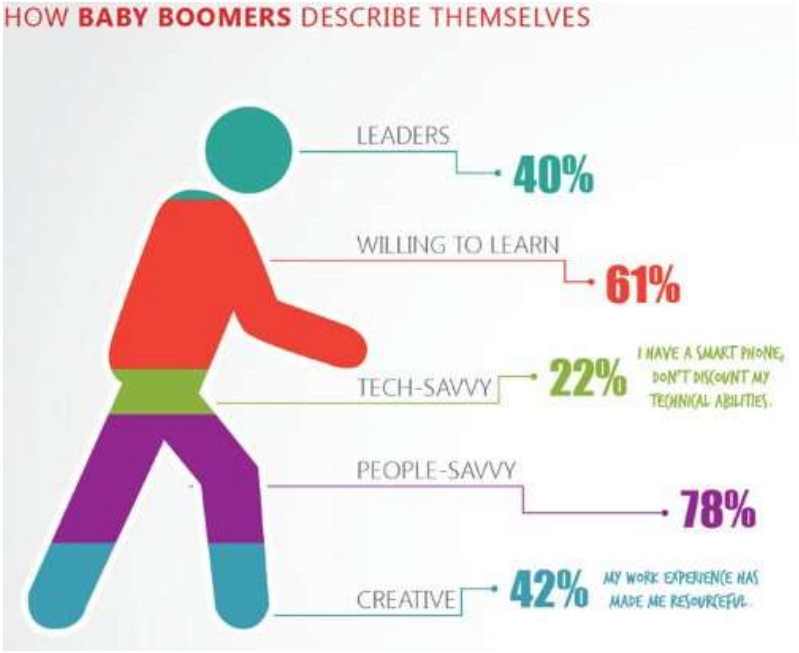




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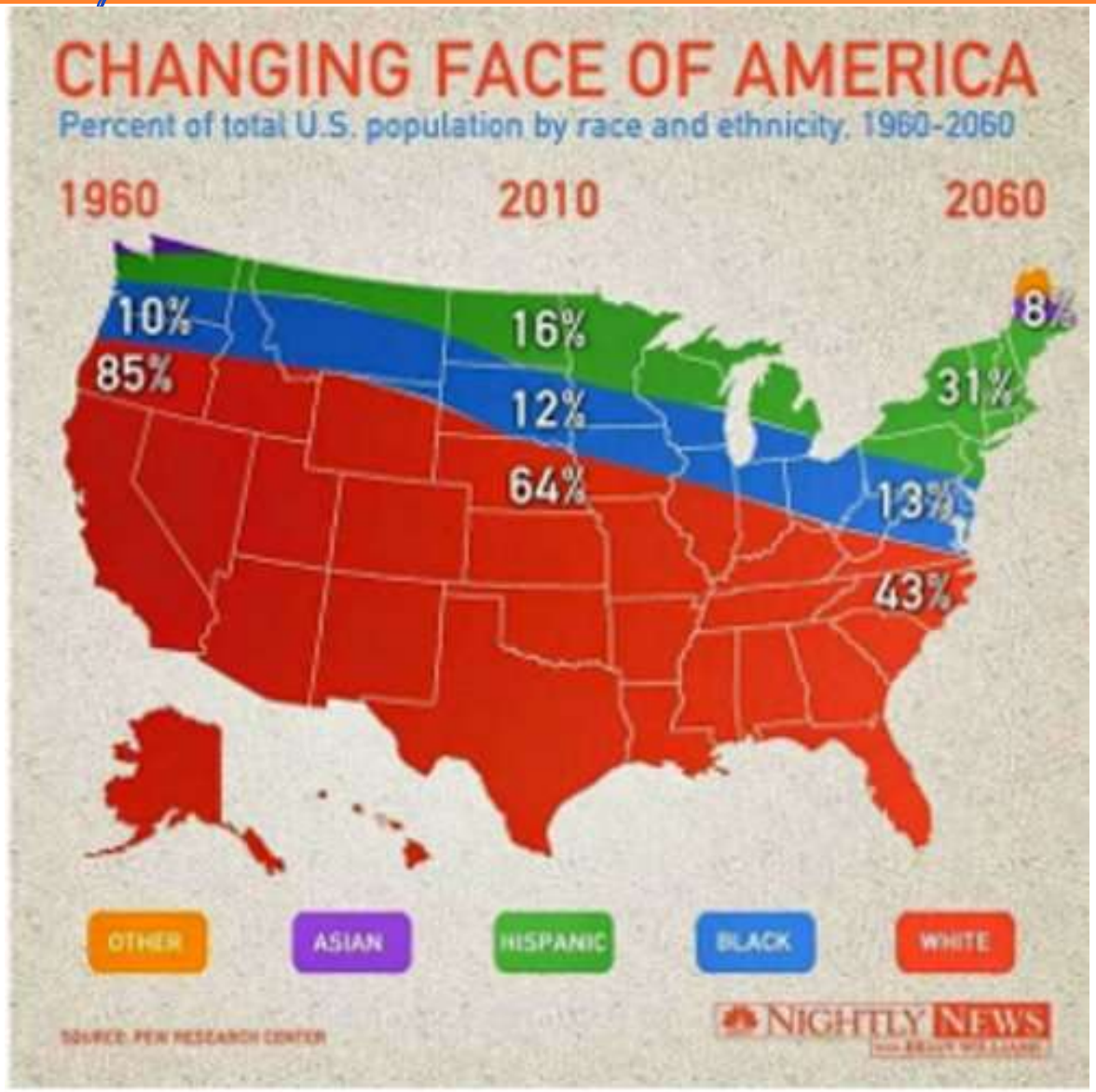


- Ovo se dešava kada date dizajneru da se bavi vizuelizacijom
- Procenti su procenti… nečega?
  Ne znamo čega.
- Prikazani su delovima oblika potpuno ne-uniformnog oblika. Da skinemo procente, ko bi mogao da proceni vrednosti?



- Oblik (dominanta vizuelni element!) ne nosi *apsolutno nikakve podatke.* Što je to figura čoveka? Što da ne!
- Koji uvid možete da dobijete iz ovoga?

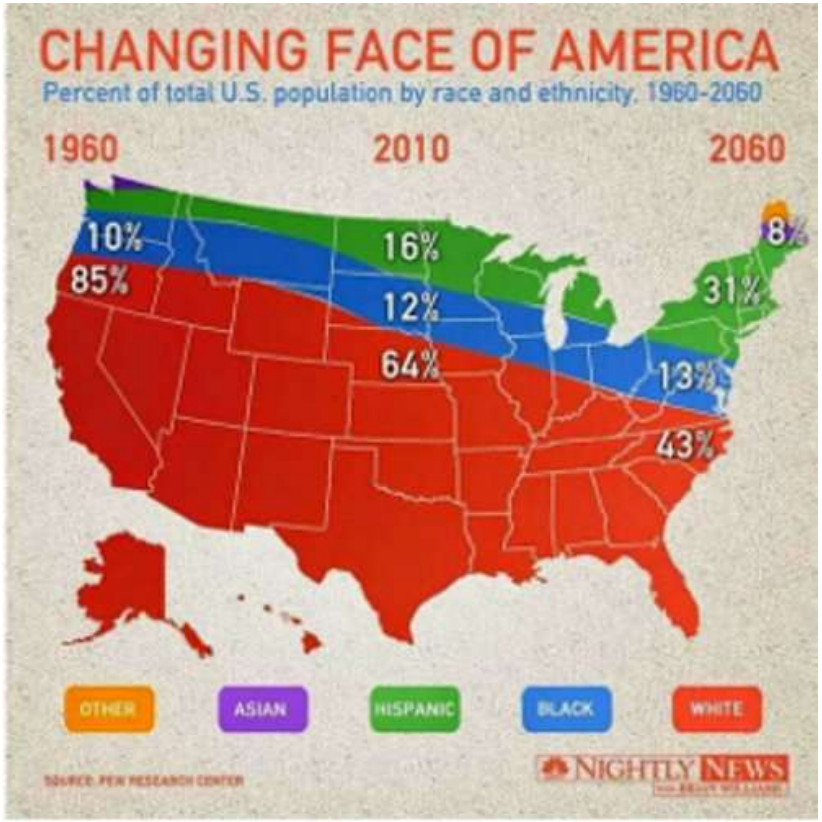
… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

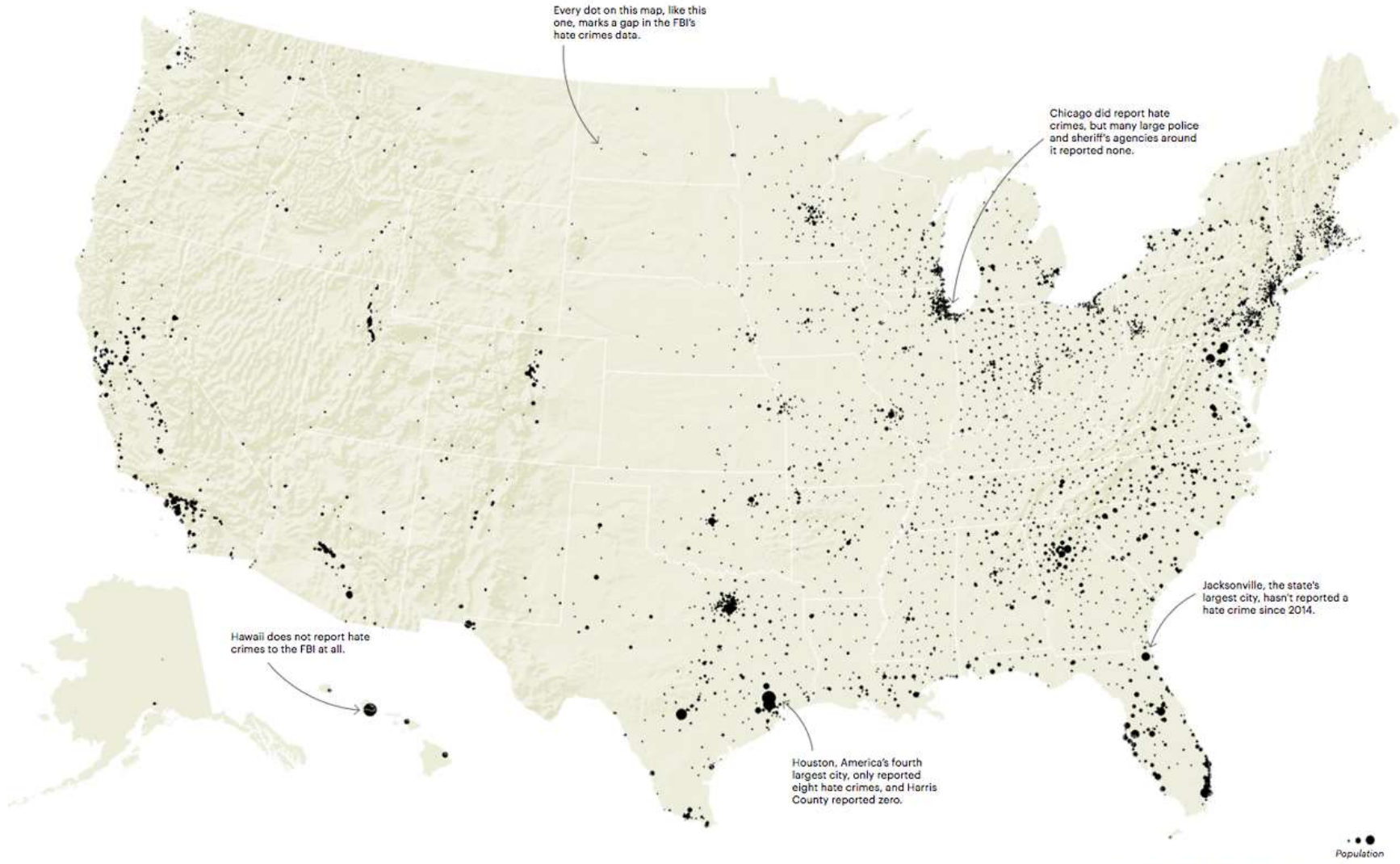


- Šta mapa znači?
- NIŠTA!
- Imamo grafikon koji je svakome jasan koji je ovde predstavlja potpuno drugačiju stvar.
- Ovo je ekvivalent toga da, kada neko nešto broji, da konstantno uzvikujete nasumične brojeve u nadi da ćete da ih zbunite.



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

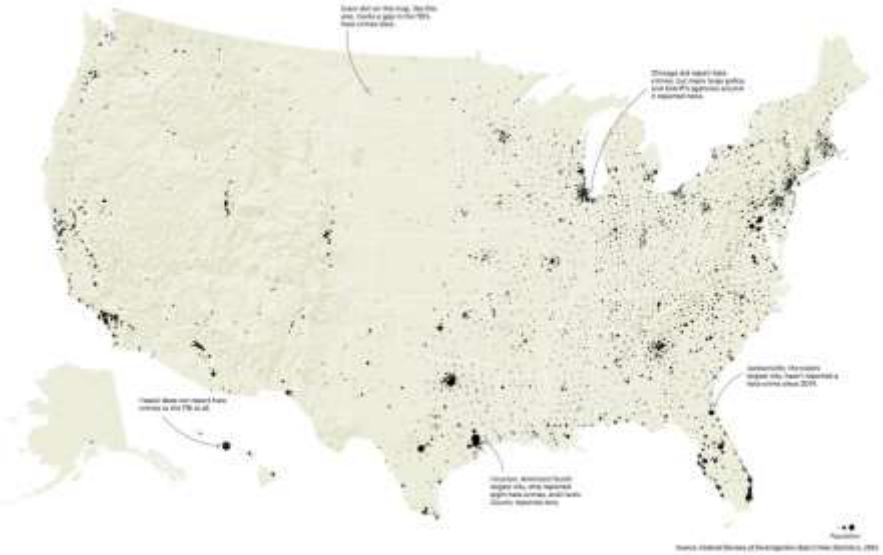




… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …



- Ok, ali *ovde* mapa nešto znači? Da?
- Ovo je neverovatno čest problem: ako napravite vizuelizaciju distribucije nekog socijalnog fenomena geografski ono što 99% vremena proizvedete je *apsolutno isti grafikon,* tj. grafikon *gustine naseljenosti*



- Podaci *moraju* da se prvo transformišu da prikazuju ono što bi nekoga, u stvari, zanimalo, a to je da li ima više šta god da je to što merimo *u odnosu na stanovništvo*

… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …





… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

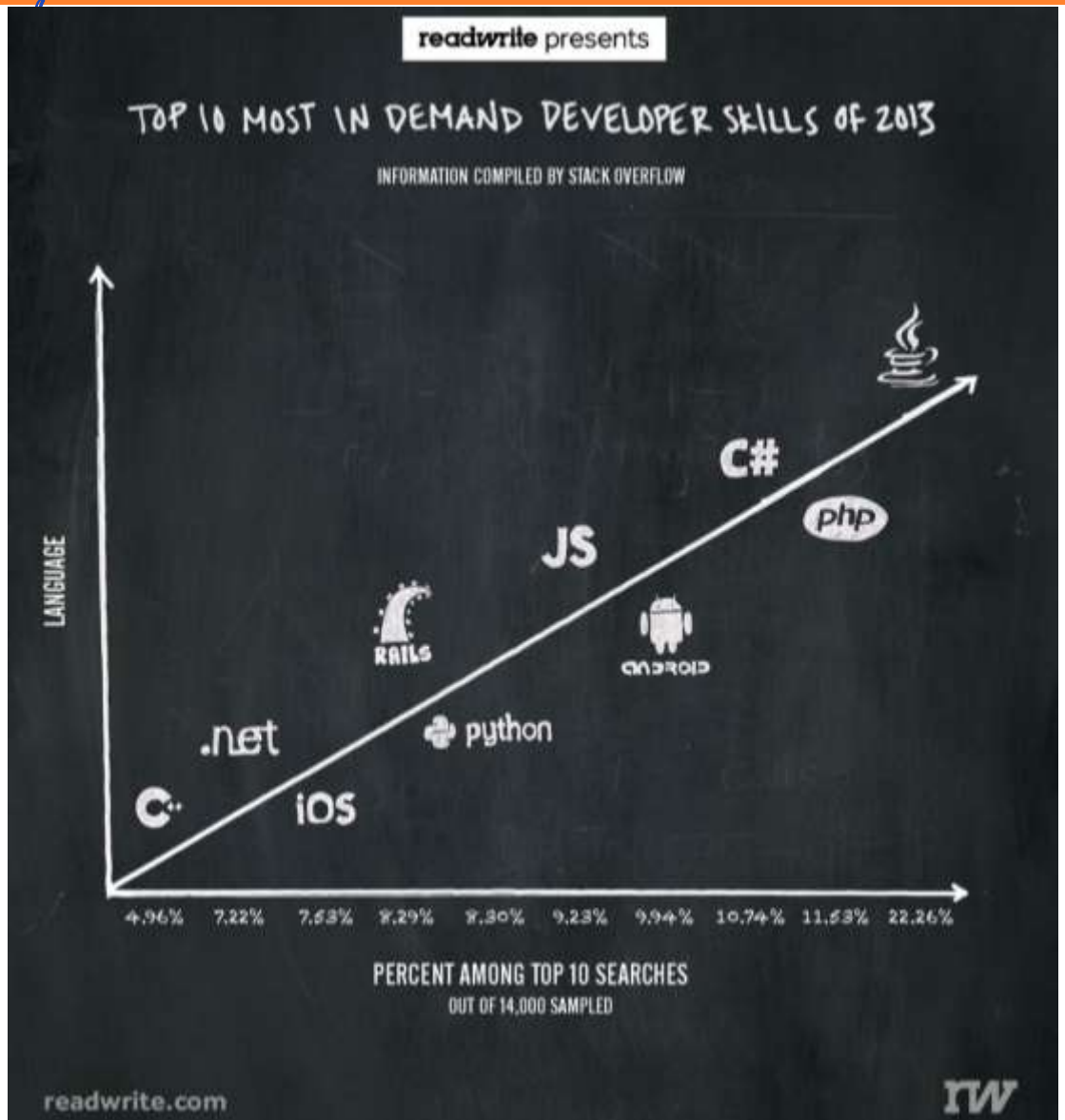


- Biću iskren, ne znam ni *šta* je ovo.
- Ali je objavljeno u NYT-u.



… KAKO NAPRAVITI LOŠU VIZUELIZACIJU …

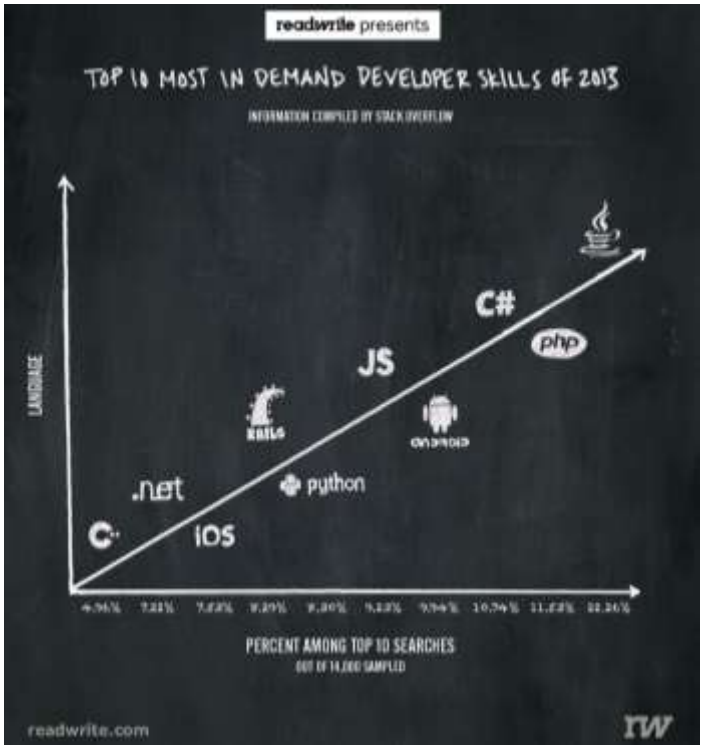




… KAKO NAPRAVITI LOŠU VIZUELIZACIJU



- GRAPHS DO NOT WORK THAT WAY!
- Prvo, ako pogledate metodologiju, naslov ne odgovara sadržaju.
- Drugo sugeriše linearni odnos između dve stvari koje ne mogu da budu u linearnom odnosu a i *nisu.*
- Šta za ime sveta je osa 'jezik?'
- Treće jedina osa sa brojevima na sebi je kalibrisana na opseg 4.96%-22.26%. Vrlo korisno.



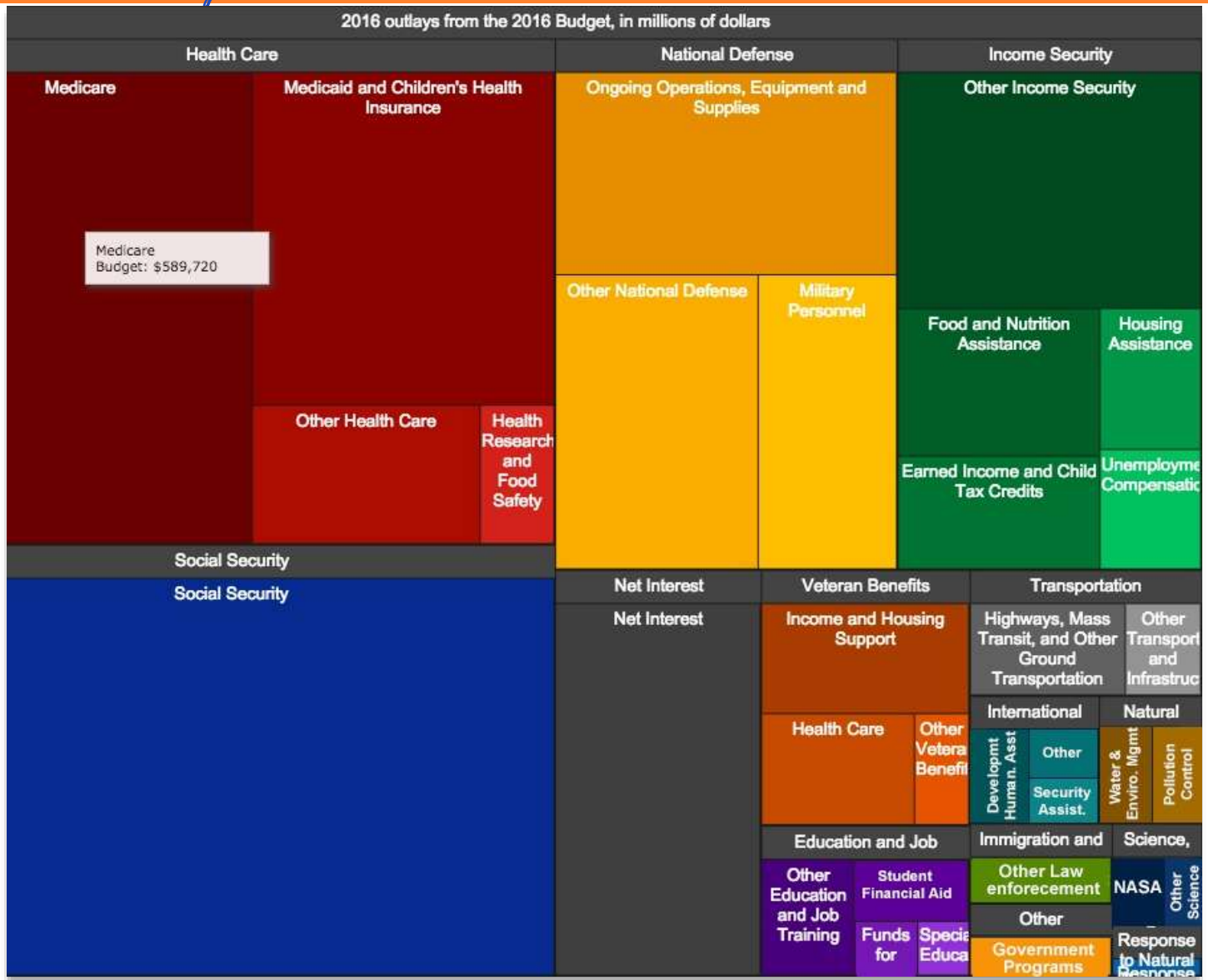
ŠTA ČINI DOBRU VIZUELIZACIJU …



- Uvid.
- Fundamentalno, ono što čini vizuelizacije korisnim jeste da iz njih ekstrahujemo uvid u nekakve podatke koji, inače, ne bi mogli da dobijemo lako.
- Sve ostalo je sekundarno.

… ŠTA ČINI DOBRU VIZUELIZACIJU …

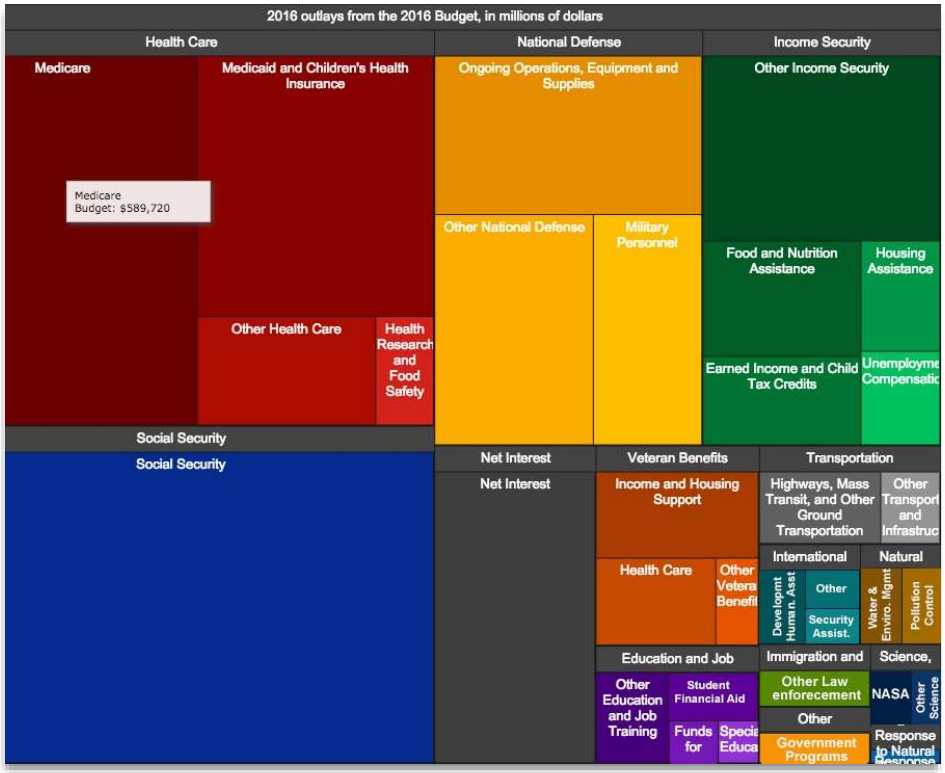




… ŠTA ČINI DOBRU VIZUELIZACIJU …

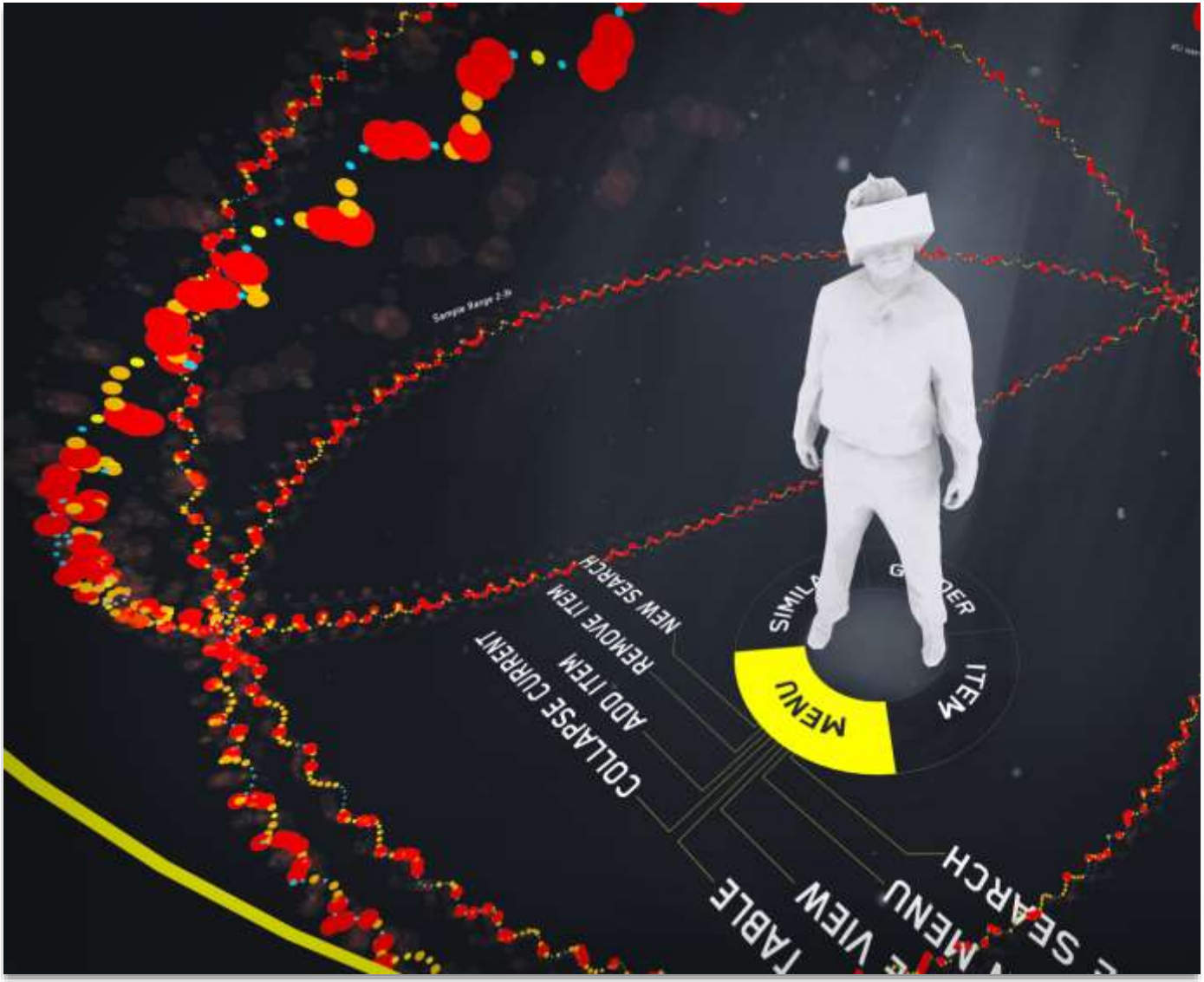


- Zašto ovaj primer?
- Zato što je ružan.
- Uprkos tome što je prilično ružan, uz samo malo zurenja, sada je apsolutno jasno šta se dešava u podacima koje prikazujemo.
- To je važna lekcija: vizuelizacije mogu da budu lepe, čak prelepe, ali to je samo bonus.
- Ono što je *bitno* jeste da nose informacije efektno.



… ŠTA ČINI DOBRU VIZUELIZACIJU





- Ako je prioritet izgled pre svega drugog?